

Instructor's Solution Manual
Introduction to Electrodynamics
Fourth Edition

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Preface

Although I wrote these solutions, much of the typesetting was done by Jonah Gollub, Christopher Lee, and James Terwilliger (any mistakes are, of course, entirely their fault). Chris also did many of the figures, and I would like to thank him particularly for all his help. If you find errors, please let me know (griffith@reed.edu).

David Griffiths

Chapter 1

Vector Analysis

Problem 1.1

- (a) From the diagram, $|\mathbf{B} + \mathbf{C}| \cos \theta_3 = |\mathbf{B}| \cos \theta_1 + |\mathbf{C}| \cos \theta_2$. Multiply by $|\mathbf{A}|$.

$$|\mathbf{A}| |\mathbf{B} + \mathbf{C}| \cos \theta_3 = |\mathbf{A}| |\mathbf{B}| \cos \theta_1 + |\mathbf{A}| |\mathbf{C}| \cos \theta_2.$$

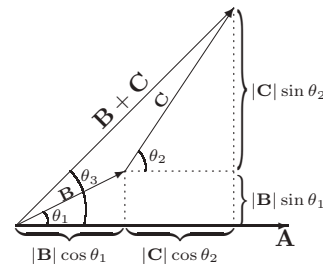
So: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$. (Dot product is distributive)

Similarly: $|\mathbf{B} + \mathbf{C}| \sin \theta_3 = |\mathbf{B}| \sin \theta_1 + |\mathbf{C}| \sin \theta_2$. Multiply by $|\mathbf{A}| \hat{\mathbf{n}}$.

$$|\mathbf{A}| |\mathbf{B} + \mathbf{C}| \sin \theta_3 \hat{\mathbf{n}} = |\mathbf{A}| |\mathbf{B}| \sin \theta_1 \hat{\mathbf{n}} + |\mathbf{A}| |\mathbf{C}| \sin \theta_2 \hat{\mathbf{n}}.$$

If $\hat{\mathbf{n}}$ is the unit vector pointing out of the page, it follows that

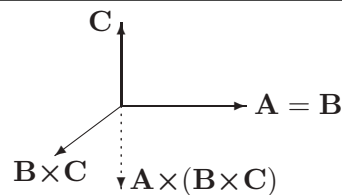
$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C}). \quad (\text{Cross product is distributive})$$



- (b) For the general case, see G. E. Hay's *Vector and Tensor Analysis*, Chapter 1, Section 7 (dot product) and Section 8 (cross product)

Problem 1.2

The triple cross-product is *not* in general associative. For example, suppose $\mathbf{A} = \mathbf{B}$ and $\mathbf{C}$ is perpendicular to $\mathbf{A}$, as in the diagram. Then $(\mathbf{B} \times \mathbf{C})$ points out-of-the-page, and $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ points *down*, and has magnitude ABC . But $(\mathbf{A} \times \mathbf{B}) = \mathbf{0}$, so $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{0} \neq \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

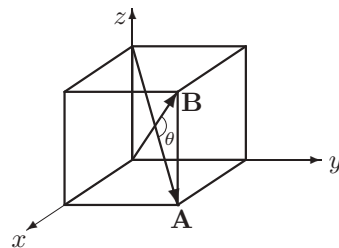


Problem 1.3

$$\mathbf{A} = +1 \hat{\mathbf{x}} + 1 \hat{\mathbf{y}} - 1 \hat{\mathbf{z}}; \quad A = \sqrt{3}; \quad \mathbf{B} = 1 \hat{\mathbf{x}} + 1 \hat{\mathbf{y}} + 1 \hat{\mathbf{z}}; \quad B = \sqrt{3}.$$

$$\mathbf{A} \cdot \mathbf{B} = +1 + 1 - 1 = 1 = AB \cos \theta = \sqrt{3} \sqrt{3} \cos \theta \Rightarrow \cos \theta = \frac{1}{3}.$$

$$\theta = \cos^{-1} \left(\frac{1}{3} \right) \approx 70.5288^\circ$$



Problem 1.4

The cross-product of any two vectors in the plane will give a vector perpendicular to the plane. For example, we might pick the base ($\mathbf{A}$) and the left side ($\mathbf{B}$):

$$\mathbf{A} = -1 \hat{\mathbf{x}} + 2 \hat{\mathbf{y}} + 0 \hat{\mathbf{z}}; \quad \mathbf{B} = -1 \hat{\mathbf{x}} + 0 \hat{\mathbf{y}} + 3 \hat{\mathbf{z}}.$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ -1 & 2 & 0 \\ -1 & 0 & 3 \end{vmatrix} = 6\hat{\mathbf{x}} + 3\hat{\mathbf{y}} + 2\hat{\mathbf{z}}.$$

This has the right *direction*, but the wrong *magnitude*. To make a *unit* vector out of it, simply divide by its length:

$$|\mathbf{A} \times \mathbf{B}| = \sqrt{36 + 9 + 4} = 7. \quad \hat{\mathbf{n}} = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} = \boxed{\frac{6}{7}\hat{\mathbf{x}} + \frac{3}{7}\hat{\mathbf{y}} + \frac{2}{7}\hat{\mathbf{z}}}.$$

Problem 1.5

$$\begin{aligned} \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ (B_y C_z - B_z C_y) & (B_z C_x - B_x C_z) & (B_x C_y - B_y C_x) \end{vmatrix} \\ &= \hat{\mathbf{x}}[A_y(B_x C_y - B_y C_x) - A_z(B_z C_x - B_x C_z)] + \hat{\mathbf{y}}(\quad) + \hat{\mathbf{z}}(\quad) \\ &\quad (\text{I'll just check the x-component; the others go the same way}) \\ &= \hat{\mathbf{x}}(A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z) + \hat{\mathbf{y}}(\quad) + \hat{\mathbf{z}}(\quad). \\ \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) &= [B_x(A_x C_x + A_y C_y + A_z C_z) - C_x(A_x B_x + A_y B_y + A_z B_z)]\hat{\mathbf{x}} + (\quad)\hat{\mathbf{y}} + (\quad)\hat{\mathbf{z}} \\ &= \hat{\mathbf{x}}(A_y B_x C_y + A_z B_x C_z - A_y B_y C_x - A_z B_z C_x) + \hat{\mathbf{y}}(\quad) + \hat{\mathbf{z}}(\quad). \text{ They agree.} \end{aligned}$$

Problem 1.6

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) + \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) + \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) - \mathbf{A}(\mathbf{C} \cdot \mathbf{B}) + \mathbf{A}(\mathbf{B} \cdot \mathbf{C}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A}) = \mathbf{0}.$$

So: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) - (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = -\mathbf{B} \times (\mathbf{C} \times \mathbf{A}) = \mathbf{A}(\mathbf{B} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}).$

If this is zero, then either $\mathbf{A}$ is parallel to $\mathbf{C}$ (including the case in which they point in *opposite* directions, or one is zero), or else $\mathbf{B} \cdot \mathbf{C} = \mathbf{B} \cdot \mathbf{A} = 0$, in which case $\mathbf{B}$ is perpendicular to $\mathbf{A}$ and $\mathbf{C}$ (including the case $\mathbf{B} = \mathbf{0}$.)

Conclusion: $\boxed{\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \iff \text{either } \mathbf{A} \text{ is parallel to } \mathbf{C}, \text{ or } \mathbf{B} \text{ is perpendicular to } \mathbf{A} \text{ and } \mathbf{C}.}$

Problem 1.7

$$\boldsymbol{\mathcal{Z}} = (4\hat{\mathbf{x}} + 6\hat{\mathbf{y}} + 8\hat{\mathbf{z}}) - (2\hat{\mathbf{x}} + 8\hat{\mathbf{y}} + 7\hat{\mathbf{z}}) = \boxed{2\hat{\mathbf{x}} - 2\hat{\mathbf{y}} + \hat{\mathbf{z}}}$$

$$\mathcal{Z} = \sqrt{4 + 4 + 1} = \boxed{3}$$

$$\hat{\boldsymbol{\mathcal{Z}}} = \frac{\boldsymbol{\mathcal{Z}}}{\mathcal{Z}} = \boxed{\frac{2}{3}\hat{\mathbf{x}} - \frac{2}{3}\hat{\mathbf{y}} + \frac{1}{3}\hat{\mathbf{z}}}$$

Problem 1.8

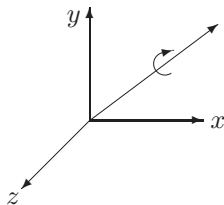
$$\begin{aligned} \text{(a)} \quad \bar{A}_y \bar{B}_y + \bar{A}_z \bar{B}_z &= (\cos \phi A_y + \sin \phi A_z)(\cos \phi B_y + \sin \phi B_z) + (-\sin \phi A_y + \cos \phi A_z)(-\sin \phi B_y + \cos \phi B_z) \\ &= \cos^2 \phi A_y B_y + \sin \phi \cos \phi (A_y B_z + A_z B_y) + \sin^2 \phi A_z B_z + \sin^2 \phi A_y B_y - \sin \phi \cos \phi (A_y B_z + A_z B_y) + \cos^2 \phi A_z B_z \\ &= (\cos^2 \phi + \sin^2 \phi) A_y B_y + (\sin^2 \phi + \cos^2 \phi) A_z B_z = A_y B_y + A_z B_z. \quad \checkmark \end{aligned}$$

$$\text{(b)} \quad (\bar{A}_x)^2 + (\bar{A}_y)^2 + (\bar{A}_z)^2 = \Sigma_{i=1}^3 \bar{A}_i \bar{A}_i = \Sigma_{i=1}^3 (\Sigma_{j=1}^3 R_{ij} A_j) (\Sigma_{k=1}^3 R_{ik} A_k) = \Sigma_{j,k} (\Sigma_i R_{ij} R_{ik}) A_j A_k.$$

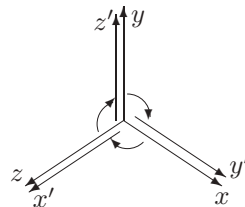
$$\text{This equals } A_x^2 + A_y^2 + A_z^2 \text{ provided } \boxed{\Sigma_{i=1}^3 R_{ij} R_{ik} = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{if } j \neq k \end{cases}}$$

Moreover, if R is to preserve lengths for *all* vectors $\mathbf{A}$, then this condition is not only *sufficient* but also *necessary*. For suppose $\mathbf{A} = (1, 0, 0)$. Then $\Sigma_{j,k} (\Sigma_i R_{ij} R_{ik}) A_j A_k = \Sigma_i R_{i1} R_{i1}$, and this must equal 1 (since we want $\bar{A}_x^2 + \bar{A}_y^2 + \bar{A}_z^2 = 1$). Likewise, $\Sigma_{i=1}^3 R_{i2} R_{i2} = \Sigma_{i=1}^3 R_{i3} R_{i3} = 1$. To check the case $j \neq k$, choose $\mathbf{A} = (1, 1, 0)$. Then we want $2 = \Sigma_{j,k} (\Sigma_i R_{ij} R_{ik}) A_j A_k = \Sigma_i R_{i1} R_{i1} + \Sigma_i R_{i2} R_{i2} + \Sigma_i R_{i1} R_{i2} + \Sigma_i R_{i2} R_{i1}$. But we already know that the first two sums are both 1; the third and fourth are *equal*, so $\Sigma_i R_{i1} R_{i2} = \Sigma_i R_{i2} R_{i1} = 0$, and so on for other unequal combinations of j, k . $\checkmark$ In matrix notation: $\bar{R}R = 1$, where $\bar{R}$ is the *transpose* of R .

Problem 1.9



Looking down the axis:



A 120° rotation carries the z axis into the y ($= \bar{z}$) axis, y into x ($= \bar{y}$), and x into z ($= \bar{x}$). So $\bar{A}_x = A_z$, $\bar{A}_y = A_x$, $\bar{A}_z = A_y$.

$$R = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Problem 1.10

(a) No change. ($\bar{A}_x = A_x$, $\bar{A}_y = A_y$, $\bar{A}_z = A_z$)

(b) $\mathbf{A} \longrightarrow -\mathbf{A}$, in the sense ($\bar{A}_x = -A_x$, $\bar{A}_y = -A_y$, $\bar{A}_z = -A_z$)

(c) $(\mathbf{A} \times \mathbf{B}) \longrightarrow (-\mathbf{A}) \times (-\mathbf{B}) = (\mathbf{A} \times \mathbf{B})$. That is, if $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, $\mathbf{C} \longrightarrow \mathbf{C}$. No minus sign, in contrast to behavior of an “ordinary” vector, as given by (b). If $\mathbf{A}$ and $\mathbf{B}$ are *pseudovectors*, then $(\mathbf{A} \times \mathbf{B}) \longrightarrow (\mathbf{A}) \times (\mathbf{B}) = (\mathbf{A} \times \mathbf{B})$. So the cross-product of two pseudovectors is again a *pseudovector*. In the cross-product of a vector and a pseudovector, one changes sign, the other doesn't, and therefore the cross-product is itself a *vector*. *Angular momentum* ($\mathbf{L} = \mathbf{r} \times \mathbf{p}$) and *torque* ($\mathbf{N} = \mathbf{r} \times \mathbf{F}$) are pseudovectors.

(d) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) \longrightarrow (-\mathbf{A}) \cdot ((-\mathbf{B}) \times (-\mathbf{C})) = -\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$. So, if $a = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$, then $a \longrightarrow -a$; a pseudoscalar *changes sign* under inversion of coordinates.

Problem 1.11

$$(a) \nabla f = 2x \hat{\mathbf{x}} + 3y^2 \hat{\mathbf{y}} + 4z^3 \hat{\mathbf{z}}$$

$$(b) \nabla f = 2xy^3z^4 \hat{\mathbf{x}} + 3x^2y^2z^4 \hat{\mathbf{y}} + 4x^2y^3z^3 \hat{\mathbf{z}}$$

$$(c) \nabla f = e^x \sin y \ln z \hat{\mathbf{x}} + e^x \cos y \ln z \hat{\mathbf{y}} + e^x \sin y (1/z) \hat{\mathbf{z}}$$

Problem 1.12

(a) $\nabla h = 10[(2y - 6x - 18) \hat{\mathbf{x}} + (2x - 8y + 28) \hat{\mathbf{y}}]$. $\nabla h = 0$ at summit, so

$$\left. \begin{aligned} 2y - 6x - 18 &= 0 \\ 2x - 8y + 28 &= 0 \implies 6x - 24y + 84 = 0 \end{aligned} \right\} 2y - 18 - 24y + 84 = 0.$$

$$22y = 66 \implies y = 3 \implies 2x - 24 + 28 = 0 \implies x = -2.$$

Top is 3 miles north, 2 miles west, of South Hadley.

(b) Putting in $x = -2$, $y = 3$:

$$h = 10(-12 - 12 - 36 + 36 + 84 + 12) = \span style="border: 1px solid black; padding: 2px;">720 \text{ ft.}$$

(c) Putting in $x = 1$, $y = 1$: $\nabla h = 10[(2 - 6 - 18) \hat{\mathbf{x}} + (2 - 8 + 28) \hat{\mathbf{y}}] = 10(-22 \hat{\mathbf{x}} + 22 \hat{\mathbf{y}}) = 220(-\hat{\mathbf{x}} + \hat{\mathbf{y}})$.

$$|\nabla h| = 220\sqrt{2} \approx \span style="border: 1px solid black; padding: 2px;">311 \text{ ft/mile;} direction: northwest.$$

Problem 1.13

$$\mathbf{r} = (x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}; \quad r = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}.$$

$$(a) \nabla(r^2) = \frac{\partial}{\partial x}[(x - x')^2 + (y - y')^2 + (z - z')^2]\hat{\mathbf{x}} + \frac{\partial}{\partial y}(\quad)\hat{\mathbf{y}} + \frac{\partial}{\partial z}(\quad)\hat{\mathbf{z}} = 2(x - x')\hat{\mathbf{x}} + 2(y - y')\hat{\mathbf{y}} + 2(z - z')\hat{\mathbf{z}} = 2\mathbf{r}.$$

$$(b) \nabla\left(\frac{1}{r}\right) = \frac{\partial}{\partial x}[(x - x')^2 + (y - y')^2 + (z - z')^2]^{-\frac{1}{2}}\hat{\mathbf{x}} + \frac{\partial}{\partial y}(\quad)^{-\frac{1}{2}}\hat{\mathbf{y}} + \frac{\partial}{\partial z}(\quad)^{-\frac{1}{2}}\hat{\mathbf{z}} \\ = -\frac{1}{2}(\quad)^{-\frac{3}{2}}2(x - x')\hat{\mathbf{x}} - \frac{1}{2}(\quad)^{-\frac{3}{2}}2(y - y')\hat{\mathbf{y}} - \frac{1}{2}(\quad)^{-\frac{3}{2}}2(z - z')\hat{\mathbf{z}} \\ = -(\quad)^{-\frac{3}{2}}[(x - x')\hat{\mathbf{x}} + (y - y')\hat{\mathbf{y}} + (z - z')\hat{\mathbf{z}}] = -(1/r^3)\mathbf{r} = -(1/r^2)\hat{\mathbf{r}}.$$

$$(c) \frac{\partial}{\partial x}(r^n) = n r^{n-1} \frac{\partial r}{\partial x} = n r^{n-1} \left(\frac{1}{2} \frac{1}{r} 2 r_x\right) = n r^{n-1} \hat{\mathbf{r}}_x, \text{ so } \boxed{\nabla(r^n) = n r^{n-1} \hat{\mathbf{r}}}$$

Problem 1.14

$$\bar{y} = +y \cos \phi + z \sin \phi; \text{ multiply by } \sin \phi: \bar{y} \sin \phi = +y \sin \phi \cos \phi + z \sin^2 \phi.$$

$$\bar{z} = -y \sin \phi + z \cos \phi; \text{ multiply by } \cos \phi: \bar{z} \cos \phi = -y \sin \phi \cos \phi + z \cos^2 \phi.$$

$$\text{Add: } \bar{y} \sin \phi + \bar{z} \cos \phi = z(\sin^2 \phi + \cos^2 \phi) = z. \text{ Likewise, } \bar{y} \cos \phi - \bar{z} \sin \phi = y.$$

$$\text{So } \frac{\partial \bar{y}}{\partial y} = \cos \phi; \frac{\partial \bar{y}}{\partial \bar{z}} = -\sin \phi; \frac{\partial \bar{z}}{\partial y} = \sin \phi; \frac{\partial \bar{z}}{\partial \bar{z}} = \cos \phi. \text{ Therefore}$$

$$\left. \begin{aligned} \overline{(\nabla f)}_y &= \frac{\partial f}{\partial \bar{y}} = \frac{\partial f}{\partial y} \frac{\partial y}{\partial \bar{y}} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial \bar{y}} = +\cos \phi (\nabla f)_y + \sin \phi (\nabla f)_z \\ \overline{(\nabla f)}_z &= \frac{\partial f}{\partial \bar{z}} = \frac{\partial f}{\partial y} \frac{\partial y}{\partial \bar{z}} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial \bar{z}} = -\sin \phi (\nabla f)_y + \cos \phi (\nabla f)_z \end{aligned} \right\} \text{ So } \nabla f \text{ transforms as a vector.} \quad \text{qed}$$

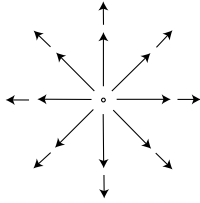
Problem 1.15

$$(a) \nabla \cdot \mathbf{v}_a = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(3xz^2) + \frac{\partial}{\partial z}(-2xz) = 2x + 0 - 2x = 0.$$

$$(b) \nabla \cdot \mathbf{v}_b = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}(3xz) = y + 2z + 3x.$$

$$(c) \nabla \cdot \mathbf{v}_c = \frac{\partial}{\partial x}(y^2) + \frac{\partial}{\partial y}(2xy + z^2) + \frac{\partial}{\partial z}(2yz) = 0 + (2x) + (2y) = 2(x + y)$$

Problem 1.16



$$\nabla \cdot \mathbf{v} = \frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r^3} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r^3} \right) = \frac{\partial}{\partial x} \left[x(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right] \\ + \frac{\partial}{\partial y} \left[y(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right] + \frac{\partial}{\partial z} \left[z(x^2 + y^2 + z^2)^{-\frac{3}{2}} \right] \\ = (-\frac{3}{2})(x^2 + y^2 + z^2)^{-\frac{5}{2}}2x + (-\frac{3}{2})(x^2 + y^2 + z^2)^{-\frac{5}{2}}2y + (-\frac{3}{2})(x^2 + y^2 + z^2)^{-\frac{5}{2}}2z \\ = -3(x^2 + y^2 + z^2)^{-\frac{5}{2}}(x + y + z) = -3r^{-5}(x^2 + y^2 + z^2) = -3r^{-3} = 0.$$

This conclusion is surprising, because, from the diagram, this vector field is obviously diverging away from the origin. How, then, can $\nabla \cdot \mathbf{v} = 0$? The answer is that $\nabla \cdot \mathbf{v} = 0$ everywhere *except* at the origin, but at the origin our calculation is no good, since $r = 0$, and the expression for $\mathbf{v}$ blows up. In fact, $\nabla \cdot \mathbf{v}$ is *infinite* at that one point, and zero elsewhere, as we shall see in Sect. 1.5.

Problem 1.17

$$\bar{v}_y = \cos \phi v_y + \sin \phi v_z; \quad \bar{v}_z = -\sin \phi v_y + \cos \phi v_z.$$

$$\frac{\partial \bar{v}_y}{\partial \bar{y}} = \frac{\partial v_y}{\partial y} \cos \phi + \frac{\partial v_z}{\partial y} \sin \phi = \left(\frac{\partial v_y}{\partial y} \frac{\partial y}{\partial \bar{y}} + \frac{\partial v_y}{\partial z} \frac{\partial z}{\partial \bar{y}} \right) \cos \phi + \left(\frac{\partial v_z}{\partial y} \frac{\partial y}{\partial \bar{y}} + \frac{\partial v_z}{\partial z} \frac{\partial z}{\partial \bar{y}} \right) \sin \phi. \text{ Use result in Prob. 1.14:}$$

$$= \left(\frac{\partial v_y}{\partial y} \cos \phi + \frac{\partial v_y}{\partial z} \sin \phi \right) \cos \phi + \left(\frac{\partial v_z}{\partial y} \cos \phi + \frac{\partial v_z}{\partial z} \sin \phi \right) \sin \phi.$$

$$\frac{\partial \bar{v}_z}{\partial \bar{z}} = -\frac{\partial v_y}{\partial z} \sin \phi + \frac{\partial v_z}{\partial z} \cos \phi = -\left(\frac{\partial v_y}{\partial y} \frac{\partial y}{\partial \bar{z}} + \frac{\partial v_y}{\partial z} \frac{\partial z}{\partial \bar{z}} \right) \sin \phi + \left(\frac{\partial v_z}{\partial y} \frac{\partial y}{\partial \bar{z}} + \frac{\partial v_z}{\partial z} \frac{\partial z}{\partial \bar{z}} \right) \cos \phi$$

$$= -\left(-\frac{\partial v_y}{\partial y} \sin \phi + \frac{\partial v_y}{\partial z} \cos \phi \right) \sin \phi + \left(-\frac{\partial v_z}{\partial y} \sin \phi + \frac{\partial v_z}{\partial z} \cos \phi \right) \cos \phi. \text{ So}$$

$$\begin{aligned}\frac{\partial \bar{v}_y}{\partial y} + \frac{\partial \bar{v}_z}{\partial z} &= \frac{\partial v_y}{\partial y} \cos^2 \phi + \frac{\partial v_y}{\partial z} \sin \phi \cos \phi + \frac{\partial v_z}{\partial y} \sin \phi \cos \phi + \frac{\partial v_z}{\partial z} \sin^2 \phi + \frac{\partial v_y}{\partial y} \sin^2 \phi - \frac{\partial v_y}{\partial z} \sin \phi \cos \phi \\ &\quad - \frac{\partial v_z}{\partial y} \sin \phi \cos \phi + \frac{\partial v_z}{\partial z} \cos^2 \phi \\ &= \frac{\partial v_y}{\partial y} (\cos^2 \phi + \sin^2 \phi) + \frac{\partial v_z}{\partial z} (\sin^2 \phi + \cos^2 \phi) = \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}. \quad \checkmark\end{aligned}$$

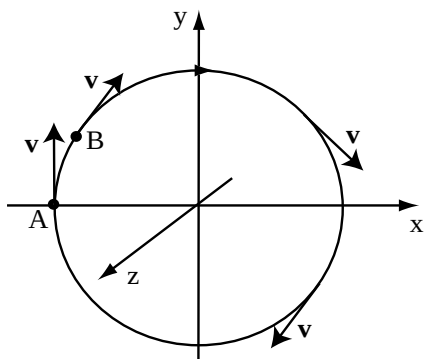
Problem 1.18

$$(a) \quad \nabla \times \mathbf{v}_a = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 3xz^2 & -2xz \end{vmatrix} = \hat{\mathbf{x}}(0 - 6xz) + \hat{\mathbf{y}}(0 + 2z) + \hat{\mathbf{z}}(3z^2 - 0) = \boxed{-6xz \hat{\mathbf{x}} + 2z \hat{\mathbf{y}} + 3z^2 \hat{\mathbf{z}}}.$$

$$(b) \quad \nabla \times \mathbf{v}_b = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 2yz & 3xz \end{vmatrix} = \hat{\mathbf{x}}(0 - 2y) + \hat{\mathbf{y}}(0 - 3z) + \hat{\mathbf{z}}(0 - x) = \boxed{-2y \hat{\mathbf{x}} - 3z \hat{\mathbf{y}} - x \hat{\mathbf{z}}}.$$

$$(c) \quad \nabla \times \mathbf{v}_c = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & (2xy + z^2) & 2yz \end{vmatrix} = \hat{\mathbf{x}}(2z - 2z) + \hat{\mathbf{y}}(0 - 0) + \hat{\mathbf{z}}(2y - 2y) = \boxed{\mathbf{0}}.$$

Problem 1.19



As we go from point A to point B (9 o'clock to 10 o'clock), x increases, y increases, v_x increases, and v_y decreases, so $\partial v_x / \partial y > 0$, while $\partial v_y / \partial y < 0$. On the circle, $v_z = 0$, and there is no dependence on z , so Eq. 1.41 says

$$\nabla \times \mathbf{v} = \hat{\mathbf{z}} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

points in the negative z direction (into the page), as the right hand rule would suggest. (Pick any other nearby points on the circle and you will come to the same conclusion.) [I'm sorry, but I cannot remember who suggested this cute illustration.]

Problem 1.20

$$\begin{aligned}\mathbf{v} &= y \hat{\mathbf{x}} + x \hat{\mathbf{y}}; \text{ or } \mathbf{v} = yz \hat{\mathbf{x}} + xz \hat{\mathbf{y}} + xy \hat{\mathbf{z}}; \text{ or } \mathbf{v} = (3x^2z - z^3) \hat{\mathbf{x}} + 3\hat{\mathbf{y}} + (x^3 - 3xz^2) \hat{\mathbf{z}}; \\ \text{or } \mathbf{v} &= (\sin x)(\cosh y) \hat{\mathbf{x}} - (\cos x)(\sinh y) \hat{\mathbf{y}}; \text{ etc.}\end{aligned}$$

Problem 1.21

$$\begin{aligned}(i) \quad \nabla(fg) &= \frac{\partial(fg)}{\partial x} \hat{\mathbf{x}} + \frac{\partial(fg)}{\partial y} \hat{\mathbf{y}} + \frac{\partial(fg)}{\partial z} \hat{\mathbf{z}} = \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} \right) \hat{\mathbf{x}} + \left(f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} \right) \hat{\mathbf{y}} + \left(f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} \right) \hat{\mathbf{z}} \\ &= f \left(\frac{\partial g}{\partial x} \hat{\mathbf{x}} + \frac{\partial g}{\partial y} \hat{\mathbf{y}} + \frac{\partial g}{\partial z} \hat{\mathbf{z}} \right) + g \left(\frac{\partial f}{\partial x} \hat{\mathbf{x}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}} \right) = f(\nabla g) + g(\nabla f). \quad \text{qed}\end{aligned}$$

$$\begin{aligned}(iv) \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \frac{\partial}{\partial x} (A_y B_z - A_z B_y) + \frac{\partial}{\partial y} (A_z B_x - A_x B_z) + \frac{\partial}{\partial z} (A_x B_y - A_y B_x) \\ &= A_y \frac{\partial B_z}{\partial x} + B_z \frac{\partial A_y}{\partial x} - A_z \frac{\partial B_y}{\partial x} - B_y \frac{\partial A_z}{\partial x} + A_z \frac{\partial B_x}{\partial y} + B_x \frac{\partial A_z}{\partial y} - A_x \frac{\partial B_z}{\partial y} - B_z \frac{\partial A_x}{\partial y} \\ &\quad + A_x \frac{\partial B_y}{\partial z} + B_y \frac{\partial A_x}{\partial z} - A_y \frac{\partial B_x}{\partial z} - B_x \frac{\partial A_y}{\partial z} \\ &= B_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + B_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + B_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - A_x \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \\ &\quad - A_y \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) - A_z \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}). \quad \text{qed}\end{aligned}$$

$$(v) \quad \nabla \times (f\mathbf{A}) = \left(\frac{\partial(fA_z)}{\partial y} - \frac{\partial(fA_y)}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial(fA_x)}{\partial z} - \frac{\partial(fA_z)}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial(fA_y)}{\partial x} - \frac{\partial(fA_x)}{\partial y} \right) \hat{\mathbf{z}}$$

$$\begin{aligned}
&= \left(f \frac{\partial A_z}{\partial y} + A_z \frac{\partial f}{\partial y} - f \frac{\partial A_y}{\partial z} - A_y \frac{\partial f}{\partial z} \right) \hat{\mathbf{x}} + \left(f \frac{\partial A_x}{\partial z} + A_x \frac{\partial f}{\partial z} - f \frac{\partial A_z}{\partial x} - A_z \frac{\partial f}{\partial x} \right) \hat{\mathbf{y}} \\
&\quad + \left(f \frac{\partial A_y}{\partial x} + A_y \frac{\partial f}{\partial x} - f \frac{\partial A_x}{\partial y} - A_x \frac{\partial f}{\partial y} \right) \hat{\mathbf{z}} \\
&= f \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{\mathbf{z}} \right] \\
&\quad - \left[\left(A_y \frac{\partial f}{\partial z} - A_z \frac{\partial f}{\partial y} \right) \hat{\mathbf{x}} + \left(A_z \frac{\partial f}{\partial x} - A_x \frac{\partial f}{\partial z} \right) \hat{\mathbf{y}} + \left(A_x \frac{\partial f}{\partial y} - A_y \frac{\partial f}{\partial x} \right) \hat{\mathbf{z}} \right] \\
&= f (\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f). \quad \text{qed}
\end{aligned}$$

Problem 1.22

(a) $(\mathbf{A} \cdot \nabla) \mathbf{B} = \left(A_x \frac{\partial B_x}{\partial x} + A_y \frac{\partial B_x}{\partial y} + A_z \frac{\partial B_x}{\partial z} \right) \hat{\mathbf{x}} + \left(A_x \frac{\partial B_y}{\partial x} + A_y \frac{\partial B_y}{\partial y} + A_z \frac{\partial B_y}{\partial z} \right) \hat{\mathbf{y}} + \left(A_x \frac{\partial B_z}{\partial x} + A_y \frac{\partial B_z}{\partial y} + A_z \frac{\partial B_z}{\partial z} \right) \hat{\mathbf{z}}.$

(b) $\hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}}.$ Let's just do the x component.

$$\begin{aligned}
[(\hat{\mathbf{r}} \cdot \nabla) \hat{\mathbf{r}}]_x &= \frac{1}{\sqrt{x^2 + y^2 + z^2}} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z} \right) \frac{x}{\sqrt{x^2 + y^2 + z^2}} \\
&= \frac{1}{r} \left\{ x \left[\frac{1}{\sqrt{}} + x \left(-\frac{1}{2} \right) \left(\frac{1}{\sqrt{}} \right)^3 2x \right] + yx \left[-\frac{1}{2} \left(\frac{1}{\sqrt{}} \right)^3 2y \right] + zx \left[-\frac{1}{2} \left(\frac{1}{\sqrt{}} \right)^3 2z \right] \right\} \\
&= \frac{1}{r} \left\{ \frac{x}{r} - \frac{1}{r^3} (x^3 + xy^2 + xz^2) \right\} = \frac{1}{r} \left\{ \frac{x}{r} - \frac{x}{r^3} (x^2 + y^2 + z^2) \right\} = \frac{1}{r} \left(\frac{x}{r} - \frac{x}{r} \right) = 0.
\end{aligned}$$

Same goes for the other components. Hence: $\boxed{(\hat{\mathbf{r}} \cdot \nabla) \hat{\mathbf{r}} = \mathbf{0}}.$

(c) $(\mathbf{v}_a \cdot \nabla) \mathbf{v}_b = \left(x^2 \frac{\partial}{\partial x} + 3xz^2 \frac{\partial}{\partial y} - 2xz \frac{\partial}{\partial z} \right) (xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3xz\hat{\mathbf{z}})$

$$\begin{aligned}
&= x^2 (y\hat{\mathbf{x}} + 0\hat{\mathbf{y}} + 3z\hat{\mathbf{z}}) + 3xz^2 (x\hat{\mathbf{x}} + 2z\hat{\mathbf{y}} + 0\hat{\mathbf{z}}) - 2xz (0\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 3x\hat{\mathbf{z}}) \\
&= (x^2y + 3x^2z^2) \hat{\mathbf{x}} + (6xz^3 - 4xyz) \hat{\mathbf{y}} + (3x^2z - 6x^2z) \hat{\mathbf{z}} \\
&= \boxed{x^2 (y + 3z^2) \hat{\mathbf{x}} + 2xz (3z^2 - 2y) \hat{\mathbf{y}} - 3x^2z \hat{\mathbf{z}}}
\end{aligned}$$

Problem 1.23

(ii) $[\nabla(\mathbf{A} \cdot \mathbf{B})]_x = \frac{\partial}{\partial x} (A_x B_x + A_y B_y + A_z B_z) = \frac{\partial A_x}{\partial x} B_x + A_x \frac{\partial B_x}{\partial x} + \frac{\partial A_y}{\partial x} B_y + A_y \frac{\partial B_y}{\partial x} + \frac{\partial A_z}{\partial x} B_z + A_z \frac{\partial B_z}{\partial x}$

$$[\mathbf{A} \times (\nabla \times \mathbf{B})]_x = A_y (\nabla \times \mathbf{B})_z - A_z (\nabla \times \mathbf{B})_y = A_y \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) - A_z \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right)$$

$$[\mathbf{B} \times (\nabla \times \mathbf{A})]_x = B_y \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - B_z \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right)$$

$$[(\mathbf{A} \cdot \nabla) \mathbf{B}]_x = \left(A_x \frac{\partial}{\partial x} + A_y \frac{\partial}{\partial y} + A_z \frac{\partial}{\partial z} \right) B_x = A_x \frac{\partial B_x}{\partial x} + A_y \frac{\partial B_x}{\partial y} + A_z \frac{\partial B_x}{\partial z}$$

$$[(\mathbf{B} \cdot \nabla) \mathbf{A}]_x = B_x \frac{\partial A_x}{\partial x} + B_y \frac{\partial A_x}{\partial y} + B_z \frac{\partial A_x}{\partial z}$$

So $[\mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A}]_x$

$$\begin{aligned}
&= A_y \frac{\partial B_y}{\partial x} - A_y \frac{\partial B_x}{\partial y} - A_z \frac{\partial B_x}{\partial z} + A_z \frac{\partial B_z}{\partial x} + B_y \frac{\partial A_y}{\partial x} - B_y \frac{\partial A_x}{\partial y} - B_z \frac{\partial A_x}{\partial z} + B_z \frac{\partial A_z}{\partial x} \\
&\quad + A_x \frac{\partial B_x}{\partial x} + A_y \frac{\partial B_x}{\partial y} + A_z \frac{\partial B_x}{\partial z} + B_x \frac{\partial A_x}{\partial x} + B_y \frac{\partial A_x}{\partial y} + B_z \frac{\partial A_x}{\partial z}
\end{aligned}$$

$$\begin{aligned}
&= B_x \frac{\partial A_x}{\partial x} + A_x \frac{\partial B_x}{\partial x} + B_y \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} + \frac{\partial A_x}{\partial y} \right) + A_y \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} + \frac{\partial B_x}{\partial y} \right) \\
&\quad + B_z \left(-\frac{\partial A_x}{\partial z} + \frac{\partial A_z}{\partial x} + \frac{\partial A_x}{\partial x} \right) + A_z \left(-\frac{\partial B_x}{\partial z} + \frac{\partial B_z}{\partial x} + \frac{\partial B_x}{\partial x} \right)
\end{aligned}$$

$$= [\nabla(\mathbf{A} \cdot \mathbf{B})]_x \text{ (same for } y \text{ and } z)$$

(vi) $[\nabla \times (\mathbf{A} \times \mathbf{B})]_x = \frac{\partial}{\partial y} (\mathbf{A} \times \mathbf{B})_z - \frac{\partial}{\partial z} (\mathbf{A} \times \mathbf{B})_y = \frac{\partial}{\partial y} (A_x B_y - A_y B_x) - \frac{\partial}{\partial z} (A_z B_x - A_x B_z)$

$$= \frac{\partial A_x}{\partial y} B_y + A_x \frac{\partial B_y}{\partial y} - \frac{\partial A_y}{\partial y} B_x - A_y \frac{\partial B_x}{\partial y} - \frac{\partial A_z}{\partial z} B_x - A_z \frac{\partial B_x}{\partial z} + \frac{\partial A_x}{\partial z} B_z + A_x \frac{\partial B_z}{\partial z}$$

$$[(\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})]_x$$

$$= B_x \frac{\partial A_x}{\partial x} + B_y \frac{\partial A_x}{\partial y} + B_z \frac{\partial A_x}{\partial z} - A_x \frac{\partial B_x}{\partial x} - A_y \frac{\partial B_x}{\partial y} - A_z \frac{\partial B_x}{\partial z} + A_x \left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right) - B_x \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right)$$

$$\begin{aligned}
&= B_y \frac{\partial A_x}{\partial y} + A_x \left(-\frac{\partial H_x}{\partial x} + \frac{\partial H_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right) + B_x \left(\frac{\partial A_x}{\partial x} - \frac{\partial A_x}{\partial x} - \frac{\partial A_y}{\partial y} - \frac{\partial A_z}{\partial z} \right) \\
&\quad + A_y \left(-\frac{\partial B_x}{\partial y} \right) + A_z \left(-\frac{\partial B_z}{\partial z} \right) + B_z \left(\frac{\partial A_x}{\partial z} \right) \\
&= [\nabla \times (\mathbf{A} \times \mathbf{B})]_x \text{ (same for } y \text{ and } z)
\end{aligned}$$

Problem 1.24

$$\begin{aligned}
\nabla(f/g) &= \frac{\partial}{\partial x}(f/g) \hat{\mathbf{x}} + \frac{\partial}{\partial y}(f/g) \hat{\mathbf{y}} + \frac{\partial}{\partial z}(f/g) \hat{\mathbf{z}} \\
&= \frac{g \frac{\partial f}{\partial x} - f \frac{\partial g}{\partial x}}{g^2} \hat{\mathbf{x}} + \frac{g \frac{\partial f}{\partial y} - f \frac{\partial g}{\partial y}}{g^2} \hat{\mathbf{y}} + \frac{g \frac{\partial f}{\partial z} - f \frac{\partial g}{\partial z}}{g^2} \hat{\mathbf{z}} \\
&= \frac{1}{g^2} \left[g \left(\frac{\partial f}{\partial x} \hat{\mathbf{x}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}} \right) - f \left(\frac{\partial g}{\partial x} \hat{\mathbf{x}} + \frac{\partial g}{\partial y} \hat{\mathbf{y}} + \frac{\partial g}{\partial z} \hat{\mathbf{z}} \right) \right] = \frac{g \nabla f - f \nabla g}{g^2}. \quad \text{qed}
\end{aligned}$$

$$\begin{aligned}
\nabla \cdot (\mathbf{A}/g) &= \frac{\partial}{\partial x}(A_x/g) + \frac{\partial}{\partial y}(A_y/g) + \frac{\partial}{\partial z}(A_z/g) \\
&= \frac{g \frac{\partial A_x}{\partial x} - A_x \frac{\partial g}{\partial x}}{g^2} + \frac{g \frac{\partial A_y}{\partial y} - A_y \frac{\partial g}{\partial y}}{g^2} + \frac{g \frac{\partial A_z}{\partial z} - A_z \frac{\partial g}{\partial z}}{g^2} \\
&= \frac{1}{g^2} \left[g \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) - \left(A_x \frac{\partial g}{\partial x} + A_y \frac{\partial g}{\partial y} + A_z \frac{\partial g}{\partial z} \right) \right] = \frac{g \nabla \cdot \mathbf{A} - \mathbf{A} \cdot \nabla g}{g^2}. \quad \text{qed}
\end{aligned}$$

$$\begin{aligned}
[\nabla \times (\mathbf{A}/g)]_x &= \frac{\partial}{\partial y}(A_z/g) - \frac{\partial}{\partial z}(A_y/g) \\
&= \frac{g \frac{\partial A_z}{\partial y} - A_z \frac{\partial g}{\partial y}}{g^2} - \frac{g \frac{\partial A_y}{\partial z} - A_y \frac{\partial g}{\partial z}}{g^2} \\
&= \frac{1}{g^2} \left[g \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - \left(A_z \frac{\partial g}{\partial y} - A_y \frac{\partial g}{\partial z} \right) \right] \\
&= \frac{g(\nabla \times \mathbf{A})_x + (\mathbf{A} \times \nabla g)_x}{g^2} \text{ (same for } y \text{ and } z). \quad \text{qed}
\end{aligned}$$

Problem 1.25

$$\text{(a) } \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ x & 2y & 3z \\ 3y & -2x & 0 \end{vmatrix} = \hat{\mathbf{x}}(6xz) + \hat{\mathbf{y}}(9zy) + \hat{\mathbf{z}}(-2x^2 - 6y^2)$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9zy) + \frac{\partial}{\partial z}(-2x^2 - 6y^2) = 6z + 9z + 0 = 15z$$

$$\nabla \times \mathbf{A} = \hat{\mathbf{x}} \left(\frac{\partial}{\partial y}(3z) - \frac{\partial}{\partial z}(2y) \right) + \hat{\mathbf{y}} \left(\frac{\partial}{\partial z}(x) - \frac{\partial}{\partial x}(3z) \right) + \hat{\mathbf{z}} \left(\frac{\partial}{\partial x}(2y) - \frac{\partial}{\partial y}(x) \right) = 0; \quad \mathbf{B} \cdot (\nabla \times \mathbf{A}) = 0$$

$$\nabla \times \mathbf{B} = \hat{\mathbf{x}} \left(\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(-2x) \right) + \hat{\mathbf{y}} \left(\frac{\partial}{\partial z}(3y) - \frac{\partial}{\partial x}(0) \right) + \hat{\mathbf{z}} \left(\frac{\partial}{\partial x}(-2x) - \frac{\partial}{\partial y}(3y) \right) = -5\hat{\mathbf{z}}; \quad \mathbf{A} \cdot (\nabla \times \mathbf{B}) = -15z$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) \stackrel{?}{=} \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) = 0 - (-15z) = 15z. \quad \checkmark$$

$$\text{(b) } \mathbf{A} \cdot \mathbf{B} = 3xy - 4xy = -xy; \quad \nabla(\mathbf{A} \cdot \mathbf{B}) = \nabla(-xy) = \hat{\mathbf{x}} \frac{\partial}{\partial x}(-xy) + \hat{\mathbf{y}} \frac{\partial}{\partial y}(-xy) = -y\hat{\mathbf{x}} - x\hat{\mathbf{y}}$$

$$\mathbf{A} \times (\nabla \times \mathbf{B}) = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ x & 2y & 3z \\ 0 & 0 & -5 \end{vmatrix} = \hat{\mathbf{x}}(-10y) + \hat{\mathbf{y}}(5x); \quad \mathbf{B} \times (\nabla \times \mathbf{A}) = \mathbf{0}$$

$$(\mathbf{A} \cdot \nabla) \mathbf{B} = \left(x \frac{\partial}{\partial x} + 2y \frac{\partial}{\partial y} + 3z \frac{\partial}{\partial z} \right) (3y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}) = \hat{\mathbf{x}}(6y) + \hat{\mathbf{y}}(-2x)$$

$$(\mathbf{B} \cdot \nabla) \mathbf{A} = \left(3y \frac{\partial}{\partial x} - 2x \frac{\partial}{\partial y} \right) (x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 3z\hat{\mathbf{z}}) = \hat{\mathbf{x}}(3y) + \hat{\mathbf{y}}(-4x)$$

$$\begin{aligned}
&\mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A} \\
&= -10y\hat{\mathbf{x}} + 5x\hat{\mathbf{y}} + 6y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}} + 3y\hat{\mathbf{x}} - 4x\hat{\mathbf{y}} = -y\hat{\mathbf{x}} - x\hat{\mathbf{y}} = \nabla \cdot (\mathbf{A} \cdot \mathbf{B}). \quad \checkmark
\end{aligned}$$

$$\begin{aligned}
\text{(c) } \nabla \times (\mathbf{A} \times \mathbf{B}) &= \hat{\mathbf{x}} \left(\frac{\partial}{\partial y}(-2x^2 - 6y^2) - \frac{\partial}{\partial z}(9zy) \right) + \hat{\mathbf{y}} \left(\frac{\partial}{\partial z}(6xz) - \frac{\partial}{\partial x}(-2x^2 - 6y^2) \right) + \hat{\mathbf{z}} \left(\frac{\partial}{\partial x}(9zy) - \frac{\partial}{\partial y}(6xz) \right) \\
&= \hat{\mathbf{x}}(-12y - 9y) + \hat{\mathbf{y}}(6x + 4x) + \hat{\mathbf{z}}(0) = -21y\hat{\mathbf{x}} + 10x\hat{\mathbf{y}}
\end{aligned}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(2y) + \frac{\partial}{\partial z}(3z) = 1 + 2 + 3 = 6; \quad \nabla \cdot \mathbf{B} = \frac{\partial}{\partial x}(3y) + \frac{\partial}{\partial y}(-2x) = 0$$

$$(\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) = 3y \hat{\mathbf{x}} - 4x \hat{\mathbf{y}} - 6y \hat{\mathbf{x}} + 2x \hat{\mathbf{y}} - 18y \hat{\mathbf{x}} + 12x \hat{\mathbf{y}} = -21y \hat{\mathbf{x}} + 10x \hat{\mathbf{y}} = \nabla \times (\mathbf{A} \times \mathbf{B}). \quad \checkmark$$

Problem 1.26

$$\begin{aligned} \text{(a)} \quad \frac{\partial^2 T_a}{\partial x^2} &= 2; \quad \frac{\partial^2 T_a}{\partial y^2} = \frac{\partial^2 T_a}{\partial z^2} = 0 \Rightarrow \boxed{\nabla^2 T_a = 2.} \\ \text{(b)} \quad \frac{\partial^2 T_b}{\partial x^2} &= \frac{\partial^2 T_b}{\partial y^2} = \frac{\partial^2 T_b}{\partial z^2} = -T_b \Rightarrow \boxed{\nabla^2 T_b = -3T_b = -3 \sin x \sin y \sin z.} \\ \text{(c)} \quad \frac{\partial^2 T_c}{\partial x^2} &= 25T_c; \quad \frac{\partial^2 T_c}{\partial y^2} = -16T_c; \quad \frac{\partial^2 T_c}{\partial z^2} = -9T_c \Rightarrow \boxed{\nabla^2 T_c = 0.} \\ \text{(d)} \quad \left. \begin{aligned} \frac{\partial^2 v_x}{\partial x^2} &= 2; \quad \frac{\partial^2 v_x}{\partial y^2} = \frac{\partial^2 v_x}{\partial z^2} = 0 \Rightarrow \nabla^2 v_x = 2 \\ \frac{\partial^2 v_y}{\partial x^2} &= \frac{\partial^2 v_y}{\partial y^2} = 0; \quad \frac{\partial^2 v_y}{\partial z^2} = 6x \Rightarrow \nabla^2 v_y = 6x \\ \frac{\partial^2 v_z}{\partial x^2} &= \frac{\partial^2 v_z}{\partial y^2} = \frac{\partial^2 v_z}{\partial z^2} = 0 \Rightarrow \nabla^2 v_z = 0 \end{aligned} \right\} \boxed{\nabla^2 \mathbf{v} = 2 \hat{\mathbf{x}} + 6x \hat{\mathbf{y}}.} \end{aligned}$$

Problem 1.27

$$\begin{aligned} \nabla \cdot (\nabla \times \mathbf{v}) &= \frac{\partial}{\partial x} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \\ &= \left(\frac{\partial^2 v_z}{\partial x \partial y} - \frac{\partial^2 v_z}{\partial y \partial x} \right) + \left(\frac{\partial^2 v_x}{\partial y \partial z} - \frac{\partial^2 v_x}{\partial z \partial y} \right) + \left(\frac{\partial^2 v_y}{\partial z \partial x} - \frac{\partial^2 v_y}{\partial x \partial z} \right) = 0, \text{ by equality of cross-derivatives.} \end{aligned}$$

$$\text{From Prob. 1.18: } \nabla \times \mathbf{v}_a = -6xz \hat{\mathbf{x}} + 2z \hat{\mathbf{y}} + 3z^2 \hat{\mathbf{z}} \Rightarrow \nabla \cdot (\nabla \times \mathbf{v}_a) = \frac{\partial}{\partial x}(-6xz) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(3z^2) = -6z + 6z = 0.$$

Problem 1.28

$$\begin{aligned} \nabla \times (\nabla t) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial t}{\partial x} & \frac{\partial t}{\partial y} & \frac{\partial t}{\partial z} \end{vmatrix} = \hat{\mathbf{x}} \left(\frac{\partial^2 t}{\partial y \partial z} - \frac{\partial^2 t}{\partial z \partial y} \right) + \hat{\mathbf{y}} \left(\frac{\partial^2 t}{\partial z \partial x} - \frac{\partial^2 t}{\partial x \partial z} \right) + \hat{\mathbf{z}} \left(\frac{\partial^2 t}{\partial x \partial y} - \frac{\partial^2 t}{\partial y \partial x} \right) \\ &= 0, \text{ by equality of cross-derivatives.} \end{aligned}$$

$$\text{In Prob. 1.11(b), } \nabla f = 2xy^3z^4 \hat{\mathbf{x}} + 3x^2y^2z^4 \hat{\mathbf{y}} + 4x^2y^3z^3 \hat{\mathbf{z}}, \text{ so}$$

$$\begin{aligned} \nabla \times (\nabla f) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy^3z^4 & 3x^2y^2z^4 & 4x^2y^3z^3 \end{vmatrix} \\ &= \hat{\mathbf{x}}(3 \cdot 4x^2y^2z^3 - 4 \cdot 3x^2y^2z^3) + \hat{\mathbf{y}}(4 \cdot 2xy^3z^3 - 2 \cdot 4xy^3z^3) + \hat{\mathbf{z}}(2 \cdot 3xy^2z^4 - 3 \cdot 2xy^2z^4) = 0. \quad \checkmark \end{aligned}$$

Problem 1.29

$$\begin{aligned} \text{(a)} \quad (0, 0, 0) &\longrightarrow (1, 0, 0). \quad x: 0 \rightarrow 1, y = z = 0; \quad d\mathbf{l} = dx \hat{\mathbf{x}}; \quad \mathbf{v} \cdot d\mathbf{l} = x^2 dx; \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 x^2 dx = (x^3/3)|_0^1 = 1/3. \\ (1, 0, 0) &\longrightarrow (1, 1, 0). \quad x = 1, y: 0 \rightarrow 1, z = 0; \quad d\mathbf{l} = dy \hat{\mathbf{y}}; \quad \mathbf{v} \cdot d\mathbf{l} = 2yz dy = 0; \quad \int \mathbf{v} \cdot d\mathbf{l} = 0. \\ (1, 1, 0) &\longrightarrow (1, 1, 1). \quad x = y = 1, z: 0 \rightarrow 1; \quad d\mathbf{l} = dz \hat{\mathbf{z}}; \quad \mathbf{v} \cdot d\mathbf{l} = y^2 dz = dz; \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 dz = z|_0^1 = 1. \\ \text{Total: } \int \mathbf{v} \cdot d\mathbf{l} &= (1/3) + 0 + 1 = \boxed{4/3.} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad (0, 0, 0) &\longrightarrow (0, 0, 1). \quad x = y = 0, z: 0 \rightarrow 1; \quad d\mathbf{l} = dz \hat{\mathbf{z}}; \quad \mathbf{v} \cdot d\mathbf{l} = y^2 dz = 0; \quad \int \mathbf{v} \cdot d\mathbf{l} = 0. \\ (0, 0, 1) &\longrightarrow (0, 1, 1). \quad x = 0, y: 0 \rightarrow 1, z = 1; \quad d\mathbf{l} = dy \hat{\mathbf{y}}; \quad \mathbf{v} \cdot d\mathbf{l} = 2yz dy = 2y dy; \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 2y dy = y^2|_0^1 = 1. \\ (0, 1, 1) &\longrightarrow (1, 1, 1). \quad x: 0 \rightarrow 1, y = z = 1; \quad d\mathbf{l} = dx \hat{\mathbf{x}}; \quad \mathbf{v} \cdot d\mathbf{l} = x^2 dx; \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_0^1 x^2 dx = (x^3/3)|_0^1 = 1/3. \\ \text{Total: } \int \mathbf{v} \cdot d\mathbf{l} &= 0 + 1 + (1/3) = \boxed{4/3.} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad x = y = z: 0 &\rightarrow 1; \quad dx = dy = dz; \quad \mathbf{v} \cdot d\mathbf{l} = x^2 dx + 2yz dy + y^2 dz = x^2 dx + 2x^2 dx + x^2 dx = 4x^2 dx; \\ \int \mathbf{v} \cdot d\mathbf{l} &= \int_0^1 4x^2 dx = (4x^3/3)|_0^1 = \boxed{4/3.} \end{aligned}$$

$$\text{(d)} \quad \oint \mathbf{v} \cdot d\mathbf{l} = (4/3) - (4/3) = \boxed{0.}$$

Problem 1.30

$x, y : 0 \rightarrow 1, z = 0; d\mathbf{a} = dx dy \hat{\mathbf{z}}; \mathbf{v} \cdot d\mathbf{a} = y(z^2 - 3) dx dy = -3y dx dy; \int \mathbf{v} \cdot d\mathbf{a} = -3 \int_0^2 dx \int_0^2 y dy = -3(x|_0^2)(\frac{y^2}{2}|_0^2) = -3(2)(2) = \boxed{-12.}$ In Ex. 1.7 we got 20, for the same boundary line (the square in the xy -plane), so the answer is no: the surface integral does *not* depend only on the boundary line. The *total* flux for the cube is $20 + 12 = \boxed{32.}$

Problem 1.31

$\int T d\tau = \int z^2 dx dy dz$. You can do the integrals in any order—here it is simplest to save z for last:

$$\int z^2 \left[\int \left(\int dx \right) dy \right] dz.$$

The sloping surface is $x + y + z = 1$, so the x integral is $\int_0^{(1-y-z)} dx = 1 - y - z$. For a given z , y ranges from 0 to $1 - z$, so the y integral is $\int_0^{(1-z)} (1 - y - z) dy = [(1 - z)y - (y^2/2)]|_0^{(1-z)} = (1 - z)^2 - [(1 - z)^2/2] = (1 - z)^2/2 = (1/2) - z + (z^2/2)$. Finally, the z integral is $\int_0^1 z^2 (\frac{1}{2} - z + \frac{z^2}{2}) dz = \int_0^1 (\frac{z^2}{2} - z^3 + \frac{z^4}{2}) dz = (\frac{z^3}{6} - \frac{z^4}{4} + \frac{z^5}{10})|_0^1 = \frac{1}{6} - \frac{1}{4} + \frac{1}{10} = \boxed{1/60.}$

Problem 1.32

$$T(\mathbf{b}) = 1 + 4 + 2 = 7; T(\mathbf{a}) = 0. \Rightarrow \boxed{T(\mathbf{b}) - T(\mathbf{a}) = 7.}$$

$$\nabla T = (2x + 4y)\hat{\mathbf{x}} + (4x + 2z^3)\hat{\mathbf{y}} + (6yz^2)\hat{\mathbf{z}}; \nabla T \cdot d\mathbf{l} = (2x + 4y)dx + (4x + 2z^3)dy + (6yz^2)dz$$

- (a) Segment 1: $x : 0 \rightarrow 1, y = z = dy = dz = 0. \int \nabla T \cdot d\mathbf{l} = \int_0^1 (2x) dx = x^2|_0^1 = 1.$
 Segment 2: $y : 0 \rightarrow 1, x = 1, z = 0, dx = dz = 0. \int \nabla T \cdot d\mathbf{l} = \int_0^1 (4) dy = 4y|_0^1 = 4.$
 Segment 3: $z : 0 \rightarrow 1, x = y = 1, dx = dy = 0. \int \nabla T \cdot d\mathbf{l} = \int_0^1 (6z^2) dz = 2z^3|_0^1 = 2.$ } $\int_{\mathbf{a}}^{\mathbf{b}} \nabla T \cdot d\mathbf{l} = 7. \checkmark$
- (b) Segment 1: $z : 0 \rightarrow 1, x = y = dx = dy = 0. \int \nabla T \cdot d\mathbf{l} = \int_0^1 (0) dz = 0.$
 Segment 2: $y : 0 \rightarrow 1, x = 0, z = 1, dx = dz = 0. \int \nabla T \cdot d\mathbf{l} = \int_0^1 (2) dy = 2y|_0^1 = 2.$
 Segment 3: $x : 0 \rightarrow 1, y = z = 1, dy = dz = 0. \int \nabla T \cdot d\mathbf{l} = \int_0^1 (2x + 4) dx = (x^2 + 4x)|_0^1 = 1 + 4 = 5.$ } $\int_{\mathbf{a}}^{\mathbf{b}} \nabla T \cdot d\mathbf{l} = 7. \checkmark$
- (c) $x : 0 \rightarrow 1, y = x, z = x^2, dy = dx, dz = 2x dx.$

$$\nabla T \cdot d\mathbf{l} = (2x + 4x)dx + (4x + 2x^6)dx + (6xx^4)2x dx = (10x + 14x^6)dx.$$

$$\int_{\mathbf{a}}^{\mathbf{b}} \nabla T \cdot d\mathbf{l} = \int_0^1 (10x + 14x^6)dx = (5x^2 + 2x^7)|_0^1 = 5 + 2 = 7. \checkmark$$

Problem 1.33

$$\nabla \cdot \mathbf{v} = y + 2z + 3x$$

$$\begin{aligned} \int (\nabla \cdot \mathbf{v}) d\tau &= \int (y + 2z + 3x) dx dy dz = \iiint \left\{ \int_0^2 (y + 2z + 3x) dx \right\} dy dz \\ &\quad \hookrightarrow [(y + 2z)x + \frac{3}{2}x^2]_0^2 = 2(y + 2z) + 6 \\ &= \int \left\{ \int_0^2 (2y + 4z + 6) dy \right\} dz \\ &\quad \hookrightarrow [y^2 + (4z + 6)y]_0^2 = 4 + 2(4z + 6) = 8z + 16 \\ &= \int_0^2 (8z + 16) dz = (4z^2 + 16z)|_0^2 = 16 + 32 = \boxed{48.} \end{aligned}$$

Numbering the surfaces as in Fig. 1.29:

- (i) $d\mathbf{a} = dy dz \hat{\mathbf{x}}, x = 2$. $\mathbf{v} \cdot d\mathbf{a} = 2y dy dz$. $\int \mathbf{v} \cdot d\mathbf{a} = \iint 2y dy dz = 2y^2 \Big|_0^2 = 8$.
(ii) $d\mathbf{a} = -dy dz \hat{\mathbf{x}}, x = 0$. $\mathbf{v} \cdot d\mathbf{a} = 0$. $\int \mathbf{v} \cdot d\mathbf{a} = 0$.
(iii) $d\mathbf{a} = dx dz \hat{\mathbf{y}}, y = 2$. $\mathbf{v} \cdot d\mathbf{a} = 4z dx dz$. $\int \mathbf{v} \cdot d\mathbf{a} = \iint 4z dx dz = 16$.
(iv) $d\mathbf{a} = -dx dz \hat{\mathbf{y}}, y = 0$. $\mathbf{v} \cdot d\mathbf{a} = 0$. $\int \mathbf{v} \cdot d\mathbf{a} = 0$.
(v) $d\mathbf{a} = dx dy \hat{\mathbf{z}}, z = 2$. $\mathbf{v} \cdot d\mathbf{a} = 6x dx dy$. $\int \mathbf{v} \cdot d\mathbf{a} = 24$.
(vi) $d\mathbf{a} = -dx dy \hat{\mathbf{z}}, z = 0$. $\mathbf{v} \cdot d\mathbf{a} = 0$. $\int \mathbf{v} \cdot d\mathbf{a} = 0$.
 $\Rightarrow \int \mathbf{v} \cdot d\mathbf{a} = 8 + 16 + 24 = 48 \checkmark$

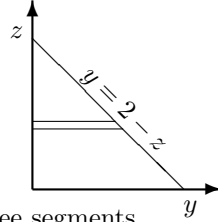
Problem 1.34

$$\nabla \times \mathbf{v} = \hat{\mathbf{x}}(0 - 2y) + \hat{\mathbf{y}}(0 - 3z) + \hat{\mathbf{z}}(0 - x) = -2y \hat{\mathbf{x}} - 3z \hat{\mathbf{y}} - x \hat{\mathbf{z}}.$$

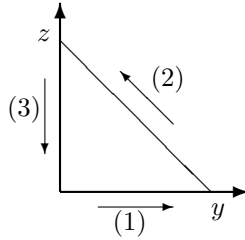
$d\mathbf{a} = dy dz \hat{\mathbf{x}}$, if we agree that the path integral shall run counterclockwise. So

$$(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = -2y dy dz.$$

$$\begin{aligned} \int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} &= \int \left\{ \int_0^{2-z} (-2y) dy \right\} dz \\ &\quad \hookrightarrow y^2 \Big|_0^{2-z} = -(2-z)^2 \\ &= - \int_0^2 (4 - 4z + z^2) dz = - \left(4z - 2z^2 + \frac{z^3}{3} \right) \Big|_0^2 \\ &= - \left(8 - 8 + \frac{8}{3} \right) = \boxed{-\frac{8}{3}} \end{aligned}$$



Meanwhile, $\mathbf{v} \cdot d\mathbf{l} = (xy)dx + (2yz)dy + (3zx)dz$. There are three segments.



- (1) $x = z = 0$; $dx = dz = 0$. $y : 0 \rightarrow 2$. $\int \mathbf{v} \cdot d\mathbf{l} = 0$.
(2) $x = 0$; $z = 2 - y$; $dx = 0$, $dz = -dy$, $y : 2 \rightarrow 0$. $\mathbf{v} \cdot d\mathbf{l} = 2yz dy$.
 $\int \mathbf{v} \cdot d\mathbf{l} = \int_2^0 2y(2 - y) dy = - \int_0^2 (4y - 2y^2) dy = - \left(2y^2 - \frac{2}{3}y^3 \right) \Big|_0^2 = - \left(8 - \frac{2}{3} \cdot 8 \right) = -\frac{8}{3}$.
(3) $x = y = 0$; $dx = dy = 0$; $z : 2 \rightarrow 0$. $\mathbf{v} \cdot d\mathbf{l} = 0$. $\int \mathbf{v} \cdot d\mathbf{l} = 0$. So $\oint \mathbf{v} \cdot d\mathbf{l} = -\frac{8}{3}$. $\checkmark$

Problem 1.35

By Corollary 1, $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ should equal $\frac{4}{3}$. $\nabla \times \mathbf{v} = (4z^2 - 2x)\hat{\mathbf{x}} + 2z\hat{\mathbf{z}}$.

- (i) $d\mathbf{a} = dy dz \hat{\mathbf{x}}, x = 1$; $y, z : 0 \rightarrow 1$. $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = (4z^2 - 2) dy dz$; $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^1 (4z^2 - 2) dz$
 $= \left(\frac{4}{3}z^3 - 2z \right) \Big|_0^1 = \frac{4}{3} - 2 = -\frac{2}{3}$.
(ii) $d\mathbf{a} = -dx dy \hat{\mathbf{z}}, z = 0$; $x, y : 0 \rightarrow 1$. $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$; $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$.
(iii) $d\mathbf{a} = dx dz \hat{\mathbf{y}}, y = 1$; $x, z : 0 \rightarrow 1$. $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$; $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$.
(iv) $d\mathbf{a} = -dx dz \hat{\mathbf{y}}, y = 0$; $x, z : 0 \rightarrow 1$. $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$; $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$.
(v) $d\mathbf{a} = dx dy \hat{\mathbf{z}}, z = 1$; $x, y : 0 \rightarrow 1$. $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 2 dx dy$; $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 2$.
 $\Rightarrow \int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = -\frac{2}{3} + 2 = \frac{4}{3}$. $\checkmark$

Problem 1.36

(a) Use the product rule $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$:

$$\int_S f(\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \int_S \nabla \times (f\mathbf{A}) \cdot d\mathbf{a} + \int_S [\mathbf{A} \times (\nabla f)] \cdot d\mathbf{a} = \oint_{\mathcal{P}} f\mathbf{A} \cdot d\mathbf{l} + \int_S [\mathbf{A} \times (\nabla f)] \cdot d\mathbf{a}. \quad \text{qed}$$

(I used Stokes' theorem in the last step.)

(b) Use the product rule $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$:

$$\int_V \mathbf{B} \cdot (\nabla \times \mathbf{A}) d\tau = \int_V \nabla \cdot (\mathbf{A} \times \mathbf{B}) d\tau + \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) d\tau = \oint_S (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a} + \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) d\tau. \quad \text{qed}$$

(I used the divergence theorem in the last step.)

Problem 1.37

$$r = \sqrt{x^2 + y^2 + z^2}; \quad \theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right); \quad \phi = \tan^{-1} \left(\frac{y}{x} \right).$$

Problem 1.38

There are many ways to do this one—probably the most illuminating way is to work it out by trigonometry from Fig. 1.36. The most systematic approach is to study the expression:

$$\mathbf{r} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}} = r \sin \theta \cos \phi \hat{\mathbf{x}} + r \sin \theta \sin \phi \hat{\mathbf{y}} + r \cos \theta \hat{\mathbf{z}}.$$

If I only vary r slightly, then $d\mathbf{r} = \frac{\partial}{\partial r}(\mathbf{r})dr$ is a short vector pointing in the direction of increase in r . To make it a unit vector, I must divide by its length. Thus:

$$\hat{\mathbf{r}} = \frac{\frac{\partial \mathbf{r}}{\partial r}}{\left| \frac{\partial \mathbf{r}}{\partial r} \right|}; \quad \hat{\boldsymbol{\theta}} = \frac{\frac{\partial \mathbf{r}}{\partial \theta}}{\left| \frac{\partial \mathbf{r}}{\partial \theta} \right|}; \quad \hat{\boldsymbol{\phi}} = \frac{\frac{\partial \mathbf{r}}{\partial \phi}}{\left| \frac{\partial \mathbf{r}}{\partial \phi} \right|}.$$

$$\frac{\partial \mathbf{r}}{\partial r} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}; \quad \left| \frac{\partial \mathbf{r}}{\partial r} \right|^2 = \sin^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \theta = 1.$$

$$\frac{\partial \mathbf{r}}{\partial \theta} = r \cos \theta \cos \phi \hat{\mathbf{x}} + r \cos \theta \sin \phi \hat{\mathbf{y}} - r \sin \theta \hat{\mathbf{z}}; \quad \left| \frac{\partial \mathbf{r}}{\partial \theta} \right|^2 = r^2 \cos^2 \theta \cos^2 \phi + r^2 \cos^2 \theta \sin^2 \phi + r^2 \sin^2 \theta = r^2.$$

$$\frac{\partial \mathbf{r}}{\partial \phi} = -r \sin \theta \sin \phi \hat{\mathbf{x}} + r \sin \theta \cos \phi \hat{\mathbf{y}}; \quad \left| \frac{\partial \mathbf{r}}{\partial \phi} \right|^2 = r^2 \sin^2 \theta \sin^2 \phi + r^2 \sin^2 \theta \cos^2 \phi = r^2 \sin^2 \theta.$$

$$\Rightarrow \begin{cases} \hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}. \\ \hat{\boldsymbol{\theta}} = \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}}. \\ \hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}. \end{cases}$$

$$\text{Check: } \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + \cos^2 \theta = \sin^2 \theta + \cos^2 \theta = 1, \quad \checkmark$$

$$\hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} = -\cos \theta \sin \phi \cos \phi + \cos \theta \sin \phi \cos \phi = 0, \quad \checkmark \quad \text{etc.}$$

$$\sin \theta \hat{\mathbf{r}} = \sin^2 \theta \cos \phi \hat{\mathbf{x}} + \sin^2 \theta \sin \phi \hat{\mathbf{y}} + \sin \theta \cos \theta \hat{\mathbf{z}}.$$

$$\cos \theta \hat{\boldsymbol{\theta}} = \cos^2 \theta \cos \phi \hat{\mathbf{x}} + \cos^2 \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \cos \theta \hat{\mathbf{z}}.$$

Add these:

$$(1) \quad \sin \theta \hat{\mathbf{r}} + \cos \theta \hat{\boldsymbol{\theta}} = +\cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}};$$

$$(2) \quad \hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}.$$

Multiply (1) by $\cos \phi$, (2) by $\sin \phi$, and subtract:

$$\hat{\mathbf{x}} = \sin \theta \cos \phi \hat{\mathbf{r}} + \cos \theta \cos \phi \hat{\boldsymbol{\theta}} - \sin \phi \hat{\boldsymbol{\phi}}.$$

Multiply (1) by $\sin \phi$, (2) by $\cos \phi$, and add:

$$\hat{\mathbf{y}} = \sin \theta \sin \phi \hat{\mathbf{r}} + \cos \theta \sin \phi \hat{\boldsymbol{\theta}} + \cos \phi \hat{\boldsymbol{\phi}}.$$

$$\cos \theta \hat{\mathbf{r}} = \sin \theta \cos \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \cos \theta \sin \phi \hat{\mathbf{y}} + \cos^2 \theta \hat{\mathbf{z}}.$$

$$\sin \theta \hat{\boldsymbol{\theta}} = \sin \theta \cos \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \cos \theta \sin \phi \hat{\mathbf{y}} - \sin^2 \theta \hat{\mathbf{z}}.$$

Subtract these:

$$\hat{\mathbf{z}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}.$$

Problem 1.39

(a) $\nabla \cdot \mathbf{v}_1 = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r^2) = \frac{1}{r^2} 4r^3 = 4r$

$$\int (\nabla \cdot \mathbf{v}_1) d\tau = \int (4r) (r^2 \sin \theta dr d\theta d\phi) = (4) \int_0^R r^3 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi = (4) \left(\frac{R^4}{4} \right) (2)(2\pi) = \boxed{4\pi R^4}$$

$$\int \mathbf{v}_1 \cdot d\mathbf{a} = \int (r^2 \hat{\mathbf{r}}) \cdot (r^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) = r^4 \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi = 4\pi R^4 \quad \checkmark \quad (\text{Note: at surface of sphere } r = R.)$$

(b) $\nabla \cdot \mathbf{v}_2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r^2} \right) = 0 \Rightarrow \boxed{\int (\nabla \cdot \mathbf{v}_2) d\tau = 0}$

$$\int \mathbf{v}_2 \cdot d\mathbf{a} = \int \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) (r^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) = \int \sin \theta d\theta d\phi = \boxed{4\pi}.$$

They *don't* agree! The point is that this divergence is zero *except at the origin*, where it blows up, so our calculation of $\int (\nabla \cdot \mathbf{v}_2)$ is *incorrect*. The right answer is 4π .

Problem 1.40

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r \cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta r \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (r \sin \theta \cos \phi) \\ &= \frac{1}{r^2} 3r^2 \cos \theta + \frac{1}{r \sin \theta} r 2 \sin \theta \cos \theta + \frac{1}{r \sin \theta} r \sin \theta (-\sin \phi) \\ &= 3 \cos \theta + 2 \cos \theta - \sin \phi = 5 \cos \theta - \sin \phi \end{aligned}$$

$$\int (\nabla \cdot \mathbf{v}) d\tau = \int (5 \cos \theta - \sin \phi) r^2 \sin \theta dr d\theta d\phi = \int_0^R r^2 dr \int_0^{\frac{\pi}{2}} \left[\int_0^{2\pi} (5 \cos \theta - \sin \phi) d\phi \right] d\theta \sin \theta$$

$$\hookrightarrow 2\pi (5 \cos \theta)$$

$$= \left(\frac{R^3}{3} \right) (10\pi) \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta d\theta$$

$$\hookrightarrow \frac{\sin^2 \theta}{2} \Big|_0^{\frac{\pi}{2}} = \frac{1}{2}$$

$$= \boxed{\frac{5\pi}{3} R^3}.$$

Two surfaces—one the hemisphere: $d\mathbf{a} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}$; $r = R$; $\phi : 0 \rightarrow 2\pi$, $\theta : 0 \rightarrow \frac{\pi}{2}$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int (r \cos \theta) R^2 \sin \theta d\theta d\phi = R^3 \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta d\theta \int_0^{2\pi} d\phi = R^3 \left(\frac{1}{2} \right) (2\pi) = \pi R^3.$$

other the flat bottom: $d\mathbf{a} = (dr)(r \sin \theta d\phi)(+\hat{\boldsymbol{\theta}}) = r dr d\phi \hat{\boldsymbol{\theta}}$ (here $\theta = \frac{\pi}{2}$). $r : 0 \rightarrow R$, $\phi : 0 \rightarrow 2\pi$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int (r \sin \theta) (r dr d\phi) = \int_0^R r^2 dr \int_0^{2\pi} d\phi = 2\pi \frac{R^3}{3}.$$

$$\text{Total: } \int \mathbf{v} \cdot d\mathbf{a} = \pi R^3 + \frac{2}{3}\pi R^3 = \frac{5}{3}\pi R^3. \quad \checkmark$$

Problem 1.41 $\boxed{\nabla t = (\cos \theta + \sin \theta \cos \phi) \hat{\mathbf{r}} + (-\sin \theta + \cos \theta \cos \phi) \hat{\boldsymbol{\theta}} + \frac{1}{\sin \theta} (-\sin \theta \sin \phi) \hat{\boldsymbol{\phi}}}$

$$\nabla^2 t = \nabla \cdot (\nabla t)$$

$$\begin{aligned} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 (\cos \theta + \sin \theta \cos \phi)) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta (-\sin \theta + \cos \theta \cos \phi)) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (-\sin \phi) \\ &= \frac{1}{r^2} 2r (\cos \theta + \sin \theta \cos \phi) + \frac{1}{r \sin \theta} (-2 \sin \theta \cos \theta + \cos^2 \theta \cos \phi - \sin^2 \theta \cos \phi) - \frac{1}{r \sin \theta} \cos \phi \\ &= \frac{1}{r \sin \theta} [2 \sin \theta \cos \theta + 2 \sin^2 \theta \cos \phi - 2 \sin \theta \cos \theta + \cos^2 \theta \cos \phi - \sin^2 \theta \cos \phi - \cos \phi] \\ &= \frac{1}{r \sin \theta} [(\sin^2 \theta + \cos^2 \theta) \cos \phi - \cos \phi] = 0. \end{aligned}$$

$$\Rightarrow \boxed{\nabla^2 t = 0}$$

Check: $r \cos \theta = z$, $r \sin \theta \cos \phi = x \Rightarrow$ in Cartesian coordinates $t = x + z$. Obviously Laplacian is zero.

Gradient Theorem: $\int_{\mathbf{a}}^{\mathbf{b}} \nabla t \cdot d\mathbf{l} = t(\mathbf{b}) - t(\mathbf{a})$

Segment 1: $\theta = \frac{\pi}{2}$, $\phi = 0$, $r : 0 \rightarrow 2$. $d\mathbf{l} = dr \hat{\mathbf{r}}$; $\nabla t \cdot d\mathbf{l} = (\cos \theta + \sin \theta \cos \phi) dr = (0 + 1) dr = dr$.
 $\int \nabla t \cdot d\mathbf{l} = \int_0^2 dr = 2$.

Segment 2: $\theta = \frac{\pi}{2}$, $r = 2$, $\phi : 0 \rightarrow \frac{\pi}{2}$. $d\mathbf{l} = r \sin \theta d\phi \hat{\phi} = 2 d\phi \hat{\phi}$.

$$\nabla t \cdot d\mathbf{l} = (-\sin \phi)(2 d\phi) = -2 \sin \phi d\phi. \quad \int \nabla t \cdot d\mathbf{l} = -\int_0^{\frac{\pi}{2}} 2 \sin \phi d\phi = 2 \cos \phi \Big|_0^{\frac{\pi}{2}} = -2.$$

Segment 3: $r = 2$, $\phi = \frac{\pi}{2}$; $\theta : \frac{\pi}{2} \rightarrow 0$.

$$d\mathbf{l} = r d\theta \hat{\theta} = 2 d\theta \hat{\theta}; \quad \nabla t \cdot d\mathbf{l} = (-\sin \theta + \cos \theta \cos \phi)(2 d\theta) = -2 \sin \theta d\theta.$$

$$\int \nabla t \cdot d\mathbf{l} = -\int_{\frac{\pi}{2}}^0 2 \sin \theta d\theta = 2 \cos \theta \Big|_{\frac{\pi}{2}}^0 = 2.$$

Total: $\int_{\mathbf{a}}^{\mathbf{b}} \nabla t \cdot d\mathbf{l} = 2 - 2 + 2 = \boxed{2}$. Meanwhile, $t(\mathbf{b}) - t(\mathbf{a}) = [2(1 + 0)] - [0(\quad)] = 2$. ✓

Problem 1.42 From Fig. 1.42, $\boxed{\hat{\mathbf{s}} = \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}; \quad \hat{\phi} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}; \quad \hat{\mathbf{z}} = \hat{\mathbf{z}}}$

Multiply first by $\cos \phi$, second by $\sin \phi$, and subtract:

$$\hat{\mathbf{s}} \cos \phi - \hat{\phi} \sin \phi = \cos^2 \phi \hat{\mathbf{x}} + \cos \phi \sin \phi \hat{\mathbf{y}} + \sin^2 \phi \hat{\mathbf{x}} - \sin \phi \cos \phi \hat{\mathbf{y}} = \hat{\mathbf{x}}(\sin^2 \phi + \cos^2 \phi) = \hat{\mathbf{x}}.$$

So $\boxed{\hat{\mathbf{x}} = \cos \phi \hat{\mathbf{s}} - \sin \phi \hat{\phi}}$.

Multiply first by $\sin \phi$, second by $\cos \phi$, and add:

$$\hat{\mathbf{s}} \sin \phi + \hat{\phi} \cos \phi = \sin \phi \cos \phi \hat{\mathbf{x}} + \sin^2 \phi \hat{\mathbf{y}} - \sin \phi \cos \phi \hat{\mathbf{x}} + \cos^2 \phi \hat{\mathbf{y}} = \hat{\mathbf{y}}(\sin^2 \phi + \cos^2 \phi) = \hat{\mathbf{y}}.$$

So $\boxed{\hat{\mathbf{y}} = \sin \phi \hat{\mathbf{s}} + \cos \phi \hat{\phi}}$.

$\boxed{\hat{\mathbf{z}} = \hat{\mathbf{z}}}$.

Problem 1.43

(a) $\nabla \cdot \mathbf{v} = \frac{1}{s} \frac{\partial}{\partial s} (s s(2 + \sin^2 \phi)) + \frac{1}{s} \frac{\partial}{\partial \phi} (s \sin \phi \cos \phi) + \frac{\partial}{\partial z} (3z)$
 $= \frac{1}{s} 2s(2 + \sin^2 \phi) + \frac{1}{s} s(\cos^2 \phi - \sin^2 \phi) + 3$
 $= 4 + 2 \sin^2 \phi + \cos^2 \phi - \sin^2 \phi + 3$
 $= 4 + \sin^2 \phi + \cos^2 \phi + 3 = \boxed{8}.$

(b) $\int (\nabla \cdot \mathbf{v}) d\tau = \int (8) s ds d\phi dz = 8 \int_0^2 s ds \int_0^{\frac{\pi}{2}} d\phi \int_0^5 dz = 8(2) \left(\frac{\pi}{2}\right) (5) = \boxed{40\pi}.$

Meanwhile, the surface integral has five parts:

top: $z = 5$, $d\mathbf{a} = s ds d\phi \hat{\mathbf{z}}$; $\mathbf{v} \cdot d\mathbf{a} = 3z s ds d\phi = 15s ds d\phi$. $\int \mathbf{v} \cdot d\mathbf{a} = 15 \int_0^2 s ds \int_0^{\frac{\pi}{2}} d\phi = 15\pi$.

bottom: $z = 0$, $d\mathbf{a} = -s ds d\phi \hat{\mathbf{z}}$; $\mathbf{v} \cdot d\mathbf{a} = -3z s ds d\phi = 0$. $\int \mathbf{v} \cdot d\mathbf{a} = 0$.

back: $\phi = \frac{\pi}{2}$, $d\mathbf{a} = ds dz \hat{\phi}$; $\mathbf{v} \cdot d\mathbf{a} = s \sin \phi \cos \phi ds dz = 0$. $\int \mathbf{v} \cdot d\mathbf{a} = 0$.

left: $\phi = 0$, $d\mathbf{a} = -ds dz \hat{\phi}$; $\mathbf{v} \cdot d\mathbf{a} = -s \sin \phi \cos \phi ds dz = 0$. $\int \mathbf{v} \cdot d\mathbf{a} = 0$.

front: $s = 2$, $d\mathbf{a} = s d\phi dz \hat{\mathbf{s}}$; $\mathbf{v} \cdot d\mathbf{a} = s(2 + \sin^2 \phi) s d\phi dz = 4(2 + \sin^2 \phi) d\phi dz$.

$$\int \mathbf{v} \cdot d\mathbf{a} = 4 \int_0^{\frac{\pi}{2}} (2 + \sin^2 \phi) d\phi \int_0^5 dz = (4)(\pi + \frac{\pi}{4})(5) = 25\pi.$$

So $\oint \mathbf{v} \cdot d\mathbf{a} = 15\pi + 25\pi = 40\pi$. ✓

(c) $\nabla \times \mathbf{v} = \left(\frac{1}{s} \frac{\partial}{\partial \phi} (3z) - \frac{\partial}{\partial z} (s \sin \phi \cos \phi) \right) \hat{\mathbf{s}} + \left(\frac{\partial}{\partial z} (s(2 + \sin^2 \phi)) - \frac{\partial}{\partial s} (3z) \right) \hat{\phi}$
 $+ \frac{1}{s} \left(\frac{\partial}{\partial s} (s^2 \sin \phi \cos \phi) - \frac{\partial}{\partial \phi} (s(2 + \sin^2 \phi)) \right) \hat{\mathbf{z}}$
 $= \frac{1}{s} (2s \sin \phi \cos \phi - s 2 \sin \phi \cos \phi) \hat{\mathbf{z}} = \boxed{\mathbf{0}}.$

Problem 1.44

(a) $3(3^2) - 2(3) - 1 = 27 - 6 - 1 = \boxed{20}.$

(b) $\cos \pi = \boxed{-1}.$

(c) $\boxed{\text{zero}}.$

(d) $\ln(-2 + 3) = \ln 1 = \boxed{\text{zero}}.$

Problem 1.45

(a) $\int_{-2}^2 (2x + 3)^{\frac{1}{3}} \delta(x) dx = \frac{1}{3}(0 + 3) = \boxed{1}.$

(b) By Eq. 1.94, $\delta(1 - x) = \delta(x - 1)$, so $1 + 3 + 2 = \boxed{6}.$

(c) $\int_{-1}^1 9x^2 \frac{1}{3} \delta(x + \frac{1}{3}) dx = 9 \left(-\frac{1}{3}\right)^2 \frac{1}{3} = \boxed{\frac{1}{3}}.$

(d) $\boxed{1 \text{ (if } a > b), 0 \text{ (if } a < b)}.$

Problem 1.46

(a) $\int_{-\infty}^{\infty} f(x) \left[x \frac{d}{dx} \delta(x)\right] dx = x f(x) \delta(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d}{dx} (x f(x)) \delta(x) dx.$

The first term is zero, since $\delta(x) = 0$ at $\pm\infty$; $\frac{d}{dx} (x f(x)) = x \frac{df}{dx} + \frac{dx}{dx} f = x \frac{df}{dx} + f.$

So the integral is $-\int_{-\infty}^{\infty} \left(x \frac{df}{dx} + f\right) \delta(x) dx = 0 - f(0) = -f(0) = -\int_{-\infty}^{\infty} f(x) \delta(x) dx.$

So, $x \frac{d}{dx} \delta(x) = -\delta(x).$ qed

(b) $\int_{-\infty}^{\infty} f(x) \frac{d\theta}{dx} dx = f(x) \theta(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{df}{dx} \theta(x) dx = f(\infty) - \int_0^{\infty} \frac{df}{dx} dx = f(\infty) - (f(\infty) - f(0))$
 $= f(0) = \int_{-\infty}^{\infty} f(x) \delta(x) dx.$ So $\frac{d\theta}{dx} = \delta(x).$ qed

Problem 1.47

(a) $\boxed{\rho(\mathbf{r}) = q\delta^3(\mathbf{r} - \mathbf{r}')}.$ Check: $\int \rho(\mathbf{r}) d\tau = q \int \delta^3(\mathbf{r} - \mathbf{r}') d\tau = q.$ ✓

(b) $\boxed{\rho(\mathbf{r}) = q\delta^3(\mathbf{r} - \mathbf{a}) - q\delta^3(\mathbf{r})}.$

(c) Evidently $\rho(r) = A\delta(r - R)$. To determine the constant A , we require

$$Q = \int \rho d\tau = \int A\delta(r - R) 4\pi r^2 dr = A 4\pi R^2. \quad \text{So } A = \frac{Q}{4\pi R^2}. \quad \boxed{\rho(r) = \frac{Q}{4\pi R^2} \delta(r - R)}.$$

Problem 1.48

(a) $a^2 + \mathbf{a} \cdot \mathbf{a} + a^2 = \boxed{3a^2}.$

(b) $\int (\mathbf{r} - \mathbf{b})^2 \frac{1}{53} \delta^3(\mathbf{r}) d\tau = \frac{1}{125} b^2 = \frac{1}{125} (4^2 + 3^2) = \boxed{\frac{1}{5}}.$

(c) $c^2 = 25 + 9 + 4 = 38 > 36 = 6^2$, so $\mathbf{c}$ is outside $\mathcal{V}$, so the integral is $\boxed{\text{zero}}.$

(d) $(\mathbf{e} - (2\hat{\mathbf{x}} + 2\hat{\mathbf{y}} + 2\hat{\mathbf{z}}))^2 = (1\hat{\mathbf{x}} + 0\hat{\mathbf{y}} + (-1)\hat{\mathbf{z}})^2 = 1 + 1 = 2 < (1.5)^2 = 2.25$, so $\mathbf{e}$ is inside $\mathcal{V}$,
and hence the integral is $\mathbf{e} \cdot (\mathbf{d} - \mathbf{e}) = (3, 2, 1) \cdot (-2, 0, 2) = -6 + 0 + 2 = \boxed{-4}.$

Problem 1.49

First method: use Eq. 1.99 to write $J = \int e^{-r} (4\pi \delta^3(\mathbf{r})) d\tau = 4\pi e^{-0} = \boxed{4\pi}.$

Second method: integrating by parts (use Eq. 1.59).

$$\begin{aligned}
J &= - \int_V \frac{\hat{\mathbf{r}}}{r^2} \cdot \nabla(e^{-r}) d\tau + \oint_S e^{-r} \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{a}. \quad \text{But} \quad \nabla(e^{-r}) = \left(\frac{\partial}{\partial r} e^{-r} \right) \hat{\mathbf{r}} = -e^{-r} \hat{\mathbf{r}}. \\
&= \int \frac{1}{r^2} e^{-r} 4\pi r^2 dr + \int e^{-r} \frac{\hat{\mathbf{r}}}{r^2} \cdot r^2 \sin\theta d\theta d\phi \hat{\mathbf{r}} = 4\pi \int_0^R e^{-r} dr + e^{-R} \int \sin\theta d\theta d\phi \\
&= 4\pi (-e^{-r}) \Big|_0^R + 4\pi e^{-R} = 4\pi (-e^{-R} + e^{-0}) + 4\pi e^{-R} = 4\pi \checkmark \quad (\text{Here } R = \infty, \text{ so } e^{-R} = 0.)
\end{aligned}$$

Problem 1.50 (a) $\nabla \cdot \mathbf{F}_1 = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(x^2) = \boxed{0}$; $\nabla \cdot \mathbf{F}_2 = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = \boxed{3}$

$$\nabla \times \mathbf{F}_1 = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & x^2 \end{vmatrix} = -\hat{\mathbf{y}} \frac{\partial}{\partial x}(x^2) = \boxed{-2x\hat{\mathbf{y}}}; \quad \nabla \times \mathbf{F}_2 = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \boxed{\mathbf{0}}$$

$\mathbf{F}_2$ is a gradient; $\mathbf{F}_1$ is a curl $\left[U_2 = \frac{1}{2}(x^3 + y^2 + z^2) \right]$ would do ($\mathbf{F}_2 = \nabla U_2$).

For $\mathbf{A}_1$, we want $\left(\frac{\partial A_y}{\partial z} - \frac{\partial A_z}{\partial y} \right) = \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) = 0$; $\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} = x^2$. $A_y = \frac{x^3}{3}$, $A_x = A_z = 0$ would do it.

$\mathbf{A}_1 = \frac{1}{3}x^2 \hat{\mathbf{y}}$ ($\mathbf{F}_1 = \nabla \times \mathbf{A}_1$). (But these are not unique.)

$$\text{(b) } \nabla \cdot \mathbf{F}_3 = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy) = 0; \quad \nabla \times \mathbf{F}_3 = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix} = \hat{\mathbf{x}}(x - x) + \hat{\mathbf{y}}(y - y) + \hat{\mathbf{z}}(z - z) = \mathbf{0}.$$

So $\mathbf{F}_3$ can be written as the gradient of a scalar ($\mathbf{F}_3 = \nabla U_3$) and as the curl of a vector ($\mathbf{F}_3 = \nabla \times \mathbf{A}_3$). In fact, $\boxed{U_3 = xyz}$ does the job. For the vector potential, we have

$$\left\{ \begin{array}{ll} \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} = yz, & \text{which suggests } A_z = \frac{1}{4}y^2z + f(x, z); \quad A_y = -\frac{1}{4}yz^2 + g(x, y) \\ \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} = xz, & \text{suggesting } A_x = \frac{1}{4}z^2x + h(x, y); \quad A_z = -\frac{1}{4}xz^2 + j(y, z) \\ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} = xy, & \text{so } A_y = \frac{1}{4}x^2y + k(y, z); \quad A_x = -\frac{1}{4}xy^2 + l(x, z) \end{array} \right\}$$

Putting this all together: $\mathbf{A}_3 = \frac{1}{4} \{ x(z^2 - y^2) \hat{\mathbf{x}} + y(x^2 - z^2) \hat{\mathbf{y}} + z(y^2 - x^2) \hat{\mathbf{z}} \}$ (again, not unique).

Problem 1.51

(d) $\Rightarrow$ (a): $\nabla \times \mathbf{F} = \nabla \times (-\nabla U) = \mathbf{0}$ (Eq. 1.44 – curl of gradient is always zero).

(a) $\Rightarrow$ (c): $\oint \mathbf{F} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{F}) \cdot d\mathbf{a} = 0$ (Eq. 1.57–Stokes' theorem).

(c) $\Rightarrow$ (b): $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} + \int_{\mathbf{b}}^{\mathbf{a}} \mathbf{F} \cdot d\mathbf{l} = \oint \mathbf{F} \cdot d\mathbf{l} = 0$, so

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l} = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l}.$$

(b) $\Rightarrow$ (c): same as (c) $\Rightarrow$ (b), only in reverse; (c) $\Rightarrow$ (a): same as (a) $\Rightarrow$ (c).

Problem 1.52

(d) $\Rightarrow$ (a): $\nabla \cdot \mathbf{F} = \nabla \cdot (\nabla \times \mathbf{W}) = 0$ (Eq 1.46—divergence of curl is always zero).

(a) $\Rightarrow$ (c): $\oint \mathbf{F} \cdot d\mathbf{a} = \int (\nabla \cdot \mathbf{F}) d\tau = 0$ (Eq. 1.56—divergence theorem).

(c) $\Rightarrow$ (b): $\int_I \mathbf{F} \cdot d\mathbf{a} - \int_{II} \mathbf{F} \cdot d\mathbf{a} = \oint \mathbf{F} \cdot d\mathbf{a} = 0$, so

$$\int_I \mathbf{F} \cdot d\mathbf{a} = \int_{II} \mathbf{F} \cdot d\mathbf{a}.$$

(Note: sign change because for $\oint \mathbf{F} \cdot d\mathbf{a}$, $d\mathbf{a}$ is *outward*, whereas for surface II it is *inward*.)

(b) $\Rightarrow$ (c): same as (c) $\Rightarrow$ (b), in reverse; (c) $\Rightarrow$ (a): same as (a) $\Rightarrow$ (c) .

Problem 1.53

In Prob. 1.15 we found that $\nabla \cdot \mathbf{v}_a = 0$; in Prob. 1.18 we found that $\nabla \times \mathbf{v}_c = \mathbf{0}$. So

$\mathbf{v}_c$ can be written as the gradient of a scalar; $\mathbf{v}_a$ can be written as the curl of a vector.

(a) To find t :

$$(1) \quad \frac{\partial t}{\partial x} = y^2 \Rightarrow t = y^2 x + f(y, z)$$

$$(2) \quad \frac{\partial t}{\partial y} = (2xy + z^2)$$

$$(3) \quad \frac{\partial t}{\partial z} = 2yz$$

From (1) & (3) we get $\frac{\partial f}{\partial z} = 2yz \Rightarrow f = yz^2 + g(y) \Rightarrow t = y^2 x + yz^2 + g(y)$, so $\frac{\partial t}{\partial y} = 2xy + z^2 + \frac{\partial g}{\partial y} = 2xy + z^2$ (from (2)) $\Rightarrow \frac{\partial g}{\partial y} = 0$. We may as well pick $g = 0$; then $t = xy^2 + yz^2$.

(b) To find $\mathbf{W}$: $\frac{\partial W_z}{\partial y} - \frac{\partial W_y}{\partial z} = x^2$; $\frac{\partial W_x}{\partial z} - \frac{\partial W_z}{\partial x} = 3z^2 x$; $\frac{\partial W_y}{\partial x} - \frac{\partial W_x}{\partial y} = -2xz$.

Pick $W_x = 0$; then

$$\frac{\partial W_z}{\partial x} = -3xz^2 \Rightarrow W_z = -\frac{3}{2}x^2 z^2 + f(y, z)$$

$$\frac{\partial W_y}{\partial x} = -2xz \Rightarrow W_y = -x^2 z + g(y, z).$$

$\frac{\partial W_z}{\partial y} - \frac{\partial W_y}{\partial z} = \frac{\partial f}{\partial y} + x^2 - \frac{\partial g}{\partial z} = x^2 \Rightarrow \frac{\partial f}{\partial y} - \frac{\partial g}{\partial z} = 0$. May as well pick $f = g = 0$.

$$\mathbf{W} = -x^2 z \hat{\mathbf{y}} - \frac{3}{2} x^2 z^2 \hat{\mathbf{z}}.$$

$$\text{Check: } \nabla \times \mathbf{W} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & -x^2 z & -\frac{3}{2} x^2 z^2 \end{vmatrix} = \hat{\mathbf{x}} (x^2) + \hat{\mathbf{y}} (3xz^2) + \hat{\mathbf{z}} (-2xz). \checkmark$$

You can add any gradient (∇t) to $\mathbf{W}$ without changing its curl, so this answer is far from unique. Some other solutions:

$$\mathbf{W} = xz^3 \hat{\mathbf{x}} - x^2 z \hat{\mathbf{y}};$$

$$\mathbf{W} = (2xyz + xz^3) \hat{\mathbf{x}} + x^2 y \hat{\mathbf{z}};$$

$$\mathbf{W} = xyz \hat{\mathbf{x}} - \frac{1}{2} x^2 z \hat{\mathbf{y}} + \frac{1}{2} x^2 (y - 3z^2) \hat{\mathbf{z}}.$$

Problem 1.54

$$\begin{aligned}
 \nabla \cdot \mathbf{v} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r^2 \cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta r^2 \cos \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (-r^2 \cos \theta \sin \phi) \\
 &= \frac{1}{r^2} 4r^3 \cos \theta + \frac{1}{r \sin \theta} \cos \theta r^2 \cos \phi + \frac{1}{r \sin \theta} (-r^2 \cos \theta \cos \phi) \\
 &= \frac{r \cos \theta}{\sin \theta} [4 \sin \theta + \cos \phi - \cos \phi] = 4r \cos \theta.
 \end{aligned}$$

$$\begin{aligned}
 \int (\nabla \cdot \mathbf{v}) d\tau &= \int (4r \cos \theta) r^2 \sin \theta dr d\theta d\phi = 4 \int_0^R r^3 dr \int_0^{\pi/2} \cos \theta \sin \theta d\theta \int_0^{\pi/2} d\phi \\
 &= (R^4) \left(\frac{1}{2} \right) \left(\frac{\pi}{2} \right) = \boxed{\frac{\pi R^4}{4}}.
 \end{aligned}$$

Surface consists of four parts:

(1) *Curved*: $d\mathbf{a} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}$; $r = R$. $\mathbf{v} \cdot d\mathbf{a} = (R^2 \cos \theta) (R^2 \sin \theta d\theta d\phi)$.

$$\int \mathbf{v} \cdot d\mathbf{a} = R^4 \int_0^{\pi/2} \cos \theta \sin \theta d\theta \int_0^{\pi/2} d\phi = R^4 \left(\frac{1}{2} \right) \left(\frac{\pi}{2} \right) = \frac{\pi R^4}{4}.$$

(2) *Left*: $d\mathbf{a} = -r dr d\theta \hat{\phi}$; $\phi = 0$. $\mathbf{v} \cdot d\mathbf{a} = (r^2 \cos \theta \sin \phi) (r dr d\theta) = 0$. $\int \mathbf{v} \cdot d\mathbf{a} = 0$.

(3) *Back*: $d\mathbf{a} = r dr d\theta \hat{\phi}$; $\phi = \pi/2$. $\mathbf{v} \cdot d\mathbf{a} = (-r^2 \cos \theta \sin \phi) (r dr d\theta) = -r^3 \cos \theta dr d\theta$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^R r^3 dr \int_0^{\pi/2} \cos \theta d\theta = -\left(\frac{1}{4} R^4 \right) (+1) = -\frac{1}{4} R^4.$$

(4) *Bottom*: $d\mathbf{a} = r \sin \theta dr d\phi \hat{\theta}$; $\theta = \pi/2$. $\mathbf{v} \cdot d\mathbf{a} = (r^2 \cos \phi) (r dr d\phi)$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^R r^3 dr \int_0^{\pi/2} \cos \phi d\phi = \frac{1}{4} R^4.$$

Total: $\oint \mathbf{v} \cdot d\mathbf{a} = \pi R^4/4 + 0 - \frac{1}{4} R^4 + \frac{1}{4} R^4 = \frac{\pi R^4}{4}$. $\checkmark$

Problem 1.55

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ay & bx & 0 \end{vmatrix} = \hat{\mathbf{z}}(b - a). \quad \text{So} \quad \int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = (b - a)\pi R^2.$$

$\mathbf{v} \cdot d\mathbf{l} = (ay \hat{\mathbf{x}} + bx \hat{\mathbf{y}}) \cdot (dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}) = ay dx + bx dy$; $x^2 + y^2 = R^2 \Rightarrow 2x dx + 2y dy = 0$, so $dy = -(x/y) dx$. So $\mathbf{v} \cdot d\mathbf{l} = ay dx + bx(-x/y) dx = \frac{1}{y} (ay^2 - bx^2) dx$.

For the “upper” semicircle, $y = \sqrt{R^2 - x^2}$, so $\mathbf{v} \cdot d\mathbf{l} = \frac{a(R^2 - x^2) - bx^2}{\sqrt{R^2 - x^2}} dx$.

$$\begin{aligned}\oint \mathbf{v} \cdot d\mathbf{l} &= \int_R^{-R} \frac{aR^2 - (a+b)x^2}{\sqrt{R^2 - x^2}} dx = \left\{ aR^2 \sin^{-1}\left(\frac{x}{R}\right) - (a+b) \left[-\frac{x}{2} \sqrt{R^2 - x^2} + \frac{R^2}{2} \sin^{-1}\left(\frac{x}{R}\right) \right] \right\} \Big|_{+R}^{-R} \\ &= \frac{1}{2} R^2 (a-b) \sin^{-1}(x/R) \Big|_{+R}^{-R} = \frac{1}{2} R^2 (a-b) (\sin^{-1}(-1) - \sin^{-1}(+1)) = \frac{1}{2} R^2 (a-b) \left(-\frac{\pi}{2} - \frac{\pi}{2} \right) \\ &= \frac{1}{2} \pi R^2 (b-a).\end{aligned}$$

And the same for the lower semicircle (y changes sign, but the limits on the integral are reversed) so

$$\oint \mathbf{v} \cdot d\mathbf{l} = \pi R^2 (b-a). \quad \checkmark$$

Problem 1.56

$$(1) \ x = z = 0; \ dx = dz = 0; \ y : 0 \rightarrow 1. \quad \mathbf{v} \cdot d\mathbf{l} = (yz^2) dy = 0; \quad \int \mathbf{v} \cdot d\mathbf{l} = 0.$$

$$(2) \ x = 0; \ z = 2 - 2y; \ dz = -2 dy; \ y : 1 \rightarrow 0. \quad \mathbf{v} \cdot d\mathbf{l} = (yz^2) dy + (3y + z) dz = y(2 - 2y)^2 dy - (3y + 2 - 2y) 2 dy;$$

$$\int \mathbf{v} \cdot d\mathbf{l} = 2 \int_1^0 (2y^3 - 4y^2 + y - 2) dy = 2 \left(\frac{y^4}{2} - \frac{4y^3}{3} + \frac{y^2}{2} - 2y \right) \Big|_1^0 = \frac{14}{3}.$$

$$(3) \ x = y = 0; \ dx = dy = 0; \ z : 2 \rightarrow 0. \quad \mathbf{v} \cdot d\mathbf{l} = (3y + z) dz = z dz;$$

$$\int \mathbf{v} \cdot d\mathbf{l} = \int_2^0 z dz = \frac{z^2}{2} \Big|_2^0 = -2.$$

$$\text{Total: } \oint \mathbf{v} \cdot d\mathbf{l} = 0 + \frac{14}{3} - 2 = \boxed{\frac{8}{3}}.$$

Meanwhile, Stokes' theorem says $\oint \mathbf{v} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$. Here $d\mathbf{a} = dy dz \hat{\mathbf{x}}$, so all we need is $(\nabla \times \mathbf{v})_x = \frac{\partial}{\partial y}(3y + z) - \frac{\partial}{\partial z}(yz^2) = 3 - 2yz$. Therefore

$$\begin{aligned}\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} &= \int \int (3 - 2yz) dy dz = \int_0^1 \left[\int_0^{2-2y} (3 - 2yz) dz \right] dy \\ &= \int_0^1 \left[3(2 - 2y) - 2y \frac{1}{2} (2 - 2y)^2 \right] dy = \int_0^1 (-4y^3 + 8y^2 - 10y + 6) dy \\ &= \left(-y^4 + \frac{8}{3}y^3 - 5y^2 + 6y \right) \Big|_0^1 = -1 + \frac{8}{3} - 5 + 6 = \frac{8}{3}. \quad \checkmark\end{aligned}$$

Problem 1.57

Start at the origin.

$$(1) \ \theta = \frac{\pi}{2}, \ \phi = 0; \ r : 0 \rightarrow 1. \quad \mathbf{v} \cdot d\mathbf{l} = (r \cos^2 \theta) (dr) = 0. \quad \int \mathbf{v} \cdot d\mathbf{l} = 0.$$

$$(2) \ r = 1, \ \theta = \frac{\pi}{2}; \ \phi : 0 \rightarrow \pi/2. \quad \mathbf{v} \cdot d\mathbf{l} = (3r)(r \sin \theta d\phi) = 3 d\phi. \quad \int \mathbf{v} \cdot d\mathbf{l} = 3 \int_0^{\pi/2} d\phi = \frac{3\pi}{2}.$$

$$(3) \quad \phi = \frac{\pi}{2}; \quad r \sin \theta = y = 1, \text{ so } r = \frac{1}{\sin \theta}, \quad dr = \frac{-1}{\sin^2 \theta} \cos \theta d\theta, \quad \theta : \frac{\pi}{2} \rightarrow \theta_0 \equiv \tan^{-1}(1/2).$$

$$\begin{aligned} \mathbf{v} \cdot d\mathbf{l} &= (r \cos^2 \theta) (dr) - (r \cos \theta \sin \theta)(r d\theta) = \frac{\cos^2 \theta}{\sin \theta} \left(-\frac{\cos \theta}{\sin^2 \theta} \right) d\theta - \frac{\cos \theta \sin \theta}{\sin^2 \theta} d\theta \\ &= -\left(\frac{\cos^3 \theta}{\sin^3 \theta} + \frac{\cos \theta}{\sin \theta} \right) d\theta = -\frac{\cos \theta}{\sin \theta} \left(\frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \theta} \right) d\theta = -\frac{\cos \theta}{\sin^3 \theta} d\theta. \end{aligned}$$

Therefore

$$\int \mathbf{v} \cdot d\mathbf{l} = - \int_{\pi/2}^{\theta_0} \frac{\cos \theta}{\sin^3 \theta} d\theta = \frac{1}{2 \sin^2 \theta} \Big|_{\pi/2}^{\theta_0} = \frac{1}{2 \cdot (1/5)} - \frac{1}{2 \cdot (1)} = \frac{5}{2} - \frac{1}{2} = 2.$$

$$(4) \quad \theta = \theta_0, \quad \phi = \frac{\pi}{2}; \quad r : \sqrt{5} \rightarrow 0. \quad \mathbf{v} \cdot d\mathbf{l} = (r \cos^2 \theta) (dr) = \frac{4}{5} r dr.$$

$$\int \mathbf{v} \cdot d\mathbf{l} = \frac{4}{5} \int_{\sqrt{5}}^0 r dr = \frac{4}{5} \frac{r^2}{2} \Big|_{\sqrt{5}}^0 = -\frac{4}{5} \cdot \frac{5}{2} = -2.$$

Total:

$$\oint \mathbf{v} \cdot d\mathbf{l} = 0 + \frac{3\pi}{2} + 2 - 2 = \boxed{\frac{3\pi}{2}}.$$

Stokes' theorem says this should equal $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$

$$\begin{aligned} \nabla \times \mathbf{v} &= \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta 3r) - \frac{\partial}{\partial \phi} (-r \sin \theta \cos \theta) \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \phi} (r \cos^2 \theta) - \frac{\partial}{\partial r} (r 3r) \right] \hat{\boldsymbol{\theta}} \\ &\quad + \frac{1}{r} \left[\frac{\partial}{\partial r} (-rr \cos \theta \sin \theta) - \frac{\partial}{\partial \theta} (r \cos^2 \theta) \right] \hat{\boldsymbol{\phi}} \\ &= \frac{1}{r \sin \theta} [3r \cos \theta] \hat{\mathbf{r}} + \frac{1}{r} [-6r] \hat{\boldsymbol{\theta}} + \frac{1}{r} [-2r \cos \theta \sin \theta + 2r \cos \theta \sin \theta] \hat{\boldsymbol{\phi}} \\ &= 3 \cot \theta \hat{\mathbf{r}} - 6 \hat{\boldsymbol{\theta}}. \end{aligned}$$

$$(1) \quad \text{Back face: } d\mathbf{a} = -r dr d\theta \hat{\boldsymbol{\phi}}; \quad (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0. \quad \int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0.$$

$$(2) \quad \text{Bottom: } d\mathbf{a} = -r \sin \theta dr d\phi \hat{\boldsymbol{\theta}}; \quad (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 6r \sin \theta dr d\phi. \quad \theta = \frac{\pi}{2}, \text{ so } (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 6r dr d\phi$$

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^1 6r dr \int_0^{\pi/2} d\phi = 6 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{2}. \quad \checkmark$$

Problem 1.58

$$\mathbf{v} \cdot d\mathbf{l} = y dz.$$

$$(1) \quad \text{Left side: } z = a - x; \quad dz = -dx; \quad y = 0. \quad \text{Therefore } \int \mathbf{v} \cdot d\mathbf{l} = 0.$$

$$(2) \quad \text{Bottom: } dz = 0. \quad \text{Therefore } \int \mathbf{v} \cdot d\mathbf{l} = 0.$$

$$(3) \text{ Back: } z = a - \frac{1}{2}y; \quad dz = -1/2 dy; \quad y : 2a \rightarrow 0. \quad \int \mathbf{v} \cdot d\mathbf{l} = \int_{2a}^0 y \left(-\frac{1}{2} dy\right) = -\frac{1}{2} \frac{y^2}{2} \Big|_{2a}^0 = \frac{4a^2}{4} = \boxed{a^2}.$$

Meanwhile, $\nabla \times \mathbf{v} = \hat{\mathbf{x}}$, so $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ is the projection of this surface on the xy plane $= \frac{1}{2} \cdot a \cdot 2a = a^2$. ✓

Problem 1.59

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r^2 \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta 4r^2 \cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (r^2 \tan \theta) \\ &= \frac{1}{r^2} 4r^3 \sin \theta + \frac{1}{r \sin \theta} 4r^2 (\cos^2 \theta - \sin^2 \theta) = \frac{4r}{\sin \theta} (\sin^2 \theta + \cos^2 \theta - \sin^2 \theta) \\ &= 4r \frac{\cos^2 \theta}{\sin \theta}. \end{aligned}$$

$$\begin{aligned} \int (\nabla \cdot \mathbf{v}) d\tau &= \int \left(4r \frac{\cos^2 \theta}{\sin \theta} \right) (r^2 \sin \theta dr d\theta d\phi) = \int_0^R 4r^3 dr \int_0^{\pi/6} \cos^2 \theta d\theta \int_0^{2\pi} d\phi = (R^4) (2\pi) \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right] \Big|_0^{\pi/6} \\ &= 2\pi R^4 \left(\frac{\pi}{12} + \frac{\sin 60^\circ}{4} \right) = \frac{\pi R^4}{6} \left(\pi + 3 \frac{\sqrt{3}}{2} \right) = \boxed{\frac{\pi R^4}{12} (2\pi + 3\sqrt{3})}. \end{aligned}$$

Surface consists of two parts:

$$(1) \text{ The ice cream: } r = R; \quad \phi : 0 \rightarrow 2\pi; \quad \theta : 0 \rightarrow \pi/6; \quad d\mathbf{a} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}; \quad \mathbf{v} \cdot d\mathbf{a} = (R^2 \sin \theta) (R^2 \sin \theta d\theta d\phi) = R^4 \sin^2 \theta d\theta d\phi.$$

$$\int \mathbf{v} \cdot d\mathbf{a} = R^4 \int_0^{\pi/6} \sin^2 \theta d\theta \int_0^{2\pi} d\phi = (R^4) (2\pi) \left[\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right]_0^{\pi/6} = 2\pi R^4 \left(\frac{\pi}{12} - \frac{1}{4} \sin 60^\circ \right) = \frac{\pi R^4}{6} \left(\pi - 3 \frac{\sqrt{3}}{2} \right)$$

$$(2) \text{ The cone: } \theta = \frac{\pi}{6}; \quad \phi : 0 \rightarrow 2\pi; \quad r : 0 \rightarrow R; \quad d\mathbf{a} = r \sin \theta d\phi dr \hat{\boldsymbol{\theta}} = \frac{\sqrt{3}}{2} r dr d\phi \hat{\boldsymbol{\theta}}; \quad \mathbf{v} \cdot d\mathbf{a} = \sqrt{3} r^3 dr d\phi$$

$$\int \mathbf{v} \cdot d\mathbf{a} = \sqrt{3} \int_0^R r^3 dr \int_0^{2\pi} d\phi = \sqrt{3} \cdot \frac{R^4}{4} \cdot 2\pi = \frac{\sqrt{3}}{2} \pi R^4.$$

$$\text{Therefore } \int \mathbf{v} \cdot d\mathbf{a} = \frac{\pi R^4}{2} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} + \sqrt{3} \right) = \frac{\pi R^4}{12} (2\pi + 3\sqrt{3}). \quad \checkmark.$$

Problem 1.60

(a) Corollary 2 says $\oint (\nabla T) \cdot d\mathbf{l} = 0$. Stokes' theorem says $\oint (\nabla T) \cdot d\mathbf{l} = \int [\nabla \times (\nabla T)] \cdot d\mathbf{a}$. So $\int [\nabla \times (\nabla T)] \cdot d\mathbf{a} = 0$, and since this is true for *any* surface, the integrand must vanish: $\nabla \times (\nabla T) = \mathbf{0}$, confirming Eq. 1.44.

(b) Corollary 2 says $\oint (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$. Divergence theorem says $\oint (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int \nabla \cdot (\nabla \times \mathbf{v}) d\tau$. So $\int \nabla \cdot (\nabla \times \mathbf{v}) d\tau = 0$, and since this is true for *any* volume, the integrand must vanish: $\nabla \cdot (\nabla \times \mathbf{v}) = 0$, confirming Eq. 1.46.

Problem 1.61

(a) Divergence theorem: $\oint \mathbf{v} \cdot d\mathbf{a} = \int (\nabla \cdot \mathbf{v}) d\tau$. Let $\mathbf{v} = \mathbf{c}T$, where $\mathbf{c}$ is a constant vector. Using product rule #5 in front cover: $\nabla \cdot \mathbf{v} = \nabla \cdot (\mathbf{c}T) = T(\nabla \cdot \mathbf{c}) + \mathbf{c} \cdot (\nabla T)$. But $\mathbf{c}$ is constant so $\nabla \cdot \mathbf{c} = 0$. Therefore we have: $\int \mathbf{c} \cdot (\nabla T) d\tau = \int T \mathbf{c} \cdot d\mathbf{a}$. Since $\mathbf{c}$ is constant, take it outside the integrals: $\mathbf{c} \cdot \int \nabla T d\tau = \mathbf{c} \cdot \int T d\mathbf{a}$. But $\mathbf{c}$

is *any* constant vector—in particular, it could be $\hat{\mathbf{x}}$, or $\hat{\mathbf{y}}$, or $\hat{\mathbf{z}}$ —so each *component* of the integral on left equals corresponding component on the right, and hence

$$\int \nabla T d\tau = \int T d\mathbf{a}. \quad \text{qed}$$

(b) Let $\mathbf{v} \rightarrow (\mathbf{v} \times \mathbf{c})$ in divergence theorem. Then $\int \nabla \cdot (\mathbf{v} \times \mathbf{c}) d\tau = \int (\mathbf{v} \times \mathbf{c}) \cdot d\mathbf{a}$. Product rule #6 $\Rightarrow \nabla \cdot (\mathbf{v} \times \mathbf{c}) = \mathbf{c} \cdot (\nabla \times \mathbf{v}) - \mathbf{v} \cdot (\nabla \times \mathbf{c}) = \mathbf{c} \cdot (\nabla \times \mathbf{v})$. (Note: $\nabla \times \mathbf{c} = \mathbf{0}$, since $\mathbf{c}$ is constant.) Meanwhile vector identity (1) says $d\mathbf{a} \cdot (\mathbf{v} \times \mathbf{c}) = \mathbf{c} \cdot (d\mathbf{a} \times \mathbf{v}) = -\mathbf{c} \cdot (\mathbf{v} \times d\mathbf{a})$. Thus $\int \mathbf{c} \cdot (\nabla \times \mathbf{v}) d\tau = -\int \mathbf{c} \cdot (\mathbf{v} \times d\mathbf{a})$. Take $\mathbf{c}$ outside, and again let $\mathbf{c}$ be $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ then:

$$\int (\nabla \times \mathbf{v}) d\tau = -\int \mathbf{v} \times d\mathbf{a}. \quad \text{qed}$$

(c) Let $\mathbf{v} = T\nabla U$ in divergence theorem: $\int \nabla \cdot (T\nabla U) d\tau = \int T\nabla U \cdot d\mathbf{a}$. Product rule #5 $\Rightarrow \nabla \cdot (T\nabla U) = T\nabla \cdot (\nabla U) + (\nabla U) \cdot (\nabla T) = T\nabla^2 U + (\nabla U) \cdot (\nabla T)$. Therefore

$$\int (T\nabla^2 U + (\nabla U) \cdot (\nabla T)) d\tau = \int (T\nabla U) \cdot d\mathbf{a}. \quad \text{qed}$$

(d) Rewrite (c) with $T \leftrightarrow U$: $\int (U\nabla^2 T + (\nabla T) \cdot (\nabla U)) d\tau = \int (U\nabla T) \cdot d\mathbf{a}$. Subtract this from (c), noting that the $(\nabla U) \cdot (\nabla T)$ terms cancel:

$$\int (T\nabla^2 U - U\nabla^2 T) d\tau = \int (T\nabla U - U\nabla T) \cdot d\mathbf{a}. \quad \text{qed}$$

(e) Stokes' theorem: $\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint \mathbf{v} \cdot d\mathbf{l}$. Let $\mathbf{v} = \mathbf{c}T$. By Product Rule #7: $\nabla \times (\mathbf{c}T) = T(\nabla \times \mathbf{c}) - \mathbf{c} \times (\nabla T) = -\mathbf{c} \times (\nabla T)$ (since $\mathbf{c}$ is constant). Therefore, $-\int (\mathbf{c} \times (\nabla T)) \cdot d\mathbf{a} = \oint T\mathbf{c} \cdot d\mathbf{l}$. Use vector identity #1 to rewrite the first term $(\mathbf{c} \times (\nabla T)) \cdot d\mathbf{a} = \mathbf{c} \cdot (\nabla T \times d\mathbf{a})$. So $-\int \mathbf{c} \cdot (\nabla T \times d\mathbf{a}) = \oint \mathbf{c} \cdot T d\mathbf{l}$. Pull $\mathbf{c}$ outside, and let $\mathbf{c} \rightarrow \hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ to prove:

$$\int \nabla T \times d\mathbf{a} = -\oint T d\mathbf{l}. \quad \text{qed}$$

Problem 1.62

(a) $d\mathbf{a} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}$. Let the surface be the northern hemisphere. The $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components clearly integrate to zero, and the $\hat{\mathbf{z}}$ component of $\hat{\mathbf{r}}$ is $\cos \theta$, so

$$\mathbf{a} = \int R^2 \sin \theta \cos \theta d\theta d\phi \hat{\mathbf{z}} = 2\pi R^2 \hat{\mathbf{z}} \int_0^{\pi/2} \sin \theta \cos \theta d\theta = 2\pi R^2 \hat{\mathbf{z}} \frac{\sin^2 \theta}{2} \Big|_0^{\pi/2} = \boxed{\pi R^2 \hat{\mathbf{z}}}.$$

(b) Let $T = 1$ in Prob. 1.61(a). Then $\nabla T = 0$, so $\oint d\mathbf{a} = 0$. qed

(c) This follows from (b). For suppose $\mathbf{a}_1 \neq \mathbf{a}_2$; then if you put them together to make a closed surface, $\oint d\mathbf{a} = \mathbf{a}_1 - \mathbf{a}_2 \neq 0$.

(d) For one such triangle, $d\mathbf{a} = \frac{1}{2}(\mathbf{r} \times d\mathbf{l})$ (since $\mathbf{r} \times d\mathbf{l}$ is the area of the parallelogram, and the direction is perpendicular to the surface), so for the entire conical surface, $\mathbf{a} = \frac{1}{2} \oint \mathbf{r} \times d\mathbf{l}$.

(e) Let $T = \mathbf{c} \cdot \mathbf{r}$, and use product rule #4: $\nabla T = \nabla(\mathbf{c} \cdot \mathbf{r}) = \mathbf{c} \times (\nabla \times \mathbf{r}) + (\mathbf{c} \cdot \nabla)\mathbf{r}$. But $\nabla \times \mathbf{r} = 0$, and $(\mathbf{c} \cdot \nabla)\mathbf{r} = (c_x \frac{\partial}{\partial x} + c_y \frac{\partial}{\partial y} + c_z \frac{\partial}{\partial z})(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}) = c_x \hat{\mathbf{x}} + c_y \hat{\mathbf{y}} + c_z \hat{\mathbf{z}} = \mathbf{c}$. So Prob. 1.61(e) says

$$\oint T d\mathbf{l} = \oint (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l} = -\int (\nabla T) \times d\mathbf{a} = -\int \mathbf{c} \times d\mathbf{a} = -\mathbf{c} \times \int d\mathbf{a} = -\mathbf{c} \times \mathbf{a} = \mathbf{a} \times \mathbf{c}. \quad \text{qed}$$

Problem 1.63

(1)

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \frac{1}{r} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} (r) = \boxed{\frac{1}{r^2}}.$$

For a sphere of radius R :

$$\left. \begin{aligned} \int \mathbf{v} \cdot d\mathbf{a} &= \int \left(\frac{1}{R} \hat{\mathbf{r}} \right) \cdot (R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) = R \int \sin \theta d\theta d\phi = 4\pi R. \\ \int (\nabla \cdot \mathbf{v}) d\tau &= \int \left(\frac{1}{r^2} \right) (r^2 \sin \theta dr d\theta d\phi) = \left(\int_0^R dr \right) (\int \sin \theta d\theta d\phi) = 4\pi R. \end{aligned} \right\} \begin{array}{l} \text{So divergence} \\ \text{theorem checks.} \end{array}$$

Evidently there is *no* delta function at the origin.

$$\nabla \times (r^n \hat{\mathbf{r}}) = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 r^n) = \frac{1}{r^2} \frac{\partial}{\partial r} (r^{n+2}) = \frac{1}{r^2} (n+2) r^{n+1} = \boxed{(n+2) r^{n-1}}$$

(except for $n = -2$, for which we already know (Eq. 1.99) that the divergence is $4\pi\delta^3(\mathbf{r})$).

- (2) *Geometrically*, it should be zero. Likewise, the curl in the spherical coordinates obviously gives zero. To be certain there is no lurking delta function here, we integrate over a sphere of radius R , using Prob. 1.61(b): If $\nabla \times (r^n \hat{\mathbf{r}}) = \mathbf{0}$, then $\int (\nabla \times \mathbf{v}) d\tau = \mathbf{0} \stackrel{?}{=} -\oint \mathbf{v} \times d\mathbf{a}$. But $\mathbf{v} = r^n \hat{\mathbf{r}}$ and $d\mathbf{a} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}$ are both in the $\hat{\mathbf{r}}$ directions, so $\mathbf{v} \times d\mathbf{a} = \mathbf{0}$. ✓

Problem 1.64

(a) Since the argument is not a function of angle, Eq. 1.73 says

$$\begin{aligned} D &= -\frac{1}{4\pi} \frac{1}{r^2} \frac{d}{dr} \left[r^2 \left(-\frac{1}{2} \right) \frac{2r}{(r^2 + \epsilon^2)^{3/2}} \right] = \frac{1}{4\pi r^2} \frac{d}{dr} \left[\frac{r^3}{(r^2 + \epsilon^2)^{3/2}} \right] \\ &= \frac{1}{4\pi r^2} \left[\frac{3r^2}{(r^2 + \epsilon^2)^{3/2}} - \frac{3}{2} \frac{r^3 2r}{(r^2 + \epsilon^2)^{5/2}} \right] = \frac{1}{4\pi r^2} \frac{3r^2}{(r^2 + \epsilon^2)^{5/2}} (r^2 + \epsilon^2 - r^2) = \frac{3\epsilon^2}{4\pi(r^2 + \epsilon^2)^{5/2}}. \quad \checkmark \end{aligned}$$

(b) Setting $r \rightarrow 0$:

$$D(0, \epsilon) = \frac{3\epsilon^2}{4\pi\epsilon^5} = \frac{3}{4\pi\epsilon^3},$$

which goes to infinity as $\epsilon \rightarrow 0$. ✓

(c) From (a) it is clear that $D(r, 0) = 0$ for $r \neq 0$. ✓

(d)

$$\int D(r, \epsilon) 4\pi r^2 dr = 3\epsilon^2 \int_0^\infty \frac{r^2}{(r^2 + \epsilon^2)^{5/2}} dr = 3\epsilon^2 \left(\frac{1}{3\epsilon^2} \right) = 1. \quad \checkmark$$

(I looked up the integral.) Note that (b), (c), and (d) are the defining conditions for $\delta^3(\mathbf{r})$.

Chapter 2

Electrostatics

Problem 2.1

(a) Zero.

(b) $F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2}$, where r is the distance from center to each numeral. $\mathbf{F}$ points *toward* the missing q .

Explanation: by superposition, this is equivalent to (a), with an extra $-q$ at 6 o'clock—since the force of all twelve is zero, the net force is that of $-q$ only.

(c) Zero.

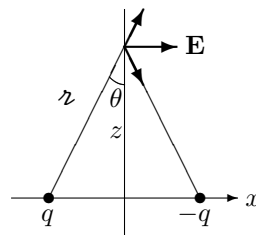
(d) $\frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2}$, pointing toward the missing q . Same reason as (b). Note, however, that if you explained (b) as a cancellation in pairs of opposite charges (1 o'clock against 7 o'clock; 2 against 8, etc.), with one unpaired q doing the job, then you'll need a *different* explanation for (d).

Problem 2.2

This time the “vertical” components cancel, leaving

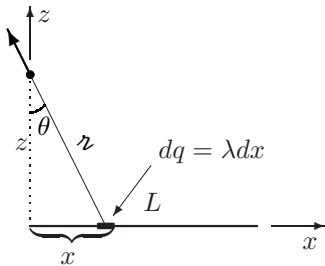
$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} 2 \frac{q}{z^2} \sin \theta \hat{\mathbf{x}}, \text{ or}$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{qd}{\left(z^2 + \left(\frac{d}{2}\right)^2\right)^{3/2}} \hat{\mathbf{x}}.$$



From far away, ($z \gg d$), the field goes like $\mathbf{E} \approx \frac{1}{4\pi\epsilon_0} \frac{qd}{z^3} \hat{\mathbf{z}}$, which, as we shall see, is the field of a *dipole*. (If we set $d \rightarrow 0$, we get $\mathbf{E} = \mathbf{0}$, as is appropriate; to the extent that this configuration looks like a single point charge from far away, the net charge is zero, so $\mathbf{E} \rightarrow \mathbf{0}$.)

Problem 2.3



$$\begin{aligned}
 E_z &= \frac{1}{4\pi\epsilon_0} \int_0^L \frac{\lambda dx}{r^2} \cos \theta; \quad (r^2 = z^2 + x^2; \cos \theta = \frac{z}{r}) \\
 &= \frac{1}{4\pi\epsilon_0} \lambda z \int_0^L \frac{1}{(z^2 + x^2)^{3/2}} dx \\
 &= \frac{1}{4\pi\epsilon_0} \lambda z \left[\frac{1}{z^2} \frac{x}{\sqrt{z^2 + x^2}} \right]_0^L = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{z} \frac{L}{\sqrt{z^2 + L^2}}. \\
 E_x &= -\frac{1}{4\pi\epsilon_0} \int_0^L \frac{\lambda dx}{r^2} \sin \theta = -\frac{1}{4\pi\epsilon_0} \lambda \int \frac{x dx}{(x^2 + z^2)^{3/2}} \\
 &= -\frac{1}{4\pi\epsilon_0} \lambda \left[-\frac{1}{\sqrt{x^2 + z^2}} \right]_0^L = -\frac{1}{4\pi\epsilon_0} \lambda \left[\frac{1}{z} - \frac{1}{\sqrt{z^2 + L^2}} \right].
 \end{aligned}$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{z} \left[\left(-1 + \frac{z}{\sqrt{z^2 + L^2}} \right) \hat{\mathbf{x}} + \left(\frac{L}{\sqrt{z^2 + L^2}} \right) \hat{\mathbf{z}} \right].$$

For $z \gg L$ you expect it to look like a point charge $q = \lambda L$: $\mathbf{E} \rightarrow \frac{1}{4\pi\epsilon_0} \frac{\lambda L}{z^2} \hat{\mathbf{z}}$. It checks, for with $z \gg L$ the $\hat{\mathbf{x}}$ term $\rightarrow 0$, and the $\hat{\mathbf{z}}$ term $\rightarrow \frac{1}{4\pi\epsilon_0} \frac{\lambda}{z} \frac{L}{z} \hat{\mathbf{z}}$.

Problem 2.4

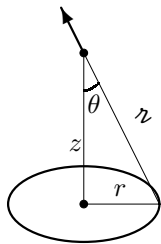
From Ex. 2.2, with $L \rightarrow \frac{a}{2}$ and $z \rightarrow \sqrt{z^2 + (\frac{a}{2})^2}$ (distance from center of edge to P), field of *one* edge is:

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{\lambda a}{\sqrt{z^2 + \frac{a^2}{4}} \sqrt{z^2 + \frac{a^2}{4} + \frac{a^2}{4}}}.$$

There are 4 sides, and we want vertical components only, so multiply by $4 \cos \theta = 4 \frac{z}{\sqrt{z^2 + \frac{a^2}{4}}}$:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{4\lambda a z}{\left(z^2 + \frac{a^2}{4}\right) \sqrt{z^2 + \frac{a^2}{2}}} \hat{\mathbf{z}}.$$

Problem 2.5



“Horizontal” components cancel, leaving: $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left\{ \int \frac{\lambda dl}{r^2} \cos \theta \right\} \hat{\mathbf{z}}$.
 Here, $r^2 = r^2 + z^2$, $\cos \theta = \frac{z}{r}$ (both constants), while $\int dl = 2\pi r$. So

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda(2\pi r)z}{(r^2 + z^2)^{3/2}} \hat{\mathbf{z}}.$$

Problem 2.6

Break it into rings of radius r , and thickness dr , and use Prob. 2.5 to express the field of each ring. Total charge of a ring is $\sigma \cdot 2\pi r \cdot dr = \lambda \cdot 2\pi r$, so $\lambda = \sigma dr$ is the “line charge” of each ring.

$$E_{\text{ring}} = \frac{1}{4\pi\epsilon_0} \frac{(\sigma dr) 2\pi r z}{(r^2 + z^2)^{3/2}}; \quad E_{\text{disk}} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma z \int_0^R \frac{r}{(r^2 + z^2)^{3/2}} dr.$$

$$\mathbf{E}_{\text{disk}} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma z \left[\frac{1}{z} - \frac{1}{\sqrt{R^2 + z^2}} \right] \hat{\mathbf{z}}.$$

For $R \gg z$ the second term $\rightarrow 0$, so $\mathbf{E}_{\text{plane}} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma \hat{\mathbf{z}} = \boxed{\frac{\sigma}{2\epsilon_0} \hat{\mathbf{z}}}$.

For $z \gg R$, $\frac{1}{\sqrt{R^2+z^2}} = \frac{1}{z} \left(1 + \frac{R^2}{z^2}\right)^{-1/2} \approx \frac{1}{z} \left(1 - \frac{1}{2} \frac{R^2}{z^2}\right)$, so $[\] \approx \frac{1}{z} - \frac{1}{z} + \frac{1}{2} \frac{R^2}{z^3} = \frac{R^2}{2z^3}$,
and $E = \frac{1}{4\pi\epsilon_0} \frac{2\pi R^2 \sigma}{2z^2} = \frac{1}{4\pi\epsilon_0} \frac{Q}{z^2}$, where $Q = \pi R^2 \sigma$. ✓

Problem 2.7

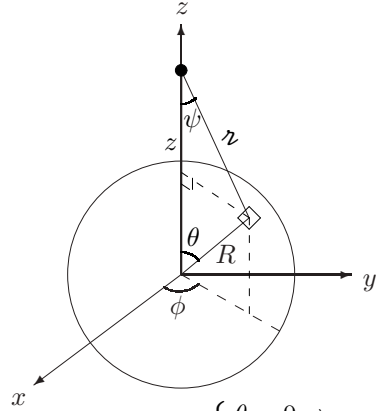
$\mathbf{E}$ is clearly in the z direction. From the diagram,

$$dq = \sigma da = \sigma R^2 \sin \theta d\theta d\phi,$$

$$r^2 = R^2 + z^2 - 2Rz \cos \theta,$$

$$\cos \psi = \frac{z - R \cos \theta}{r}.$$

So



$$\begin{aligned} E_z &= \frac{1}{4\pi\epsilon_0} \int \frac{\sigma R^2 \sin \theta d\theta d\phi (z - R \cos \theta)}{(R^2 + z^2 - 2Rz \cos \theta)^{3/2}}. \quad \int d\phi = 2\pi. \\ &= \frac{1}{4\pi\epsilon_0} (2\pi R^2 \sigma) \int_0^\pi \frac{(z - R \cos \theta) \sin \theta}{(R^2 + z^2 - 2Rz \cos \theta)^{3/2}} d\theta. \quad \text{Let } u = \cos \theta; \quad du = -\sin \theta d\theta; \quad \begin{cases} \theta = 0 \Rightarrow u = +1 \\ \theta = \pi \Rightarrow u = -1 \end{cases} \\ &= \frac{1}{4\pi\epsilon_0} (2\pi R^2 \sigma) \int_{-1}^1 \frac{z - Ru}{(R^2 + z^2 - 2Rzu)^{3/2}} du. \quad \text{Integral can be done by partial fractions—or look it up.} \\ &= \frac{1}{4\pi\epsilon_0} (2\pi R^2 \sigma) \left[\frac{1}{z^2} \frac{zu - R}{\sqrt{R^2 + z^2 - 2Rzu}} \right]_{-1}^1 = \frac{1}{4\pi\epsilon_0} \frac{2\pi R^2 \sigma}{z^2} \left\{ \frac{(z - R)}{|z - R|} - \frac{(-z - R)}{|z + R|} \right\}. \end{aligned}$$

For $z > R$ (outside the sphere), $E_z = \frac{1}{4\pi\epsilon_0} \frac{4\pi R^2 \sigma}{z^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2}$, so $\mathbf{E} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \hat{\mathbf{z}}}$.

For $z < R$ (inside), $E_z = 0$, so $\mathbf{E} = \mathbf{0}$.

Problem 2.8

According to Prob. 2.7, all shells *interior* to the point (i.e. at smaller r) contribute as though their charge were concentrated at the center, while all exterior shells contribute nothing. Therefore:

$$\mathbf{E}(r) = \frac{1}{4\pi\epsilon_0} \frac{Q_{\text{int}}}{r^2} \hat{\mathbf{r}},$$

where Q_{int} is the total charge interior to the point. *Outside* the sphere, *all* the charge is interior, so

$$\mathbf{E} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}}.$$

Inside the sphere, only that fraction of the total which is interior to the point counts:

$$Q_{\text{int}} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} Q = \frac{r^3}{R^3} Q, \quad \text{so} \quad \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{r^3}{R^3} Q \frac{1}{r^2} \hat{\mathbf{r}} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \mathbf{r}}.$$

Problem 2.9

(a) $\rho = \epsilon_0 \nabla \cdot \mathbf{E} = \epsilon_0 \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot kr^3) = \epsilon_0 \frac{1}{r^2} k(5r^4) = \boxed{5\epsilon_0 kr^2}$.

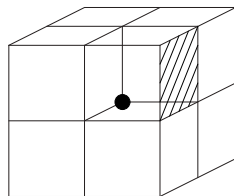
(b) By Gauss's law: $Q_{\text{enc}} = \epsilon_0 \oint \mathbf{E} \cdot d\mathbf{a} = \epsilon_0 (kR^3)(4\pi R^2) = \boxed{4\pi\epsilon_0 kR^5}.$

By direct integration: $Q_{\text{enc}} = \int \rho d\tau = \int_0^R (5\epsilon_0 k r^2)(4\pi r^2 dr) = 20\pi\epsilon_0 k \int_0^R r^4 dr = 4\pi\epsilon_0 k R^5. \checkmark$

Problem 2.10

Think of this cube as one of 8 surrounding the charge. Each of the 24 squares which make up the surface of this larger cube gets the same flux as every other one, so:

$$\int_{\text{one face}} \mathbf{E} \cdot d\mathbf{a} = \frac{1}{24} \int_{\text{whole large cube}} \mathbf{E} \cdot d\mathbf{a}.$$

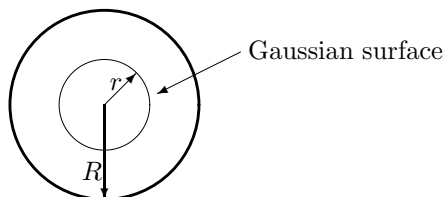


The latter is $\frac{1}{\epsilon_0} q$, by Gauss's law. Therefore

$$\int_{\text{one face}} \mathbf{E} \cdot d\mathbf{a} = \frac{q}{24\epsilon_0}.$$

Problem 2.11

Problem 2.12

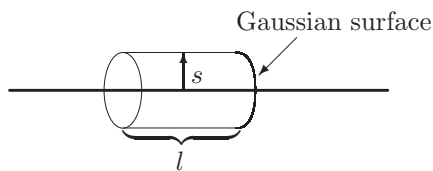


$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \frac{4}{3} \pi r^3 \rho. \quad \text{So}$$

$$\boxed{\mathbf{E} = \frac{1}{3\epsilon_0} \rho r \hat{\mathbf{r}}.}$$

Since $Q_{\text{tot}} = \frac{4}{3} \pi R^3 \rho$, $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \hat{\mathbf{r}}$ (as in Prob. 2.8).

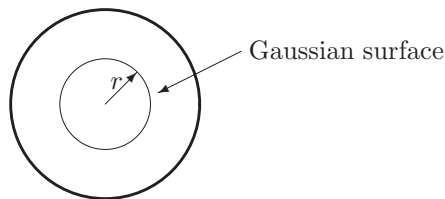
Problem 2.13



$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 2\pi s \cdot l = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \lambda l. \quad \text{So}$$

$$\boxed{\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}} \quad (\text{same as Eq. 2.9}).$$

Problem 2.14



$$\begin{aligned} \oint \mathbf{E} \cdot d\mathbf{a} &= E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \int \rho d\tau = \frac{1}{\epsilon_0} \int (k\bar{r}) (\bar{r}^2 \sin\theta d\bar{r} d\theta d\phi) \\ &= \frac{1}{\epsilon_0} k 4\pi \int_0^r \bar{r}^3 d\bar{r} = \frac{4\pi k}{\epsilon_0} \frac{r^4}{4} = \frac{\pi k}{\epsilon_0} r^4. \end{aligned}$$

$$\therefore \boxed{\mathbf{E} = \frac{1}{4\pi\epsilon_0} \pi k r^2 \hat{\mathbf{r}}.}$$

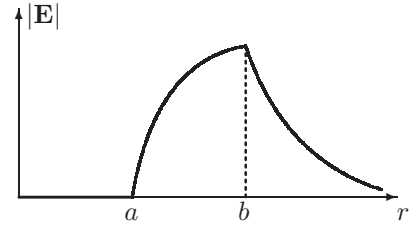
Problem 2.15

(i) $Q_{\text{enc}} = 0$, so $\mathbf{E} = \mathbf{0}$.

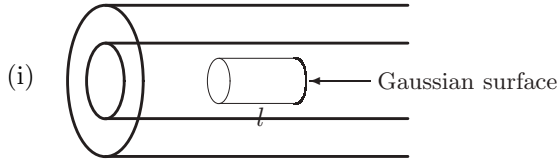
(ii) $\oint \mathbf{E} \cdot d\mathbf{a} = E(4\pi r^2) = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \int \rho d\tau = \frac{1}{\epsilon_0} \int \frac{k}{r^2} \tilde{r}^2 \sin\theta d\tilde{r} d\theta d\phi$
 $= \frac{4\pi k}{\epsilon_0} \int_a^r d\tilde{r} = \frac{4\pi k}{\epsilon_0} (r - a) \therefore \mathbf{E} = \frac{k}{\epsilon_0} \left(\frac{r - a}{r^2} \right) \hat{\mathbf{r}}.$

(iii) $E(4\pi r^2) = \frac{4\pi k}{\epsilon_0} \int_a^b d\tilde{r} = \frac{4\pi k}{\epsilon_0} (b - a)$, so

$$\mathbf{E} = \frac{k}{\epsilon_0} \left(\frac{b - a}{r^2} \right) \hat{\mathbf{r}}.$$

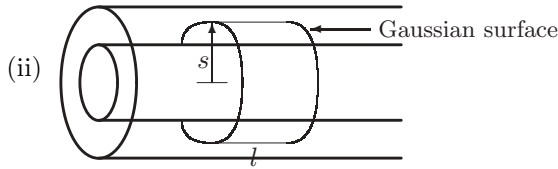


Problem 2.16



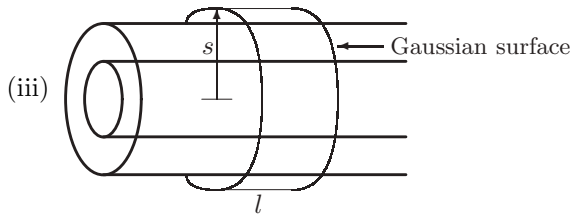
$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 2\pi s \cdot l = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \rho \pi s^2 l;$$

$$\mathbf{E} = \frac{\rho s}{2\epsilon_0} \hat{\mathbf{s}}.$$



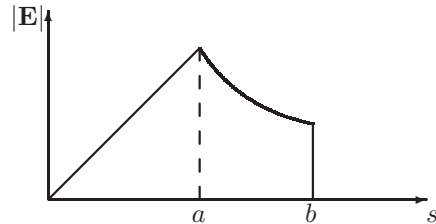
$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 2\pi s \cdot l = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \rho \pi a^2 l;$$

$$\mathbf{E} = \frac{\rho a^2}{2\epsilon_0 s} \hat{\mathbf{s}}.$$

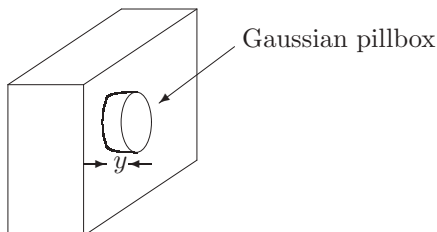


$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 2\pi s \cdot l = \frac{1}{\epsilon_0} Q_{\text{enc}} = 0;$$

$$\mathbf{E} = \mathbf{0}.$$



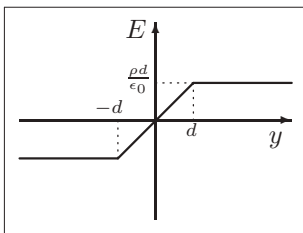
Problem 2.17 On the xz plane $E = 0$ by symmetry. Set up a Gaussian “pillbox” with one face in this plane and the other at y .



$$\int \mathbf{E} \cdot d\mathbf{a} = E \cdot A = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} A y \rho;$$

$$\mathbf{E} = \frac{\rho}{\epsilon_0} y \hat{\mathbf{y}} \quad (\text{for } |y| < d).$$

$$Q_{\text{enc}} = \frac{1}{\epsilon_0} A d \rho \Rightarrow \boxed{\mathbf{E} = \frac{\rho}{\epsilon_0} d \hat{\mathbf{y}}} \quad (\text{for } y > d).$$

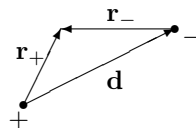


Problem 2.18

From Prob. 2.12, the field inside the positive sphere is $\mathbf{E}_+ = \frac{\rho}{3\epsilon_0} \mathbf{r}_+$, where $\mathbf{r}_+$ is the vector from the positive center to the point in question. Likewise, the field of the negative sphere is $-\frac{\rho}{3\epsilon_0} \mathbf{r}_-$. So the *total* field is

$$\mathbf{E} = \frac{\rho}{3\epsilon_0} (\mathbf{r}_+ - \mathbf{r}_-)$$

But (see diagram) $\mathbf{r}_+ - \mathbf{r}_- = \mathbf{d}$. So $\boxed{\mathbf{E} = \frac{\rho}{3\epsilon_0} \mathbf{d}}$.



Problem 2.19

$$\begin{aligned} \nabla \times \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \nabla \times \int \frac{\hat{\mathbf{r}}}{r^2} \rho d\tau = \frac{1}{4\pi\epsilon_0} \int \left[\nabla \times \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \right] \rho d\tau \quad (\text{since } \rho \text{ depends on } \mathbf{r}', \text{ not } \mathbf{r}) \\ &= \mathbf{0} \quad (\text{since } \nabla \times \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = \mathbf{0}, \text{ from Prob. 1.63}). \end{aligned}$$

Problem 2.20

$$(1) \nabla \times \mathbf{E}_1 = k \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 2yz & 3zx \end{vmatrix} = k [\hat{\mathbf{x}}(0 - 2y) + \hat{\mathbf{y}}(0 - 3z) + \hat{\mathbf{z}}(0 - x)] \neq \mathbf{0},$$

so $\mathbf{E}_1$ is an *impossible* electrostatic field.

$$(2) \nabla \times \mathbf{E}_2 = k \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & 2xy + z^2 & 2yz \end{vmatrix} = k [\hat{\mathbf{x}}(2z - 2z) + \hat{\mathbf{y}}(0 - 0) + \hat{\mathbf{z}}(2y - 2y)] = \mathbf{0},$$

so $\mathbf{E}_2$ is a *possible* electrostatic field.

Let's go by the indicated path:

$$\mathbf{E} \cdot d\mathbf{l} = (y^2 dx + (2xy + z^2) dy + 2yz dz) k$$

Step I: $y = z = 0$; $dy = dz = 0$. $\mathbf{E} \cdot d\mathbf{l} = ky^2 dx = 0$.

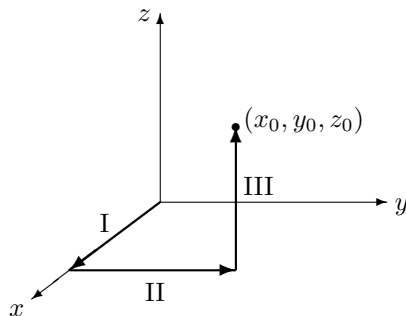
Step II: $x = x_0$, $y : 0 \rightarrow y_0$, $z = 0$. $dx = dz = 0$.

$$\mathbf{E} \cdot d\mathbf{l} = k(2xy + z^2) dy = 2kx_0 y dy.$$

$$\int_{II} \mathbf{E} \cdot d\mathbf{l} = 2kx_0 \int_0^{y_0} y dy = kx_0 y_0^2.$$

Step III: $x = x_0$, $y = y_0$, $z : 0 \rightarrow z_0$; $dx = dy = 0$.

$$\mathbf{E} \cdot d\mathbf{l} = 2ky_0 z dz = 2ky_0 z dz.$$



$$\int_{III} \mathbf{E} \cdot d\mathbf{l} = 2y_0 k \int_0^{z_0} z \, dz = ky_0 z_0^2.$$

$$V(x_0, y_0, z_0) = - \int_0^{(x_0, y_0, z_0)} \mathbf{E} \cdot d\mathbf{l} = -k(x_0 y_0^2 + y_0 z_0^2), \text{ or } \boxed{V(x, y, z) = -k(xy^2 + yz^2)}.$$

$$\text{Check: } -\nabla V = k \left[\frac{\partial}{\partial x}(xy^2 + yz^2) \hat{\mathbf{x}} + \frac{\partial}{\partial y}(xy^2 + yz^2) \hat{\mathbf{y}} + \frac{\partial}{\partial z}(xy^2 + yz^2) \hat{\mathbf{z}} \right] = k[y^2 \hat{\mathbf{x}} + (2xy + z^2) \hat{\mathbf{y}} + 2yz \hat{\mathbf{z}}] = \mathbf{E}. \checkmark$$

Problem 2.21

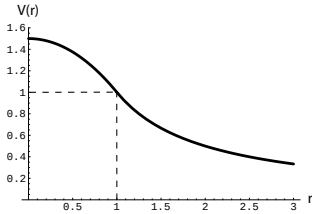
$$V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l}. \quad \begin{cases} \text{Outside the sphere } (r > R) : \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}. \\ \text{Inside the sphere } (r < R) : \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{\mathbf{r}}. \end{cases}$$

$$\text{So for } r > R: V(r) = - \int_{\infty}^r \left(\frac{1}{4\pi\epsilon_0} \frac{q}{\bar{r}^2} \right) d\bar{r} = \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{\bar{r}} \right) \Big|_{\infty}^r = \boxed{\frac{q}{4\pi\epsilon_0} \frac{1}{r}},$$

$$\begin{aligned} \text{and for } r < R: V(r) &= - \int_{\infty}^R \left(\frac{1}{4\pi\epsilon_0} \frac{q}{\bar{r}^2} \right) d\bar{r} - \int_R^r \left(\frac{1}{4\pi\epsilon_0} \frac{q}{R^3} \bar{r} \right) d\bar{r} = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{R^3} \left(\frac{r^2 - R^2}{2} \right) \right] \\ &= \boxed{\frac{q}{4\pi\epsilon_0} \frac{1}{2R} \left(3 - \frac{r^2}{R^2} \right)}. \end{aligned}$$

$$\text{When } r > R, \nabla V = \frac{q}{4\pi\epsilon_0} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \hat{\mathbf{r}} = -\frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{r}}, \text{ so } \mathbf{E} = -\nabla V = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{r}}. \checkmark$$

$$\text{When } r < R, \nabla V = \frac{q}{4\pi\epsilon_0} \frac{1}{2R} \frac{\partial}{\partial r} \left(3 - \frac{r^2}{R^2} \right) \hat{\mathbf{r}} = \frac{q}{4\pi\epsilon_0} \frac{1}{2R} \left(-\frac{2r}{R^2} \right) \hat{\mathbf{r}} = -\frac{q}{4\pi\epsilon_0} \frac{r}{R^3} \hat{\mathbf{r}}; \text{ so } \mathbf{E} = -\nabla V = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{\mathbf{r}}. \checkmark$$



(In the figure, r is in units of R , and $V(r)$ is in units of $q/4\pi\epsilon_0 R$.)

Problem 2.22

$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{s} \hat{\mathbf{s}}$ (Prob. 2.13). In this case we cannot set the reference point at ∞ , since the charge itself extends to ∞ . Let's set it at $s = a$. Then

$$V(s) = - \int_a^s \left(\frac{1}{4\pi\epsilon_0} \frac{2\lambda}{\bar{s}} \right) d\bar{s} = \boxed{-\frac{1}{4\pi\epsilon_0} 2\lambda \ln \left(\frac{s}{a} \right)}.$$

(In this form it is clear why $a = \infty$ would be no good—likewise the other “natural” point, $a = 0$.)

$$\nabla V = -\frac{1}{4\pi\epsilon_0} 2\lambda \frac{\partial}{\partial s} \left(\ln \left(\frac{s}{a} \right) \right) \hat{\mathbf{s}} = -\frac{1}{4\pi\epsilon_0} 2\lambda \frac{1}{s} \hat{\mathbf{s}} = -\mathbf{E}. \checkmark$$

Problem 2.23

$$\begin{aligned} V(0) &= - \int_{\infty}^0 \mathbf{E} \cdot d\mathbf{l} = - \int_{\infty}^b \left(\frac{k}{\epsilon_0} \frac{(b-a)}{r^2} \right) dr - \int_b^a \left(\frac{k}{\epsilon_0} \frac{(r-a)}{r^2} \right) dr - \int_a^0 (0) dr = \frac{k}{\epsilon_0} \frac{(b-a)}{b} - \frac{k}{\epsilon_0} \left(\ln \left(\frac{a}{b} \right) + a \left(\frac{1}{a} - \frac{1}{b} \right) \right) \\ &= \frac{k}{\epsilon_0} \left\{ 1 - \frac{a}{b} - \ln \left(\frac{a}{b} \right) - 1 + \frac{a}{b} \right\} = \boxed{\frac{k}{\epsilon_0} \ln \left(\frac{b}{a} \right)}. \end{aligned}$$

Problem 2.24

Using Eq. 2.22 and the fields from Prob. 2.16:

$$V(b) - V(0) = - \int_0^b \mathbf{E} \cdot d\mathbf{l} = - \int_0^a \mathbf{E} \cdot d\mathbf{l} - \int_a^b \mathbf{E} \cdot d\mathbf{l} = -\frac{\rho}{2\epsilon_0} \int_0^a s \, ds - \frac{\rho a^2}{2\epsilon_0} \int_a^b \frac{1}{s} \, ds$$

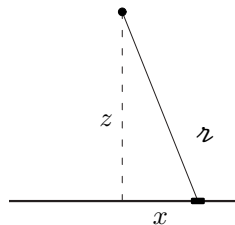
$$= - \left(\frac{\rho}{2\epsilon_0} \right) \frac{s^2}{2} \Big|_0^a + \frac{\rho a^2}{2\epsilon_0} \ln s \Big|_a^b = \boxed{-\frac{\rho a^2}{4\epsilon_0} \left(1 + 2 \ln \left(\frac{b}{a} \right) \right)}.$$

Problem 2.25

$$(a) \quad V = \frac{1}{4\pi\epsilon_0} \frac{2q}{\sqrt{z^2 + \left(\frac{d}{2}\right)^2}}.$$

$$(b) \quad V = \frac{1}{4\pi\epsilon_0} \int_{-L}^L \frac{\lambda dx}{\sqrt{z^2 + x^2}} = \frac{\lambda}{4\pi\epsilon_0} \ln(x + \sqrt{z^2 + x^2}) \Big|_{-L}^L$$

$$= \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{L + \sqrt{z^2 + L^2}}{-L + \sqrt{z^2 + L^2}} \right] = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{L + \sqrt{z^2 + L^2}}{z} \right).$$



$$(c) \quad V = \frac{1}{4\pi\epsilon_0} \int_0^R \frac{\sigma 2\pi r dr}{\sqrt{r^2 + z^2}} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma (\sqrt{r^2 + z^2}) \Big|_0^R = \boxed{\frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + z^2} - z)}.$$

In each case, by symmetry $\frac{\partial V}{\partial y} = \frac{\partial V}{\partial x} = 0$. $\therefore \mathbf{E} = -\frac{\partial V}{\partial z} \hat{\mathbf{z}}$.

$$(a) \quad \mathbf{E} = -\frac{1}{4\pi\epsilon_0} 2q \left(-\frac{1}{2} \right) \frac{2z}{\left(z^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}} \hat{\mathbf{z}} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{2qz}{\left(z^2 + \left(\frac{d}{2} \right)^2 \right)^{3/2}} \hat{\mathbf{z}}} \text{ (agrees with Ex. 2.1).}$$

$$(b) \quad \mathbf{E} = -\frac{\lambda}{4\pi\epsilon_0} \left\{ \frac{1}{(L + \sqrt{z^2 + L^2})} \frac{1}{2} \frac{1}{\sqrt{z^2 + L^2}} 2z - \frac{1}{(-L + \sqrt{z^2 + L^2})} \frac{1}{2} \frac{1}{\sqrt{z^2 + L^2}} 2z \right\} \hat{\mathbf{z}}$$

$$= -\frac{\lambda}{4\pi\epsilon_0} \frac{z}{\sqrt{z^2 + L^2}} \left\{ \frac{-L + \sqrt{z^2 + L^2} - L - \sqrt{z^2 + L^2}}{(z^2 + L^2) - L^2} \right\} \hat{\mathbf{z}} = \boxed{\frac{2L\lambda}{4\pi\epsilon_0} \frac{1}{z\sqrt{z^2 + L^2}} \hat{\mathbf{z}}} \text{ (agrees with Ex. 2.2).}$$

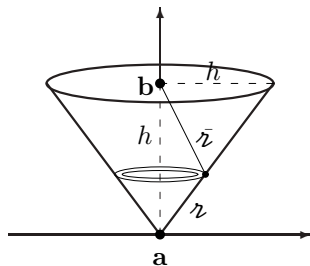
$$(c) \quad \mathbf{E} = -\frac{\sigma}{2\epsilon_0} \left\{ \frac{1}{2} \frac{1}{\sqrt{R^2 + z^2}} 2z - 1 \right\} \hat{\mathbf{z}} = \boxed{\frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R^2 + z^2}} \right] \hat{\mathbf{z}}} \text{ (agrees with Prob. 2.6).}$$

If the right-hand charge in (a) is $-q$, then $V = 0$, which, naively, suggests $\mathbf{E} = -\nabla V = \mathbf{0}$, in contradiction with the answer to Prob. 2.2. The point is that we only know V on the z axis, and from this we cannot hope to compute $E_x = -\frac{\partial V}{\partial x}$ or $E_y = -\frac{\partial V}{\partial y}$. That was OK in part (a), because we knew from symmetry that $E_x = E_y = 0$. But now $\mathbf{E}$ points in the x direction, so knowing V on the z axis is insufficient to determine $\mathbf{E}$.

Problem 2.26

$$V(a) = \frac{1}{4\pi\epsilon_0} \int_0^{\sqrt{2}h} \left(\frac{\sigma 2\pi r}{z} \right) dz = \frac{2\pi\sigma}{4\pi\epsilon_0} \frac{1}{\sqrt{2}} (\sqrt{2}h) = \frac{\sigma h}{2\epsilon_0}$$

(where $r = z / \sqrt{2}$)



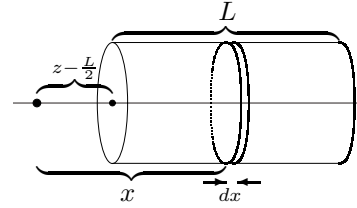
$$\begin{aligned}
V(\mathbf{b}) &= \frac{1}{4\pi\epsilon_0} \int_0^{\sqrt{2}h} \left(\frac{\sigma 2\pi r}{\bar{z}} \right) d\bar{z} \quad (\text{where } \bar{z} = \sqrt{h^2 + z^2} - \sqrt{2}hz) \\
&= \frac{2\pi\sigma}{4\pi\epsilon_0} \frac{1}{\sqrt{2}} \int_0^{\sqrt{2}h} \frac{z}{\sqrt{h^2 + z^2} - \sqrt{2}hz} dz \\
&= \frac{\sigma}{2\sqrt{2}\epsilon_0} \left[\sqrt{h^2 + z^2} - \sqrt{2}hz} + \frac{h}{\sqrt{2}} \ln(2\sqrt{h^2 + z^2} - \sqrt{2}hz} + 2z - \sqrt{2}h) \right] \Big|_0^{\sqrt{2}h} \\
&= \frac{\sigma}{2\sqrt{2}\epsilon_0} \left[h + \frac{h}{\sqrt{2}} \ln(2h + 2\sqrt{2}h - \sqrt{2}h) - h - \frac{h}{\sqrt{2}} \ln(2h - \sqrt{2}h) \right] \\
&= \frac{\sigma}{2\sqrt{2}\epsilon_0} \frac{h}{\sqrt{2}} \left[\ln(2h + \sqrt{2}h) - \ln(2h - \sqrt{2}h) \right] = \frac{\sigma h}{4\epsilon_0} \ln \left(\frac{2 + \sqrt{2}}{2 - \sqrt{2}} \right) = \frac{\sigma h}{4\epsilon_0} \ln \left(\frac{(2 + \sqrt{2})^2}{2} \right) \\
&= \frac{\sigma h}{2\epsilon_0} \ln(1 + \sqrt{2}). \quad \therefore \boxed{V(\mathbf{a}) - V(\mathbf{b}) = \frac{\sigma h}{2\epsilon_0} \left[1 - \ln(1 + \sqrt{2}) \right]}.
\end{aligned}$$

Problem 2.27

Cut the cylinder into slabs, as shown in the figure, and use result of Prob. 2.25c, with $z \rightarrow x$ and $\sigma \rightarrow \rho dx$:

$$\begin{aligned}
V &= \frac{\rho}{2\epsilon_0} \int_{z=L/2}^{z+L/2} (\sqrt{R^2 + x^2} - x) dx \\
&= \frac{\rho}{2\epsilon_0} \frac{1}{2} \left[x\sqrt{R^2 + x^2} + R^2 \ln(x + \sqrt{R^2 + x^2}) - x^2 \right] \Big|_{z=L/2}^{z+L/2} \\
&= \boxed{\frac{\rho}{4\epsilon_0} \left\{ \left(z + \frac{L}{2} \right) \sqrt{R^2 + \left(z + \frac{L}{2} \right)^2} - \left(z - \frac{L}{2} \right) \sqrt{R^2 + \left(z - \frac{L}{2} \right)^2} + R^2 \ln \left[\frac{z + \frac{L}{2} + \sqrt{R^2 + \left(z + \frac{L}{2} \right)^2}}{z - \frac{L}{2} + \sqrt{R^2 + \left(z - \frac{L}{2} \right)^2}} \right] - 2zL \right\}}.
\end{aligned}$$

$$(Note: -\left(z + \frac{L}{2}\right)^2 + \left(z - \frac{L}{2}\right)^2 = -z^2 - zL - \frac{L^2}{4} + z^2 - zL + \frac{L^2}{4} = -2zL.)$$



$$\begin{aligned}
\mathbf{E} &= -\nabla V = -\hat{\mathbf{z}} \frac{\partial V}{\partial z} = -\frac{\hat{\mathbf{z}}\rho}{4\epsilon_0} \left\{ \sqrt{R^2 + \left(z + \frac{L}{2} \right)^2} + \frac{\left(z + \frac{L}{2} \right)^2}{\sqrt{R^2 + \left(z + \frac{L}{2} \right)^2}} - \sqrt{R^2 + \left(z - \frac{L}{2} \right)^2} - \frac{\left(z - \frac{L}{2} \right)^2}{\sqrt{R^2 + \left(z - \frac{L}{2} \right)^2}} \right. \\
&\quad \left. + R^2 \left[\frac{1 + \frac{z + \frac{L}{2}}{\sqrt{R^2 + \left(z + \frac{L}{2} \right)^2}}}{z + \frac{L}{2} + \sqrt{R^2 + \left(z + \frac{L}{2} \right)^2}} - \frac{1 + \frac{z - \frac{L}{2}}{\sqrt{R^2 + \left(z - \frac{L}{2} \right)^2}}}{z - \frac{L}{2} + \sqrt{R^2 + \left(z - \frac{L}{2} \right)^2}} \right] - 2L \right\} \\
&\quad \underbrace{\frac{1}{\sqrt{R^2 + \left(z + \frac{L}{2} \right)^2}} - \frac{1}{\sqrt{R^2 + \left(z - \frac{L}{2} \right)^2}}}
\end{aligned}$$

$$\mathbf{E} = -\frac{\hat{\mathbf{z}}\rho}{4\epsilon_0} \left\{ 2\sqrt{R^2 + \left(z + \frac{L}{2} \right)^2} - 2\sqrt{R^2 + \left(z - \frac{L}{2} \right)^2} - 2L \right\}$$

$$= \left[\frac{\rho}{2\epsilon_0} \left[L - \sqrt{R^2 + \left(z + \frac{L}{2}\right)^2} + \sqrt{R^2 + \left(z - \frac{L}{2}\right)^2} \right] \right] \hat{\mathbf{z}}.$$

Problem 2.28

Orient axes so P is on z axis.

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{\mathcal{Z}} d\tau. \quad \begin{cases} \text{Here } \rho \text{ is constant, } d\tau = r^2 \sin\theta dr d\theta d\phi, \\ \mathcal{Z} = \sqrt{z^2 + r^2 - 2rz \cos\theta}. \end{cases}$$

$$V = \frac{\rho}{4\pi\epsilon_0} \int \frac{r^2 \sin\theta dr d\theta d\phi}{\sqrt{z^2 + r^2 - 2rz \cos\theta}}; \int_0^{2\pi} d\phi = 2\pi.$$

$$\begin{aligned} \int_0^\pi \frac{\sin\theta}{\sqrt{z^2 + r^2 - 2rz \cos\theta}} d\theta &= \frac{1}{rz} \left(\sqrt{r^2 + z^2 - 2rz \cos\theta} \right) \Big|_0^\pi = \frac{1}{rz} \left(\sqrt{r^2 + z^2 + 2rz} - \sqrt{r^2 + z^2 - 2rz} \right) \\ &= \frac{1}{rz} (r + z - |r - z|) = \begin{cases} 2/z, & \text{if } r < z, \\ 2/r, & \text{if } r > z. \end{cases} \end{aligned}$$

$$\therefore V = \frac{\rho}{4\pi\epsilon_0} \cdot 2\pi \cdot 2 \left\{ \int_0^z \frac{1}{z} r^2 dr + \int_z^R \frac{1}{r} r^2 dr \right\} = \frac{\rho}{\epsilon_0} \left\{ \frac{1}{z} \frac{z^3}{3} + \frac{R^2 - z^2}{2} \right\} = \frac{\rho}{2\epsilon_0} \left(R^2 - \frac{z^2}{3} \right).$$

But $\rho = \frac{q}{\frac{4}{3}\pi R^3}$, so $V(z) = \frac{1}{2\epsilon_0} \frac{3q}{4\pi R^3} \left(R^2 - \frac{z^2}{3} \right) = \frac{q}{8\pi\epsilon_0 R} \left(3 - \frac{z^2}{R^2} \right); \quad V(r) = \frac{q}{8\pi\epsilon_0 R} \left(3 - \frac{r^2}{R^2} \right). \quad \checkmark$

Problem 2.29

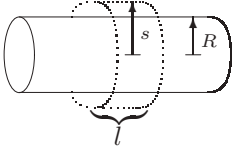
$$\begin{aligned} \nabla^2 V &= \frac{1}{4\pi\epsilon_0} \nabla^2 \int \left(\frac{\rho}{\mathcal{Z}} \right) d\tau = \frac{1}{4\pi\epsilon_0} \int \rho(\mathbf{r}') \left(\nabla^2 \frac{1}{\mathcal{Z}} \right) d\tau \quad (\text{since } \rho \text{ is a function of } \mathbf{r}', \text{ not } \mathbf{r}) \\ &= \frac{1}{4\pi\epsilon_0} \int \rho(\mathbf{r}') [-4\pi\delta^3(\mathbf{r} - \mathbf{r}')] d\tau = -\frac{1}{\epsilon_0} \rho(\mathbf{r}). \quad \checkmark \end{aligned}$$

Problem 2.30.

(a) Ex. 2.5: $\mathbf{E}_{\text{above}} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}; \mathbf{E}_{\text{below}} = -\frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}$ ($\hat{\mathbf{n}}$ always pointing *up*); $\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}. \quad \checkmark$

Ex. 2.6: At each surface, $E = 0$ one side and $E = \frac{\sigma}{\epsilon_0}$ other side, so $\Delta E = \frac{\sigma}{\epsilon_0}. \quad \checkmark$

Prob. 2.11: $\mathbf{E}_{\text{out}} = \frac{\sigma R^2}{\epsilon_0 r^2} \hat{\mathbf{r}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{r}}; \mathbf{E}_{\text{in}} = 0$; so $\Delta \mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{r}}. \quad \checkmark$

(b)  *Outside:* $\oint \mathbf{E} \cdot d\mathbf{a} = E(2\pi s)l = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{\sigma}{\epsilon_0} (2\pi R)l \Rightarrow \mathbf{E} = \frac{\sigma}{\epsilon_0} \frac{R}{s} \hat{\mathbf{s}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{s}}$ (at surface).
Inside: $Q_{\text{enc}} = 0$, so $\mathbf{E} = 0. \therefore \Delta \mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{s}}. \quad \checkmark$

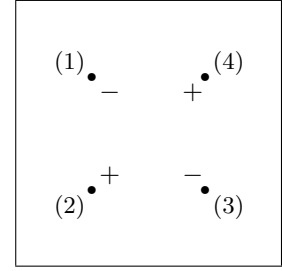
(c) $V_{\text{out}} = \frac{R^2 \sigma}{\epsilon_0 r} = \frac{R\sigma}{\epsilon_0}$ (at surface); $V_{\text{in}} = \frac{R\sigma}{\epsilon_0}$; so $V_{\text{out}} = V_{\text{in}}. \quad \checkmark$

$$\frac{\partial V_{\text{out}}}{\partial r} = -\frac{R^2 \sigma}{\epsilon_0 r^2} = -\frac{\sigma}{\epsilon_0} \quad (\text{at surface}); \quad \frac{\partial V_{\text{in}}}{\partial r} = 0; \quad \text{so } \frac{\partial V_{\text{out}}}{\partial r} - \frac{\partial V_{\text{in}}}{\partial r} = -\frac{\sigma}{\epsilon_0}. \quad \checkmark$$

Problem 2.31

$$(a) \quad V = \frac{1}{4\pi\epsilon_0} \sum \frac{q_i}{r_{ij}} = \frac{1}{4\pi\epsilon_0} \left\{ \frac{-q}{a} + \frac{q}{\sqrt{2}a} + \frac{-q}{a} \right\} = \frac{q}{4\pi\epsilon_0 a} \left(-2 + \frac{1}{\sqrt{2}} \right).$$

$$\therefore W_4 = qV = \boxed{\frac{q^2}{4\pi\epsilon_0 a} \left(-2 + \frac{1}{\sqrt{2}} \right)}.$$



$$(b) \quad W_1 = 0, \quad W_2 = \frac{1}{4\pi\epsilon_0} \left(\frac{-q^2}{a} \right); \quad W_3 = \frac{1}{4\pi\epsilon_0} \left(\frac{q^2}{\sqrt{2}a} - \frac{q^2}{a} \right); \quad W_4 = (\text{see (a)}).$$

$$W_{\text{tot}} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left\{ -1 + \frac{1}{\sqrt{2}} - 1 - 2 + \frac{1}{\sqrt{2}} \right\} = \boxed{\frac{1}{4\pi\epsilon_0} \frac{2q^2}{a} \left(-2 + \frac{1}{\sqrt{2}} \right)}.$$

Problem 2.32

Conservation of energy (kinetic plus potential):

$$\frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 + \frac{1}{4\pi\epsilon_0} \frac{q_A q_B}{r} = E.$$

At release $v_A = v_B = 0$, $r = a$, so

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_A q_B}{a}.$$

When they are very far apart ($r \rightarrow \infty$) the potential energy is zero, so

$$\frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 = \frac{1}{4\pi\epsilon_0} \frac{q_A q_B}{a}.$$

Meanwhile, conservation of momentum says $m_A v_A = m_B v_B$, or $v_B = (m_A/m_B) v_A$. So

$$\frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B \left(\frac{m_A}{m_B} \right)^2 v_A^2 = \frac{1}{2} \left(\frac{m_A}{m_B} \right) (m_A + m_B) v_A^2 = \frac{1}{4\pi\epsilon_0} \frac{q_A q_B}{a}.$$

$$\boxed{v_A = \sqrt{\frac{1}{2\pi\epsilon_0} \frac{q_A q_B}{(m_A + m_B)a} \left(\frac{m_A}{m_B} \right)}; \quad v_B = \sqrt{\frac{1}{2\pi\epsilon_0} \frac{q_A q_B}{(m_A + m_B)a} \left(\frac{m_B}{m_A} \right)}}.$$

Problem 2.33

From Eq. 2.42, the energy of one charge is

$$W = \frac{1}{2} qV = \frac{1}{2} (2) \sum_{n=1}^{\infty} \frac{1}{4\pi\epsilon_0} \frac{(-1)^n q^2}{na} = \frac{q^2}{4\pi\epsilon_0 a} \sum_{n=1}^{\infty} \frac{(-1)^n}{n}.$$

(The factor of 2 out front counts the charges to the left as well as to the right of q .) The sum is $-\ln 2$ (you can get it from the Taylor expansion of $\ln(1+x)$):

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

with $x = 1$. Evidently $\boxed{\alpha = \ln 2}$.

Problem 2.34

(a) $W = \frac{1}{2} \int \rho V d\tau$. From Prob. 2.21 (or Prob. 2.28): $V = \frac{\rho}{2\epsilon_0} \left(R^2 - \frac{r^2}{3} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{2R} \left(3 - \frac{r^2}{R^2} \right)$

$$\begin{aligned} W &= \frac{1}{2} \rho \frac{1}{4\pi\epsilon_0} \frac{q}{2R} \int_0^R \left(3 - \frac{r^2}{R^2} \right) 4\pi r^2 dr = \frac{q\rho}{4\epsilon_0 R} \left[3 \frac{r^3}{3} - \frac{1}{R^2} \frac{r^5}{5} \right]_0^R = \frac{q\rho}{4\epsilon_0 R} \left(R^3 - \frac{R^3}{5} \right) \\ &= \frac{q\rho}{5\epsilon_0} R^2 = \frac{qR^2}{5\epsilon_0} \frac{q}{\frac{4}{3}\pi R^3} = \boxed{\frac{1}{4\pi\epsilon_0} \left(\frac{3}{5} \frac{q^2}{R} \right)}. \end{aligned}$$

(b) $W = \frac{\epsilon_0}{2} \int E^2 d\tau$. *Outside* ($r > R$) $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$; *Inside* ($r < R$) $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{\mathbf{r}}$.

$$\begin{aligned} \therefore W &= \frac{\epsilon_0}{2} \frac{1}{(4\pi\epsilon_0)^2} q^2 \left\{ \int_R^\infty \frac{1}{r^4} (r^2 4\pi dr) + \int_0^R \left(\frac{r}{R^3} \right)^2 (4\pi r^2 dr) \right\} \\ &= \frac{1}{4\pi\epsilon_0} \frac{q^2}{2} \left\{ \left(-\frac{1}{r} \right) \Big|_R^\infty + \frac{1}{R^6} \left(\frac{r^5}{5} \right) \Big|_0^R \right\} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{2} \left(\frac{1}{R} + \frac{1}{5R} \right) = \frac{1}{4\pi\epsilon_0} \frac{3}{5} \frac{q^2}{R}. \checkmark \end{aligned}$$

(c) $W = \frac{\epsilon_0}{2} \left\{ \oint_S \mathbf{V} \mathbf{E} \cdot d\mathbf{a} + \int_V E^2 d\tau \right\}$, where $\mathcal{V}$ is large enough to enclose all the charge, but otherwise arbitrary. Let's use a sphere of radius $a > R$. Here $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$.

$$\begin{aligned} W &= \frac{\epsilon_0}{2} \left\{ \int_{r=a} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) r^2 \sin\theta d\theta d\phi + \int_0^R E^2 d\tau + \int_R^a \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right)^2 (4\pi r^2 dr) \right\} \\ &= \frac{\epsilon_0}{2} \left\{ \frac{q^2}{(4\pi\epsilon_0)^2} \frac{1}{a} 4\pi + \frac{q^2}{(4\pi\epsilon_0)^2} \frac{4\pi}{5R} + \frac{1}{(4\pi\epsilon_0)^2} 4\pi q^2 \left(-\frac{1}{r} \right) \Big|_R^a \right\} \\ &= \frac{1}{4\pi\epsilon_0} \frac{q^2}{2} \left\{ \frac{1}{a} + \frac{1}{5R} - \frac{1}{a} + \frac{1}{R} \right\} = \frac{1}{4\pi\epsilon_0} \frac{3}{5} \frac{q^2}{R}. \checkmark \end{aligned}$$

As $a \rightarrow \infty$, the contribution from the surface integral $\left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{2a} \right)$ goes to zero, while the volume integral $\left(\frac{1}{4\pi\epsilon_0} \frac{q^2}{2a} \left(\frac{6a}{5R} - 1 \right) \right)$ picks up the slack.

Problem 2.35

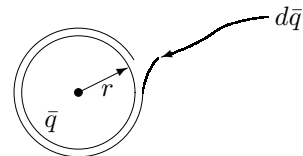
$$dW = d\bar{q} V = d\bar{q} \left(\frac{1}{4\pi\epsilon_0} \right) \frac{\bar{q}}{r}, \quad (\bar{q} = \text{charge on sphere of radius } r).$$

$$\bar{q} = \frac{4}{3} \pi r^3 \rho = q \frac{r^3}{R^3} \quad (q = \text{total charge on sphere}).$$

$$d\bar{q} = 4\pi r^2 dr \rho = \frac{4\pi r^2}{\frac{4}{3}\pi R^3} q dr = \frac{3q}{R^3} r^2 dr.$$

$$dW = \frac{1}{4\pi\epsilon_0} \left(\frac{qr^3}{R^3} \right) \frac{1}{r} \left(\frac{3q}{R^3} r^2 dr \right) = \frac{1}{4\pi\epsilon_0} \frac{3q^2}{R^6} r^4 dr$$

$$W = \frac{1}{4\pi\epsilon_0} \frac{3q^2}{R^6} \int_0^R r^4 dr = \frac{1}{4\pi\epsilon_0} \frac{3q^2}{R^6} \frac{R^5}{5} = \frac{1}{4\pi\epsilon_0} \left(\frac{3}{5} \frac{q^2}{R} \right). \checkmark$$

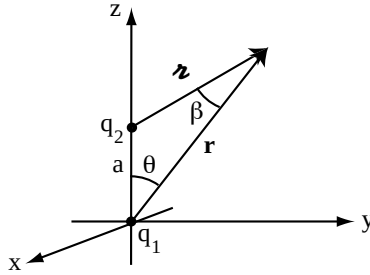


Problem 2.36

(a) $W = \frac{\epsilon_0}{2} \int E^2 d\tau$. $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$ ($a < r < b$), zero elsewhere.

$$W = \frac{\epsilon_0}{2} \left(\frac{q}{4\pi\epsilon_0} \right)^2 \int_a^b \left(\frac{1}{r^2} \right)^2 4\pi r^2 dr = \frac{q^2}{8\pi\epsilon_0} \int_a^b \frac{1}{r^2} = \boxed{\frac{q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)}.$$

(b) $W_1 = \frac{1}{8\pi\epsilon_0} \frac{q^2}{a}$, $W_2 = \frac{1}{8\pi\epsilon_0} \frac{q^2}{b}$, $\mathbf{E}_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$ ($r > a$), $\mathbf{E}_2 = \frac{1}{4\pi\epsilon_0} \frac{-q}{r^2} \hat{\mathbf{r}}$ ($r > b$). So
 $\mathbf{E}_1 \cdot \mathbf{E}_2 = \left(\frac{1}{4\pi\epsilon_0} \right)^2 \frac{-q^2}{r^4}$, ($r > b$), and hence $\int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau = - \left(\frac{1}{4\pi\epsilon_0} \right)^2 q^2 \int_b^\infty \frac{1}{r^4} 4\pi r^2 dr = - \frac{q^2}{4\pi\epsilon_0 b}$.
 $W_{\text{tot}} = W_1 + W_2 + \epsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau = \frac{1}{8\pi\epsilon_0} q^2 \left(\frac{1}{a} + \frac{1}{b} - \frac{2}{b} \right) = \frac{q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$. ✓

Problem 2.37


$$\mathbf{E}_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2} \hat{\mathbf{r}}; \quad \mathbf{E}_2 = \frac{1}{4\pi\epsilon_0} \frac{q_2}{z^2} \hat{\mathbf{z}}; \quad W_i = \epsilon_0 \frac{q_1 q_2}{(4\pi\epsilon_0)^2} \int \frac{1}{r^2 z^2} \cos \beta r^2 \sin \theta dr d\theta d\phi,$$

where (from the figure)

$$z = \sqrt{r^2 + a^2 - 2ra \cos \theta}, \quad \cos \beta = \frac{(r - a \cos \theta)}{z}.$$

Therefore

$$W_i = \frac{q_1 q_2}{(4\pi)^2 \epsilon_0} 2\pi \int \frac{(r - a \cos \theta)}{z^3} \sin \theta dr d\theta.$$

It's simplest to do the r integral first, changing variables to z :

$$2z dz = (2r - 2a \cos \theta) dr \quad \Rightarrow \quad (r - a \cos \theta) dr = z dz.$$

As $r : 0 \rightarrow \infty$, $z : a \rightarrow \infty$, so

$$W_i = \frac{q_1 q_2}{8\pi\epsilon_0} \int_0^\pi \left(\int_a^\infty \frac{1}{z^2} dz \right) \sin \theta d\theta.$$

The z integral is $1/a$, so

$$W_i = \frac{q_1 q_2}{8\pi\epsilon_0 a} \int_0^\pi \sin \theta d\theta = \boxed{\frac{q_1 q_2}{4\pi\epsilon_0 a}}.$$

Of course, this is precisely the interaction energy of two point charges.

Problem 2.38

(a) $\sigma_R = \frac{q}{4\pi R^2}; \sigma_a = \frac{-q}{4\pi a^2}; \sigma_b = \frac{q}{4\pi b^2}.$

$$(b) V(0) = - \int_{\infty}^0 \mathbf{E} \cdot d\mathbf{l} = - \int_{\infty}^b \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) dr - \int_b^a (0) dr - \int_a^R \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) dr - \int_R^0 (0) dr = \boxed{\frac{1}{4\pi\epsilon_0} \left(\frac{q}{b} + \frac{q}{R} - \frac{q}{a} \right)}.$$

$$(c) \boxed{\sigma_b \rightarrow 0} \text{ (the charge "drains off"); } V(0) = - \int_{\infty}^a (0) dr - \int_a^R \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right) dr - \int_R^0 (0) dr = \boxed{\frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \frac{q}{a} \right)}.$$

Problem 2.39

$$(a) \boxed{\sigma_a = -\frac{q_a}{4\pi a^2}; \quad \sigma_b = -\frac{q_b}{4\pi b^2}; \quad \sigma_R = \frac{q_a + q_b}{4\pi R^2}}.$$

$$(b) \boxed{\mathbf{E}_{\text{out}} = \frac{1}{4\pi\epsilon_0} \frac{q_a + q_b}{r^2} \hat{\mathbf{r}}}, \text{ where } \mathbf{r} = \text{vector from center of large sphere}.$$

$$(c) \boxed{\mathbf{E}_a = \frac{1}{4\pi\epsilon_0} \frac{q_a}{r_a^2} \hat{\mathbf{r}}_a, \quad \mathbf{E}_b = \frac{1}{4\pi\epsilon_0} \frac{q_b}{r_b^2} \hat{\mathbf{r}}_b}, \text{ where } \mathbf{r}_a \text{ (} \mathbf{r}_b \text{) is the vector from center of cavity } a \text{ (} b \text{)}.$$

$$(d) \boxed{\text{Zero.}}$$

$$(e) \sigma_R \text{ changes (but not } \sigma_a \text{ or } \sigma_b \text{); } \mathbf{E}_{\text{outside}} \text{ changes (but not } \mathbf{E}_a \text{ or } \mathbf{E}_b \text{); force on } q_a \text{ and } q_b \text{ still zero.}$$

Problem 2.40

(a) $\boxed{\text{No.}}$ For example, if it is very close to the wall, it will induce charge of the opposite sign on the wall, and it will be attracted.

(b) $\boxed{\text{No.}}$ *Typically* it will be attractive, but see footnote 12 for an extraordinary counterexample.

Problem 2.41

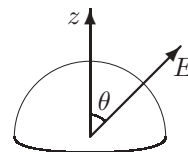
Between the plates, $E = 0$; outside the plates $E = \sigma/\epsilon_0 = Q/\epsilon_0 A$. So

$$P = \frac{\epsilon_0}{2} E^2 = \frac{\epsilon_0}{2} \frac{Q^2}{\epsilon_0^2 A^2} = \boxed{\frac{Q^2}{2\epsilon_0 A^2}}.$$

Problem 2.42

Inside, $\mathbf{E} = \mathbf{0}$; outside, $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$; so

$$\mathbf{E}_{\text{ave}} = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \hat{\mathbf{r}}; \quad f_z = \sigma(E_{\text{ave}})_z; \quad \sigma = \frac{Q}{4\pi R^2}.$$



$$F_z = \int f_z da = \int \left(\frac{Q}{4\pi R^2} \right) \frac{1}{2} \left(\frac{1}{4\pi\epsilon_0} \frac{Q}{R^2} \right) \cos \theta R^2 \sin \theta d\theta d\phi$$

$$= \frac{1}{2\epsilon_0} \left(\frac{Q}{4\pi R} \right)^2 2\pi \int_0^{\pi/2} \sin \theta \cos \theta d\theta = \frac{1}{\pi\epsilon_0} \left(\frac{Q}{4R} \right)^2 \left(\frac{1}{2} \sin^2 \theta \right) \Big|_0^{\pi/2} = \frac{1}{2\pi\epsilon_0} \left(\frac{Q}{4R} \right)^2 = \boxed{\frac{Q^2}{32\pi R^2 \epsilon_0}}.$$

Problem 2.43

Say the charge on the inner cylinder is Q , for a length L . The field is given by Gauss's law:

$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 2\pi s \cdot L = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} Q \Rightarrow \mathbf{E} = \frac{Q}{2\pi\epsilon_0 L} \frac{1}{s} \hat{\mathbf{s}}$. Potential difference between the cylinders is

$$V(b) - V(a) = - \int_a^b \mathbf{E} \cdot d\mathbf{l} = - \frac{Q}{2\pi\epsilon_0 L} \int_a^b \frac{1}{s} ds = - \frac{Q}{2\pi\epsilon_0 L} \ln \left(\frac{b}{a} \right).$$

As set up here, a is at the higher potential, so $V = V(a) - V(b) = \frac{Q}{2\pi\epsilon_0 L} \ln \left(\frac{b}{a} \right)$.

$$C = \frac{Q}{V} = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{b}{a}\right)}, \text{ so capacitance per unit length is } \boxed{\frac{2\pi\epsilon_0}{\ln\left(\frac{b}{a}\right)}}.$$

Problem 2.44

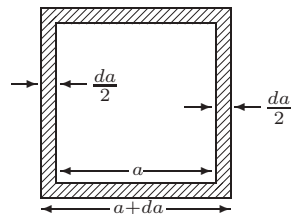
(a) $W = (\text{force}) \times (\text{distance}) = (\text{pressure}) \times (\text{area}) \times (\text{distance}) = \boxed{\frac{\epsilon_0}{2} E^2 A \epsilon.}$

(b) $W = (\text{energy per unit volume}) \times (\text{decrease in volume}) = \left(\epsilon_0 \frac{E^2}{2}\right) (A\epsilon).$ Same as (a), confirming that the energy lost is equal to the work done.

Problem 2.45

From Prob. 2.4, the field at height z above the center of a square loop (side a) is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{4\lambda a z}{\left(z^2 + \frac{a^2}{4}\right) \sqrt{z^2 + \frac{a^2}{2}}} \hat{\mathbf{z}}.$$



Here $\lambda \rightarrow \sigma \frac{da}{2}$ (see figure), and we integrate over a from 0 to $\bar{a}$:

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} 2\sigma z \int_0^{\bar{a}} \frac{a da}{\left(z^2 + \frac{a^2}{4}\right) \sqrt{z^2 + \frac{a^2}{2}}}. \text{ Let } u = \frac{a^2}{4}, \text{ so } a da = 2 du. \\ &= \frac{1}{4\pi\epsilon_0} 4\sigma z \int_0^{\bar{a}^2/4} \frac{du}{(u + z^2) \sqrt{2u + z^2}} = \frac{\sigma z}{\pi\epsilon_0} \left[\frac{2}{z} \tan^{-1} \left(\frac{\sqrt{2u + z^2}}{z} \right) \right]_0^{\bar{a}^2/4} \\ &= \frac{2\sigma}{\pi\epsilon_0} \left\{ \tan^{-1} \left(\frac{\sqrt{\frac{\bar{a}^2}{2} + z^2}}{z} \right) - \tan^{-1}(1) \right\}; \end{aligned}$$

$$\boxed{\mathbf{E} = \frac{2\sigma}{\pi\epsilon_0} \left[\tan^{-1} \sqrt{1 + \frac{a^2}{2z^2}} - \frac{\pi}{4} \right] \hat{\mathbf{z}}} = \frac{\sigma}{\pi\epsilon_0} \tan^{-1} \left(\frac{a^2}{4z\sqrt{z^2 + (a^2/2)}} \right) \hat{\mathbf{z}}.$$

$a \rightarrow \infty$ (infinite plane): $E = \frac{2\sigma}{\pi\epsilon_0} \left[\tan^{-1}(\infty) - \frac{\pi}{4} \right] = \frac{2\sigma}{\pi\epsilon_0} \left(\frac{\pi}{2} - \frac{\pi}{4} \right) = \frac{\sigma}{2\epsilon_0} \cdot \checkmark$

$z \gg a$ (point charge): Let $f(x) = \tan^{-1} \sqrt{1+x} - \frac{\pi}{4}$, and expand as a Taylor series:

$$f(x) = f(0) + x f'(0) + \frac{1}{2} x^2 f''(0) + \dots$$

Here $f(0) = \tan^{-1}(1) - \frac{\pi}{4} = \frac{\pi}{4} - \frac{\pi}{4} = 0$; $f'(x) = \frac{1}{1+(1+x)} \frac{1}{2} \frac{1}{\sqrt{1+x}} = \frac{1}{2(2+x)\sqrt{1+x}}$, so $f'(0) = \frac{1}{4}$, so

$$f(x) = \frac{1}{4} x + () x^2 + () x^3 + \dots$$

Thus (since $\frac{a^2}{2z^2} = x \ll 1$), $E \approx \frac{2\sigma}{\pi\epsilon_0} \left(\frac{1}{4} \frac{a^2}{2z^2} \right) = \frac{1}{4\pi\epsilon_0} \frac{\sigma a^2}{z^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \cdot \checkmark$

Problem 2.46

$$\begin{aligned}
 \rho &= \epsilon_0 \nabla \cdot \mathbf{E} = \epsilon_0 \left\{ \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{3k}{r} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{2k \sin \theta \cos \theta \sin \phi}{r} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\frac{k \sin \theta \cos \phi}{r} \right) \right\} \\
 &= \epsilon_0 \left[\frac{1}{r^2} 3k + \frac{1}{r \sin \theta} \frac{2k \sin \phi (2 \sin \theta \cos^2 \theta - \sin^3 \theta)}{r} + \frac{1}{r \sin \theta} \frac{(-k \sin \theta \sin \phi)}{r} \right] \\
 &= \frac{k \epsilon_0}{r^2} [3 + 2 \sin \phi (2 \cos^2 \theta - \sin^2 \theta) - \sin \phi] = \frac{k \epsilon_0}{r^2} [3 + \sin \phi (4 \cos^2 \theta - 2 + 2 \cos^2 \theta - 1)] \\
 &= \frac{3k \epsilon_0}{r^2} [1 + \sin \phi (2 \cos^2 \theta - 1)] = \boxed{\frac{3k \epsilon_0}{r^2} (1 + \sin \phi \cos 2\theta)}.
 \end{aligned}$$

Problem 2.47

From Prob. 2.12, the field inside a uniformly charged sphere is: $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \mathbf{r}$. So the force per unit volume is $\mathbf{f} = \rho \mathbf{E} = \left(\frac{Q}{\frac{4}{3}\pi R^3} \right) \left(\frac{Q}{4\pi\epsilon_0 R^3} \right) \mathbf{r} = \frac{3}{\epsilon_0} \left(\frac{Q}{4\pi R^3} \right)^2 \mathbf{r}$, and the force in the z direction on $d\tau$ is:

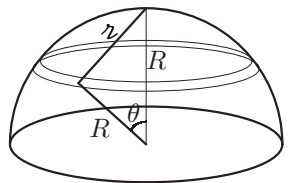
$$dF_z = f_z d\tau = \frac{3}{\epsilon_0} \left(\frac{Q}{4\pi R^3} \right)^2 r \cos \theta (r^2 \sin \theta dr d\theta d\phi).$$

The total force on the “northern” hemisphere is:

$$\begin{aligned}
 F_z &= \int f_z d\tau = \frac{3}{\epsilon_0} \left(\frac{Q}{4\pi R^3} \right)^2 \int_0^R r^3 dr \int_0^{\pi/2} \cos \theta \sin \theta d\theta \int_0^{2\pi} d\phi \\
 &= \frac{3}{\epsilon_0} \left(\frac{Q}{4\pi R^3} \right)^2 \left(\frac{R^4}{4} \right) \left(\frac{\sin^2 \theta}{2} \Big|_0^{\pi/2} \right) (2\pi) = \boxed{\frac{3Q^2}{64\pi\epsilon_0 R^2}}.
 \end{aligned}$$

Problem 2.48

$$\begin{aligned}
 V_{\text{center}} &= \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{z} da = \frac{1}{4\pi\epsilon_0} \frac{\sigma}{R} \int da = \frac{1}{4\pi\epsilon_0} \frac{\sigma}{R} (2\pi R^2) = \frac{\sigma R}{2\epsilon_0} \\
 V_{\text{pole}} &= \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{z} da, \text{ with } \begin{cases} da = 2\pi R^2 \sin \theta d\theta, \\ z^2 = R^2 + R^2 - 2R^2 \cos \theta = 2R^2(1 - \cos \theta). \end{cases} \\
 &= \frac{1}{4\pi\epsilon_0} \frac{\sigma (2\pi R^2)}{R\sqrt{2}} \int_0^{\pi/2} \frac{\sin \theta d\theta}{\sqrt{1 - \cos \theta}} = \frac{\sigma R}{2\sqrt{2}\epsilon_0} (2\sqrt{1 - \cos \theta}) \Big|_0^{\pi/2} \\
 &= \frac{\sigma R}{\sqrt{2}\epsilon_0} (1 - 0) = \frac{\sigma R}{\sqrt{2}\epsilon_0}. \quad \therefore V_{\text{pole}} - V_{\text{center}} = \boxed{\frac{\sigma R}{2\epsilon_0} (\sqrt{2} - 1)}.
 \end{aligned}$$



Problem 2.49

First let's determine the electric field inside and outside the sphere, using Gauss's law:

$$\epsilon_0 \oint \mathbf{E} \cdot d\mathbf{a} = \epsilon_0 4\pi r^2 E = Q_{\text{enc}} = \int \rho d\tau = \int (k\bar{r}) \bar{r}^2 \sin \theta d\bar{r} d\theta d\phi = 4\pi k \int_0^r \bar{r}^3 d\bar{r} = \begin{cases} \pi k r^4 & (r < R), \\ \pi k R^4 & (r > R). \end{cases}$$

$$\text{So } \mathbf{E} = \frac{k}{4\epsilon_0} r^2 \hat{\mathbf{r}} \quad (r < R); \quad \mathbf{E} = \frac{kR^4}{4\epsilon_0 r^2} \hat{\mathbf{r}} \quad (r > R).$$

Method I:

$$\begin{aligned}
 W &= \frac{\epsilon_0}{2} \int E^2 d\tau \text{ (Eq. 2.45)} = \frac{\epsilon_0}{2} \int_0^R \left(\frac{kr^2}{4\epsilon_0} \right)^2 4\pi r^2 dr + \frac{\epsilon_0}{2} \int_R^\infty \left(\frac{kR^4}{4\epsilon_0 r^2} \right)^2 4\pi r^2 dr \\
 &= 4\pi \frac{\epsilon_0}{2} \left(\frac{k}{4\epsilon_0} \right)^2 \left\{ \int_0^R r^6 dr + R^8 \int_R^\infty \frac{1}{r^2} dr \right\} = \frac{\pi k^2}{8\epsilon_0} \left\{ \frac{R^7}{7} + R^8 \left(-\frac{1}{r} \right) \Big|_R^\infty \right\} = \frac{\pi k^2}{8\epsilon_0} \left(\frac{R^7}{7} + R^7 \right) \\
 &= \boxed{\frac{\pi k^2 R^7}{7\epsilon_0}}.
 \end{aligned}$$

Method II:

$$W = \frac{1}{2} \int \rho V d\tau \text{ (Eq. 2.43).}$$

$$\begin{aligned}
 \text{For } r < R, \quad V(r) &= - \int_\infty^r \mathbf{E} \cdot d\mathbf{l} = - \int_\infty^R \left(\frac{kR^4}{4\epsilon_0 r^2} \right) dr - \int_R^r \left(\frac{kr^2}{4\epsilon_0} \right) dr = - \frac{k}{4\epsilon_0} \left\{ R^4 \left(-\frac{1}{r} \right) \Big|_\infty^R + \frac{r^3}{3} \Big|_R^r \right\} \\
 &= - \frac{k}{4\epsilon_0} \left(-R^3 + \frac{r^3}{3} - \frac{R^3}{3} \right) = \frac{k}{3\epsilon_0} \left(R^3 - \frac{r^3}{4} \right). \\
 \therefore W &= \frac{1}{2} \int_0^R (kr) \left[\frac{k}{3\epsilon_0} \left(R^3 - \frac{r^3}{4} \right) \right] 4\pi r^2 dr = \frac{2\pi k^2}{3\epsilon_0} \int_0^R \left(R^3 r^3 - \frac{1}{4} r^6 \right) dr \\
 &= \frac{2\pi k^2}{3\epsilon_0} \left\{ R^3 \frac{R^4}{4} - \frac{1}{4} \frac{R^7}{7} \right\} = \frac{\pi k^2 R^7}{2 \cdot 3\epsilon_0} \left(\frac{6}{7} \right) = \frac{\pi k^2 R^7}{7\epsilon_0}. \checkmark
 \end{aligned}$$

Problem 2.50

$$\mathbf{E} = -\nabla V = -A \frac{\partial}{\partial r} \left(\frac{e^{-\lambda r}}{r} \right) \hat{\mathbf{r}} = -A \left\{ \frac{r(-\lambda)e^{-\lambda r} - e^{-\lambda r}}{r^2} \right\} \hat{\mathbf{r}} = \boxed{Ae^{-\lambda r}(1 + \lambda r) \frac{\hat{\mathbf{r}}}{r^2}}.$$

$\rho = \epsilon_0 \nabla \cdot \mathbf{E} = \epsilon_0 A \left\{ e^{-\lambda r}(1 + \lambda r) \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) + \frac{\hat{\mathbf{r}}}{r^2} \cdot \nabla (e^{-\lambda r}(1 + \lambda r)) \right\}$. But $\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r})$ (Eq. 1.99), and $e^{-\lambda r}(1 + \lambda r) \delta^3(\mathbf{r}) = \delta^3(\mathbf{r})$ (Eq. 1.88). Meanwhile,

$$\nabla (e^{-\lambda r}(1 + \lambda r)) = \hat{\mathbf{r}} \frac{\partial}{\partial r} (e^{-\lambda r}(1 + \lambda r)) = \hat{\mathbf{r}} \{-\lambda e^{-\lambda r}(1 + \lambda r) + e^{-\lambda r} \lambda\} = \hat{\mathbf{r}}(-\lambda^2 r e^{-\lambda r}).$$

$$\text{So } \frac{\hat{\mathbf{r}}}{r^2} \cdot \nabla (e^{-\lambda r}(1 + \lambda r)) = -\frac{\lambda^2}{r} e^{-\lambda r}, \text{ and } \boxed{\rho = \epsilon_0 A \left[4\pi \delta^3(\mathbf{r}) - \frac{\lambda^2}{r} e^{-\lambda r} \right]}.$$

$$Q = \int \rho d\tau = \epsilon_0 A \left\{ 4\pi \int \delta^3(\mathbf{r}) d\tau - \lambda^2 \int \frac{e^{-\lambda r}}{r} 4\pi r^2 dr \right\} = \epsilon_0 A \left(4\pi - \lambda^2 4\pi \int_0^\infty r e^{-\lambda r} dr \right).$$

$$\text{But } \int_0^\infty r e^{-\lambda r} dr = \frac{1}{\lambda^2}, \text{ so } Q = 4\pi \epsilon_0 A \left(1 - \frac{\lambda^2}{\lambda^2} \right) = \boxed{\text{zero.}}$$

Problem 2.51

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{\mathcal{Z}} da = \frac{\sigma}{4\pi\epsilon_0} \int_0^R \int_0^{2\pi} \frac{1}{\sqrt{R^2 + s^2 - 2Rs \cos \phi}} s ds d\phi.$$

Let $u \equiv s/R$. Then

$$V = \frac{2\sigma R}{4\pi\epsilon_0} \int_0^1 \left(\int_0^\pi \frac{u}{\sqrt{1 + u^2 - 2u \cos \phi}} d\phi \right) du.$$

The (double) integral is a pure number; Mathematica says it is 2. So

$$V = \frac{\sigma R}{\pi \epsilon_0}.$$

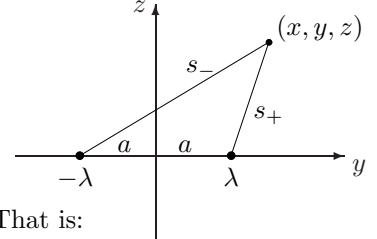
Problem 2.52

- (a) Potential of $+\lambda$ is $V_+ = -\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_+}{a}\right)$, where s_+ is distance from λ_+ (Prob. 2.22).
 Potential of $-\lambda$ is $V_- = +\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_-}{a}\right)$, where s_- is distance from λ_- .

$$\therefore \text{Total } V = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{s_-}{s_+}\right).$$

Now $s_+ = \sqrt{(y-a)^2 + z^2}$, and $s_- = \sqrt{(y+a)^2 + z^2}$, so

$$V(x, y, z) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{\sqrt{(y+a)^2 + z^2}}{\sqrt{(y-a)^2 + z^2}}\right) = \frac{\lambda}{4\pi\epsilon_0} \ln\left[\frac{(y+a)^2 + z^2}{(y-a)^2 + z^2}\right].$$



- (b) Equipotentials are given by $\frac{(y+a)^2 + z^2}{(y-a)^2 + z^2} = e^{(4\pi\epsilon_0 V_0/\lambda)} = k = \text{constant}$. That is:
 $y^2 + 2ay + a^2 + z^2 = k(y^2 - 2ay + a^2 + z^2) \Rightarrow y^2(k-1) + z^2(k-1) + a^2(k-1) - 2ay(k+1) = 0$, or
 $y^2 + z^2 + a^2 - 2ay\left(\frac{k+1}{k-1}\right) = 0$. The equation for a *circle*, with center at $(y_0, 0)$ and radius R , is
 $(y - y_0)^2 + z^2 = R^2$, or $y^2 + z^2 + (y_0^2 - R^2) - 2yy_0 = 0$.

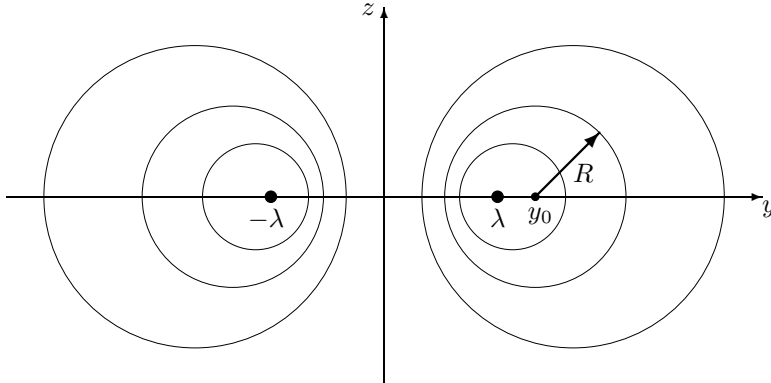
Evidently the equipotentials *are* circles, with $y_0 = a\left(\frac{k+1}{k-1}\right)$ and

$$a^2 = y_0^2 - R^2 \Rightarrow R^2 = y_0^2 - a^2 = a^2\left(\frac{k+1}{k-1}\right)^2 - a^2 = a^2\frac{(k^2+2k+1-k^2-2k-1)}{(k-1)^2} = a^2\frac{4k}{(k-1)^2}, \text{ or}$$

$$R = \frac{2a\sqrt{k}}{|k-1|}; \text{ or, in terms of } V_0:$$

$$y_0 = a\frac{e^{4\pi\epsilon_0 V_0/\lambda} + 1}{e^{4\pi\epsilon_0 V_0/\lambda} - 1} = a\frac{e^{2\pi\epsilon_0 V_0/\lambda} + e^{-2\pi\epsilon_0 V_0/\lambda}}{e^{2\pi\epsilon_0 V_0/\lambda} - e^{-2\pi\epsilon_0 V_0/\lambda}} = a \coth\left(\frac{2\pi\epsilon_0 V_0}{\lambda}\right).$$

$$R = 2a\frac{e^{2\pi\epsilon_0 V_0/\lambda}}{e^{4\pi\epsilon_0 V_0/\lambda} - 1} = a\frac{2}{(e^{2\pi\epsilon_0 V_0/\lambda} - e^{-2\pi\epsilon_0 V_0/\lambda})} = \frac{a}{\sinh\left(\frac{2\pi\epsilon_0 V_0}{\lambda}\right)} = a \operatorname{csch}\left(\frac{2\pi\epsilon_0 V_0}{\lambda}\right).$$



Problem 2.53

(a) $\nabla^2 V = -\frac{\rho}{\epsilon_0}$ (Eq. 2.24), so $\boxed{\frac{d^2 V}{dx^2} = -\frac{1}{\epsilon_0} \rho}$.

(b) $qV = \frac{1}{2}mv^2 \rightarrow \boxed{v = \sqrt{\frac{2qV}{m}}}$.

(c) $dq = A\rho dx$; $\frac{dq}{dt} = a\rho \frac{dx}{dt} = \boxed{A\rho v = I}$ (constant). (Note: ρ , hence also I , is negative.)

(d) $\frac{d^2 V}{dx^2} = -\frac{1}{\epsilon_0} \rho = -\frac{1}{\epsilon_0} \frac{I}{Av} = -\frac{I}{\epsilon_0 A} \sqrt{\frac{m}{2qV}} \Rightarrow \boxed{\frac{d^2 V}{dx^2} = \beta V^{-1/2}}$, where $\beta = -\frac{I}{\epsilon_0 A} \sqrt{\frac{m}{2q}}$.

(Note: I is negative, so β is positive; q is positive.)

(e) Multiply by $V' = \frac{dV}{dx}$:

$$V' \frac{dV'}{dx} = \beta V^{-1/2} \frac{dV}{dx} \Rightarrow \int V' dV' = \beta \int V^{-1/2} dV \Rightarrow \frac{1}{2} V'^2 = 2\beta V^{1/2} + \text{constant}.$$

But $V(0) = V'(0) = 0$ (cathode is at potential zero, and field at cathode is zero), so the constant is zero, and

$$V'^2 = 4\beta V^{1/2} \Rightarrow \frac{dV}{dx} = 2\sqrt{\beta} V^{1/4} \Rightarrow V^{-1/4} dV = 2\sqrt{\beta} dx;$$

$$\int V^{-1/4} dV = 2\sqrt{\beta} \int dx \Rightarrow \frac{4}{3} V^{3/4} = 2\sqrt{\beta} x + \text{constant}.$$

But $V(0) = 0$, so this constant is also zero.

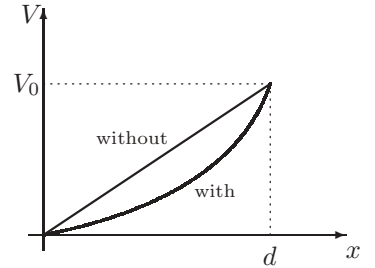
$$V^{3/4} = \frac{3}{2} \sqrt{\beta} x, \text{ so } V(x) = \left(\frac{3}{2} \sqrt{\beta} \right)^{4/3} x^{4/3}, \text{ or } V(x) = \left(\frac{9}{4} \beta \right)^{2/3} x^{4/3} = \left(\frac{81 I^2 m}{32 \epsilon_0^2 A^2 q} \right)^{1/3} x^{4/3}.$$

In terms of V_0 (instead of I): $\boxed{V(x) = V_0 \left(\frac{x}{d} \right)^{4/3}}$ (see graph).

Without space-charge, V would increase linearly: $V(x) = V_0 \left(\frac{x}{d} \right)$.

$$\rho = -\epsilon_0 \frac{d^2 V}{dx^2} = -\epsilon_0 V_0 \frac{1}{d^{4/3}} \frac{4}{3} \cdot \frac{1}{3} x^{-2/3} = \boxed{-\frac{4\epsilon_0 V_0}{9(d^2 x)^{2/3}}}.$$

$$v = \sqrt{\frac{2q}{m}} \sqrt{V} = \boxed{\sqrt{2qV_0/m} \left(\frac{x}{d} \right)^{2/3}}.$$



(f) $V(d) = V_0 = \left(\frac{81 I^2 m}{32 \epsilon_0^2 A^2 q} \right)^{1/3} d^{4/3} \Rightarrow V_0^3 = \frac{81 m d^4}{32 \epsilon_0^2 A^2 q} I^2$; $I^2 = \frac{32 \epsilon_0^2 A^2 q}{81 m d^4} V_0^3$;

$$I = \frac{4\sqrt{2} \epsilon_0 A \sqrt{q}}{9\sqrt{m} d^2} V_0^{3/2} = K V_0^{3/2}, \text{ where } \boxed{K = \frac{4\epsilon_0 A}{9d^2} \sqrt{\frac{2q}{m}}}.$$

Problem 2.54

(a) $\boxed{\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho \hat{\mathbf{z}}}{z^2} \left(1 + \frac{z}{\lambda} \right) e^{-z/\lambda} d\tau}$.

(b) Yes. The field of a point charge at the origin is radial and symmetric, so $\nabla \times \mathbf{E} = \mathbf{0}$, and hence this is also true (by superposition) for any *collection* of charges.

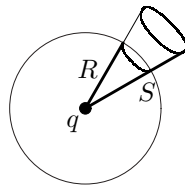
$$\begin{aligned} \text{(c)} \quad V &= - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = - \frac{1}{4\pi\epsilon_0} q \int_{\infty}^r \frac{1}{r^2} \left(1 + \frac{r}{\lambda}\right) e^{-r/\lambda} dr \\ &= \frac{1}{4\pi\epsilon_0} q \int_r^{\infty} \frac{1}{r^2} \left(1 + \frac{r}{\lambda}\right) e^{-r/\lambda} dr = \frac{q}{4\pi\epsilon_0} \left\{ \int_r^{\infty} \frac{1}{r^2} e^{-r/\lambda} dr + \frac{1}{\lambda} \int_r^{\infty} \frac{1}{r} e^{-r/\lambda} dr \right\}. \end{aligned}$$

Now $\int \frac{1}{r^2} e^{-r/\lambda} dr = -\frac{e^{-r/\lambda}}{r} - \frac{1}{\lambda} \int \frac{e^{-r/\lambda}}{r} dr \leftarrow$ exactly right to kill the last term. Therefore

$$V(r) = \frac{q}{4\pi\epsilon_0} \left\{ -\frac{e^{-r/\lambda}}{r} \Big|_r^{\infty} \right\} = \boxed{\frac{q}{4\pi\epsilon_0} \frac{e^{-r/\lambda}}{r}}.$$

$$\begin{aligned} \text{(d)} \quad \oint_S \mathbf{E} \cdot d\mathbf{a} &= \frac{1}{4\pi\epsilon_0} q \frac{1}{R^2} \left(1 + \frac{R}{\lambda}\right) e^{-R/\lambda} 4\pi R^2 = \frac{q}{\epsilon_0} \left(1 + \frac{R}{\lambda}\right) e^{-R/\lambda}. \\ \int_V V d\tau &= \frac{q}{4\pi\epsilon_0} \int_0^R \frac{e^{-r/\lambda}}{r} r^2 4\pi dr = \frac{q}{\epsilon_0} \int_0^R r e^{-r/\lambda} dr = \frac{q}{\epsilon_0} \left[\frac{e^{-r/\lambda}}{(1/\lambda)^2} \left(-\frac{r}{\lambda} - 1\right) \right]_0^R \\ &= \lambda^2 \frac{q}{\epsilon_0} \left\{ -e^{-R/\lambda} \left(1 + \frac{R}{\lambda}\right) + 1 \right\}. \\ \therefore \oint_S \mathbf{E} \cdot d\mathbf{a} + \frac{1}{\lambda^2} \int_V V d\tau &= \frac{q}{\epsilon_0} \left\{ \left(1 + \frac{R}{\lambda}\right) e^{-R/\lambda} - \left(1 + \frac{R}{\lambda}\right) e^{-R/\lambda} + 1 \right\} = \frac{q}{\epsilon_0}. \quad \text{qed} \end{aligned}$$

(e) Does the result in (d) hold for a *nonspherical* surface? Suppose we make a “dent” in the sphere—pushing a patch (area $R^2 \sin \theta d\theta d\phi$) from radius R out to radius S (area $S^2 \sin \theta d\theta d\phi$).



$$\begin{aligned} \Delta \oint \mathbf{E} \cdot d\mathbf{a} &= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{S^2} \left(1 + \frac{S}{\lambda}\right) e^{-S/\lambda} (S^2 \sin \theta d\theta d\phi) - \frac{1}{R^2} \left(1 + \frac{R}{\lambda}\right) e^{-R/\lambda} (R^2 \sin \theta d\theta d\phi) \right\} \\ &= \frac{q}{4\pi\epsilon_0} \left[\left(1 + \frac{S}{\lambda}\right) e^{-S/\lambda} - \left(1 + \frac{R}{\lambda}\right) e^{-R/\lambda} \right] \sin \theta d\theta d\phi. \end{aligned}$$

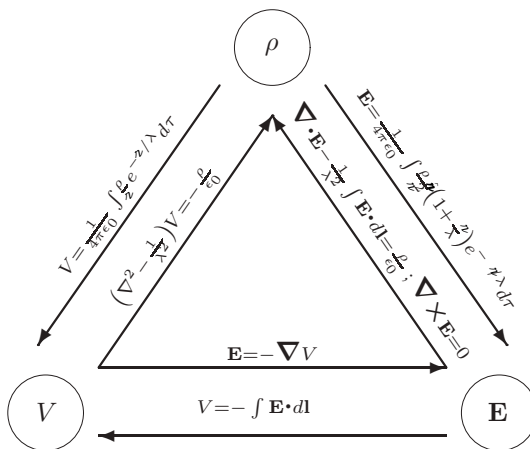
$$\begin{aligned} \Delta \frac{1}{\lambda^2} \int V d\tau &= \frac{1}{\lambda^2} \frac{q}{4\pi\epsilon_0} \int \frac{e^{-r/\lambda}}{r} r^2 \sin \theta dr d\theta d\phi = \frac{1}{\lambda^2} \frac{q}{4\pi\epsilon_0} \sin \theta d\theta d\phi \int_R^S r e^{-r/\lambda} dr \\ &= -\frac{q}{4\pi\epsilon_0} \sin \theta d\theta d\phi \left(e^{-r/\lambda} \left(1 + \frac{r}{\lambda}\right) \right) \Big|_R^S \\ &= -\frac{q}{4\pi\epsilon_0} \left[\left(1 + \frac{S}{\lambda}\right) e^{-S/\lambda} - \left(1 + \frac{R}{\lambda}\right) e^{-R/\lambda} \right] \sin \theta d\theta d\phi. \end{aligned}$$

So the change in $\frac{1}{\lambda^2} \int V d\tau$ exactly compensates for the change in $\oint \mathbf{E} \cdot d\mathbf{a}$, and we get $\frac{1}{\epsilon_0} q$ for the total using the dented sphere, just as we did with the perfect sphere. Any closed surface can be built up by successive distortions of the sphere, so the result holds for all shapes. By superposition, if there are many charges inside, the total is $\frac{1}{\epsilon_0} Q_{\text{enc}}$. Charges *outside* do not contribute (in the argument above we found that $\oint \mathbf{E} \cdot d\mathbf{a} = 0$ for this volume).

for this volume $\oint \mathbf{E} \cdot d\mathbf{a} + \frac{1}{\lambda^2} \int V d\tau = 0$ —and, again, the sum is not changed by distortions of the surface, as long as q remains outside). So the new “Gauss’s Law” holds for *any* charge configuration.

(f) In differential form, “Gauss’s law” reads: $\nabla \cdot \mathbf{E} + \frac{1}{\lambda^2} V = \frac{1}{\epsilon_0} \rho$, or, putting it all in terms of $\mathbf{E}$:

$$\nabla \cdot \mathbf{E} - \frac{1}{\lambda^2} \int \mathbf{E} \cdot d\mathbf{l} = \frac{1}{\epsilon_0} \rho. \text{ Since } \mathbf{E} = -\nabla V, \text{ this also yields “Poisson’s equation”}: -\nabla^2 V + \frac{1}{\lambda^2} V = \frac{1}{\epsilon_0} \rho.$$



(g) Refer to “Gauss’s law” in differential form (f). Since $\mathbf{E}$ is zero, inside a conductor (otherwise charge would move, and in such a direction as to cancel the field), V is constant (inside), and hence ρ is uniform, throughout the volume. Any “extra” charge must reside on the surface. (The fraction at the surface depends on λ , and on the shape of the conductor.)

Problem 2.55

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} = \epsilon_0 \frac{\partial}{\partial x}(ax) = \boxed{\epsilon_0 a} \text{ (constant everywhere).}$$

The same charge density would be compatible (as far as Gauss’s law is concerned) with $\mathbf{E} = ay\hat{\mathbf{y}}$, for instance, or $\mathbf{E} = (\frac{a}{3})\mathbf{r}$, etc. The point is that Gauss’s law (and $\nabla \times \mathbf{E} = \mathbf{0}$) by themselves *do not determine the field*—like any differential equations, they must be supplemented by appropriate *boundary conditions*. Ordinarily, these are so “obvious” that we impose them almost subconsciously (“ E must go to zero far from the source charges”)—or we appeal to symmetry to resolve the ambiguity (“the field must be the same—in magnitude—on both sides of an infinite plane of surface charge”). But in this case there are *no* natural boundary conditions, and no persuasive symmetry conditions, to fix the answer. The question “What is the electric field produced by a uniform charge density filling all of space?” is simply *ill-posed*: it does not give us sufficient information to determine the answer. (Incidentally, it won’t help to appeal to Coulomb’s law ($\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \rho \frac{\hat{\mathbf{r}}}{r^2} d\tau$)—the integral is hopelessly indefinite, in this case.)

Problem 2.56

Compare Newton’s law of universal gravitation to Coulomb’s law:

$$\mathbf{F} = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}}; \quad \mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}.$$

Evidently $\frac{1}{4\pi\epsilon_0} \rightarrow G$ and $q \rightarrow m$. The gravitational energy of a sphere (translating Prob. 2.34) is therefore

$$\boxed{W_{\text{grav}} = \frac{3}{5} G \frac{M^2}{R}}.$$

Now, $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$, and for the sun $M = 1.99 \times 10^{30} \text{ kg}$, $R = 6.96 \times 10^8 \text{ m}$, so the sun's gravitational energy is $W = 2.28 \times 10^{41} \text{ J}$. At the current rate this energy would be dissipated in a time

$$t = \frac{W}{P} = \frac{2.28 \times 10^{41}}{3.86 \times 10^{26}} = 5.90 \times 10^{14} \text{ s} = \boxed{1.87 \times 10^7 \text{ years.}}$$

Problem 2.57

First eliminate z , using the formula for the ellipsoid:

$$\sigma(x, y) = \frac{Q}{4\pi ab} \frac{1}{\sqrt{c^2(x^2/a^4) + c^2(y^2/b^4) + 1 - (x^2/a^2) - (y^2/b^2)}}.$$

Now (for parts (a) and (b)) set $c \rightarrow 0$, “squashing” the ellipsoid down to an ellipse in the xy plane:

$$\sigma(x, y) = \frac{Q}{2\pi ab} \frac{1}{\sqrt{1 - (x/a)^2 - (y/b)^2}}.$$

(I multiplied by 2 to count both surfaces.)

(a) For the circular disk, set $a = b = R$ and let $r \equiv \sqrt{x^2 + y^2}$. $\sigma(r) = \frac{Q}{2\pi R} \frac{1}{\sqrt{R^2 - r^2}}.$

(b) For the ribbon, let $Q/2b \equiv \Lambda$, and then take the limit $b \rightarrow \infty$: $\sigma(x) = \frac{\Lambda}{2\pi} \frac{1}{\sqrt{a^2 - x^2}}.$

(c) Let $b = c$, $r \equiv \sqrt{y^2 + z^2}$, making an ellipsoid of revolution:

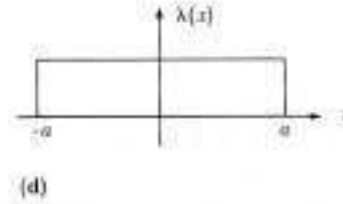
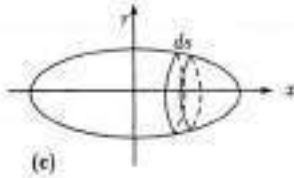
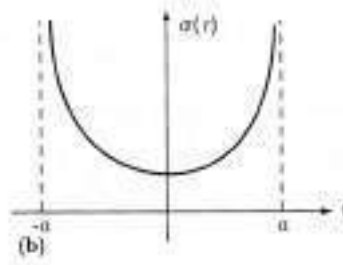
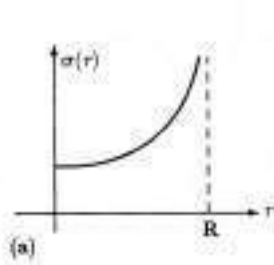
$$\frac{x^2}{a^2} + \frac{r^2}{c^2} = 1, \quad \text{with } \sigma = \frac{Q}{4\pi ac^2} \frac{1}{\sqrt{x^2/a^4 + r^2/c^4}}.$$

The charge on a ring of width dx is

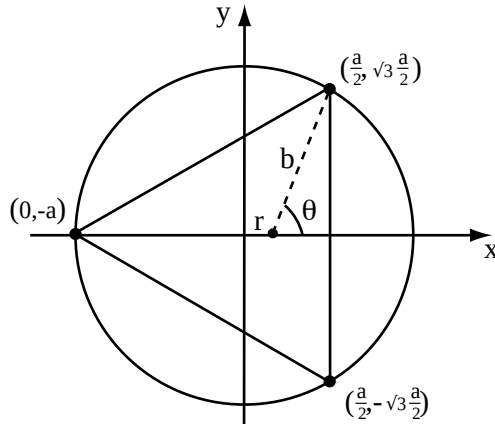
$$dq = \sigma 2\pi r ds, \quad \text{where } ds = \sqrt{dx^2 + dr^2} = dx \sqrt{1 + (dr/dx)^2}.$$

Now $\frac{2x dx}{a^2} + \frac{2r dr}{c^2} = 0 \Rightarrow \frac{dr}{dx} = -\frac{c^2 x}{a^2 r}$, so $ds = dx \sqrt{1 + \frac{c^4 x^2}{a^4 r^2}} = dx \frac{c^2}{r} \sqrt{x^2/a^4 + r^2/c^4}$. Thus

$$\lambda(x) = \frac{dq}{dx} = 2\pi r \frac{Q}{4\pi ac^2} \frac{1}{\sqrt{x^2/a^4 + r^2/c^4}} \frac{c^2}{r} \sqrt{x^2/a^4 + r^2/c^4} = \boxed{\frac{Q}{2a}. \quad (\text{Constant!})}$$



Problem 2.58



(a) One such point is on the x axis (see diagram) at $x = r$. Here the field is

$$E_x = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(a+r)^2} - 2 \frac{\cos \theta}{b^2} \right] = 0, \quad \text{or} \quad \frac{2 \cos \theta}{b^2} = \frac{1}{(a+r)^2}.$$

Now,

$$\cos \theta = \frac{(a/2) - r}{b}; \quad b^2 = \left(\frac{a}{2} - r \right)^2 + \left(\frac{\sqrt{3}}{2} a \right)^2 = (a^2 - ar + r^2).$$

Therefore

$$\frac{2[(a/2) - r]}{(a^2 - ar + r^2)^{3/2}} = \frac{1}{(a+r)^2}. \quad \text{To simplify, let } \frac{r}{a} \equiv u :$$

$$\frac{(1-2u)}{(1-u+u^2)^{3/2}} = \frac{1}{(1+u)^2}, \quad \text{or} \quad (1-2u)^2(1+u)^4 = (1-u+u^2)^3.$$

Multiplying out each side:

$$1 - 6u^2 - 4u^3 + 9u^4 + 12u^5 + 4u^6 = 1 - 3u + 6u^2 - 7u^3 + 6u^4 - 3u^5 + u^6,$$

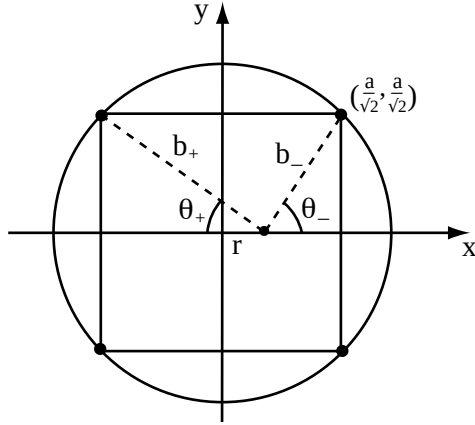
or

$$3u - 12u^2 + 3u^3 + 3u^4 + 15u^5 + 3u^6 = 0.$$

$u = 0$ is a solution (of course—the center of the triangle); factoring out $3u$ we are left with a quintic equation:

$$1 - 4u + u^2 + u^3 + 5u^4 + u^5 = 0.$$

According to Mathematica, this has two complex roots, and one negative root. The two remaining solutions are $u = 0.284718$ and $u = 0.626691$. The latter is outside the triangle, and clearly spurious. So $\boxed{r = 0.284718 a}$. (The other two places where $\mathbf{E} = \mathbf{0}$ are at the symmetrically located points, of course.)



(b) For the square:

$$E_x = \frac{q}{4\pi\epsilon_0} \left(2 \frac{\cos \theta_+}{b_+^2} - 2 \frac{\cos \theta_-}{b_-^2} \right) = 0 \quad \Rightarrow \quad \frac{\cos \theta_+}{b_+^2} = \frac{\cos \theta_-}{b_-^2},$$

where

$$\cos \theta_{\pm} = \frac{(a/\sqrt{2}) \pm r}{b_{\pm}}; \quad b_{\pm}^2 = \left(\frac{a}{\sqrt{2}} \right)^2 + \left(\frac{a}{\sqrt{2}} \pm r \right)^2 = a^2 \pm \sqrt{2} ar + r^2.$$

Thus

$$\frac{(a/\sqrt{2}) + r}{(a^2 + \sqrt{2} ar + r^2)^{3/2}} = \frac{(a/\sqrt{2}) - r}{(a^2 - \sqrt{2} ar + r^2)^{3/2}}.$$

To simplify, let $w \equiv \sqrt{2} r/a$; then

$$\frac{1+w}{(2+2w+w^2)^{3/2}} = \frac{1-w}{(2-2w+w^2)^{3/2}}, \quad \text{or} \quad (1+w)^2(2-2w+w^2)^3 = (1-w)^2(2+2w+w^2)^3.$$

Multiplying out the left side:

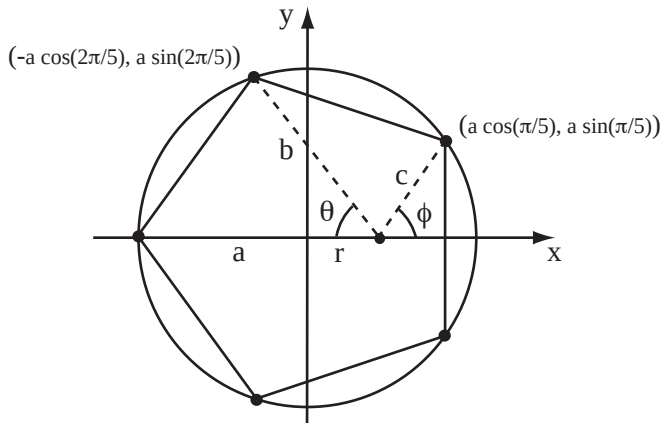
$$8 - 8w - 4w^2 + 16w^3 - 10w^4 - 2w^5 + 7w^6 - 4w^7 + w^8 = (\text{same thing with } w \rightarrow -w).$$

The even powers cancel, leaving

$$8w - 16w^3 + 2w^5 + 4w^7 = 0, \quad \text{or} \quad 4 - 8v + v^2 + 2v^3 = 0,$$

where $v \equiv w^2$. According to Mathematica, this cubic equation has one negative root, one root that is spurious (the point lies outside the square), and $v = 0.598279$, which yields

$$r = \sqrt{\frac{v}{2}} a = \boxed{0.546936 a}.$$



For the pentagon:

$$E_x = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{(a+r)^2} + 2\frac{\cos\theta}{b^2} - 2\frac{\cos\phi}{c^2} \right) = 0,$$

where

$$\cos\theta = \frac{a \cos(2\pi/5) + r}{b}, \quad \cos\phi = \frac{a \cos(\pi/5) - r}{c};$$

$$b^2 = [a \cos(2\pi/5) + r]^2 + [a \sin(2\pi/5)]^2 = a^2 + r^2 + 2ar \cos(2\pi/5),$$

$$c^2 = [a \cos(\pi/5) - r]^2 + [a \sin(\pi/5)]^2 = a^2 + r^2 - 2ar \cos(\pi/5).$$

$$\frac{1}{(a+r)^2} + 2\frac{r + a \cos(2\pi/5)}{[a^2 + r^2 + 2ar \cos(2\pi/5)]^{3/2}} + 2\frac{r - a \cos(\pi/5)}{[a^2 + r^2 - 2ar \cos(\pi/5)]^{3/2}} = 0.$$

Mathematica gives the solution $\boxed{r = 0.688917 a}.$

For an n -sided regular polygon there are evidently n such points, lying on the radial spokes that bisect the sides; their distance from the center appears to grow monotonically with n : $r(3) = 0.285$, $r(4) = 0.547$, $r(5) = 0.689$, $\dots$. As $n \rightarrow \infty$ they fill out a circle that (in the limit) coincides with the ring of charge itself.

Problem 2.59 The theorem is *false*. For example, suppose the conductor is a neutral sphere and the external field is due to a nearby positive point charge q . A negative charge will be induced on the near side of the sphere (and a positive charge on the far side), so the force will be *attractive* (toward q). If we now reverse the sign of q , the induced charges will also reverse, but the force will still be attractive.

If the external field is *uniform*, then the net force on the induced charges is zero, and the total force on the conductor is $Q\mathbf{E}_e$, which *does* switch signs if $\mathbf{E}_e$ is reversed. So the “theorem” is valid in this very special case.

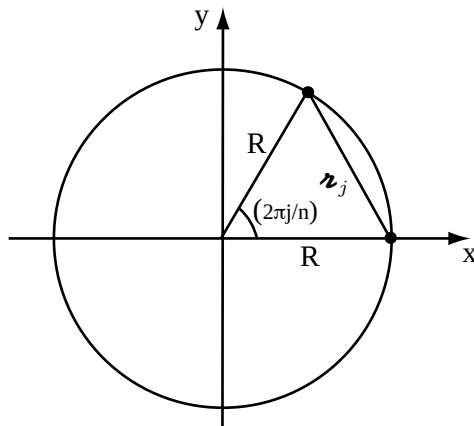
Problem 2.60 The initial configuration consists of a point charge q at the center, $-q$ induced on the inner surface, and $+q$ on the outer surface. What is the energy of this configuration? Imagine assembling it piece-by-piece. First bring in q and place it at the origin—this takes no work. Now bring in $-q$ and spread it over the surface at a —using the method in Prob. 2.35, this takes work $-q^2/(8\pi\epsilon_0 a)$. Finally, bring in $+q$ and spread it over the surface at b —this costs $q^2/(8\pi\epsilon_0 b)$. Thus the energy of the initial configuration is

$$W_i = -\frac{q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right).$$

The final configuration is a neutral shell and a distant point charge—the energy is zero. Thus the work necessary to go from the initial to the final state is

$$W = W_f - W_i = \boxed{\frac{q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)}.$$

Problem 2.61



Suppose the n point charges are evenly spaced around the circle, with the j th particle at angle $j(2\pi/n)$. According to Eq. 2.42, the energy of the configuration is

$$W_n = n \frac{1}{2} q V,$$

where V is the potential due to the $(n-1)$ other charges, at charge $\neq n$ (on the x axis).

$$V = \frac{1}{4\pi\epsilon_0} q \sum_{j=1}^{n-1} \frac{1}{r_j}, \quad r_j = 2R \sin \left(\frac{j\pi}{n} \right)$$

(see the figure). So

$$W_n = \frac{q^2}{4\pi\epsilon_0 R} \frac{n}{4} \sum_{j=1}^{n-1} \frac{1}{\sin(j\pi/n)} = \frac{q^2}{4\pi\epsilon_0 R} \Omega_n.$$

Mathematica says

$$\Omega_{10} = \frac{10}{4} \sum_{j=1}^9 \frac{1}{\sin(j\pi/10)} = 38.6245$$

$$\Omega_{11} = \frac{11}{4} \sum_{j=1}^{10} \frac{1}{\sin(j\pi/11)} = \boxed{48.5757}$$

$$\Omega_{12} = \frac{12}{4} \sum_{j=1}^{11} \frac{1}{\sin(j\pi/12)} = \boxed{59.8074}$$

If $(n-1)$ charges are on the circle (energy $\Omega_{n-1}q^2/4\pi\epsilon_0 R$), and the n th is at the center, the total energy is

$$W_n = [\Omega_{n-1} + (n-1)] \frac{q^2}{4\pi\epsilon_0 R}.$$

For

$$n = 11 : \quad \Omega_{10} + 10 = 38.6245 + 10 = \boxed{48.6245} > \Omega_{11}$$

$$n = 12 : \quad \Omega_{11} + 11 = 48.5757 + 11 = \boxed{59.5757} < \Omega_{12}$$

Thus a lower energy is achieved for 11 charges if they are all at the rim, but for 12 it is better to put one at the center.

Chapter 3

Potential

Problem 3.1

The argument is exactly the same as in Sect. 3.1.4, except that since $z < R$, $\sqrt{z^2 + R^2 - 2zR} = (R - z)$, instead of $(z - R)$. Hence $V_{\text{ave}} = \frac{q}{4\pi\epsilon_0} \frac{1}{2zR} [(z + R) - (R - z)] = \boxed{\frac{1}{4\pi\epsilon_0} \frac{q}{R}}$. If there is more than one charge inside the sphere, the average potential due to interior charges is $\frac{1}{4\pi\epsilon_0} \frac{Q_{\text{enc}}}{R}$, and the average due to exterior charges is V_{center} , so $V_{\text{ave}} = V_{\text{center}} + \frac{Q_{\text{enc}}}{4\pi\epsilon_0 R}$. ✓

Problem 3.2

A stable equilibrium is a point of local minimum in the potential energy. Here the potential energy is qV . But we know that Laplace's equation allows no local minima for V . What *looks* like a minimum, in the figure, must in fact be a saddle point, and the box "leaks" through the center of each face.

Problem 3.3

Laplace's equation in *spherical* coordinates, for V dependent only on r , reads:

$$\nabla^2 V = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = 0 \Rightarrow r^2 \frac{dV}{dr} = c \text{ (constant)} \Rightarrow \frac{dV}{dr} = \frac{c}{r^2} \Rightarrow \boxed{V = -\frac{c}{r} + k.}$$

Example: potential of a uniformly charged sphere.

In *cylindrical* coordinates: $\nabla^2 V = \frac{1}{s} \frac{d}{ds} \left(s \frac{dV}{ds} \right) = 0 \Rightarrow s \frac{dV}{ds} = c \Rightarrow \frac{dV}{ds} = \frac{c}{s} \Rightarrow \boxed{V = c \ln s + k.}$

Example: potential of a long wire.

Problem 3.4

Refer to Fig. 3.3, letting α be the angle between $\mathbf{r}$ and the z axis. Obviously, $\mathbf{E}_{\text{ave}}$ points in the $-\hat{\mathbf{z}}$ direction, so

$$\mathbf{E}_{\text{ave}} = \frac{1}{4\pi R^2} \oint \mathbf{E} da = -\hat{\mathbf{z}} \frac{1}{4\pi R^2} \frac{q}{4\pi\epsilon_0} \int \frac{1}{r^2} \cos \alpha da.$$

By the law of cosines,

$$\begin{aligned} R^2 = z^2 + r^2 - 2zr \cos \alpha &\Rightarrow \cos \alpha = \frac{z^2 + r^2 - R^2}{2zr}, \\ r^2 = R^2 + z^2 - 2Rz \cos \theta &\Rightarrow \frac{\cos \alpha}{r^2} = \frac{z^2 + r^2 - R^2}{2zr^3} = \frac{z - R \cos \theta}{(R^2 + z^2 - 2Rz \cos \theta)^{3/2}}. \end{aligned}$$

$$\begin{aligned}\mathbf{E}_{\text{ave}} &= -\hat{\mathbf{z}} \frac{q}{16\pi^2 R^2 \epsilon_0} \int \frac{z - R \cos \theta}{(R^2 + z^2 - 2Rz \cos \theta)^{3/2}} R^2 \sin \theta d\theta d\phi \\ &= -\frac{q\hat{\mathbf{z}}}{8\pi\epsilon_0} \int_0^\pi \frac{z - R \cos \theta}{(R^2 + z^2 - 2Rz \cos \theta)^{3/2}} \sin \theta d\theta = -\frac{q\hat{\mathbf{z}}}{8\pi\epsilon_0} \int_{-1}^1 \frac{z - Ru}{(R^2 + z^2 - 2Rzu)^{3/2}} du\end{aligned}$$

(where $u \equiv \cos \theta$). The integral is

$$\begin{aligned}I &= \left. \frac{1}{R\sqrt{R^2 + z^2 - 2Rzu}} \right|_{-1}^1 - \frac{1}{2Rz^2} \left(\sqrt{R^2 + z^2 - 2Rzu} + \frac{R^2 + z^2}{\sqrt{R^2 + z^2 - 2Rzu}} \right) \Big|_{-1}^1 \\ &= \frac{1}{R} \left(\frac{1}{|z - R|} - \frac{1}{z + R} \right) - \frac{1}{2Rz^2} \left[|z - R| - (z + R) + (R^2 + z^2) \left(\frac{1}{|z - R|} - \frac{1}{z + R} \right) \right].\end{aligned}$$

(a) If $z > R$,

$$\begin{aligned}I &= \frac{1}{R} \left(\frac{1}{z - R} - \frac{1}{z + R} \right) - \frac{1}{2Rz^2} \left[(z - R) - (z + R) + (R^2 + z^2) \left(\frac{1}{z - R} - \frac{1}{z + R} \right) \right] \\ &= \frac{1}{R} \left(\frac{2R}{z^2 - R^2} \right) - \frac{1}{2Rz^2} \left[-2R + (R^2 + z^2) \frac{2R}{z^2 - R^2} \right] = \frac{2}{z^2}.\end{aligned}$$

So

$$\mathbf{E}_{\text{ave}} = -\frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \hat{\mathbf{z}},$$

the same as the field at the center. By superposition the same holds for any *collection* of charges outside the sphere.

(b) If $z < R$,

$$\begin{aligned}I &= \frac{1}{R} \left(\frac{1}{R - z} - \frac{1}{z + R} \right) - \frac{1}{2Rz^2} \left[(R - z) - (z + R) + (R^2 + z^2) \left(\frac{1}{R - z} - \frac{1}{z + R} \right) \right] \\ &= \frac{1}{R} \left(\frac{2z}{R^2 - z^2} \right) - \frac{1}{2Rz^2} \left[-2z + (R^2 + z^2) \frac{2z}{R^2 - z^2} \right] = 0.\end{aligned}$$

So

$$\mathbf{E}_{\text{ave}} = \mathbf{0}.$$

By superposition the same holds for any *collection* of charges inside the sphere.

Problem 3.5

Same as proof of second uniqueness theorem, up to the equation $\oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{a} = -\int_V (E_3)^2 d\tau$. But on each surface, either $V_3 = 0$ (if V is specified on the surface), or else $E_{3\perp} = 0$ (if $\frac{\partial V}{\partial n} = -E_\perp$ is specified). So $\int_V (E_3)^2 d\tau = 0$, and hence $\mathbf{E}_2 = \mathbf{E}_1$. $\quad \text{qed}$

Problem 3.6

Putting $U = T = V_3$ into Green's identity:

$$\int_V [V_3 \nabla^2 V_3 + \nabla V_3 \cdot \nabla V_3] d\tau = \oint_S V_3 \nabla V_3 \cdot d\mathbf{a}. \quad \text{But } \nabla^2 V_3 = \nabla^2 V_1 - \nabla^2 V_2 = -\frac{\rho}{\epsilon_0} + \frac{\rho}{\epsilon_0} = 0, \text{ and } \nabla V_3 = -\mathbf{E}_3.$$

So $\int_V E_3^2 d\tau = -\oint_S V_3 \mathbf{E}_3 \cdot d\mathbf{a}$, and the rest is the same as before.

Problem 3.7

Place image charges $+2q$ at $z = -d$ and $-q$ at $z = -3d$. Total force on $+q$ is

$$\mathbf{F} = \frac{q}{4\pi\epsilon_0} \left[\frac{-2q}{(2d)^2} + \frac{2q}{(4d)^2} + \frac{-q}{(6d)^2} \right] \hat{\mathbf{z}} = \frac{q^2}{4\pi\epsilon_0 d^2} \left(-\frac{1}{2} + \frac{1}{8} - \frac{1}{36} \right) \hat{\mathbf{z}} = \boxed{-\frac{1}{4\pi\epsilon_0} \left(\frac{29q^2}{72d^2} \right) \hat{\mathbf{z}}}.$$

Problem 3.8

(a) From Fig. 3.13: $z = \sqrt{r^2 + a^2 - 2ra \cos \theta}$; $z' = \sqrt{r^2 + b^2 - 2rb \cos \theta}$. Therefore:

$$\begin{aligned} \frac{q'}{z'} &= -\frac{R}{a} \frac{q}{\sqrt{r^2 + b^2 - 2rb \cos \theta}} \quad (\text{Eq. 3.15}), \text{ while } b = \frac{R^2}{a} \quad (\text{Eq. 3.16}). \\ &= -\frac{q}{\left(\frac{a}{R}\right) \sqrt{r^2 + \frac{R^4}{a^2} - 2r \frac{R^2}{a} \cos \theta}} = -\frac{q}{\sqrt{\left(\frac{ar}{R}\right)^2 + R^2 - 2ra \cos \theta}}. \end{aligned}$$

Therefore:

$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{z} + \frac{q'}{z'} \right) = \boxed{\frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{\sqrt{r^2 + a^2 - 2ra \cos \theta}} - \frac{1}{\sqrt{R^2 + (ra/R)^2 - 2ra \cos \theta}} \right\}}.$$

Clearly, when $r = R$, $V \rightarrow 0$.

(b) $\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$ (Eq. 2.49). In this case, $\frac{\partial V}{\partial n} = \frac{\partial V}{\partial r}$ at the point $r = R$. Therefore,

$$\begin{aligned} \sigma(\theta) &= -\epsilon_0 \left(\frac{q}{4\pi\epsilon_0} \right) \left\{ -\frac{1}{2} (r^2 + a^2 - 2ra \cos \theta)^{-3/2} (2r - 2a \cos \theta) \right. \\ &\quad \left. + \frac{1}{2} (R^2 + (ra/R)^2 - 2ra \cos \theta)^{-3/2} \left(\frac{a^2}{R^2} 2r - 2a \cos \theta \right) \right\} \Big|_{r=R} \\ &= -\frac{q}{4\pi} \left\{ -(R^2 + a^2 - 2Ra \cos \theta)^{-3/2} (R - a \cos \theta) + (R^2 + a^2 - 2Ra \cos \theta)^{-3/2} \left(\frac{a^2}{R} - a \cos \theta \right) \right\} \\ &= \frac{q}{4\pi} (R^2 + a^2 - 2Ra \cos \theta)^{-3/2} \left[R - a \cos \theta - \frac{a^2}{R} + a \cos \theta \right] \\ &= \boxed{\frac{q}{4\pi R} (R^2 - a^2) (R^2 + a^2 - 2Ra \cos \theta)^{-3/2}}. \end{aligned}$$

$$\begin{aligned} q_{\text{induced}} &= \int \sigma da = \frac{q}{4\pi R} (R^2 - a^2) \int (R^2 + a^2 - 2Ra \cos \theta)^{-3/2} R^2 \sin \theta d\theta d\phi \\ &= \frac{q}{4\pi R} (R^2 - a^2) 2\pi R^2 \left[-\frac{1}{Ra} (R^2 + a^2 - 2Ra \cos \theta)^{-1/2} \right] \Big|_0^\pi \\ &= \frac{q}{2a} (a^2 - R^2) \left[\frac{1}{\sqrt{R^2 + a^2 + 2Ra}} - \frac{1}{\sqrt{R^2 + a^2 - 2Ra}} \right]. \\ &\quad \text{But } a > R \text{ (else } q \text{ would be inside), so } \sqrt{R^2 + a^2 - 2Ra} = a - R. \\ &= \frac{q}{2a} (a^2 - R^2) \left[\frac{1}{(a+R)} - \frac{1}{(a-R)} \right] = \frac{q}{2a} [(a-R) - (a+R)] = \frac{q}{2a} (-2R) \\ &= \boxed{-\frac{qR}{a} = q'}. \end{aligned}$$

(c) The force on q , due to the sphere, is the same as the force of the image charge q' , to wit:

$$F = \frac{1}{4\pi\epsilon_0} \frac{qq'}{(a-b)^2} = \frac{1}{4\pi\epsilon_0} \left(-\frac{R}{a} q^2 \right) \frac{1}{(a - R^2/a)^2} = -\frac{1}{4\pi\epsilon_0} \frac{q^2 Ra}{(a^2 - R^2)^2}.$$

To bring q in from infinity to a , then, we do work

$$W = \frac{q^2 R}{4\pi\epsilon_0} \int_{\infty}^a \frac{\bar{a}}{(\bar{a}^2 - R^2)^2} d\bar{a} = \frac{q^2 R}{4\pi\epsilon_0} \left[-\frac{1}{2} \frac{1}{(\bar{a}^2 - R^2)} \right]_{\infty}^a = \boxed{-\frac{1}{4\pi\epsilon_0} \frac{q^2 R}{2(a^2 - R^2)}}.$$

Problem 3.9

Place a second image charge, q'' , at the *center* of the sphere; this will not alter the fact that the sphere is an *equipotential*, but merely *increase* that potential from zero to $V_0 = \frac{1}{4\pi\epsilon_0} \frac{q''}{R}$;

$$\boxed{q'' = 4\pi\epsilon_0 V_0 R \text{ at center of sphere.}}$$

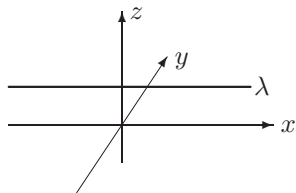
For a *neutral* sphere, $q' + q'' = 0$.

$$\begin{aligned} F &= \frac{1}{4\pi\epsilon_0} q \left(\frac{q''}{a^2} + \frac{q'}{(a-b)^2} \right) = \frac{qq'}{4\pi\epsilon_0} \left(-\frac{1}{a^2} + \frac{1}{(a-b)^2} \right) \\ &= \frac{qq'}{4\pi\epsilon_0} \frac{b(2a-b)}{a^2(a-b)^2} = \frac{q(-Rq/a)}{4\pi\epsilon_0} \frac{(R^2/a)(2a-R^2/a)}{a^2(a-R^2/a)^2} \\ &= -\boxed{\frac{q^2}{4\pi\epsilon_0} \left(\frac{R}{a} \right)^3 \frac{(2a^2 - R^2)}{(a^2 - R^2)^2}}. \end{aligned}$$

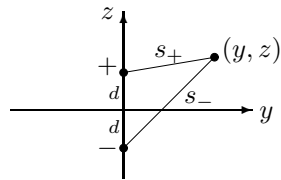
(Drop the minus sign, because the problem asks for the force of *attraction*.)

Problem 3.10

(a) Image problem: λ above, $-\lambda$ below. Potential was found in Prob. 2.52:



$$\begin{aligned} V(y, z) &= \frac{2\lambda}{4\pi\epsilon_0} \ln(s_-/s_+) = \frac{\lambda}{4\pi\epsilon_0} \ln(s_-^2/s_+^2) \\ &= \boxed{\frac{\lambda}{4\pi\epsilon_0} \ln \left\{ \frac{y^2 + (z+d)^2}{y^2 + (z-d)^2} \right\}} \end{aligned}$$



(b) $\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$. Here $\frac{\partial V}{\partial n} = \frac{\partial V}{\partial z}$, evaluated at $z = 0$.

$$\begin{aligned} \sigma(y) &= -\epsilon_0 \frac{\lambda}{4\pi\epsilon_0} \left\{ \frac{1}{y^2 + (z+d)^2} 2(z+d) - \frac{1}{y^2 + (z-d)^2} 2(z-d) \right\} \Big|_{z=0} \\ &= -\frac{2\lambda}{4\pi} \left\{ \frac{d}{y^2 + d^2} - \frac{-d}{y^2 + d^2} \right\} = \boxed{-\frac{\lambda d}{\pi(y^2 + d^2)}}. \end{aligned}$$

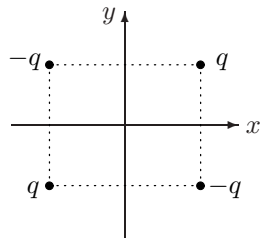
Check: Total charge induced on a strip of width l parallel to the y axis:

$$\begin{aligned} q_{\text{ind}} &= -\frac{l\lambda d}{\pi} \int_{-\infty}^{\infty} \frac{1}{y^2 + d^2} dy = -\frac{l\lambda d}{\pi} \left[\frac{1}{d} \tan^{-1} \left(\frac{y}{d} \right) \right]_{-\infty}^{\infty} = -\frac{l\lambda d}{\pi} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] \\ &= -\lambda l. \quad \text{Therefore } \lambda_{\text{ind}} = -\lambda, \text{ as it should be.} \end{aligned}$$

Problem 3.11

The image configuration is as shown.

$$V(x, y) = \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{\sqrt{(x-a)^2 + (y-b)^2 + z^2}} + \frac{1}{\sqrt{(x+a)^2 + (y+b)^2 + z^2}} - \frac{1}{\sqrt{(x+a)^2 + (y-b)^2 + z^2}} - \frac{1}{\sqrt{(x-a)^2 + (y+b)^2 + z^2}} \right\}.$$



$$\mathbf{F} = \frac{q^2}{4\pi\epsilon_0} \left\{ -\frac{1}{(2a)^2} \hat{\mathbf{x}} - \frac{1}{(2b)^2} \hat{\mathbf{y}} + \frac{1}{(2\sqrt{a^2 + b^2})^2} [\cos\theta \hat{\mathbf{x}} + \sin\theta \hat{\mathbf{y}}] \right\},$$

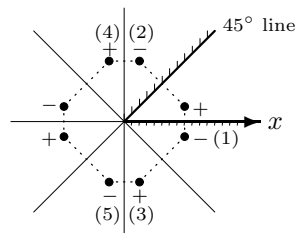
where $\cos\theta = a/\sqrt{a^2 + b^2}$, $\sin\theta = b/\sqrt{a^2 + b^2}$.

$$\mathbf{F} = \frac{q^2}{16\pi\epsilon_0} \left\{ \left[\frac{a}{(a^2 + b^2)^{3/2}} - \frac{1}{a^2} \right] \hat{\mathbf{x}} + \left[\frac{b}{(a^2 + b^2)^{3/2}} - \frac{1}{b^2} \right] \hat{\mathbf{y}} \right\}.$$

$$W = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \left[\frac{-q^2}{(2a)} + \frac{-q^2}{(2b)} + \frac{q^2}{(2\sqrt{a^2 + b^2})} \right] = \frac{q^2}{16\pi\epsilon_0} \left[\frac{1}{\sqrt{a^2 + b^2}} - \frac{1}{a} - \frac{1}{b} \right].$$

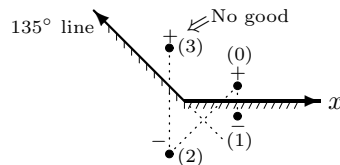
For this to work, θ must be an integer divisor of 180° . Thus 180° , 90° , 60° , 45° , etc., are OK, but no others. It works for 45° , say, with the charges as shown.

(Note the strategy: to make the x axis an equipotential ($V = 0$), you place the image charge (1) in the reflection point. To make the 45° line an equipotential, you place charge (2) at the image point. But that screws up the x axis, so you must now insert image (3) to balance (2). Moreover, to make the 45° line $V = 0$ you also need (4), to balance (1). But now, to restore the x axis to $V = 0$ you need (5) to balance (4), and so on.



why it works for $\theta = 45^\circ$

The reason this doesn't work for *arbitrary* angles is that you are eventually forced to place an image charge *within the original region of interest*, and that's not allowed—all images must go *outside* the region, or you're no longer dealing with the same problem at all.)



why it *doesn't* work for $\theta = 135^\circ$

Problem 3.12

From Prob. 2.52 (with $y_0 \rightarrow d$): $V = \frac{\lambda}{4\pi\epsilon_0} \ln \left[\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2} \right]$, where $a^2 = y_0^2 - R^2 \Rightarrow a = \sqrt{d^2 - R^2}$,

and

$$\left\{ \begin{array}{l} a \coth(2\pi\epsilon_0 V_0/\lambda) = d \\ a \operatorname{csch}(2\pi\epsilon_0 V_0/\lambda) = R \end{array} \right\} \Rightarrow (\text{dividing}) \quad \frac{d}{R} = \cosh \left(\frac{2\pi\epsilon_0 V_0}{\lambda} \right), \text{ or } \lambda = \frac{2\pi\epsilon_0 V_0}{\cosh^{-1}(d/R)}.$$

Problem 3.13

$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin(n\pi y/a) \quad (\text{Eq. 3.30}), \quad \text{where} \quad C_n = \frac{2}{a} \int_0^a V_0(y) \sin(n\pi y/a) dy \quad (\text{Eq. 3.34}).$$

In this case $V_0(y) = \begin{cases} +V_0, & \text{for } 0 < y < a/2 \\ -V_0, & \text{for } a/2 < y < a \end{cases}$. Therefore,

$$\begin{aligned} C_n &= \frac{2}{a} V_0 \left\{ \int_0^{a/2} \sin(n\pi y/a) dy - \int_{a/2}^a \sin(n\pi y/a) dy \right\} = \frac{2V_0}{a} \left\{ -\frac{\cos(n\pi y/a)}{(n\pi/a)} \Big|_0^{a/2} + \frac{\cos(n\pi y/a)}{(n\pi/a)} \Big|_{a/2}^a \right\} \\ &= \frac{2V_0}{n\pi} \left\{ -\cos\left(\frac{n\pi}{2}\right) + \cos(0) + \cos(n\pi) - \cos\left(\frac{n\pi}{2}\right) \right\} = \frac{2V_0}{n\pi} \left\{ 1 + (-1)^n - 2\cos\left(\frac{n\pi}{2}\right) \right\}. \end{aligned}$$

The term in curly brackets is:

$$\left\{ \begin{array}{ll} n = 1 & : 1 - 1 - 2\cos(\pi/2) = 0, \\ n = 2 & : 1 + 1 - 2\cos(\pi) = 4, \\ n = 3 & : 1 - 1 - 2\cos(3\pi/2) = 0, \\ n = 4 & : 1 + 1 - 2\cos(2\pi) = 0, \end{array} \right\} \text{ etc. (Zero if } n \text{ is odd or divisible by 4, otherwise 4.)}$$

Therefore

$$C_n = \begin{cases} 8V_0/n\pi, & n = 2, 6, 10, 14, \text{ etc. (in general, } 4j + 2, \text{ for } j = 0, 1, 2, \dots), \\ 0, & \text{otherwise.} \end{cases}$$

So

$$\boxed{V(x, y) = \frac{8V_0}{\pi} \sum_{n=2,6,10,\dots} \frac{e^{-n\pi x/a} \sin(n\pi y/a)}{n}} = \frac{8V_0}{\pi} \sum_{j=0}^{\infty} \frac{e^{-(4j+2)\pi x/a} \sin[(4j+2)\pi y/a]}{(4j+2)}.$$

Problem 3.14

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} e^{-n\pi x/a} \sin(n\pi y/a) \quad (\text{Eq. 3.36}); \quad \sigma = -\epsilon_0 \frac{\partial V}{\partial n} \quad (\text{Eq. 2.49}).$$

So

$$\begin{aligned} \sigma(y) &= -\epsilon_0 \frac{\partial}{\partial x} \left\{ \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} e^{-n\pi x/a} \sin(n\pi y/a) \right\} \Big|_{x=0} = -\epsilon_0 \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} \left(-\frac{n\pi}{a}\right) e^{-n\pi x/a} \sin(n\pi y/a) \Big|_{x=0} \\ &= \boxed{\frac{4\epsilon_0 V_0}{a} \sum_{n=1,3,5,\dots} \sin(n\pi y/a)}. \end{aligned}$$

Or, using the closed form 3.37:

$$\begin{aligned} V(x, y) &= \frac{2V_0}{\pi} \tan^{-1} \left(\frac{\sin(\pi y/a)}{\sinh(\pi x/a)} \right) \Rightarrow \sigma = -\epsilon_0 \frac{2V_0}{\pi} \frac{1}{1 + \frac{\sin^2(\pi y/a)}{\sinh^2(\pi x/a)}} \left(\frac{-\sin(\pi y/a)}{\sinh^2(\pi x/a)} \right) \frac{\pi}{a} \cosh(\pi x/a) \Big|_{x=0} \\ &= \frac{2\epsilon_0 V_0}{a} \frac{\sin(\pi y/a) \cosh(\pi x/a)}{\sin^2(\pi y/a) + \sinh^2(\pi x/a)} \Big|_{x=0} = \boxed{\frac{2\epsilon_0 V_0}{a} \frac{1}{\sin(\pi y/a)}}. \end{aligned}$$

[*Comment:* Technically, the series solution for σ is defective, since term-by-term differentiation has produced a (naively) non-convergent sum. More sophisticated definitions of convergence permit one to work with series of this form, but it is better to sum the series *first* and *then* differentiate (the second method.)]

Summation of series Eq. 3.36

$$V(x, y) = \frac{4V_0}{\pi} I, \text{ where } I \equiv \sum_{n=1,3,5,\dots} \frac{1}{n} e^{-n\pi x/a} \sin(n\pi y/a).$$

Now $\sin w = \mathcal{I}m(e^{iw})$, so

$$I = \mathcal{I}m \sum_n \frac{1}{n} e^{-n\pi x/a} e^{in\pi y/a} = \mathcal{I}m \sum_n \frac{1}{n} \mathcal{Z}^n,$$

where $\mathcal{Z} \equiv e^{-\pi(x-iy)/a}$. Now

$$\begin{aligned} \sum_{1,3,5,\dots} \frac{1}{n} \mathcal{Z}^n &= \sum_{j=0}^{\infty} \frac{1}{(2j+1)} \mathcal{Z}^{(2j+1)} = \int_0^{\mathcal{Z}} \left\{ \sum_{j=0}^{\infty} u^{2j} \right\} du \\ &= \int_0^{\mathcal{Z}} \frac{1}{1-u^2} du = \frac{1}{2} \ln \left(\frac{1+\mathcal{Z}}{1-\mathcal{Z}} \right) = \frac{1}{2} \ln(R e^{i\theta}) = \frac{1}{2} (\ln R + i\theta), \end{aligned}$$

where $R e^{i\theta} = \frac{1+\mathcal{Z}}{1-\mathcal{Z}}$. Therefore

$$\begin{aligned} I &= \mathcal{I}m \left\{ \frac{1}{2} (\ln R + i\theta) \right\} = \frac{1}{2} \theta. \quad \text{But } \frac{1+\mathcal{Z}}{1-\mathcal{Z}} = \frac{1+e^{-\pi(x-iy)/a}}{1-e^{-\pi(x-iy)/a}} = \frac{(1+e^{-\pi(x-iy)/a})(1-e^{-\pi(x+iy)/a})}{(1-e^{-\pi(x-iy)/a})(1-e^{-\pi(x+iy)/a})} \\ &= \frac{1+e^{-\pi x/a} (e^{i\pi y/a} - e^{-i\pi y/a}) - e^{-2\pi x/a}}{|1-e^{-\pi(x-iy)/a}|^2} = \frac{1+2ie^{-\pi x/a} \sin(\pi y/a) - e^{-2\pi x/a}}{|1-e^{-\pi(x-iy)/a}|^2}, \end{aligned}$$

so

$$\tan \theta = \frac{2e^{-\pi x/a} \sin(\pi y/a)}{1-e^{-2\pi x/a}} = \frac{2 \sin(\pi y/a)}{e^{\pi x/a} - e^{-\pi x/a}} = \frac{\sin(\pi y/a)}{\sinh(\pi x/a)}.$$

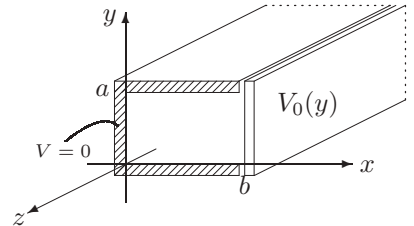
Therefore

$$I = \frac{1}{2} \tan^{-1} \left(\frac{\sin(\pi y/a)}{\sinh(\pi x/a)} \right), \text{ and } \boxed{V(x, y) = \frac{2V_0}{\pi} \tan^{-1} \left(\frac{\sin(\pi y/a)}{\sinh(\pi x/a)} \right)}.$$

Problem 3.15

(a) $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$, with boundary conditions

$$\left\{ \begin{array}{l} \text{(i)} \quad V(x, 0) = 0, \\ \text{(ii)} \quad V(x, a) = 0, \\ \text{(iii)} \quad V(0, y) = 0, \\ \text{(iv)} \quad V(b, y) = V_0(y). \end{array} \right\}$$



As in Ex. 3.4, separation of variables yields

$$V(x, y) = (Ae^{kx} + Be^{-kx}) (C \sin ky + D \cos ky).$$

Here (i) $\Rightarrow D = 0$, (iii) $\Rightarrow B = -A$, (ii) $\Rightarrow ka$ is an integer multiple of π :

$$V(x, y) = AC \left(e^{n\pi x/a} - e^{-n\pi x/a} \right) \sin(n\pi y/a) = (2AC) \sinh(n\pi x/a) \sin(n\pi y/a).$$

But $(2AC)$ is a constant, and the most general linear combination of separable solutions consistent with (i), (ii), (iii) is

$$V(x, y) = \sum_{n=1}^{\infty} C_n \sinh(n\pi x/a) \sin(n\pi y/a).$$

It remains to determine the coefficients C_n so as to fit boundary condition (iv):

$$\sum C_n \sinh(n\pi b/a) \sin(n\pi y/a) = V_0(y). \text{ Fourier's trick } \Rightarrow C_n \sinh(n\pi b/a) = \frac{2}{a} \int_0^a V_0(y) \sin(n\pi y/a) dy.$$

Therefore

$$C_n = \frac{2}{a \sinh(n\pi b/a)} \int_0^a V_0(y) \sin(n\pi y/a) dy.$$

$$(b) \ C_n = \frac{2}{a \sinh(n\pi b/a)} V_0 \int_0^a \sin(n\pi y/a) dy = \frac{2V_0}{a \sinh(n\pi b/a)} \times \begin{cases} 0, & \text{if } n \text{ is even,} \\ \frac{2a}{n\pi}, & \text{if } n \text{ is odd.} \end{cases}$$

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots} \frac{\sinh(n\pi x/a) \sin(n\pi y/a)}{n \sinh(n\pi b/a)}.$$

Problem 3.16

Same format as Ex. 3.5, only the boundary conditions are:

$$\begin{cases} \text{(i)} & V = 0 & \text{when } x = 0, \\ \text{(ii)} & V = 0 & \text{when } x = a, \\ \text{(iii)} & V = 0 & \text{when } y = 0, \\ \text{(iv)} & V = 0 & \text{when } y = a, \\ \text{(v)} & V = 0 & \text{when } z = 0, \\ \text{(vi)} & V = V_0 & \text{when } z = a. \end{cases}$$

This time we want sinusoidal functions in x and y , exponential in z :

$$X(x) = A \sin(kx) + B \cos(kx), \quad Y(y) = C \sin(l y) + D \cos(l y), \quad Z(z) = E e^{\sqrt{k^2 + l^2} z} + G e^{-\sqrt{k^2 + l^2} z}.$$

(i) $\Rightarrow B = 0$; (ii) $\Rightarrow k = n\pi/a$; (iii) $\Rightarrow D = 0$; (iv) $\Rightarrow l = m\pi/a$; (v) $\Rightarrow E + G = 0$. Therefore

$$Z(z) = 2E \sinh(\pi \sqrt{n^2 + m^2} z/a).$$

Putting this all together, and combining the constants, we have:

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{n,m} \sin(n\pi x/a) \sin(m\pi y/a) \sinh(\pi \sqrt{n^2 + m^2} z/a).$$

It remains to evaluate the constants $C_{n,m}$, by imposing boundary condition (vi):

$$V_0 = \sum \sum \left[C_{n,m} \sinh(\pi \sqrt{n^2 + m^2}) \right] \sin(n\pi x/a) \sin(m\pi y/a).$$

According to Eqs. 3.50 and 3.51:

$$C_{n,m} \sinh \left(\pi \sqrt{n^2 + m^2} \right) = \left(\frac{2}{a} \right)^2 V_0 \int_0^a \int_0^a \sin(n\pi x/a) \sin(m\pi y/a) dx dy = \begin{cases} 0, & \text{if } n \text{ or } m \text{ is even,} \\ \frac{16V_0}{\pi^2 nm}, & \text{if both are odd.} \end{cases}$$

Therefore

$$V(x, y, z) = \frac{16V_0}{\pi^2} \sum_{n=1,3,5,\dots} \sum_{m=1,3,5,\dots} \frac{1}{nm} \sin(n\pi x/a) \sin(m\pi y/a) \frac{\sinh \left(\pi \sqrt{n^2 + m^2} z/a \right)}{\sinh \left(\pi \sqrt{n^2 + m^2} \right)}.$$

Consider the superposition of *six* such cubes, one with V_0 on each of the six faces. The result is a cube with V_0 on its entire surface, so the potential at the center is V_0 . Evidently the potential at the center of the original cube (with V_0 on just one face) is one sixth of this: $\boxed{V_0/6}$. To check it, put in $x = y = z = a/2$:

$$V(a/2, a/2, a/2) = \frac{16V_0}{\pi^2} \sum_{n=1,3,5,\dots} \sum_{m=1,3,5,\dots} \frac{1}{nm} \sin(n\pi/2) \sin(m\pi/2) \frac{\sinh \left(\pi \sqrt{n^2 + m^2} /2 \right)}{\sinh \left(\pi \sqrt{n^2 + m^2} \right)}.$$

Let $n \equiv 2i + 1$, $m \equiv 2j + 1$, and note that $\sinh(2u) = 2 \sinh(u) \cosh(u)$. The double sum is then

$$S = \frac{1}{2} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{(-1)^{i+j}}{(2i+1)(2j+1)} \operatorname{sech} \left[\pi \sqrt{(2i+1)^2 + (2j+1)^2} /2 \right].$$

Setting the upper limits at $i = 3$, $j = 3$ (or above) Mathematica returns $S = 0.102808$, which (to 6 digits) is equal to $\pi^2/96$, confirming (at least, numerically) that $V(a/2, a/2, a/2) = V_0/6$.

Problem 3.17

$$\begin{aligned} P_3(x) &= \frac{1}{8 \cdot 6} \frac{d^3}{dx^3} (x^2 - 1)^3 = \frac{1}{48} \frac{d^2}{dx^2} 3 (x^2 - 1)^2 2x = \frac{1}{8} \frac{d^2}{dx^2} x (x^2 - 1)^2 \\ &= \frac{1}{8} \frac{d}{dx} \left[(x^2 - 1)^2 + 2x (x^2 - 1) 2x \right] = \frac{1}{8} \frac{d}{dx} \left[(x^2 - 1) (x^2 - 1 + 4x^2) \right] \\ &= \frac{1}{8} \frac{d}{dx} \left[(x^2 - 1) (5x^2 - 1) \right] = \frac{1}{8} \left[2x (5x^2 - 1) + (x^2 - 1) 10x \right] \\ &= \frac{1}{4} (5x^3 - x + 5x^3 - 5x) = \frac{1}{4} (10x^3 - 6x) = \boxed{\frac{5}{2}x^3 - \frac{3}{2}x}. \end{aligned}$$

We need to show that $P_3(\cos \theta)$ satisfies

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) = -l(l+1)P, \text{ with } l = 3,$$

where $P_3(\cos \theta) = \frac{1}{2} \cos \theta (5 \cos^2 \theta - 3)$.

$$\begin{aligned} \frac{dP_3}{d\theta} &= \frac{1}{2} \left[-\sin \theta (5 \cos^2 \theta - 3) + \cos \theta (10 \cos \theta (-\sin \theta)) \right] = -\frac{1}{2} \sin \theta (5 \cos^2 \theta - 3 + 10 \cos^2 \theta) \\ &= -\frac{3}{2} \sin \theta (5 \cos^2 \theta - 1). \end{aligned}$$

$$\begin{aligned}\frac{\partial}{\partial \theta} \left(\sin \theta \frac{dP_3}{d\theta} \right) &= -\frac{3}{2} \frac{d}{d\theta} [\sin^2 \theta (5 \cos^2 \theta - 1)] = -\frac{3}{2} [2 \sin \theta \cos \theta (5 \cos^2 \theta - 1) + \sin^2 \theta (-10 \cos \theta \sin \theta)] \\ &= -3 \sin \theta \cos \theta [5 \cos^2 \theta - 1 - 5 \sin^2 \theta].\end{aligned}$$

$$\begin{aligned}\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) &= -3 \cos \theta [5 \cos^2 - 1 - 5 (1 - \cos^2 \theta)] = -3 \cos \theta (10 \cos^2 \theta - 6) \\ &= -3 \cdot 4 \cdot \frac{1}{2} \cos \theta (5 \cos^2 \theta - 3) = -l(l+1)P_3. \quad \text{qed}\end{aligned}$$

$$\int_{-1}^1 P_1(x) P_3(x) dx = \int_{-1}^1 (x) \frac{1}{2} (5x^3 - 3x) dx = \frac{1}{2} (x^5 - x^3) \Big|_{-1}^1 = \frac{1}{2} (1 - 1 + 1 - 1) = 0. \quad \checkmark$$

Problem 3.18

(a) *Inside:* $V(r, \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$ (Eq. 3.66) where

$$A_l = \frac{(2l+1)}{2R^l} \int_0^{\pi} V_0(\theta) P_l(\cos \theta) \sin \theta d\theta \quad (\text{Eq. 3.69}).$$

In this case $V_0(\theta) = V_0$ comes outside the integral, so

$$A_l = \frac{(2l+1)V_0}{2R^l} \int_0^{\pi} P_l(\cos \theta) \sin \theta d\theta.$$

But $P_0(\cos \theta) = 1$, so the integral can be written

$$\int_0^{\pi} P_0(\cos \theta) P_l(\cos \theta) \sin \theta d\theta = \begin{cases} 0, & \text{if } l \neq 0 \\ 2, & \text{if } l = 0 \end{cases} \quad (\text{Eq. 3.68}).$$

Therefore

$$A_l = \begin{cases} 0, & \text{if } l \neq 0 \\ V_0, & \text{if } l = 0 \end{cases}.$$

Plugging this into the general form:

$$V(r, \theta) = A_0 r^0 P_0(\cos \theta) = \boxed{V_0}.$$

The potential is *constant throughout the sphere*.

Outside: $V(r, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta)$ (Eq. 3.72), where

$$\begin{aligned}B_l &= \frac{(2l+1)}{2} R^{l+1} \int_0^{\pi} V_0(\theta) P_l(\cos \theta) \sin \theta d\theta \quad (\text{Eq. 3.73}). \\ &= \frac{(2l+1)}{2} R^{l+1} V_0 \int_0^{\pi} P_l(\cos \theta) \sin \theta d\theta = \begin{cases} 0, & \text{if } l \neq 0 \\ R V_0, & \text{if } l = 0 \end{cases}.\end{aligned}$$

Therefore $\boxed{V(r, \theta) = V_0 \frac{R}{r}}$ (i.e. equals V_0 at $r = R$, then falls off like $\frac{1}{r}$).

(b)

$$V(r, \theta) = \left\{ \begin{array}{l} \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta), \text{ for } r \leq R \text{ (Eq. 3.78)} \\ \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta), \text{ for } r \geq R \text{ (Eq. 3.79)} \end{array} \right\},$$

where

$$B_l = R^{2l+1} A_l \quad (\text{Eq. 3.81})$$

and

$$\begin{aligned} A_l &= \frac{1}{2\epsilon_0 R^{l-1}} \int_0^\pi \sigma_0(\theta) P_l(\cos \theta) \sin \theta d\theta \quad (\text{Eq. 3.84}) \\ &= \frac{1}{2\epsilon_0 R^{l-1}} \sigma_0 \int_0^\pi P_l(\cos \theta) \sin \theta d\theta = \begin{cases} 0, & \text{if } l \neq 0 \\ R\sigma_0/\epsilon_0, & \text{if } l = 0 \end{cases}. \end{aligned}$$

Therefore

$$\boxed{V(r, \theta) = \begin{cases} \frac{R\sigma_0}{\epsilon_0}, & \text{for } r \leq R \\ \frac{R^2\sigma_0}{\epsilon_0} \frac{1}{r}, & \text{for } r \geq R \end{cases}}.$$

Note: in terms of the total charge $Q = 4\pi R^2 \sigma_0$,

$$V(r, \theta) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{Q}{R}, & \text{for } r \leq R \\ \frac{1}{4\pi\epsilon_0} \frac{Q}{r}, & \text{for } r \geq R \end{cases}.$$

Problem 3.19

$$V_0(\theta) = k \cos(3\theta) = k [4 \cos^3 \theta - 3 \cos \theta] = k [\alpha P_3(\cos \theta) + \beta P_1(\cos \theta)].$$

(I know that any 3rd order polynomial can be expressed as a linear combination of the first four Legendre polynomials; in this case, since the polynomial is *odd*, I only need P_1 and P_3 .)

$$4 \cos^3 \theta - 3 \cos \theta = \alpha \left[\frac{1}{2} (5 \cos^3 \theta - 3 \cos \theta) \right] + \beta \cos \theta = \frac{5\alpha}{2} \cos^3 \theta + \left(\beta - \frac{3}{2}\alpha \right) \cos \theta,$$

so

$$4 = \frac{5\alpha}{2} \Rightarrow \alpha = \frac{8}{5}; \quad -3 = \beta - \frac{3}{2}\alpha = \beta - \frac{3}{2} \cdot \frac{8}{5} = \beta - \frac{12}{5} \Rightarrow \beta = \frac{12}{5} - 3 = -\frac{3}{5}.$$

Therefore

$$V_0(\theta) = \frac{k}{5} [8P_3(\cos \theta) - 3P_1(\cos \theta)].$$

Now

$$V(r, \theta) = \left\{ \begin{array}{l} \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta), \text{ for } r \leq R \text{ (Eq. 3.66)} \\ \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta), \text{ for } r \geq R \text{ (Eq. 3.72)} \end{array} \right\},$$

where

$$\begin{aligned} A_l &= \frac{(2l+1)}{2R^l} \int_0^\pi V_0(\theta) P_l(\cos \theta) \sin \theta d\theta \quad (\text{Eq. 3.69}) \\ &= \frac{(2l+1)}{2R^l} \frac{k}{5} \left\{ 8 \int_0^\pi P_3(\cos \theta) P_l(\cos \theta) \sin \theta d\theta - 3 \int_0^\pi P_1(\cos \theta) P_l(\cos \theta) \sin \theta d\theta \right\} \\ &= \frac{k}{5} \frac{(2l+1)}{2R^l} \left\{ 8 \frac{2}{(2l+1)} \delta_{l3} - 3 \frac{2}{(2l+1)} \delta_{l1} \right\} = \frac{k}{5} \frac{1}{R^l} [8 \delta_{l3} - 3 \delta_{l1}] \\ &= \left\{ \begin{array}{l} 8k/5R^3, \text{ if } l = 3 \\ -3k/5R, \text{ if } l = 1 \end{array} \right\} \text{ (zero otherwise).} \end{aligned}$$

Therefore

$$V(r, \theta) = -\frac{3k}{5R} r P_1(\cos \theta) + \frac{8k}{5R^3} r^3 P_3(\cos \theta) = \boxed{\frac{k}{5} \left[8 \left(\frac{r}{R} \right)^3 P_3(\cos \theta) - 3 \left(\frac{r}{R} \right) P_1(\cos \theta) \right]},$$

or

$$\frac{k}{5} \left\{ 8 \left(\frac{r}{R} \right)^3 \frac{1}{2} [5 \cos^3 \theta - 3 \cos \theta] - 3 \left(\frac{r}{R} \right) \cos \theta \right\} \Rightarrow \boxed{V(r, \theta) = \frac{k}{5} \frac{r}{R} \cos \theta \left\{ 4 \left(\frac{r}{R} \right)^2 [5 \cos^2 \theta - 3] - 3 \right\}}$$

(for $r \leq R$). Meanwhile, $B_l = A_l R^{2l+1}$ (Eq. 3.81—this follows from the continuity of V at R). Therefore

$$B_l = \left\{ \begin{array}{l} 8kR^4/5, \text{ if } l = 3 \\ -3kR^2/5, \text{ if } l = 1 \end{array} \right\} \quad (\text{zero otherwise}).$$

So

$$V(r, \theta) = \frac{-3kR^2}{5} \frac{1}{r^2} P_1(\cos \theta) + \frac{8kR^4}{5} \frac{1}{r^4} P_3(\cos \theta) = \boxed{\frac{k}{5} \left[8 \left(\frac{R}{r} \right)^4 P_3(\cos \theta) - 3 \left(\frac{R}{r} \right)^2 P_1(\cos \theta) \right]},$$

or

$$\boxed{V(r, \theta) = \frac{k}{5} \left(\frac{R}{r} \right)^2 \cos \theta \left\{ 4 \left(\frac{R}{r} \right)^2 [5 \cos^2 \theta - 3] - 3 \right\}}$$

(for $r \geq R$). Finally, using Eq. 3.83:

$$\begin{aligned}
\sigma(\theta) &= \epsilon_0 \sum_{l=0}^{\infty} (2l+1) A_l R^{l-1} P_l(\cos \theta) = \epsilon_0 [3A_1 P_1 + 7A_3 R^2 P_3] \\
&= \epsilon_0 \left[3 \left(-\frac{3k}{5R} \right) P_1 + 7 \left(\frac{8k}{5R^3} \right) R^2 P_3 \right] = \boxed{\frac{\epsilon_0 k}{5R} [-9P_1(\cos \theta) + 56P_3(\cos \theta)]} \\
&= \frac{\epsilon_0 k}{5R} \left[-9 \cos \theta + \frac{56}{2} (5 \cos^3 \theta - 3 \cos \theta) \right] = \frac{\epsilon_0 k}{5R} \cos \theta [-9 + 28 \cdot 5 \cos^2 \theta - 28 \cdot 3] \\
&= \boxed{\frac{\epsilon_0 k}{5R} \cos \theta [140 \cos^2 \theta - 93]}.
\end{aligned}$$

Problem 3.20

Use Eq. 3.83: $\sigma(\theta) = \epsilon_0 \sum_{l=0}^{\infty} (2l+1) A_l R^{l-1} P_l(\cos \theta)$. But Eq. 3.69 says: $A_l = \frac{2l+1}{2R^l} \int_0^\pi V_0(\theta) P_l(\cos \theta) \sin \theta d\theta$.

Putting them together:

$$\sigma(\theta) = \frac{\epsilon_0}{2R} \sum_{l=0}^{\infty} (2l+1)^2 C_l P_l(\cos \theta), \quad \text{with } C_l = \int_0^\pi V_0(\theta) P_l(\cos \theta) \sin \theta d\theta. \quad \text{qed}$$

Problem 3.21

Set $V = 0$ on the equatorial plane, far from the sphere. Then the potential is the same as Ex. 3.8 *plus* the potential of a uniformly charged spherical shell:

$$\boxed{V(r, \theta) = -E_0 \left(r - \frac{R^3}{r^2} \right) \cos \theta + \frac{1}{4\pi\epsilon_0} \frac{Q}{r}.$$

Problem 3.22

$$(a) \quad V(r, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta) \quad (r > R), \quad \text{so } V(r, 0) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(1) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} = \frac{\sigma}{2\epsilon_0} \left[\sqrt{r^2 + R^2} - r \right].$$

Since $r > R$ in this region, $\sqrt{r^2 + R^2} = r\sqrt{1 + (R/r)^2} = r \left[1 + \frac{1}{2}(R/r)^2 - \frac{1}{8}(R/r)^4 + \dots \right]$, so

$$\sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} = \frac{\sigma}{2\epsilon_0} r \left[1 + \frac{1}{2} \frac{R^2}{r^2} - \frac{1}{8} \frac{R^4}{r^4} + \dots - 1 \right] = \frac{\sigma}{2\epsilon_0} \left(\frac{R^2}{2r} - \frac{R^4}{8r^3} + \dots \right).$$

Comparing like powers of r , I see that $B_0 = \frac{\sigma R^2}{4\epsilon_0}$, $B_1 = 0$, $B_2 = -\frac{\sigma R^4}{16\epsilon_0}, \dots$. Therefore

$$\begin{aligned}
V(r, \theta) &= \frac{\sigma R^2}{4\epsilon_0} \left[\frac{1}{r} - \frac{R^2}{4r^3} P_2(\cos \theta) + \dots \right], \\
&= \frac{\sigma R^2}{4\epsilon_0 r} \left[1 - \frac{1}{8} \left(\frac{R}{r} \right)^2 (3 \cos^2 \theta - 1) + \dots \right],
\end{aligned} \quad (\text{for } r > R).$$

(b) $V(r, \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$ ($r < R$). In the northern hemisphere, $0 \leq \theta \leq \pi/2$,

$$V(r, 0) = \sum_{l=0}^{\infty} A_l r^l = \frac{\sigma}{2\epsilon_0} \left[\sqrt{r^2 + R^2} - r \right].$$

Since $r < R$ in this region, $\sqrt{r^2 + R^2} = R\sqrt{1 + (r/R)^2} = R \left[1 + \frac{1}{2}(r/R)^2 - \frac{1}{8}(r/R)^4 + \dots \right]$. Therefore

$$\sum_{l=0}^{\infty} A_l r^l = \frac{\sigma}{2\epsilon_0} \left[R + \frac{1}{2} \frac{r^2}{R} - \frac{1}{8} \frac{r^4}{R^3} + \dots - r \right].$$

Comparing like powers: $A_0 = \frac{\sigma}{2\epsilon_0} R$, $A_1 = -\frac{\sigma}{2\epsilon_0}$, $A_2 = \frac{\sigma}{4\epsilon_0 R}$, $\dots$, so

$$\boxed{\begin{aligned} V(r, \theta) &= \frac{\sigma}{2\epsilon_0} \left[R - r P_1(\cos \theta) + \frac{1}{2R} r^2 P_2(\cos \theta) + \dots \right], \\ &= \frac{\sigma R}{2\epsilon_0} \left[1 - \left(\frac{r}{R} \right) \cos \theta + \frac{1}{4} \left(\frac{r}{R} \right)^2 (3 \cos^2 \theta - 1) + \dots \right], \end{aligned}} \quad (\text{for } r < R, \text{ northern hemisphere}).$$

In the southern hemisphere we'll have to go for $\theta = \pi$, using $P_l(-1) = (-1)^l$.

$$V(r, \pi) = \sum_{l=0}^{\infty} (-1)^l \bar{A}_l r^l = \frac{\sigma}{2\epsilon_0} \left[\sqrt{r^2 + R^2} - r \right].$$

(I put an overbar on $\bar{A}_l$ to distinguish it from the northern A_l). The only difference is the sign of $\bar{A}_1$: $\bar{A}_1 = +(\sigma/2\epsilon_0)$, $\bar{A}_0 = A_0$, $\bar{A}_2 = A_2$. So:

$$\boxed{\begin{aligned} V(r, \theta) &= \frac{\sigma}{2\epsilon_0} \left[R + r P_1(\cos \theta) + \frac{1}{2R} r^2 P_2(\cos \theta) + \dots \right], \\ &= \frac{\sigma R}{2\epsilon_0} \left[1 + \left(\frac{r}{R} \right) \cos \theta + \frac{1}{4} \left(\frac{r}{R} \right)^2 (3 \cos^2 \theta - 1) + \dots \right], \end{aligned}} \quad (\text{for } r < R, \text{ southern hemisphere}).$$

Problem 3.23

$$V(r, \theta) = \begin{cases} \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta), & (r \leq R) \text{ (Eq. 3.78),} \\ \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta), & (r \geq R) \text{ (Eq. 3.79),} \end{cases}$$

where $B_l = A_l R^{2l+1}$ (Eq. 3.81) and

$$\begin{aligned}
A_l &= \frac{1}{2\epsilon_0 R^{l-1}} \int_0^\pi \sigma_0(\theta) P_l(\cos \theta) \sin \theta d\theta \quad (\text{Eq. 3.84}) \\
&= \frac{1}{2\epsilon_0 R^{l-1}} \sigma_0 \left\{ \int_0^{\pi/2} P_l(\cos \theta) \sin \theta d\theta - \int_{\pi/2}^\pi P_l(\cos \theta) \sin \theta d\theta \right\} \quad (\text{let } x = \cos \theta) \\
&= \frac{\sigma_0}{2\epsilon_0 R^{l-1}} \left\{ \int_0^1 P_l(x) dx - \int_{-1}^0 P_l(x) dx \right\}.
\end{aligned}$$

Now $P_l(-x) = (-1)^l P_l(x)$, since $P_l(x)$ is even, for even l , and odd, for odd l . Therefore

$$\int_{-1}^0 P_l(x) dx = \int_1^0 P_l(-x) d(-x) = (-1)^l \int_0^1 P_l(x) dx,$$

and hence

$$A_l = \frac{\sigma_0}{2\epsilon_0 R^{l-1}} [1 - (-1)^l] \int_0^1 P_l(x) dx = \begin{cases} 0, & \text{if } l \text{ is even} \\ \frac{\sigma_0}{\epsilon_0 R^{l-1}} \int_0^1 P_l(x) dx, & \text{if } l \text{ is odd} \end{cases}.$$

So $A_0 = A_2 = A_4 = A_6 = 0$, and all we need are A_1 , A_3 , and A_5 .

$$\begin{aligned}
\int_0^1 P_1(x) dx &= \int_0^1 x dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}. \\
\int_0^1 P_3(x) dx &= \frac{1}{2} \int_0^1 (5x^3 - 3x) dx = \frac{1}{2} \left(5 \frac{x^4}{4} - 3 \frac{x^2}{2} \right) \Big|_0^1 = \frac{1}{2} \left(\frac{5}{4} - \frac{3}{2} \right) = -\frac{1}{8}. \\
\int_0^1 P_5(x) dx &= \frac{1}{8} \int_0^1 (63x^5 - 70x^3 + 15x) dx = \frac{1}{8} \left(63 \frac{x^6}{6} - 70 \frac{x^4}{4} + 15 \frac{x^2}{2} \right) \Big|_0^1 \\
&= \frac{1}{8} \left(\frac{21}{2} - \frac{35}{2} + \frac{15}{2} \right) = \frac{1}{16} (36 - 35) = \frac{1}{16}.
\end{aligned}$$

Therefore

$$A_1 = \frac{\sigma_0}{\epsilon_0} \left(\frac{1}{2} \right); \quad A_3 = \frac{\sigma_0}{\epsilon_0 R^2} \left(-\frac{1}{8} \right); \quad A_5 = \frac{\sigma_0}{\epsilon_0 R^4} \left(\frac{1}{16} \right); \quad \text{etc.}$$

and

$$B_1 = \frac{\sigma_0}{\epsilon_0} R^3 \left(\frac{1}{2} \right); \quad B_3 = \frac{\sigma_0}{\epsilon_0} R^5 \left(-\frac{1}{8} \right); \quad B_5 = \frac{\sigma_0}{\epsilon_0} R^7 \left(\frac{1}{16} \right); \quad \text{etc.}$$

Thus

$$V(r, \theta) = \left\{ \begin{array}{ll} \frac{\sigma_0 r}{2\epsilon_0} \left[P_1(\cos \theta) - \frac{1}{4} \left(\frac{r}{R} \right)^2 P_3(\cos \theta) + \frac{1}{8} \left(\frac{r}{R} \right)^4 P_5(\cos \theta) + \dots \right], & (r \leq R), \\ \frac{\sigma_0 R^3}{2\epsilon_0 r^2} \left[P_1(\cos \theta) - \frac{1}{4} \left(\frac{R}{r} \right)^2 P_3(\cos \theta) + \frac{1}{8} \left(\frac{R}{r} \right)^4 P_5(\cos \theta) + \dots \right], & (r \geq R). \end{array} \right\}$$

Problem 3.24

$$\frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial V}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 V}{\partial \phi^2} = 0.$$

Look for solutions of the form $V(s, \phi) = S(s)\Phi(\phi)$:

$$\frac{1}{s} \Phi \frac{d}{ds} \left(s \frac{dS}{ds} \right) + \frac{1}{s^2} S \frac{d^2 \Phi}{d\phi^2} = 0.$$

Multiply by s^2 and divide by $V = S\Phi$:

$$\frac{s}{S} \frac{d}{ds} \left(s \frac{dS}{ds} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = 0.$$

Since the first term involves s only, and the second ϕ only, each is a constant:

$$\frac{s}{S} \frac{d}{ds} \left(s \frac{dS}{ds} \right) = C_1, \quad \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = C_2, \quad \text{with } C_1 + C_2 = 0.$$

Now C_2 must be negative (else we get exponentials for Φ , which do not return to their original value—as geometrically they *must*—when ϕ is increased by 2π).

$$C_2 = -k^2. \quad \text{Then } \frac{d^2 \Phi}{d\phi^2} = -k^2 \Phi \Rightarrow \Phi = A \cos k\phi + B \sin k\phi.$$

Moreover, since $\Phi(\phi + 2\pi) = \Phi(\phi)$, k *must be an integer*: $k = 0, 1, 2, 3, \dots$ (negative integers are just repeats, but $k = 0$ must be included, since $\Phi = A$ (a constant) is OK).

$s \frac{d}{ds} \left(s \frac{dS}{ds} \right) = k^2 S$ can be solved by $S = s^n$, provided n is chosen right:

$$s \frac{d}{ds} (s n s^{n-1}) = n s \frac{d}{ds} (s^n) = n^2 s s^{n-1} = n^2 s^n = k^2 S \Rightarrow n = \pm k.$$

Evidently the general solution is $S(s) = C s^k + D s^{-k}$, *unless* $k = 0$, in which case we have only *one* solution to a *second-order* equation—namely, $S = \text{constant}$. So we must treat $k = 0$ separately. *One* solution is a constant—but what's the other? Go back to the differential equation for S , and put in $k = 0$:

$$s \frac{d}{ds} \left(s \frac{dS}{ds} \right) = 0 \Rightarrow s \frac{dS}{ds} = \text{constant} = C \Rightarrow \frac{dS}{ds} = \frac{C}{s} \Rightarrow dS = C \frac{ds}{s} \Rightarrow S = C \ln s + D \text{ (another constant)}.$$

So the second solution in this case is $\ln s$. [How about Φ ? That *too* reduces to a single solution, $\Phi = A$, in the case $k = 0$. What's the second solution here? Well, putting $k = 0$ into the Φ equation:

$$\frac{d^2 \Phi}{d\phi^2} = 0 \Rightarrow \frac{d\Phi}{d\phi} = \text{constant} = B \Rightarrow \Phi = B\phi + A.$$

But a term of the form $B\phi$ is unacceptable, since it does not return to its initial value when ϕ is augmented by 2π .] *Conclusion:* The general solution with cylindrical symmetry is

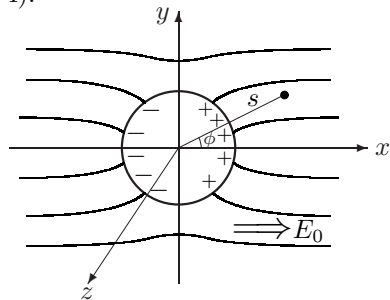
$$V(s, \phi) = a_0 + b_0 \ln s + \sum_{k=1}^{\infty} [s^k (a_k \cos k\phi + b_k \sin k\phi) + s^{-k} (c_k \cos k\phi + d_k \sin k\phi)].$$

Yes: the potential of a line charge goes like $\ln s$, which *is* included.

Problem 3.25

Picking $V = 0$ on the yz plane, with $\mathbf{E}_0$ in the x direction, we have (Eq. 3.74):

$$\left\{ \begin{array}{ll} \text{(i)} & V = 0, \\ \text{(ii)} & V \rightarrow -E_0 x = -E_0 s \cos \phi, \text{ for } s \gg R. \end{array} \right\} \quad \text{when } s = R,$$



Evidently $a_0 = b_0 = b_k = d_k = 0$, and $a_k = c_k = 0$ except for $k = 1$:

$$V(s, \phi) = \left(a_1 s + \frac{c_1}{s} \right) \cos \phi.$$

(i) $\Rightarrow c_1 = -a_1 R^2$; (ii) $\rightarrow a_1 = -E_0$. Therefore

$$V(s, \phi) = \left(-E_0 s + \frac{E_0 R^2}{s} \right) \cos \phi, \quad \text{or} \quad V(s, \phi) = E_0 s \left[\left(\frac{R}{s} \right)^2 - 1 \right] \cos \phi.$$

$$\sigma = -\epsilon_0 \left. \frac{\partial V}{\partial s} \right|_{s=R} = -\epsilon_0 E_0 \left(-\frac{R^2}{s^2} - 1 \right) \cos \phi \Big|_{s=R} = \boxed{2\epsilon_0 E_0 \cos \phi}.$$

Problem 3.26

Inside: $V(s, \phi) = a_0 + \sum_{k=1}^{\infty} s^k (a_k \cos k\phi + b_k \sin k\phi)$. (In this region $\ln s$ and s^{-k} are no good—they blow up at $s = 0$.)

Outside: $V(s, \phi) = \bar{a}_0 + \sum_{k=1}^{\infty} \frac{1}{s^k} (c_k \cos k\phi + d_k \sin k\phi)$. (Here $\ln s$ and s^k are no good at $s \rightarrow \infty$).

$$\sigma = -\epsilon_0 \left(\frac{\partial V_{\text{out}}}{\partial s} - \frac{\partial V_{\text{in}}}{\partial s} \right) \Big|_{s=R} \quad (\text{Eq. 2.36}).$$

Thus

$$a \sin 5\phi = -\epsilon_0 \sum_{k=1}^{\infty} \left\{ -\frac{k}{R^{k+1}} (c_k \cos k\phi + d_k \sin k\phi) - k R^{k-1} (a_k \cos k\phi + b_k \sin k\phi) \right\}.$$

Evidently $a_k = c_k = 0$; $b_k = d_k = 0$ except $k = 5$; $a = 5\epsilon_0 \left(\frac{1}{R^6} d_5 + R^4 b_5 \right)$. Also, V is continuous at $s = R$:

$a_0 + R^5 b_5 \sin 5\phi = \bar{a}_0 + \frac{1}{R^5} d_5 \sin 5\phi$. So $a_0 = \bar{a}_0$ (might as well choose both zero); $R^5 b_5 = R^{-5} d_5$, or $d_5 = R^{10} b_5$.

Combining these results: $a = 5\epsilon_0 (R^4 b_5 + R^4 b_5) = 10\epsilon_0 R^4 b_5$; $b_5 = \frac{a}{10\epsilon_0 R^4}$; $d_5 = \frac{aR^6}{10\epsilon_0}$. Therefore

$$V(s, \phi) = \frac{a \sin 5\phi}{10\epsilon_0} \begin{cases} s^5/R^4, & \text{for } s < R, \\ R^6/s^5, & \text{for } s > R. \end{cases}$$

Problem 3.27 Since $\mathbf{r}$ is on the z axis, the angle α is just the polar angle θ (I'll drop the primes, for simplicity).

Monopole term:

$$\int \rho \, d\tau = kR \int \left[\frac{1}{r^2} (R - 2r) \sin \theta \right] r^2 \sin \theta \, dr \, d\theta \, d\phi.$$

But the r integral is

$$\int_0^R (R - 2r) \, dr = (Rr - r^2) \Big|_0^R = R^2 - R^2 = 0.$$

So the monopole term is zero.

Dipole term:

$$\int r \cos \theta \rho \, d\tau = kR \int (r \cos \theta) \left[\frac{1}{r^2} (R - 2r) \sin \theta \right] r^2 \sin \theta \, dr \, d\theta \, d\phi.$$

But the θ integral is

$$\int_0^\pi \sin^2 \theta \cos \theta \, d\theta = \frac{\sin^3 \theta}{3} \Big|_0^\pi = \frac{1}{3}(0 - 0) = 0.$$

So the dipole contribution is likewise zero.

Quadrupole term:

$$\int r^2 \left(\frac{3}{2} \cos^2 \theta - \frac{1}{2} \right) \rho \, d\tau = \frac{1}{2} kR \int r^2 (3 \cos^2 \theta - 1) \left[\frac{1}{r^2} (R - 2r) \sin \theta \right] r^2 \sin \theta \, dr \, d\theta \, d\phi.$$

r integral:

$$\int_0^R r^2 (R - 2r) \, dr = \left(\frac{r^3}{3} R - \frac{r^4}{2} \right) \Big|_0^R = \frac{R^4}{3} - \frac{R^4}{2} = -\frac{R^4}{6}.$$

θ integral:

$$\begin{aligned} \int_0^\pi \underbrace{(3 \cos^2 \theta - 1)}_{3(1 - \sin^2 \theta) - 1 = 2 - 3 \sin^2 \theta} \sin^2 \theta \, d\theta &= 2 \int_0^\pi \sin^2 \theta \, d\theta - 3 \int_0^\pi \sin^4 \theta \, d\theta \\ &= 2 \left(\frac{\pi}{2} \right) - 3 \left(\frac{3\pi}{8} \right) = \pi \left(1 - \frac{9}{8} \right) = -\frac{\pi}{8}. \end{aligned}$$

ϕ integral:

$$\int_0^{2\pi} d\phi = 2\pi.$$

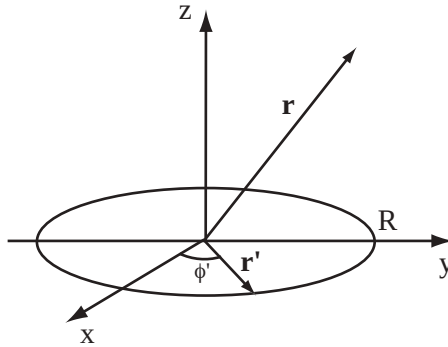
The whole integral is:

$$\frac{1}{2} kR \left(-\frac{R^4}{6} \right) \left(-\frac{\pi}{8} \right) (2\pi) = \frac{k\pi^2 R^5}{48}.$$

For point P on the z axis ($r \rightarrow z$ in Eq. 3.95) the approximate potential is

$$V(z) \cong \frac{1}{4\pi\epsilon_0} \frac{k\pi^2 R^5}{48z^3}. \quad (\text{Quadrupole.})$$

Problem 3.28



For a line charge, $\rho(\mathbf{r}') d\tau' \rightarrow \lambda(\mathbf{r}') dl'$, which in this case becomes $\lambda R d\phi'$.

$$\mathbf{r} = r \sin \theta \cos \phi \hat{\mathbf{x}} + r \sin \theta \sin \phi \hat{\mathbf{y}} + r \cos \theta \hat{\mathbf{z}},$$

$$\mathbf{r}' = R \cos \phi' + R \sin \phi' \hat{\mathbf{y}}, \quad \text{so}$$

$$\mathbf{r} \cdot \mathbf{r}' = rR \sin \theta \cos \phi \cos \phi' + rR \sin \theta \sin \phi \sin \phi' = rR \cos \alpha,$$

$$\cos \alpha = \sin \theta (\cos \phi \cos \phi' + \sin \phi \sin \phi').$$

$n = 0$:

$$\int \rho(\mathbf{r}') d\tau' \rightarrow \lambda R \int_0^{2\pi} d\phi' = 2\pi R \lambda; \quad V_0 = \frac{1}{4\pi\epsilon_0} \frac{2\pi R \lambda}{r} = \boxed{\frac{\lambda}{2\epsilon_0} \frac{R}{r}}.$$

$n = 1$:

$$\int r' \cos \alpha \rho(\mathbf{r}') d\tau' \rightarrow \int R \cos \alpha \lambda R d\phi' = \lambda R^2 \sin \theta \int_0^{2\pi} (\cos \phi \cos \phi' + \sin \phi \sin \phi') d\phi' = 0; \quad V_1 = \boxed{0}.$$

$n = 2$:

$$\begin{aligned} \int (r')^2 P_2(\cos \alpha) \rho(\mathbf{r}') d\tau' &\rightarrow \int R^2 \left(\frac{3}{2} \cos^2 \alpha - \frac{1}{2} \right) \lambda R d\phi' = \frac{\lambda R^3}{2} \int \left[3 \sin^2 \theta (\cos \phi \cos \phi' + \sin \phi \sin \phi')^2 - 1 \right] d\phi' \\ &= \frac{\lambda R^3}{2} \left[3 \sin^2 \theta \left(\cos^2 \phi \int_0^{2\pi} \cos^2 \phi' d\phi' + \sin^2 \phi \int_0^{2\pi} \sin^2 \phi' d\phi' + 2 \sin \phi \cos \phi \int_0^{2\pi} \sin \phi' \cos \phi' d\phi' \right) - \int_0^{2\pi} d\phi' \right] \\ &= \frac{\lambda R^3}{2} [3 \sin^2 \theta (\pi \cos^2 \phi + \pi \sin^2 \phi + 0) - 2\pi] = \frac{\pi \lambda R^3}{2} (3 \sin^2 \theta - 2) = -\pi \lambda R^3 \left(\frac{3}{2} \cos^2 \theta - \frac{1}{2} \right). \end{aligned}$$

So

$$V_2 = \boxed{-\frac{\lambda}{8\epsilon_0} \frac{R^3}{r^3} (3 \cos^2 \theta - 1) = -\frac{\lambda}{4\epsilon_0} \frac{R^3}{r^3} P_2(\cos \theta)}.$$

Problem 3.29

$\mathbf{p} = (3qa - qa)\hat{\mathbf{z}} + (-2qa - 2q(-a))\hat{\mathbf{y}} = 2qa\hat{\mathbf{z}}$. Therefore

$$V \cong \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2},$$

and $\mathbf{p} \cdot \hat{\mathbf{r}} = 2qa\hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = 2qa \cos \theta$, so

$$V \cong \boxed{\frac{1}{4\pi\epsilon_0} \frac{2qa \cos \theta}{r^2}}. \quad (\text{Dipole.})$$

Problem 3.30

(a) By symmetry, $\mathbf{p}$ is clearly in the z direction: $\mathbf{p} = p\hat{\mathbf{z}}$; $p = \int z\rho d\tau \Rightarrow \int z\sigma da$.

$$\begin{aligned} p &= \int (R \cos \theta)(k \cos \theta)R^2 \sin \theta d\theta d\phi = 2\pi R^3 k \int_0^\pi \cos^2 \theta \sin \theta d\theta = 2\pi R^3 k \left(-\frac{\cos^3 \theta}{3} \right) \bigg|_0^\pi \\ &= \frac{2}{3}\pi R^3 k [1 - (-1)] = \frac{4\pi R^3 k}{3}; \quad \boxed{\mathbf{p} = \frac{4\pi R^3 k}{3} \hat{\mathbf{z}}}. \end{aligned}$$

(b)

$$V \cong \frac{1}{4\pi\epsilon_0} \frac{4\pi R^3 k}{3} \frac{\cos \theta}{r^2} = \boxed{\frac{kR^3}{3\epsilon_0} \frac{\cos \theta}{r^2}}. \quad (\text{Dipole.})$$

This is *also* the *exact* potential. *Conclusion:* all multiple moments of this distribution (except the dipole) are exactly zero.

Problem 3.31

Using Eq. 3.94 with $r' = d/2$ and $\alpha = \theta$ (Fig. 3.26):

$$\frac{1}{r_+} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{d}{2r} \right)^n P_n(\cos \theta);$$

for r_- , we let $\theta \rightarrow 180^\circ + \theta$, so $\cos \theta \rightarrow -\cos \theta$:

$$\frac{1}{r_-} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{d}{2r} \right)^n P_n(-\cos \theta).$$

But $P_n(-x) = (-1)^n P_n(x)$, so

$$V = \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{r_+} - \frac{1}{r_-} \right) = \frac{1}{4\pi\epsilon_0} q \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{d}{2r} \right)^n [P_n(\cos \theta) - P_n(-\cos \theta)] = \frac{2q}{4\pi\epsilon_0 r} \sum_{n=1,3,5,\dots} \left(\frac{d}{2r} \right)^n P_n(\cos \theta).$$

Therefore

$$V_{\text{dip}} = \frac{2q}{4\pi\epsilon_0} \frac{1}{r} \frac{d}{2r} P_1(\cos \theta) = \frac{qd \cos \theta}{4\pi\epsilon_0 r^2}, \quad \text{while} \quad \boxed{V_{\text{quad}} = 0}.$$

$$V_{\text{oct}} = \frac{2q}{4\pi\epsilon_0 r} \left(\frac{d}{2r} \right)^3 P_3(\cos \theta) = \frac{2q}{4\pi\epsilon_0} \frac{d^3}{8r^4} \frac{1}{2} (5 \cos^3 \theta - 3 \cos \theta) = \boxed{\frac{qd^3}{4\pi\epsilon_0} \frac{1}{8r^4} (5 \cos^3 \theta - 3 \cos \theta)}.$$

Problem 3.32

$$\begin{aligned}
 \text{(a) (i) } Q &= \boxed{2q}, & \text{(ii) } \mathbf{p} &= \boxed{3qa \hat{\mathbf{z}}}, & \text{(iii) } V &\cong \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r} + \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} \right] = \boxed{\frac{1}{4\pi\epsilon_0} \left[\frac{2q}{r} + \frac{3qa \cos \theta}{r^2} \right]}. \\
 \text{(b) (i) } Q &= \boxed{2q}, & \text{(ii) } \mathbf{p} &= \boxed{qa \hat{\mathbf{z}}}, & \text{(iii) } V &\cong \boxed{\frac{1}{4\pi\epsilon_0} \left[\frac{2q}{r} + \frac{qa \cos \theta}{r^2} \right]}. \\
 \text{(c) (i) } Q &= \boxed{2q}, & \text{(ii) } \mathbf{p} &= \boxed{3qa \hat{\mathbf{y}}}, & \text{(iii) } V &\cong \boxed{\frac{1}{4\pi\epsilon_0} \left[\frac{2q}{r} + \frac{3qa \sin \theta \sin \phi}{r^2} \right]} \text{ (from Eq. 1.64, } \hat{\mathbf{y}} \cdot \hat{\mathbf{r}} = \sin \theta \sin \phi \text{)}.
 \end{aligned}$$

Problem 3.33

$$\begin{aligned}
 \text{(a) This point is at } r &= a, \theta = \frac{\pi}{2}, \phi = 0, \text{ so } \mathbf{E} = \frac{p}{4\pi\epsilon_0 a^3} \hat{\boldsymbol{\theta}} = \frac{p}{4\pi\epsilon_0 a^3} (-\hat{\mathbf{z}}); \mathbf{F} = q\mathbf{E} = \boxed{-\frac{pq}{4\pi\epsilon_0 a^3} \hat{\mathbf{z}}}. \\
 \text{(b) Here } r &= a, \theta = 0, \text{ so } \mathbf{E} = \frac{p}{4\pi\epsilon_0 a^3} (2\hat{\mathbf{r}}) = \frac{2p}{4\pi\epsilon_0 a^3} \hat{\mathbf{z}}. \quad \boxed{\mathbf{F} = \frac{2pq}{4\pi\epsilon_0 a^3} \hat{\mathbf{z}}}. \\
 \text{(c) } W &= q [V(0, 0, a) - V(a, 0, 0)] = \frac{qp}{4\pi\epsilon_0 a^2} \left[\cos(0) - \cos\left(\frac{\pi}{2}\right) \right] = \boxed{\frac{pq}{4\pi\epsilon_0 a^2}}.
 \end{aligned}$$

Problem 3.34

$$Q = -q, \text{ so } V_{\text{mono}} = \frac{1}{4\pi\epsilon_0} \frac{-q}{r}; \quad \mathbf{p} = qa \hat{\mathbf{z}}, \quad \text{so } V_{\text{dip}} = \frac{1}{4\pi\epsilon_0} \frac{qa \cos \theta}{r^2}. \text{ Therefore}$$

$$\boxed{V(r, \theta) \cong \frac{q}{4\pi\epsilon_0} \left(-\frac{1}{r} + \frac{a \cos \theta}{r^2} \right)}. \quad \boxed{\mathbf{E}(r, \theta) \cong \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r^2} \hat{\mathbf{r}} + \frac{a}{r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}) \right]}.$$

Problem 3.35 The total charge is zero, so the dominant term is the dipole. We need the dipole moment of this configuration. It obviously points in the z direction, and for the southern hemisphere ($\theta : \frac{\pi}{2} \rightarrow \pi$) ρ switches sign but so does z , so

$$\begin{aligned}
 p &= \int z \rho \, d\tau = 2\rho_0 \int_{\theta=0}^{\pi/2} r \cos \theta \, r^2 \sin \theta \, dr \, d\theta \, d\phi = 2\rho_0 (2\pi) \int_0^R r^3 \, dr \int_0^{\pi/2} \cos \theta \sin \theta \, d\theta \\
 &= 4\pi\rho_0 \frac{R^4}{4} \frac{\sin^2 \theta}{2} \Big|_0^{\pi/2} = \frac{\pi\rho_0 R^4}{2}.
 \end{aligned}$$

Therefore (Eq. 3.103)

$$\mathbf{E} \approx \boxed{\frac{\pi\rho_0 R^4}{8\pi\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}})}.$$

Problem 3.36

$\mathbf{p} = (\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} + (\mathbf{p} \cdot \hat{\boldsymbol{\theta}}) \hat{\boldsymbol{\theta}} = p \cos \theta \hat{\mathbf{r}} - p \sin \theta \hat{\boldsymbol{\theta}}$ (Fig. 3.36). So $3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p} = 3p \cos \theta \hat{\mathbf{r}} - p \cos \theta \hat{\mathbf{r}} + p \sin \theta \hat{\boldsymbol{\theta}} = 2p \cos \theta \hat{\mathbf{r}} + p \sin \theta \hat{\boldsymbol{\theta}}$. So Eq. 3.104 $\equiv$ Eq. 3.103. $\checkmark$

Problem 3.37

$$V_{\text{ave}}(R) = \frac{1}{4\pi R^2} \int V(\mathbf{r}) \, da, \text{ where the integral is over the surface of a sphere of radius } R. \text{ Now } da =$$

$R^2 \sin \theta d\theta d\phi$, so $V_{\text{ave}}(R) = \frac{1}{4\pi} \int V(R, \theta, \phi) \sin \theta d\theta d\phi$.

$$\begin{aligned} \frac{dV_{\text{ave}}}{dR} &= \frac{1}{4\pi} \int \frac{\partial V}{\partial R} \sin \theta d\theta d\phi = \frac{1}{4\pi} \int (\nabla V \cdot \hat{\mathbf{r}}) \sin \theta d\theta d\phi = \frac{1}{4\pi R^2} \int (\nabla V) \cdot (R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) \\ &= \frac{1}{4\pi R^2} \int (\nabla V) \cdot d\mathbf{a} = \frac{1}{4\pi R^2} \int (\nabla^2 V) d\tau = 0. \end{aligned}$$

(The final integral, from the divergence theorem, is over the *volume* of the sphere, where by assumption the Laplacian of V is zero.) So V_{ave} is independent of R —the same for all spheres, regardless of their radius—and hence (taking the limit as $R \rightarrow 0$), $V_{\text{ave}}(R) = V(0)$. $\checkmark$

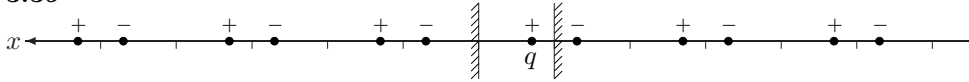
Problem 3.38 At a point (x, y) on the plane the field of q is

$$\mathbf{E}_q = \frac{1}{4\pi\epsilon_0} \frac{q}{z^3} \hat{\mathbf{z}}, \quad \text{and} \quad \hat{\mathbf{z}} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} - d \hat{\mathbf{z}},$$

so its z component is $-\frac{q}{4\pi\epsilon_0} \frac{d}{(x^2 + y^2 + d^2)^{3/2}}$. Meanwhile, the field of σ (just below the surface) is $-\frac{\sigma}{2\epsilon_0}$, (Eq. 2.17). (Of course, this is for a *uniform* surface charge, but as long as we are infinitesimally far away σ is *effectively* uniform.) The total field inside the conductor is zero, so

$$-\frac{q}{4\pi\epsilon_0} \frac{d}{(x^2 + y^2 + d^2)^{3/2}} - \frac{\sigma}{2\epsilon_0} = 0 \quad \Rightarrow \quad \sigma(x, y) = -\frac{qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}. \quad \checkmark$$

Problem 3.39



The image configuration is shown in the figure; the positive image charge forces cancel in pairs. The net force of the negative image charges is:

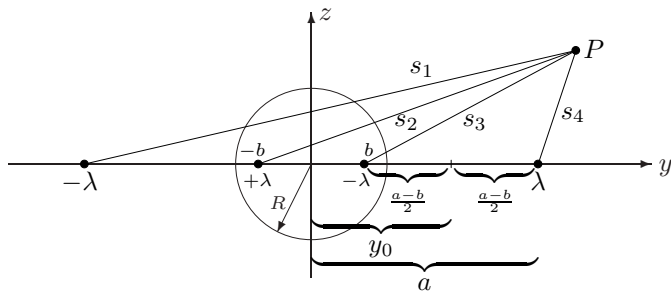
$$\begin{aligned} F &= \frac{1}{4\pi\epsilon_0} q^2 \left\{ \frac{1}{[2(a-x)]^2} + \frac{1}{[2a+2(a-x)]^2} + \frac{1}{[4a+2(a-x)]^2} + \dots \right. \\ &\quad \left. - \frac{1}{(2x)^2} - \frac{1}{(2a+2x)^2} - \frac{1}{(4a+2x)^2} - \dots \right\} \\ &= \boxed{\frac{1}{4\pi\epsilon_0} \frac{q^2}{4} \left\{ \left[\frac{1}{(a-x)^2} + \frac{1}{(2a-x)^2} + \frac{1}{(3a-x)^2} + \dots \right] - \left[\frac{1}{x^2} + \frac{1}{(a+x)^2} + \frac{1}{(2a+x)^2} + \dots \right] \right\}}. \end{aligned}$$

When $a \rightarrow \infty$ (i.e. $a \gg x$) only the $\frac{1}{x^2}$ term survives: $F = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2x)^2} \checkmark$ (same as for only *one* plane—Eq. 3.12). When $x = a/2$,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q^2}{4} \left\{ \left[\frac{1}{(a/2)^2} + \frac{1}{(3a/2)^2} + \frac{1}{(5a/2)^2} + \dots \right] - \left[\frac{1}{(a/2)^2} + \frac{1}{(3a/2)^2} + \frac{1}{(5a/2)^2} + \dots \right] \right\} = 0. \quad \checkmark$$

Problem 3.40

Following Prob. 2.52, we place image line charges $-\lambda$ at $y = b$ and $+\lambda$ at $y = -b$ (here y is the horizontal axis, z vertical).



In the solution to Prob. 2.52 substitute:

$$a \rightarrow \frac{a-b}{2}, \quad y_0 \rightarrow \frac{a+b}{2} \quad \text{so} \quad \left(\frac{a-b}{2}\right)^2 = \left(\frac{a+b}{2}\right)^2 - R^2 \Rightarrow b = \frac{R^2}{a}.$$

$$\begin{aligned} V &= \frac{\lambda}{4\pi\epsilon_0} \left[\ln\left(\frac{s_3^2}{s_4^2}\right) + \ln\left(\frac{s_1^2}{s_2^2}\right) \right] = \frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{s_1^2 s_3^2}{s_4^2 s_2^2}\right) \\ &= \frac{\lambda}{4\pi\epsilon_0} \ln \left\{ \frac{[(y+a)^2 + z^2][(y-b)^2 + z^2]}{[(y-a)^2 + z^2][(y+b)^2 + z^2]} \right\}, \quad \text{or, using } y = s \cos \phi, \quad z = s \sin \phi, \\ &= \boxed{\frac{\lambda}{4\pi\epsilon_0} \ln \left\{ \frac{(s^2 + a^2 + 2as \cos \phi)[(as/R)^2 + R^2 - 2as \cos \phi]}{(a^2 + a^2 - 2as \cos \phi)[(as/R)^2 + R^2 + 2as \cos \phi]} \right\}}. \end{aligned}$$

Problem 3.41 Same as Problem 3.9, only this time we want $q' + q'' = q$, so $q'' = q - q'$:

$$F = \frac{q}{4\pi\epsilon_0} \left(\frac{q''}{a^2} + \frac{q'}{(a-b)^2} \right) = \frac{q^2}{4\pi\epsilon_0 a^2} + \frac{qq'}{4\pi\epsilon_0} \left(-\frac{1}{a^2} + \frac{1}{(a-b)^2} \right).$$

The second term is identical to Problem 3.9, and I'll just quote the answer from there:

$$F = \frac{q^2}{4\pi\epsilon_0 a^3} \left[a - R^3 \frac{(2a^2 - R^2)}{(a^2 - R^2)^2} \right].$$

(a) $F = 0 \Rightarrow a(a^2 - R^2)^2 = R^3(2a^2 - R^2)$, or (letting $x \equiv a/R$), $x(x^2 - 1)^2 - 2x^2 + 1 = 0$. We want a real root greater than 1; Mathematica delivers $x = (1 + \sqrt{5})/2 = 1.61803$, so $a = 1.61803R = \boxed{5.66311 \text{ \AA}}$.

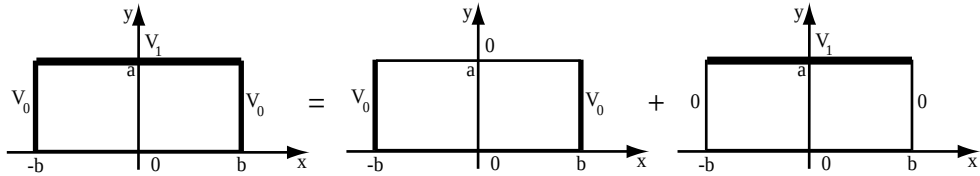
(b) Let $a_0 = x_0 R$ be the minimum value of a . The work necessary is

$$\begin{aligned} W &= - \int_{\infty}^{a_0} F da = \frac{q^2}{4\pi\epsilon_0} \int_{a_0}^{\infty} \frac{1}{a^3} \left[a - R^3 \frac{(2a^2 - R^2)}{(a^2 - R^2)^2} \right] da = \frac{q^2}{4\pi\epsilon_0 R} \int_{x_0}^{\infty} \left[\frac{1}{x^2} - \frac{(2x^2 - 1)}{x^3(x^2 - 1)^2} \right] dx \\ &= \frac{q^2}{4\pi\epsilon_0 R} \left[\frac{1 + 2x_0 - 2x_0^3}{2x_0^2(1 - x_0^2)} \right]. \end{aligned}$$

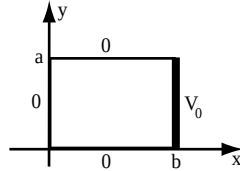
Putting in $x_0 = (1 + \sqrt{5})/2$, Mathematica says the term in square brackets is $1/2$ (this is not an accident; see footnote 6 on page 127), so $W = \frac{q^2}{8\pi\epsilon_0 R}$. Numerically,

$$W = \frac{(1.60 \times 10^{-19})^2}{8\pi(8.85 \times 10^{-12})(5.66 \times 10^{-10})} \text{ J} = 2.03 \times 10^{-19} \text{ J} = \boxed{1.27 \text{ eV}}.$$

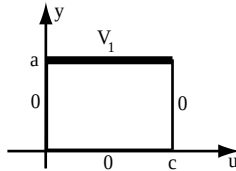
Problem 3.42



The first configuration on the right is precisely Example 3.4, but unfortunately the second configuration is not the same as Problem 3.15:



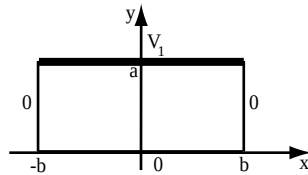
We could reconstruct Problem 3.15 with the modified boundaries, but let's see if we can't twist it around by an astute change of variables. Suppose we let $x \rightarrow y$, $y \rightarrow u$, $a \rightarrow c$, $b \rightarrow a$, and $V_0 \rightarrow V_1$:



This is closer; making the changes in the solution to Problem 3.15 we have (for this configuration)

$$V(u, y) = \frac{4V_1}{\pi} \sum_{n=1,3,5,\dots} \frac{\sinh(n\pi y/c) \sin(n\pi u/c)}{n \sinh(n\pi a/c)}.$$

Now let $c \rightarrow 2b$ and $u \rightarrow x + b$, and the configuration is just what we want:



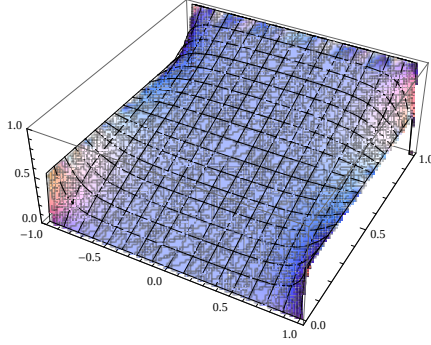
The potential for this configuration is

$$V(x, y) = \frac{4V_1}{\pi} \sum_{n=1,3,5,\dots} \frac{\sinh(n\pi y/2b) \sin(n\pi(x+b)/2b)}{n \sinh(n\pi a/2b)}.$$

(If you like, write $\sin(n\pi(x+b)/2b)$ as $(-1)^{(n-1)/2} \cos(n\pi x/2b)$.) Combining this with Eq. 3.42,

$$V(x, y) = \left[\frac{4}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} \left[V_0 \frac{\cosh(n\pi x/a) \sin(n\pi y/a)}{\cosh(n\pi b/a)} + V_1 \frac{\sinh(n\pi y/2b) \sin(n\pi(x+b)/2b)}{\sinh(n\pi a/2b)} \right] \right].$$

Here's a plot of this function, for the case $a = b = 1$, $V_0 = 1/2$, $V_1 = 1$:



Problem 3.43

Since the configuration is azimuthally symmetric, $V(r, \theta) = \sum \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$.

(a) $r > b$: $A_l = 0$ for all l , since $V \rightarrow 0$ at ∞ . Therefore $V(r, \theta) = \sum \frac{B_l}{r^{l+1}} P_l(\cos \theta)$.

$a < r < b$: $V(r, \theta) = \sum \left(C_l r^l + \frac{D_l}{r^{l+1}} \right) P_l(\cos \theta)$. $r < a$: $V(r, \theta) = V_0$.

We need to determine B_l, C_l, D_l , and V_0 . To do this, invoke boundary conditions as follows: (i) V is continuous at a , (ii) V is continuous at b , (iii) $\Delta \left(\frac{\partial V}{\partial r} \right) = -\frac{1}{\epsilon_0} \sigma(\theta)$ at b .

$$(ii) \Rightarrow \sum \frac{B_l}{b^{l+1}} P_l(\cos \theta) = \sum \left(C_l b^l + \frac{D_l}{b^{l+1}} \right) P_l(\cos \theta); \quad \frac{B_l}{b^{l+1}} = C_l b^l + \frac{D_l}{b^{l+1}} \Rightarrow \boxed{B_l = b^{2l+1} C_l + D_l}. \quad (1)$$

$$(i) \Rightarrow \sum \left(C_l a^l + \frac{D_l}{a^{l+1}} \right) P_l(\cos \theta) = V_0; \quad \left\{ \begin{array}{l} C_l a^l + \frac{D_l}{a^{l+1}} = 0, \text{ if } l \neq 0, \\ C_0 a^0 + \frac{D_0}{a^1} = V_0, \text{ if } l = 0; \end{array} \right\} \quad \boxed{\begin{array}{l} D_l = -a^{2l+1} C_l, \text{ } l \neq 0, \\ D_0 = aV_0 - aC_0. \end{array}} \quad (2)$$

Putting (2) into (1) gives $B_l = b^{2l+1} C_l - a^{2l+1} C_l$, $l \neq 0$, $B_0 = bC_0 + aV_0 - aC_0$. Therefore

$$\boxed{\begin{array}{l} B_l = (b^{2l+1} - a^{2l+1}) C_l, \text{ } l \neq 0, \\ B_0 = (b - a)C_0 + aV_0. \end{array}} \quad (1')$$

(iii) $\Rightarrow \sum B_l [-(l+1)] \frac{1}{b^{l+2}} P_l(\cos \theta) - \sum \left(C_l l b^{l-1} + D_l \frac{-(l+1)}{b^{l+2}} \right) P_l(\cos \theta) = \frac{-k}{\epsilon_0} P_1(\cos \theta)$. So

$$-\frac{(l+1)}{b^{l+2}} B_l - \left(C_l l b^{l-1} + D_l \frac{-(l+1)}{b^{l+2}} \right) = 0, \text{ if } l \neq 1;$$

or

$$-(l+1)B_l - lC_l b^{2l+1} + (l+1)D_l = 0; \quad (l+1)(B_l - D_l) = -lb^{2l+1}C_l.$$

$$B_1(+2) \frac{1}{b^2} + \left(C_1 + D_1 \frac{-2}{b^2} \right) = \frac{k}{\epsilon_0}, \text{ for } l = 1; \quad C_1 + \frac{2}{b^3} (B_1 - D_1) = k.$$

Therefore

$$\left[\begin{array}{l} (l+1)(B_l - D_l) + lb^{2l+1}C_l = 0, \text{ for } l \neq 1, \\ C_1 + \frac{2}{b^3}(B_1 - D_1) = \frac{k}{\epsilon_0}. \end{array} \right] \quad (3)$$

Plug (2) and (1') into (3):

For $l \neq 0$ or 1:

$$(l+1) \left[(b^{2l+1} - a^{2l+1}) C_l + a^{2l+1} C_l \right] + lb^{2l+1} C_l = 0; \quad (l+1)b^{2l+1} C_l + lb^{2l+1} C_l = 0; \quad (2l+1)C_l = 0 \Rightarrow C_l = 0.$$

Therefore (1') and (2) $\Rightarrow$ $B_l = C_l = D_l = 0$ for $l > 1$.

$$\text{For } l = 1: \quad C_1 + \frac{2}{b^3} \left[(b^3 - a^3) C_1 + a^3 C_1 \right] = k; \quad C_1 + 2C_1 = k \Rightarrow \boxed{C_1 = k/3\epsilon_0}; \quad D_1 = -a^3 C_1 \Rightarrow$$

$$\boxed{D_1 = -a^3 k/3\epsilon_0}; \quad B_1 = (b^3 - a^3) C_1 \Rightarrow \boxed{B_1 = (b^3 - a^3) k/3\epsilon_0}.$$

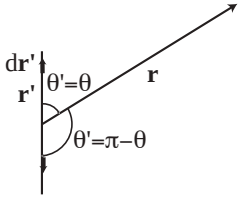
$$\text{For } l = 0: B_0 - D_0 = 0 \Rightarrow B_0 = D_0 \Rightarrow (b-a)C_0 + aV_0 = aV_0 - aC_0, \text{ so } bC_0 = 0 \Rightarrow \boxed{C_0 = 0; D_0 = aV_0 = B_0}.$$

$$\text{Conclusion: } V(r, \theta) = \frac{aV_0}{r} + \frac{(b^3 - a^3)k}{3r^2\epsilon_0} \cos \theta, \quad r \geq b. \quad V(r, \theta) = \frac{aV_0}{r} + \frac{k}{3\epsilon_0} \left(r - \frac{a^3}{r^2} \right) \cos \theta, \quad a \leq r \leq b.$$

$$(b) \sigma_i(\theta) = -\epsilon_0 \left. \frac{\partial V}{\partial r} \right|_a = -\epsilon_0 \left[-\frac{aV_0}{a^2} + \frac{k}{3\epsilon_0} \left(1 + 2\frac{a^3}{a^3} \right) \cos \theta \right] = -\epsilon_0 \left(-\frac{V_0}{a} + \frac{k}{\epsilon_0} \cos \theta \right) = \boxed{-k \cos \theta + V_0 \frac{\epsilon_0}{a}}.$$

$$(c) q_i = \int \sigma_i da = \frac{V_0 \epsilon_0}{a} 4\pi a^2 = \boxed{4\pi a \epsilon_0 V_0 = Q_{\text{tot.}}} \quad \text{At large } r: \quad V \approx \frac{aV_0}{r} \stackrel{?}{=} \frac{1}{4\pi\epsilon_0} \frac{Q}{r} = \frac{1}{4\pi\epsilon_0} \frac{4\pi a \epsilon_0 V_0}{r} = \frac{aV_0}{r}. \quad \checkmark$$

Problem 3.44



Use multipole expansion (Eq. 3.95): $\rho d\tau' \rightarrow \lambda dr'$;

$\lambda = \frac{Q}{2a}$; the r' integral breaks into two pieces:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \left[\int_0^a (r')^n P_n(\cos \theta') \lambda dr' + \int_0^a (r')^n P_n(\cos \theta') \lambda dr' \right].$$

In the first integral $\theta' = \theta$ (see diagram); in the second integral $\theta' = \pi - \theta$, so $\cos \theta' = -\cos \theta$. But $P_n(-z) = (-1)^n P_n(z)$, so the integrals cancel when n is odd, and add when n is even.

$$V(\mathbf{r}) = 2 \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \sum_{n=0,2,4,\dots}^{\infty} \frac{1}{r^{n+1}} P_n(\cos \theta) \int_0^a x^n dx.$$

The integral is $\frac{a^{n+1}}{n+1}$, so

$$V = \frac{Q}{4\pi\epsilon_0} \frac{1}{r} \sum_{n=0,2,4,\dots}^{\infty} \left[\frac{1}{n+1} \left(\frac{a}{r} \right)^n P_n(\cos \theta) \right].$$

Problem 3.45

Use separation of variables in cylindrical coordinates (Prob. 3.24):

$$V(s, \phi) = a_0 + b_0 \ln s + \sum_{k=1}^{\infty} [s^k (a_k \cos k\phi + b_k \sin k\phi) + s^{-k} (c_k \cos k\phi + d_k \sin k\phi)].$$

$$s < R: V(s, \phi) = \sum_{k=1}^{\infty} s^k (a_k \cos k\phi + b_k \sin k\phi) \quad (\ln s \text{ and } s^{-k} \text{ blow up at } s = 0);$$

$$s > R: V(s, \phi) = \sum_{k=1}^{\infty} s^{-k} (c_k \cos k\phi + d_k \sin k\phi) \quad (\ln s \text{ and } s^k \text{ blow up as } s \rightarrow \infty).$$

(We may as well pick constants so $V \rightarrow 0$ as $s \rightarrow \infty$, and hence $a_0 = 0$.) Continuity at $s = R \Rightarrow \sum R^k (a_k \cos k\phi + b_k \sin k\phi) = \sum R^{-k} (c_k \cos k\phi + d_k \sin k\phi)$, so $c_k = R^{2k} a_k$, $d_k = R^{2k} b_k$. Eq. 2.36 says:

$$\left. \frac{\partial V}{\partial s} \right|_{R^+} - \left. \frac{\partial V}{\partial s} \right|_{R^-} = -\frac{1}{\epsilon_0} \sigma. \text{ Therefore}$$

$$\sum \frac{-k}{R^{k+1}} (c_k \cos k\phi + d_k \sin k\phi) - \sum k R^{k-1} (a_k \cos k\phi + b_k \sin k\phi) = -\frac{1}{\epsilon_0} \sigma,$$

or:

$$\sum 2k R^{k-1} (a_k \cos k\phi + b_k \sin k\phi) = \begin{cases} \sigma_0 / \epsilon_0 & (0 < \phi < \pi) \\ -\sigma_0 / \epsilon_0 & (\pi < \phi < 2\pi) \end{cases}.$$

Fourier's trick: multiply by $(\cos l\phi) d\phi$ and integrate from 0 to 2π , using

$$\int_0^{2\pi} \sin k\phi \cos l\phi d\phi = 0; \quad \int_0^{2\pi} \cos k\phi \cos l\phi d\phi = \begin{cases} 0, & k \neq l \\ \pi, & k = l \end{cases}.$$

Then

$$2l R^{l-1} \pi a_l = \frac{\sigma_0}{\epsilon_0} \left[\int_0^{\pi} \cos l\phi d\phi - \int_{\pi}^{2\pi} \cos l\phi d\phi \right] = \frac{\sigma_0}{\epsilon_0} \left\{ \frac{\sin l\phi}{l} \Big|_0^{\pi} - \frac{\sin l\phi}{l} \Big|_{\pi}^{2\pi} \right\} = 0; \quad a_l = 0.$$

Multiply by $(\sin l\phi) d\phi$ and integrate, using $\int_0^{2\pi} \sin k\phi \sin l\phi d\phi = \begin{cases} 0, & k \neq l \\ \pi, & k = l \end{cases}$:

$$\begin{aligned} 2l R^{l-1} \pi b_l &= \frac{\sigma_0}{\epsilon_0} \left[\int_0^{\pi} \sin l\phi d\phi - \int_{\pi}^{2\pi} \sin l\phi d\phi \right] = \frac{\sigma_0}{\epsilon_0} \left\{ -\frac{\cos l\phi}{l} \Big|_0^{\pi} + \frac{\cos l\phi}{l} \Big|_{\pi}^{2\pi} \right\} = \frac{\sigma_0}{l\epsilon_0} (2 - 2 \cos l\pi) \\ &= \begin{cases} 0, & \text{if } l \text{ is even} \\ 4\sigma_0 / l\epsilon_0, & \text{if } l \text{ is odd} \end{cases} \Rightarrow b_l = \begin{cases} 0, & \text{if } l \text{ is even} \\ 2\sigma_0 / \pi\epsilon_0 l^2 R^{l-1}, & \text{if } l \text{ is odd} \end{cases}. \end{aligned}$$

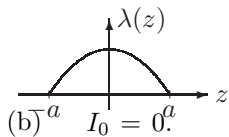
Conclusion:

$$V(s, \phi) = \frac{2\sigma_0 R}{\pi\epsilon_0} \sum_{k=1,3,5,\dots} \frac{1}{k^2} \sin k\phi \begin{cases} (s/R)^k & (s < R) \\ (R/s)^k & (s > R) \end{cases}.$$

Problem 3.46

Use Eq. 3.95, in the form $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{P_n(\cos \theta)}{r^{n+1}} I_n$; $I_n = \int_{-a}^a z^n \lambda(z) dz$.

(a) $I_0 = k \int_{-a}^a \cos\left(\frac{\pi z}{2a}\right) dz = k \left[\frac{2a}{\pi} \sin\left(\frac{\pi z}{2a}\right) \right]_{-a}^a = \frac{2ak}{\pi} \left[\sin\left(\frac{\pi}{2}\right) - \sin\left(-\frac{\pi}{2}\right) \right] = \frac{4ak}{\pi}$. Therefore:

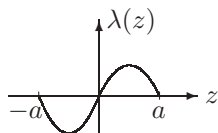


$$V(r, \theta) \cong \frac{1}{4\pi\epsilon_0} \left(\frac{4ak}{\pi} \right) \frac{1}{r}. \quad (\text{Monopole.})$$

(b) $I_0 = 0$.

$$I_1 = k \int_{-a}^a z \sin(\pi z/a) dz = k \left\{ \left(\frac{a}{\pi} \right)^2 \sin\left(\frac{\pi z}{a}\right) - \frac{az}{\pi} \cos\left(\frac{\pi z}{a}\right) \right\} \Big|_{-a}^a$$

$$= k \left\{ \left(\frac{a}{\pi} \right)^2 [\sin(\pi) - \sin(-\pi)] - \frac{a^2}{\pi} \cos(\pi) - \frac{a^2}{\pi} \cos(-\pi) \right\} = k \frac{2a^2}{\pi};$$

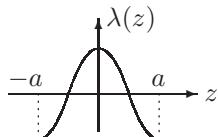


$$V(r, \theta) \cong \frac{1}{4\pi\epsilon_0} \left(\frac{2a^2 k}{\pi} \right) \frac{1}{r^2} \cos \theta. \quad (\text{Dipole.})$$

(c) $I_0 = I_1 = 0$.

$$I_2 = k \int_{-a}^a z^2 \cos\left(\frac{\pi z}{a}\right) dz = k \left\{ \frac{2z \cos(\pi z/a)}{(\pi/a)^2} + \frac{(\pi z/a)^2 - 2}{(\pi/a)^3} \sin\left(\frac{\pi z}{a}\right) \right\} \Big|_{-a}^a$$

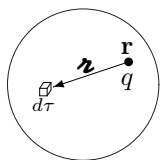
$$= 2k \left(\frac{a}{\pi} \right)^2 [a \cos(\pi) + a \cos(-\pi)] = -\frac{4a^3 k}{\pi^2}.$$



$$V(r, \theta) \cong \frac{1}{4\pi\epsilon_0} \left(-\frac{4a^3 k}{\pi^2} \right) \frac{1}{2r^3} (3 \cos^2 \theta - 1). \quad (\text{Quadrupole.})$$

Problem 3.47

(a) The average field due to a point charge q at $\mathbf{r}$ is



$$\mathbf{E}_{\text{ave}} = \frac{1}{\left(\frac{4}{3}\pi R^3\right)} \int \mathbf{E} d\tau, \quad \text{where } \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \hat{\mathbf{z}},$$

so $\mathbf{E}_{\text{ave}} = \frac{1}{\left(\frac{4}{3}\pi R^3\right)} \frac{1}{4\pi\epsilon_0} \int q \frac{\hat{\mathbf{z}}}{z^2} d\tau.$

(Here $\mathbf{r}$ is the source point, $d\tau$ is the field point, so $\mathbf{z}$ goes from $\mathbf{r}$ to $d\tau$.) The field at $\mathbf{r}$ due to uniform charge ρ over the sphere is $\mathbf{E}_\rho = \frac{1}{4\pi\epsilon_0} \int \rho \frac{\hat{\mathbf{z}}}{z^2} d\tau$. This time $d\tau$ is the source point and $\mathbf{r}$ is the field point, so $\mathbf{z}$ goes from $d\tau$ to $\mathbf{r}$, and hence carries the opposite sign. So with $\rho = -q/\left(\frac{4}{3}\pi R^3\right)$, the two expressions agree: $\mathbf{E}_{\text{ave}} = \mathbf{E}_\rho$.

(b) From Prob. 2.12:

$$\mathbf{E}_\rho = \frac{1}{3\epsilon_0} \rho \mathbf{r} = -\frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{R^3} = -\frac{\mathbf{p}}{4\pi\epsilon_0 R^3}.$$

- (c) If there are many charges inside the sphere, $\mathbf{E}_{\text{ave}}$ is the sum of the individual averages, and $\mathbf{p}_{\text{tot}}$ is the sum of the individual dipole moments. So $\mathbf{E}_{\text{ave}} = -\frac{\mathbf{P}}{4\pi\epsilon_0 R^3}$. qed
- (d) The same argument, only with q placed at $\mathbf{r}$ *outside* the sphere, gives

$$\mathbf{E}_{\text{ave}} = \mathbf{E}_\rho = \frac{1}{4\pi\epsilon_0} \frac{\left(\frac{4}{3}\pi R^3 \rho\right)}{r^2} \hat{\mathbf{r}} \quad (\text{field at } \mathbf{r} \text{ due to uniformly charged sphere}) = \frac{1}{4\pi\epsilon_0} \frac{-q}{r^2} \hat{\mathbf{r}}.$$

But this is precisely the field produced by q (at $\mathbf{r}$) at the *center* of the sphere. So the average field (over the sphere) due to a point charge *outside* the sphere is the same as the field that same charge produces at the center. And by superposition, this holds for any *collection* of exterior charges.

Problem 3.48

(a)

$$\begin{aligned} \mathbf{E}_{\text{dip}} &= \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}) \\ &= \frac{p}{4\pi\epsilon_0 r^3} [2 \cos \theta (\sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}) \\ &\quad + \sin \theta (\cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}})] \\ &= \frac{p}{4\pi\epsilon_0 r^3} \left[3 \sin \theta \cos \theta \cos \phi \hat{\mathbf{x}} + 3 \sin \theta \cos \theta \sin \phi \hat{\mathbf{y}} + \underbrace{(2 \cos^2 \theta - \sin^2 \theta)}_{=3 \cos^2 \theta - 1} \hat{\mathbf{z}} \right]. \\ \mathbf{E}_{\text{ave}} &= \frac{1}{\left(\frac{4}{3}\pi R^3\right)} \int \mathbf{E}_{\text{dip}} d\tau \\ &= \frac{1}{\left(\frac{4}{3}\pi R^3\right)} \left(\frac{p}{4\pi\epsilon_0} \right) \int \frac{1}{r^3} [3 \sin \theta \cos \theta (\cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}) + (3 \cos^2 \theta - 1) \hat{\mathbf{z}}] r^2 \sin \theta dr d\theta d\phi. \end{aligned}$$

But $\int_0^{2\pi} \cos \phi d\phi = \int_0^{2\pi} \sin \phi d\phi = 0$, so the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ terms drop out, and $\int_0^{2\pi} d\phi = 2\pi$, so

$$\mathbf{E}_{\text{ave}} = \frac{1}{\left(\frac{4}{3}\pi R^3\right)} \left(\frac{p}{4\pi\epsilon_0} \right) 2\pi \int_0^R \frac{1}{r} dr \underbrace{\int_0^\pi (3 \cos^2 \theta - 1) \sin \theta d\theta}_{(-\cos^3 \theta + \cos \theta)|_0^\pi = 1 - 1 + 1 - 1 = 0}.$$

Evidently $\boxed{\mathbf{E}_{\text{ave}} = \mathbf{0}}$, which contradicts the result of Prob. 3.47. [Note, however, that the r integral, $\int_0^R \frac{1}{r} dr$, blows up, since $\ln r \rightarrow -\infty$ as $r \rightarrow 0$. If, as suggested, we truncate the r integral at $r = \epsilon$, then it is finite, and the θ integral gives $\mathbf{E}_{\text{ave}} = \mathbf{0}$.]

(b) We want $\mathbf{E}$ within the ϵ -sphere to be a delta function: $\mathbf{E} = \mathbf{A} \delta^3(\mathbf{r})$, with $\mathbf{A}$ selected so that the *average* field is consistent with the general theorem in Prob. 3.47:

$$\mathbf{E}_{\text{ave}} = \frac{1}{\left(\frac{4}{3}\pi R^3\right)} \int \mathbf{A} \delta^3(\mathbf{r}) d\tau = \frac{\mathbf{A}}{\left(\frac{4}{3}\pi R^3\right)} = -\frac{\mathbf{P}}{4\pi\epsilon_0 R^3} \Rightarrow \mathbf{A} = -\frac{\mathbf{P}}{3\epsilon_0}, \text{ and hence } \boxed{\mathbf{E} = -\frac{\mathbf{P}}{3\epsilon_0} \delta^3(\mathbf{r})}.$$

Problem 3.49 We need to show that the field inside the sphere approaches a delta-function with the right coefficient (Eq. 3.106) in the limit as $R \rightarrow 0$. From Eq. 3.86, the potential inside is

$$V = \frac{k}{3\epsilon_0} r \cos \theta = \frac{k}{3\epsilon_0} z, \quad \text{so} \quad \mathbf{E} = -\nabla V = -\frac{k}{3\epsilon_0} \hat{\mathbf{z}}.$$

From Prob. 3.30, the dipole moment of this configuration is $\mathbf{p} = (4\pi R^3 k/3) \hat{\mathbf{z}}$, so $k \hat{\mathbf{z}} = 3\mathbf{p}/(4\pi R^3)$, and hence the field inside is

$$\mathbf{E} = -\frac{1}{4\pi\epsilon_0 R^3} \mathbf{p}.$$

Clearly $E \rightarrow \infty$ as $R \rightarrow 0$ (if $\mathbf{p}$ is held constant); its volume integral is

$$\int \mathbf{E} d\tau = -\frac{1}{4\pi\epsilon_0 R^3} \mathbf{p} \frac{4}{3} \pi R^3 = -\frac{1}{3\epsilon_0} \mathbf{p},$$

which matches the delta-function term in Eq. 3.106. ✓

Problem 3.50

(a) $I = \int (\nabla V_1) \cdot (\nabla V_2) d\tau$. But $\nabla \cdot (V_1 \nabla V_2) = (\nabla V_1) \cdot (\nabla V_2) + V_1 (\nabla^2 V_2)$, so

$$I = \int \nabla \cdot (V_1 \nabla V_2) d\tau - \int V_1 (\nabla^2 V_2) = \oint_S V_1 (\nabla V_2) \cdot d\mathbf{a} + \frac{1}{\epsilon_0} \int V_1 \rho_2 d\tau.$$

But the surface integral is over a huge sphere “at infinity”, where V_1 and $V_2 \rightarrow 0$. So $I = \frac{1}{\epsilon_0} \int V_1 \rho_2 d\tau$. By the same argument, with 1 and 2 reversed, $I = \frac{1}{\epsilon_0} \int V_2 \rho_1 d\tau$. So $\int V_1 \rho_2 d\tau = \int V_2 \rho_1 d\tau$. qed

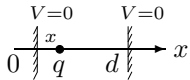
$$(b) \left\{ \begin{array}{l} \text{Situation (1): } Q_a = \int_a \rho_1 d\tau = Q; \quad Q_b = \int_b \rho_1 d\tau = 0; \quad V_{1b} \equiv V_{ab}. \\ \text{Situation (2): } Q_a = \int_a \rho_2 d\tau = 0; \quad Q_b = \int_b \rho_2 d\tau = Q; \quad V_{2a} \equiv V_{ba}. \end{array} \right\}$$

$$\left\{ \begin{array}{l} \int V_1 \rho_2 d\tau = V_{1a} \int_a \rho_2 d\tau + V_{1b} \int_b \rho_2 d\tau = V_{ab} Q. \\ \int V_2 \rho_1 d\tau = V_{2a} \int_a \rho_1 d\tau + V_{2b} \int_b \rho_1 d\tau = V_{ba} Q. \end{array} \right\}$$

Green’s reciprocity theorem says $QV_{ab} = QV_{ba}$, so $V_{ab} = V_{ba}$. qed

Problem 3.51

(a) *Situation (1)*: actual. *Situation (2)*: right plate at V_0 , left plate at $V = 0$, no charge at x .



$$\int V_1 \rho_2 d\tau = V_{l_1} Q_{l_2} + V_{x_1} Q_{x_2} + V_{r_1} Q_{r_2}.$$

But $V_{l_1} = V_{r_1} = 0$ and $Q_{x_2} = 0$, so $\int V_1 \rho_2 d\tau = 0$.

$$\int V_2 \rho_1 d\tau = V_{l_2} Q_{l_1} + V_{x_2} Q_{x_1} + V_{r_2} Q_{r_1}.$$

But $V_{l_2} = 0$, $Q_{x_1} = q$, $V_{r_2} = V_0$, $Q_{r_1} = Q_2$, and $V_{x_2} = V_0(x/d)$. So $0 = V_0(x/d)q + V_0 Q_2$, and hence

$$Q_2 = -qx/d.$$

Situation (1): actual. *Situation (2):* left plate at V_0 , right plate at $V = 0$, no charge at x .

$$\int V_1 \rho_2 d\tau = 0 = \int V_2 \rho_1 d\tau = V_{l_2} Q_{l_1} + V_{x_2} Q_{x_1} + V_{r_2} Q_{r_1} = V_0 Q_1 + q V_{x_2} + 0.$$

But $V_{x_2} = V_0 \left(1 - \frac{x}{d}\right)$, so

$$Q_1 = -q(1 - x/d).$$

(b) *Situation (1):* actual. *Situation (2):* inner sphere at V_0 , outer sphere at zero, no charge at r .

$$\int V_1 \rho_2 d\tau = V_{a_1} Q_{a_2} + V_{r_1} Q_{r_2} + V_{b_1} Q_{b_2}.$$

But $V_{a_1} = V_{b_1} = 0$, $Q_{r_2} = 0$. So $\int V_1 \rho_2 d\tau = 0$.

$$\int V_2 \rho_1 d\tau = V_{a_2} Q_{a_1} + V_{r_2} Q_{r_1} + V_{b_2} Q_{b_1} = Q_a V_0 + q V_{r_2} + 0.$$

But V_{r_2} is the potential at r in configuration 2: $V(r) = A + B/r$, with $V(a) = V_0 \Rightarrow A + B/a = V_0$, or $aA + B = aV_0$, and $V(b) = 0 \Rightarrow A + B/b = 0$, or $bA + B = 0$. Subtract: $(b-a)A = -aV_0 \Rightarrow A = -aV_0/(b-a)$; $B \left(\frac{1}{a} - \frac{1}{b}\right) = V_0 = B \frac{(b-a)}{ab} \Rightarrow B = abV_0/(b-a)$. So $V(r) = \frac{aV_0}{(b-a)} \left(\frac{b}{r} - 1\right)$. Therefore

$$Q_a V_0 + q \frac{aV_0}{(b-a)} \left(\frac{b}{r} - 1\right) = 0; \quad Q_a = -\frac{qa}{(b-a)} \left(\frac{b}{r} - 1\right).$$

Now let *Situation (2)* be: inner sphere at zero, outer at V_0 , no charge at r .

$$\int V_1 \rho_2 d\tau = 0 = \int V_2 \rho_1 d\tau = V_{a_2} Q_{a_1} + V_{r_2} Q_{r_1} + V_{b_2} Q_{b_1} = 0 + q V_{r_2} + Q_b V_0.$$

This time $V(r) = A + \frac{B}{r}$ with $V(a) = 0 \Rightarrow A + B/a = 0$; $V(b) = V_0 \Rightarrow A + B/b = V_0$, so

$$V(r) = \frac{bV_0}{(b-a)} \left(1 - \frac{a}{r}\right). \text{ Therefore } q \frac{bV_0}{(b-a)} \left(1 - \frac{a}{r}\right) + Q_b V_0 = 0; \quad Q_b = -\frac{qb}{(b-a)} \left(1 - \frac{a}{r}\right).$$

Problem 3.52

$$(a) \quad \sum_{i,j=1}^3 \hat{\mathbf{r}}_i \hat{\mathbf{r}}_j Q_{ij} = \frac{1}{2} \int \left\{ 3 \sum_{i=1}^3 \hat{\mathbf{r}}_i r'_i \sum_{j=1}^3 \hat{\mathbf{r}}_j r'_j - (r')^2 \sum_{i,j} \hat{\mathbf{r}}_i \hat{\mathbf{r}}_j \delta_{ij} \right\} \rho d\tau'$$

$$\text{But } \sum_{i=1}^3 \hat{\mathbf{r}}_i r'_i = \hat{\mathbf{r}} \cdot \mathbf{r}' = r' \cos \alpha = \sum_{j=1}^3 \hat{\mathbf{r}}_j r'_j; \quad \sum_{i,j} \hat{\mathbf{r}}_i \hat{\mathbf{r}}_j \delta_{ij} = \sum \hat{\mathbf{r}}_j \hat{\mathbf{r}}_j = \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = 1. \quad \text{So}$$

$$V_{\text{quad}} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \int \frac{1}{2} \left(3r'^2 \cos^2 \alpha - r'^2 \right) \rho d\tau' = \text{the third term in Eq. 3.96.} \quad \checkmark$$

(b) Because $x^2 = y^2 = (a/2)^2$ for all four charges, $Q_{xx} = Q_{yy} = \frac{1}{2} [3(a/2)^2 - (\sqrt{2}a/2)^2] (q - q - q + q) = 0$. Because $z = 0$ for all four charges, $Q_{zz} = \frac{1}{2} [-(\sqrt{2}a/2)^2] (q - q - q + q) = 0$ and $Q_{xz} = Q_{yz} = Q_{zx} = Q_{zy} = 0$. This leaves only

$$Q_{xy} = Q_{yx} = \frac{3}{2} \left[\left(\frac{a}{2}\right) \left(\frac{a}{2}\right) q + \left(\frac{a}{2}\right) \left(-\frac{a}{2}\right) (-q) + \left(-\frac{a}{2}\right) \left(\frac{a}{2}\right) (-q) + \left(-\frac{a}{2}\right) \left(-\frac{a}{2}\right) q \right] = \boxed{\frac{3}{2} a^2 q}.$$

(c)

$$\begin{aligned}
2\overline{Q}_{ij} &= \int [3(r_i - d_i)(r_j - d_j) - (\mathbf{r} - \mathbf{d})^2 \delta_{ij}] \rho d\tau \quad (\text{I'll drop the primes, for simplicity.}) \\
&= \int [3r_i r_j - r^2 \delta_{ij}] \rho d\tau - 3d_i \int r_j \rho d\tau - 3d_j \int r_i \rho d\tau + 3d_i d_j \int \rho d\tau + 2\mathbf{d} \cdot \int \mathbf{r} \rho d\tau \delta_{ij} \\
&\quad - d^2 \delta_{ij} \int \rho d\tau = Q_{ij} - 3(d_i p_j + d_j p_i) + 3d_i d_j Q + 2\delta_{ij} \mathbf{d} \cdot \mathbf{p} - d^2 \delta_{ij} Q.
\end{aligned}$$

So if $\mathbf{p} = 0$ and $Q = 0$ then $\overline{Q}_{ij} = Q_{ij}$. qed

(d) Eq. 3.95 with $n = 3$:

$$V_{\text{oct}} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^4} \int (r')^3 P_3(\cos \alpha) \rho d\tau'; \quad P_3(\cos \theta) = \frac{1}{2} (5 \cos^3 \theta - 3 \cos \theta).$$

$$V_{\text{oct}} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^4} \sum_{i,j,k} \hat{\mathbf{r}}_i \hat{\mathbf{r}}_j \hat{\mathbf{r}}_k Q_{ijk}.$$

Define the “octopole moment” as

$$Q_{ijk} \equiv \frac{1}{2} \int [5r'_i r'_j r'_k - (r')^2 (r'_i \delta_{jk} + r'_j \delta_{ik} + r'_k \delta_{ij})] \rho(\mathbf{r}') d\tau'.$$

Problem 3.53

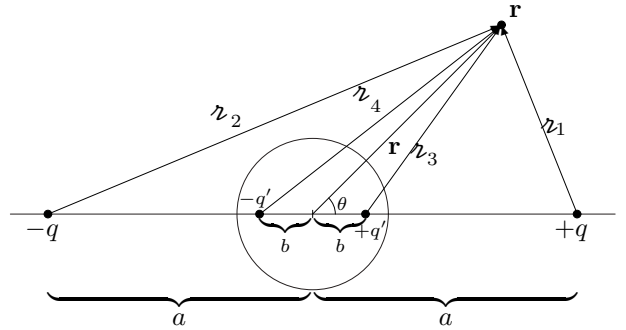
$$V = \frac{1}{4\pi\epsilon_0} \left\{ q \left(\frac{1}{z_1} - \frac{1}{z_2} \right) + q' \left(\frac{1}{z_3} - \frac{1}{z_4} \right) \right\}$$

$$z_1 = \sqrt{r^2 + a^2 - 2ra \cos \theta},$$

$$z_2 = \sqrt{r^2 + a^2 + 2ra \cos \theta},$$

$$z_3 = \sqrt{r^2 + b^2 - 2rb \cos \theta},$$

$$z_4 = \sqrt{r^2 + b^2 + 2rb \cos \theta}.$$



Expanding as in Ex. 3.10: $\left(\frac{1}{z_1} - \frac{1}{z_2} \right) \cong \frac{2r}{a^2} \cos \theta$ (we want $a \gg r$, not $r \gg a$, this time).

$$\begin{aligned}
\left(\frac{1}{z_3} - \frac{1}{z_4} \right) &\cong \frac{2b}{r^2} \cos \theta \quad (\text{here we want } b \ll r, \text{ because } b = R^2/a, \text{ Eq. 3.16}) \\
&= \frac{2}{a} \frac{R^2}{r^2} \cos \theta.
\end{aligned}$$

But $q' = -\frac{R}{a}q$ (Eq. 3.15), so

$$V(r, \theta) \cong \frac{1}{4\pi\epsilon_0} \left[q \frac{2r}{a^2} \cos \theta - \frac{R}{a} q \frac{2}{a} \frac{R^2}{r^2} \cos \theta \right] = \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{a^2} \right) \left(r - \frac{R^3}{r^2} \right) \cos \theta.$$

Set $E_0 = -\frac{1}{4\pi\epsilon_0} \frac{2q}{a^2}$ (field in the vicinity of the sphere produced by $\pm q$):

$$\boxed{V(r, \theta) = -E_0 \left(r - \frac{R^3}{r^2} \right) \cos \theta} \quad (\text{agrees with Eq. 3.76}).$$

Problem 3.54

The boundary conditions are

$$\left. \begin{array}{l} \text{(i)} \quad V = 0 \text{ when } y = 0, \\ \text{(ii)} \quad V = V_0 \text{ when } y = a, \\ \text{(iii)} \quad V = 0 \text{ when } x = b, \\ \text{(iv)} \quad V = 0 \text{ when } x = -b. \end{array} \right\}$$

Go back to Eq. 3.26 and examine the case $k = 0$: $d^2X/dx^2 = d^2Y/dy^2 = 0$, so $X(x) = Ax + B$, $Y(y) = Cy + D$. But this configuration is symmetric in x , so $A = 0$, and hence the $k = 0$ solution is $V(x, y) = Cy + D$. Pick $D = 0$, $C = V_0/a$, and subtract off this part:

$$V(x, y) = V_0 \frac{y}{a} + \bar{V}(x, y).$$

The *remainder* ($\bar{V}(x, y)$) satisfies boundary conditions similar to Ex. 3.4:

$$\left. \begin{array}{l} \text{(i)} \quad \bar{V} = 0 \text{ when } y = 0, \\ \text{(ii)} \quad \bar{V} = 0 \text{ when } y = a, \\ \text{(iii)} \quad \bar{V} = -V_0(y/a) \text{ when } x = b, \\ \text{(iv)} \quad \bar{V} = -V_0(y/a) \text{ when } x = -b. \end{array} \right\}$$

(The point of peeling off $V_0(y/a)$ was to recover (ii), on which the constraint $k = n\pi/a$ depends.)

The solution (following Ex. 3.4) is

$$\bar{V}(x, y) = \sum_{n=1}^{\infty} C_n \cosh(n\pi x/a) \sin(n\pi y/a),$$

and it remains to fit condition (iii):

$$\bar{V}(b, y) = \sum C_n \cosh(n\pi b/a) \sin(n\pi y/a) = -V_0(y/a).$$

Invoke Fourier's trick:

$$\sum C_n \cosh(n\pi b/a) \int_0^a \sin(n\pi y/a) \sin(n'\pi y/a) dy = -\frac{V_0}{a} \int_0^a y \sin(n'\pi y/a) dy,$$

$$\frac{a}{2} C_n \cosh(n\pi b/a) = -\frac{V_0}{a} \int_0^a y \sin(n\pi y/a) dy.$$

$$\begin{aligned} C_n &= -\frac{2V_0}{a^2 \cosh(n\pi b/a)} \left[\left(\frac{a}{n\pi} \right)^2 \sin(n\pi y/a) - \left(\frac{ay}{n\pi} \right) \cos(n\pi y/a) \right] \Big|_0^a \\ &= \frac{2V_0}{a^2 \cosh(n\pi b/a)} \left(\frac{a^2}{n\pi} \right) \cos(n\pi) = \frac{2V_0}{n\pi} \frac{(-1)^n}{\cosh(n\pi b/a)}. \end{aligned}$$

$$V(x, y) = V_0 \left[\frac{y}{a} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \frac{\cosh(n\pi x/a)}{\cosh(n\pi b/a)} \sin(n\pi y/a) \right].$$

Alternatively, start with the separable solution

$$V(x, y) = (C \sin kx + D \cos kx) (Ae^{ky} + Be^{-ky}).$$

Note that the configuration is symmetric in x , so $C = 0$, and $V(x, 0) = 0 \Rightarrow B = -A$, so (combining the constants)

$$V(x, y) = A \cos kx \sinh ky.$$

But $V(b, y) = 0$, so $\cos kb = 0$, which means that $kb = \pm\pi/2, \pm3\pi/2, \dots$, or $k = (2n-1)\pi/2b \equiv \alpha_n$, with $n = 1, 2, 3, \dots$ (negative k does not yield a different solution—the sign can be absorbed into A). The general linear combination is

$$V(x, y) = \sum_{n=1}^{\infty} A_n \cos \alpha_n x \sinh \alpha_n y,$$

and it remains to fit the final boundary condition:

$$V(x, a) = V_0 = \sum_{n=1}^{\infty} A_n \cos \alpha_n x \sinh \alpha_n a.$$

Use Fourier's trick, multiplying by $\cos \alpha_{n'} x$ and integrating:

$$V_0 \int_{-b}^b \cos \alpha_{n'} x dx = \sum_{n=1}^{\infty} A_n \sinh \alpha_n a \int_{-b}^b \cos \alpha_{n'} x \cos \alpha_n x dx,$$

$$V_0 \frac{2 \sin \alpha_{n'} b}{\alpha_{n'}} = \sum_{n=1}^{\infty} A_n \sinh \alpha_n a (b \delta_{n' n}) = b A_{n'} \sinh \alpha_{n'} a.$$

So $A_n = \frac{2V_0}{b} \frac{\sin \alpha_n b}{\alpha_n \sinh \alpha_n a}$. But $\sin \alpha_n b = \sin \left(\frac{2n-1}{2} \pi \right) = -(-1)^n$, so

$$V(x, y) = -\frac{2V_0}{b} \sum_{n=1}^{\infty} (-1)^n \frac{\sinh \alpha_n y}{\alpha_n \sinh \alpha_n a} \cos \alpha_n x.$$

Problem 3.55

(a) Using Prob. 3.15b (with $b = a$):

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n \text{ odd}} \frac{\sinh(n\pi x/a) \sin(n\pi y/a)}{n \sinh(n\pi)}.$$

$$\begin{aligned}
\sigma(y) &= -\epsilon_0 \frac{\partial V}{\partial x} \Big|_{x=0} = -\epsilon_0 \frac{4V_0}{\pi} \sum_{n \text{ odd}} \left(\frac{n\pi}{a} \right) \frac{\cosh(n\pi x/a) \sin(n\pi y/a)}{n \sinh(n\pi)} \Big|_{x=0} \\
&= -\frac{4\epsilon_0 V_0}{a} \sum_{n \text{ odd}} \frac{\sin(n\pi y/a)}{\sinh(n\pi)}. \\
\lambda &= \int_0^a \sigma(y) dy = -\frac{4\epsilon_0 V_0}{a} \sum_{n \text{ odd}} \frac{1}{\sinh(n\pi)} \int_0^a \sin(n\pi y/a) dy. \\
\text{But } \int_0^a \sin(n\pi y/a) dy &= -\frac{a}{n\pi} \cos(n\pi y/a) \Big|_0^a = \frac{a}{n\pi} [1 - \cos(n\pi)] = \frac{2a}{n\pi} \text{ (since } n \text{ is odd).} \\
&= -\frac{8\epsilon_0 V_0}{\pi} \sum_{n \text{ odd}} \frac{1}{n \sinh(n\pi)} = \boxed{-\frac{\epsilon_0 V_0}{\pi} \ln 2.}
\end{aligned}$$

[Summing the series numerically (using Mathematica) gives 0.0866434, which agrees precisely with $\ln 2/8$. The series can be summed analytically, by manipulation of elliptic integrals—see “Integrals and Series, Vol. I: Elementary Functions,” by A. P. Prudnikov, Yu. A. Brychkov, and O. I. Marichev (Gordon and Breach, New York, 1986), p. 721. I thank Ram Valluri for calling this to my attention.]

Using Prob. 3.54 (with $b = a/2$):

$$\begin{aligned}
V(x, y) &= V_0 \left[\frac{y}{a} + \frac{2}{\pi} \sum_n \frac{(-1)^n \cosh(n\pi x/a) \sin(n\pi y/a)}{n \cosh(n\pi/2)} \right]. \\
\sigma(x) &= -\epsilon_0 \frac{\partial V}{\partial y} \Big|_{y=0} = -\epsilon_0 V_0 \left[\frac{1}{a} + \frac{2}{\pi} \sum_n \left(\frac{n\pi}{a} \right) \frac{(-1)^n \cosh(n\pi x/a) \cos(n\pi y/a)}{n \cosh(n\pi/2)} \right] \Big|_{y=0} \\
&= -\epsilon_0 V_0 \left[\frac{1}{a} + \frac{2}{a} \sum_n \frac{(-1)^n \cosh(n\pi x/a)}{\cosh(n\pi/2)} \right] = -\frac{\epsilon_0 V_0}{a} \left[1 + 2 \sum_n \frac{(-1)^n \cosh(n\pi x/a)}{\cosh(n\pi/2)} \right]. \\
\lambda &= \int_{-a/2}^{a/2} \sigma(x) dx = -\frac{\epsilon_0 V_0}{a} \left[a + 2 \sum_n \frac{(-1)^n}{\cosh(n\pi/2)} \int_{-a/2}^{a/2} \cosh(n\pi x/a) dx \right]. \\
\text{But } \int_{-a/2}^{a/2} \cosh(n\pi x/a) dx &= \frac{a}{n\pi} \sinh(n\pi x/a) \Big|_{-a/2}^{a/2} = \frac{2a}{n\pi} \sinh(n\pi/2). \\
&= -\frac{\epsilon_0 V_0}{a} \left[a + \frac{4a}{\pi} \sum_n \frac{(-1)^n \tanh(n\pi/2)}{n} \right] = -\epsilon_0 V_0 \left[1 + \frac{4}{\pi} \sum_n \frac{(-1)^n \tanh(n\pi/2)}{n} \right] \\
&= \boxed{-\frac{\epsilon_0 V_0}{\pi} \ln 2.}
\end{aligned}$$

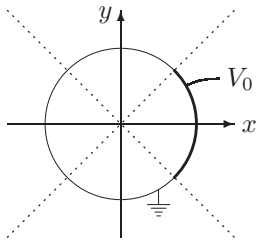
[The numerical value is -0.612111, which agrees with the expected value $(\ln 2 - \pi)/4$.]

(b) From Prob. 3.24:

$$V(s, \phi) = a_0 + b_0 \ln s + \sum_{k=1}^{\infty} \left(a_k s^k + b_k \frac{1}{s^k} \right) [c_k \cos(k\phi) + d_k \sin(k\phi)].$$

In the interior ($s < R$) b_0 and b_k must be zero ($\ln s$ and $1/s$ blow up at the origin). Symmetry $\Rightarrow d_k = 0$. So

$$V(s, \phi) = a_0 + \sum_{k=1}^{\infty} a_k s^k \cos(k\phi).$$



At the surface:

$$V(R, \phi) = \sum_{k=0} a_k R^k \cos(k\phi) = \begin{cases} V_0, & \text{if } -\pi/4 < \phi < \pi/4, \\ 0, & \text{otherwise.} \end{cases}$$

Fourier's trick: multiply by $\cos(k'\phi)$ and integrate from $-\pi$ to π :

$$\sum_{k=0}^{\infty} a_k R^k \int_{-\pi}^{\pi} \cos(k\phi) \cos(k'\phi) d\phi = V_0 \int_{-\pi/4}^{\pi/4} \cos(k'\phi) d\phi = \begin{cases} V_0 \sin(k'\phi)/k' \Big|_{-\pi/4}^{\pi/4} = (V_0/k') \sin(k'\pi/4), & \text{if } k' \neq 0, \\ V_0\pi/2, & \text{if } k' = 0. \end{cases}$$

But

$$\int_{-\pi}^{\pi} \cos(k\phi) \cos(k'\phi) d\phi = \begin{cases} 0, & \text{if } k \neq k' \\ 2\pi, & \text{if } k = k' = 0, \\ \pi, & \text{if } k = k' \neq 0. \end{cases}$$

So $2\pi a_0 = V_0\pi/2 \Rightarrow a_0 = V_0/4$; $\pi a_k R^k = (2V_0/k) \sin(k\pi/4) \Rightarrow a_k = (2V_0/\pi k R^k) \sin(k\pi/4)$ ($k \neq 0$); hence

$$V(s, \phi) = V_0 \left[\frac{1}{4} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{\sin(k\pi/4)}{k} \left(\frac{s}{R} \right)^k \cos(k\phi) \right].$$

Using Eq. 2.49, and noting that in this case $\hat{\mathbf{n}} = -\hat{\mathbf{s}}$:

$$\sigma(\phi) = \epsilon_0 \frac{\partial V}{\partial s} \Big|_{s=R} = \epsilon_0 V_0 \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{\sin(k\pi/4)}{k R^k} k s^{k-1} \cos(k\phi) \Big|_{s=R} = \frac{2\epsilon_0 V_0}{\pi R} \sum_{k=1}^{\infty} \sin(k\pi/4) \cos(k\phi).$$

We want the net (line) charge on the segment opposite to V_0 ($-\pi < \phi < -3\pi/4$ and $3\pi/4 < \phi < \pi$):

$$\begin{aligned} \lambda &= \int \sigma(\phi) R d\phi = 2R \int_{3\pi/4}^{\pi} \sigma(\phi) d\phi = \frac{4\epsilon_0 V_0}{\pi} \sum_{k=1}^{\infty} \sin(k\pi/4) \int_{3\pi/4}^{\pi} \cos(k\phi) d\phi \\ &= \frac{4\epsilon_0 V_0}{\pi} \sum_{k=1}^{\infty} \sin(k\pi/4) \left[\frac{\sin(k\phi)}{k} \Big|_{3\pi/4}^{\pi} \right] = -\frac{4\epsilon_0 V_0}{\pi} \sum_{k=1}^{\infty} \frac{\sin(k\pi/4) \sin(3k\pi/4)}{k}. \end{aligned}$$

k	$\frac{\sin(k\pi/4)}{k}$	$\frac{\sin(3k\pi/4)}{k}$	product
1	$1/\sqrt{2}$	$1/\sqrt{2}$	$1/2$
2	1	-1	-1
3	$1/\sqrt{2}$	$1/\sqrt{2}$	$1/2$
4	0	0	0
5	$-1/\sqrt{2}$	$-1/\sqrt{2}$	$1/2$
6	-1	1	-1
7	$-1/\sqrt{2}$	$-1/\sqrt{2}$	$1/2$
8	0	0	0

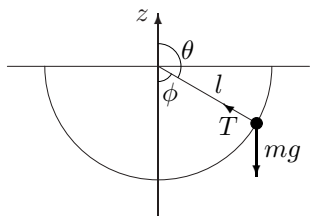
$$\lambda = -\frac{4\epsilon_0 V_0}{\pi} \left[\frac{1}{2} \sum_{1,3,5,\dots} \frac{1}{k} - \sum_{2,6,10,\dots} \frac{1}{k} \right] = -\frac{4\epsilon_0 V_0}{\pi} \left[\frac{1}{2} \sum_{1,3,5,\dots} \frac{1}{k} - \frac{1}{2} \sum_{1,3,5,\dots} \frac{1}{k} \right] = 0.$$

Ouch! What went wrong? The problem is that the series $\sum(1/k)$ is divergent, so the “subtraction” $\infty - \infty$ is suspect. One way to avoid this is to go back to $V(s, \phi)$, calculate $\epsilon_0(\partial V/\partial s)$ at $s \neq R$, and save the limit $s \rightarrow R$ until the end:

$$\begin{aligned} \sigma(\phi, s) &\equiv \epsilon_0 \frac{\partial V}{\partial s} = \frac{2\epsilon_0 V_0}{\pi} \sum_{k=1}^{\infty} \frac{\sin(k\pi/4)}{k} \frac{k s^{k-1}}{R^k} \cos(k\phi) \\ &= \frac{2\epsilon_0 V_0}{\pi R} \sum_{k=1}^{\infty} x^{k-1} \sin(k\pi/4) \cos(k\phi) \quad (\text{where } x \equiv s/R \rightarrow 1 \text{ at the end}). \end{aligned}$$

$$\begin{aligned} \lambda(x) &\equiv \sigma(\phi, s) R d\phi = -\frac{4\epsilon_0 V_0}{\pi} \sum_{k=1}^{\infty} \frac{1}{k} x^{k-1} \sin(k\pi/4) \sin(3k\pi/4) \\ &= -\frac{4\epsilon_0 V_0}{\pi} \left[\frac{1}{2x} \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right) - \frac{1}{x} \left(\frac{x^2}{2} + \frac{x^6}{6} + \frac{x^{10}}{10} + \dots \right) \right] \\ &= -\frac{2\epsilon_0 V_0}{\pi x} \left[\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right) - \left(x^2 + \frac{x^6}{3} + \frac{x^{10}}{5} + \dots \right) \right]. \\ \text{But (see math tables): } \ln \left(\frac{1+x}{1-x} \right) &= 2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right). \\ &= -\frac{2\epsilon_0 V_0}{\pi x} \left[\frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) - \frac{1}{2} \ln \left(\frac{1+x^2}{1-x^2} \right) \right] = -\frac{\epsilon_0 V_0}{\pi x} \ln \left[\left(\frac{1+x}{1-x} \right) \left(\frac{1+x^2}{1-x^2} \right) \right] \\ &= -\frac{\epsilon_0 V_0}{\pi x} \ln \left[\frac{(1+x)^2}{1+x^2} \right]; \quad \lambda = \lim_{x \rightarrow 1} \lambda(x) = \boxed{\frac{-\epsilon_0 V_0}{\pi} \ln 2}. \end{aligned}$$

Problem 3.56



$$\mathbf{F} = q\mathbf{E} = \frac{qp}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}).$$

Now consider the pendulum: $\mathbf{F} = -mg\hat{\mathbf{z}} - T\hat{\mathbf{r}}$, where $T - mg \cos \phi = mv^2/l$ and (by conservation of energy) $mg l \cos \phi = (1/2)mv^2 \Rightarrow v^2 = 2gl \cos \phi$ (assuming it started from rest at $\phi = 90^\circ$, as stipulated). But $\cos \phi = -\cos \theta$, so $T = mg(-\cos \theta) + (m/l)(-2gl \cos \theta) = -3mg \cos \theta$, and hence

$$\mathbf{F} = -mg(\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}) + 3mg \cos \theta \hat{\mathbf{r}} = mg(2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}).$$

This total force is such as to keep the pendulum on a circular arc, and it is identical to the force on q in the field of a dipole, with $mg \leftrightarrow qp/4\pi\epsilon_0 l^3$. Evidently q also executes semicircular motion, as though it were on a tether of fixed length l .

Problem 3.57 Symmetry suggests that the plane of the orbit is perpendicular to the z axis, and since we need a centripetal force, pointing in toward the axis, the orbit must lie at the bottom of the field loops (Fig. 3.37a), where the z component of the field is zero. Referring to Eq. 3.104,

$\mathbf{E} \cdot \hat{\mathbf{z}} = 0 \Rightarrow 3(\mathbf{p} \cdot \hat{\mathbf{r}})(\hat{\mathbf{r}} \cdot \hat{\mathbf{z}}) - \mathbf{p} \cdot \hat{\mathbf{z}} = 0$, or $3 \cos^2 \theta - 1 = 0$. So $\cos^2 \theta = 1/3$, $\cos \theta = -1/\sqrt{3}$, $\sin \theta = \sqrt{2/3}$, $z/s = \tan \theta \Rightarrow \boxed{z = -\sqrt{2} s}$. The field at the orbit is (Eq. 3.103)

$$\begin{aligned} \mathbf{E} &= \frac{p}{4\pi\epsilon_0 r^3} \left(-2\frac{1}{\sqrt{3}} \hat{\mathbf{r}} + \sqrt{\frac{2}{3}} \hat{\boldsymbol{\theta}} \right) \\ &= \frac{p}{4\pi\epsilon_0 r^3} \sqrt{\frac{2}{3}} \left[-\sqrt{2}(\sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}) + (\cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}}) \right] \\ &= \frac{p}{4\pi\epsilon_0 r^3} \sqrt{\frac{2}{3}} \left[\left(-\sqrt{2} \sin \theta + \cos \theta \right) \cos \phi \hat{\mathbf{x}} + \left(-\sqrt{2} \sin \theta + \cos \theta \right) \sin \phi \hat{\mathbf{y}} + \left(-\sqrt{2} \cos \theta - \sin \theta \right) \hat{\mathbf{z}} \right] \\ &= \frac{p}{4\pi\epsilon_0 r^3} \sqrt{\frac{2}{3}} \left[\left(-\sqrt{2} \sqrt{\frac{2}{3}} - \frac{1}{\sqrt{3}} \right) (\cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}) + \left(\sqrt{2} \frac{1}{\sqrt{3}} - \sqrt{\frac{2}{3}} \right) \hat{\mathbf{z}} \right] \\ &= \frac{p}{4\pi\epsilon_0 r^3} \sqrt{\frac{2}{3}} \left[-\sqrt{3} (\cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}) \right] = -\frac{p}{4\pi\epsilon_0 r^3} \sqrt{2} \hat{\mathbf{s}} = -\frac{p}{3\sqrt{3}\pi\epsilon_0 s^3} \hat{\mathbf{s}}. \end{aligned}$$

(I used $s = r \sin \theta = r \sqrt{2/3}$, in the last step.)

The centripetal force is

$$F = qE = -\frac{qp}{3\sqrt{3}\pi\epsilon_0 s^3} = -\frac{mv^2}{s} \Rightarrow v^2 = \frac{qp}{3\sqrt{3}\pi\epsilon_0 m s^2} \Rightarrow v = \boxed{\frac{1}{s} \sqrt{\frac{qp}{3\sqrt{3}\pi\epsilon_0 m}}}.$$

The angular momentum is

$$L = smv = \boxed{\sqrt{\frac{qpm}{3\sqrt{3}\pi\epsilon_0}}},$$

the same for all orbits, regardless of their radius (!), and the energy is

$$W = \frac{1}{2}mv^2 + qV = \frac{1}{2} \frac{qp}{3\sqrt{3}\pi\epsilon_0 s^2} + \frac{q}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} = \frac{qp}{6\sqrt{3}\pi\epsilon_0 s^2} - \frac{qp}{4\pi\epsilon_0 \sqrt{3}(3/2)s^2} = \boxed{0}.$$

Problem 3.58

Potential of q : $V_q(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$, where $r^2 = a^2 + r^2 - 2ar \cos \theta$.

Equation 3.94, with $r' \rightarrow a$ and $\alpha \rightarrow \theta$: $\frac{1}{r} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n P_n(\cos \theta)$. So

$$V_q(r, \theta) = \frac{q}{4\pi\epsilon_0} \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n P_n(\cos \theta).$$

Meanwhile, the potential of σ is (Eq. 3.79) $V_\sigma(r, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta)$.

Comparing the two ($V_q = V_\sigma$) we see that $B_l = (q/4\pi\epsilon_0)a^l$, and hence (Eq. 3.81) $A_l = (q/4\pi\epsilon_0)a^l/R^{2l+1}$. Then (Eq. 3.83)

$$\sigma(\theta) = \frac{q}{4\pi R^2} \sum_{l=0}^{\infty} (2l+1) \left(\frac{a}{R}\right)^l P_l(\cos \theta) = \frac{q}{4\pi R^2} \left[2 \sum_{l=0}^{\infty} l \left(\frac{a}{R}\right)^l P_l(\cos \theta) + \sum_{l=0}^{\infty} \left(\frac{a}{R}\right)^l P_l(\cos \theta) \right].$$

Now (second line above, with $r \rightarrow R$)

$$\frac{1}{\sqrt{a^2 + R^2 - 2aR \cos \theta}} = \frac{1}{R} \sum_{l=0}^{\infty} \left(\frac{a}{R}\right)^l P_l(\cos \theta).$$

Differentiating with respect to a :

$$\frac{d}{da} \left(\frac{1}{\sqrt{a^2 + R^2 - 2aR \cos \theta}} \right) = -\frac{(a - R \cos \theta)}{(a^2 + R^2 - 2aR \cos \theta)^{3/2}} = \frac{1}{aR} \sum_{l=0}^{\infty} l \left(\frac{a}{R}\right)^l P_l(\cos \theta).$$

Thus

$$\begin{aligned} \sigma(\theta) &= \frac{q}{4\pi R^2} \left[-2aR \frac{(a - R \cos \theta)}{(a^2 + R^2 - 2aR \cos \theta)^{3/2}} + \frac{R}{(a^2 + R^2 - 2aR \cos \theta)^{1/2}} \right] \\ &= \frac{q}{4\pi R} \frac{[-2a(a - R \cos \theta) + (a^2 + R^2 - 2aR \cos \theta)]}{(a^2 + R^2 - 2aR \cos \theta)^{3/2}} = \boxed{\frac{q}{4\pi R} \frac{(R^2 - a^2)}{(a^2 + R^2 - 2aR \cos \theta)^{3/2}}}. \end{aligned}$$

Chapter 4

Electric Fields in Matter

Problem 4.1

$E = V/x = 500/10^{-3} = 5 \times 10^5$. Table 4.1: $\alpha/4\pi\epsilon_0 = 0.66 \times 10^{-30}$, so $\alpha = 4\pi(8.85 \times 10^{-12})(0.66 \times 10^{-30}) = 7.34 \times 10^{-41}$. $p = \alpha E = ed \Rightarrow d = \alpha E/e = (7.34 \times 10^{-41})(5 \times 10^5)/(1.6 \times 10^{-19}) = 2.29 \times 10^{-16}$ m.
 $d/R = (2.29 \times 10^{-16})/(0.5 \times 10^{-10}) = \boxed{4.6 \times 10^{-6}}$. To ionize, say $d = R$. Then $R = \alpha E/e = \alpha V/ex \Rightarrow V = Rex/\alpha = (0.5 \times 10^{-10})(1.6 \times 10^{-19})(10^{-3})/(7.34 \times 10^{-41}) = \boxed{10^8 \text{ V.}}$

Problem 4.2

First find the field, at radius r , using Gauss' law: $\int \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$, or $E = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} Q_{\text{enc}}$.

$$\begin{aligned} Q_{\text{enc}} &= \int_0^r \rho d\tau = \frac{4\pi q}{\pi a^3} \int_0^r e^{-2\bar{r}/a} \bar{r}^2 d\bar{r} = \frac{4q}{a^3} \left[-\frac{a}{2} e^{-2\bar{r}/a} \left(\bar{r}^2 + a\bar{r} + \frac{a^2}{2} \right) \right] \Big|_0^r \\ &= -\frac{2q}{a^2} \left[e^{-2r/a} \left(r^2 + ar + \frac{a^2}{2} \right) - \frac{a^2}{2} \right] = q \left[1 - e^{-2r/a} \left(1 + 2\frac{r}{a} + 2\frac{r^2}{a^2} \right) \right]. \end{aligned}$$

[Note: $Q_{\text{enc}}(r \rightarrow \infty) = q$.] So the field of the electron cloud is $E_e = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left[1 - e^{-2r/a} \left(1 + 2\frac{r}{a} + 2\frac{r^2}{a^2} \right) \right]$. The proton will be shifted from $r = 0$ to the point d where $E_e = E$ (the external field):

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{d^2} \left[1 - e^{-2d/a} \left(1 + 2\frac{d}{a} + 2\frac{d^2}{a^2} \right) \right].$$

Expanding in powers of (d/a) :

$$\begin{aligned} e^{-2d/a} &= 1 - \left(\frac{2d}{a} \right) + \frac{1}{2} \left(\frac{2d}{a} \right)^2 - \frac{1}{3!} \left(\frac{2d}{a} \right)^3 + \cdots = 1 - 2\frac{d}{a} + 2\left(\frac{d}{a} \right)^2 - \frac{4}{3} \left(\frac{d}{a} \right)^3 + \cdots \\ 1 - e^{-2d/a} \left(1 + 2\frac{d}{a} + 2\frac{d^2}{a^2} \right) &= 1 - \left(1 - 2\frac{d}{a} + 2\left(\frac{d}{a} \right)^2 - \frac{4}{3} \left(\frac{d}{a} \right)^3 + \cdots \right) \left(1 + 2\frac{d}{a} + 2\frac{d^2}{a^2} \right) \\ &= \cancel{1} - \cancel{1} - 2\frac{d}{a} - 2\frac{d^2}{a^2} + 2\frac{d}{a} + 4\frac{d^2}{a^2} + 4\frac{d^3}{a^3} - 2\frac{d^2}{a^2} - 4\frac{d^3}{a^3} + \frac{4}{3}\frac{d^3}{a^3} + \cdots \\ &= \frac{4}{3} \left(\frac{d}{a} \right)^3 + \text{higher order terms.} \end{aligned}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{d^2} \left(\frac{4}{3} \frac{d^3}{a^3} \right) = \frac{1}{4\pi\epsilon_0} \frac{4}{3a^3} (qd) = \frac{1}{3\pi\epsilon_0 a^3} p. \quad \boxed{\alpha = 3\pi\epsilon_0 a^3.}$$

[Not so different from the *uniform* sphere model of Ex. 4.1 (see Eq. 4.2). Note that this result predicts $\frac{1}{4\pi\epsilon_0} \alpha = \frac{3}{4} a^3 = \frac{3}{4} (0.5 \times 10^{-10})^3 = 0.09 \times 10^{-30} \text{ m}^3$, compared with an experimental value (Table 4.1) of $0.66 \times 10^{-30} \text{ m}^3$. Ironically the “classical” formula (Eq. 4.2) is slightly *closer* to the empirical value.]

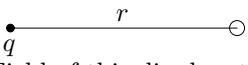
Problem 4.3

$\rho(r) = Ar$. Electric field (by Gauss’s Law): $\oint \mathbf{E} \cdot d\mathbf{a} = E(4\pi r^2) = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \int_0^r A\bar{r} 4\pi \bar{r}^2 d\bar{r}$, or $E = \frac{1}{4\pi r^2} \frac{4\pi A}{\epsilon_0} \frac{r^4}{4} = \frac{Ar^2}{4\epsilon_0}$. This “internal” field balances the external field $\mathbf{E}$ when nucleus is “off-center” an amount d : $ad^2/4\epsilon_0 = E \Rightarrow d = \sqrt{4\epsilon_0 E/A}$. So the induced dipole moment is $p = ed = 2e\sqrt{\epsilon_0/A}\sqrt{E}$. Evidently p is proportional to $E^{1/2}$.

For Eq. 4.1 to hold in the weak-field limit, E must be proportional to r , for small r , which means that ρ must go to a constant (not zero) at the origin: $\boxed{\rho(0) \neq 0}$ (nor infinite).

Problem 4.4

Field of q : $\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$. Induced dipole moment of atom: $\mathbf{p} = \alpha \mathbf{E} = \frac{\alpha q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$.



Field of this dipole, at location of q ($\theta = \pi$, in Eq. 3.103): $E = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \left(\frac{2\alpha q}{4\pi\epsilon_0 r^2} \right)$ (to the right).

Force on q due to this field: $\boxed{F = 2\alpha \left(\frac{q}{4\pi\epsilon_0} \right)^2 \frac{1}{r^5}}$ (attractive).

Problem 4.5

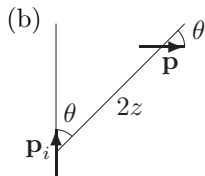
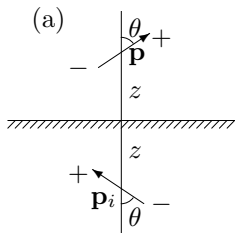
Field of $\mathbf{p}_1$ at $\mathbf{p}_2$ ($\theta = \pi/2$ in Eq. 3.103): $\mathbf{E}_1 = \frac{p_1}{4\pi\epsilon_0 r^3} \hat{\boldsymbol{\theta}}$ (points *down*).

Torque on $\mathbf{p}_2$: $\mathbf{N}_2 = \mathbf{p}_2 \times \mathbf{E}_1 = p_2 E_1 \sin 90^\circ = p_2 E_1 = \boxed{\frac{p_1 p_2}{4\pi\epsilon_0 r^3}}$ (points *into* the page).

Field of $\mathbf{p}_2$ at $\mathbf{p}_1$ ($\theta = \pi$ in Eq. 3.103): $\mathbf{E}_2 = \frac{p_2}{4\pi\epsilon_0 r^3} (-2\hat{\mathbf{r}})$ (points to the *right*).

Torque on $\mathbf{p}_1$: $\mathbf{N}_1 = \mathbf{p}_1 \times \mathbf{E}_2 = \boxed{\frac{2p_1 p_2}{4\pi\epsilon_0 r^3}}$ (points *into* the page).

Problem 4.6



Use image dipole as shown in Fig. (a).
Redraw, placing $\mathbf{p}_i$ at the origin, Fig. (b).

$$\mathbf{E}_i = \frac{p}{4\pi\epsilon_0 (2z)^3} (2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\boldsymbol{\theta}}); \quad \mathbf{p} = p\cos\theta \hat{\mathbf{r}} + p\sin\theta \hat{\boldsymbol{\theta}}.$$

$$\begin{aligned} \mathbf{N} &= \mathbf{p} \times \mathbf{E}_i = \frac{p^2}{4\pi\epsilon_0 (2z)^3} \left[(\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\boldsymbol{\theta}}) \times (2\cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\boldsymbol{\theta}}) \right] \\ &= \frac{p^2}{4\pi\epsilon_0 (2z)^3} \left[\cos\theta \sin\theta \hat{\boldsymbol{\phi}} + 2\sin\theta \cos\theta (-\hat{\boldsymbol{\phi}}) \right] \\ &= \frac{p^2 \sin\theta \cos\theta}{4\pi\epsilon_0 (2z)^3} (-\hat{\boldsymbol{\phi}}) \quad (\text{out of the page}). \end{aligned}$$

But $\sin \theta \cos \theta = (1/2) \sin 2\theta$, so $N = \frac{p^2 \sin 2\theta}{4\pi\epsilon_0(16z^3)}$ (out of the page).

For $0 < \theta < \pi/2$, $\mathbf{N}$ tends to rotate $\mathbf{p}$ counterclockwise; for $\pi/2 < \theta < \pi$, $\mathbf{N}$ rotates $\mathbf{p}$ clockwise. Thus the stable orientation is perpendicular to the surface—either $\uparrow$ or $\downarrow$.

Problem 4.7

If the potential is zero at infinity, the energy of a point charge Q is (Eq. 2.39) $W = QV(\mathbf{r})$. For a physical dipole, with $-q$ at $\mathbf{r}$ and $+q$ at $\mathbf{r} + \mathbf{d}$,

$$U = qV(\mathbf{r} + \mathbf{d}) - qV(\mathbf{r}) = q[V(\mathbf{r} + \mathbf{d}) - V(\mathbf{r})] = q \left[- \int_{\mathbf{r}}^{\mathbf{r}+\mathbf{d}} \mathbf{E} \cdot d\mathbf{l} \right].$$

For an ideal dipole the integral reduces to $\mathbf{E} \cdot \mathbf{d}$, and

$$U = -q\mathbf{E} \cdot \mathbf{d} = -\mathbf{p} \cdot \mathbf{E},$$

since $\mathbf{p} = q\mathbf{d}$. If you do not (or cannot) use infinity as the reference point, the result still holds, as long as you bring the two charges in from the *same point*, $\mathbf{r}_0$ (or two points at the same potential). In that case $W = Q[V(\mathbf{r}) - V(\mathbf{r}_0)]$, and

$$U = q[V(\mathbf{r} + \mathbf{d}) - V(\mathbf{r}_0)] - q[V(\mathbf{r}) - V(\mathbf{r}_0)] = q[V(\mathbf{r} + \mathbf{d}) - V(\mathbf{r})],$$

as before.

Problem 4.8

$$U = -\mathbf{p}_1 \cdot \mathbf{E}_2, \text{ but } \mathbf{E}_2 = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p}_2 \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}_2]. \text{ So } U = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{p}_1 \cdot \hat{\mathbf{r}})(\mathbf{p}_2 \cdot \hat{\mathbf{r}})]. \quad \text{qed}$$

Problem 4.9

$$(a) \mathbf{F} = (\mathbf{p} \cdot \nabla) \mathbf{E} \text{ (Eq. 4.5); } \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} = \frac{q}{4\pi\epsilon_0} \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{(x^2 + y^2 + z^2)^{3/2}}.$$

$$\begin{aligned} F_x &= \left(p_x \frac{\partial}{\partial x} + p_y \frac{\partial}{\partial y} + p_z \frac{\partial}{\partial z} \right) \frac{q}{4\pi\epsilon_0} \frac{x}{(x^2 + y^2 + z^2)^{3/2}} \\ &= \frac{q}{4\pi\epsilon_0} \left\{ p_x \left[\frac{1}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3}{2} x \frac{2x}{(x^2 + y^2 + z^2)^{5/2}} \right] + p_y \left[-\frac{3}{2} x \frac{2y}{(x^2 + y^2 + z^2)^{5/2}} \right] \right. \\ &\quad \left. + p_z \left[-\frac{3}{2} x \frac{2z}{(x^2 + y^2 + z^2)^{5/2}} \right] \right\} = \frac{q}{4\pi\epsilon_0} \left[\frac{p_x}{r^3} - \frac{3x}{r^5} (p_x x + p_y y + p_z z) \right] = \frac{q}{4\pi\epsilon_0} \left[\frac{\mathbf{p}}{r^3} - \frac{3\mathbf{r}(\mathbf{p} \cdot \mathbf{r})}{r^5} \right]_x. \end{aligned}$$

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} [\mathbf{p} - 3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}}].$$

(b) $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \{3[\mathbf{p} \cdot (-\hat{\mathbf{r}})](-\hat{\mathbf{r}}) - \mathbf{p}\} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}].$ (This is from Eq. 3.104; the minus signs are because $\mathbf{r}$ points *toward* $\mathbf{p}$, in this problem.)

$$\mathbf{F} = q\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}].$$

[Note that the forces are equal and opposite, as you would expect from Newton's third law.]

Problem 4.10

$$(a) \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = \boxed{kR}; \quad \rho_b = -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{\partial}{\partial r}(r^2 k r) = -\frac{1}{r^2} 3kr^2 = \boxed{-3k}.$$

$$(b) \text{ For } r < R, \mathbf{E} = \frac{1}{3\epsilon_0} \rho r \hat{\mathbf{r}} \text{ (Prob. 2.12), so } \mathbf{E} = \boxed{-(k/\epsilon_0) \mathbf{r}}.$$

For $r > R$, same as if all charge at center; but $Q_{\text{tot}} = (kR)(4\pi R^2) + (-3k)(\frac{4}{3}\pi R^3) = 0$, so $\boxed{\mathbf{E} = \mathbf{0}}$.

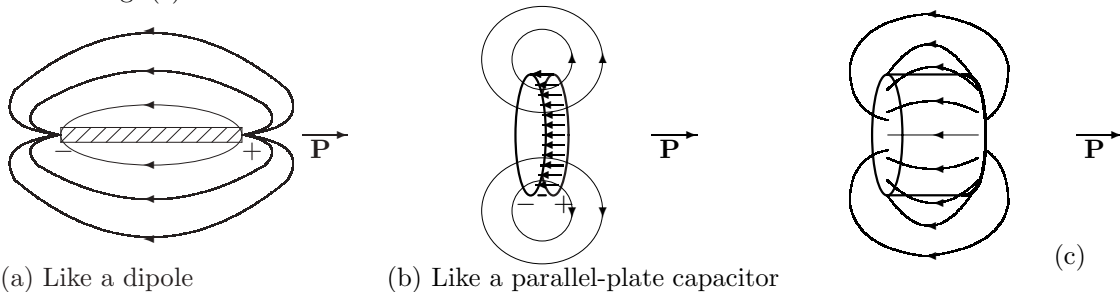
Problem 4.11

$\rho_b = 0$; $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = \pm P$ (plus sign at one end—the one $\mathbf{P}$ points *toward*; minus sign at the other—the one $\mathbf{P}$ points *away* from).

(i) $L \gg a$. Then the ends look like point charges, and the whole thing is like a physical dipole, of length L and charge $P\pi a^2$. See Fig. (a).

(ii) $L \ll a$. Then it's like a circular parallel-plate capacitor. Field is nearly uniform inside; nonuniform “fringing field” at the edges. See Fig. (b).

(iii) $L \approx a$. See Fig. (c).



Problem 4.12

$V = \frac{1}{4\pi\epsilon_0} \int \frac{\mathbf{P} \cdot \hat{\mathbf{z}}}{z^2} d\tau = \mathbf{P} \cdot \left\{ \frac{1}{4\pi\epsilon_0} \int \frac{\hat{\mathbf{z}}}{z^2} d\tau \right\}$. But the term in curly brackets is precisely the *field* of a uniformly charged sphere, divided by ρ . The integral was done explicitly in Probs. 2.7 and 2.8:

$$\frac{1}{4\pi\epsilon_0} \int \frac{\hat{\mathbf{z}}}{z^2} d\tau = \frac{1}{\rho} \left\{ \begin{array}{l} \frac{1}{4\pi\epsilon_0} \frac{(4/3)\pi R^3 \rho}{r^2} \hat{\mathbf{r}}, \quad (r > R), \\ \frac{1}{4\pi\epsilon_0} \frac{(4/3)\pi R^3 \rho}{R^3} \mathbf{r}, \quad (r < R). \end{array} \right\} \quad \text{So } V(r, \theta) = \left\{ \begin{array}{l} \frac{R^3}{3\epsilon_0 r^2} \mathbf{P} \cdot \hat{\mathbf{r}} = \boxed{\frac{R^3 P \cos \theta}{3\epsilon_0 r^2}}, \quad (r > R), \\ \frac{1}{3\epsilon_0} \mathbf{P} \cdot \mathbf{r} = \boxed{\frac{Pr \cos \theta}{3\epsilon_0}}, \quad (r < R). \end{array} \right\}$$

Problem 4.13

Think of it as two cylinders of opposite uniform charge density $\pm\rho$. *Inside*, the field at a distance s from the axis of a uniformly charge cylinder is given by Gauss's law: $E2\pi s\ell = \frac{1}{\epsilon_0} \rho\pi s^2\ell \Rightarrow \mathbf{E} = (\rho/2\epsilon_0)\mathbf{s}$. For *two* such cylinders, one plus and one minus, the net field (inside) is $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = (\rho/2\epsilon_0)(\mathbf{s}_+ - \mathbf{s}_-)$. But $\mathbf{s}_+ - \mathbf{s}_- = -\mathbf{d}$, so $\mathbf{E} = \boxed{-\rho\mathbf{d}/(2\epsilon_0)}$, where $\mathbf{d}$ is the vector from the negative axis to positive axis. In this case the total dipole moment of a chunk of length ℓ is $\mathbf{P}(\pi a^2\ell) = (\rho\pi a^2\ell)\mathbf{d}$. So $\rho\mathbf{d} = \mathbf{P}$, and $\boxed{\mathbf{E} = -\mathbf{P}/(2\epsilon_0)}$, for $s < a$.

Outside, Gauss's law gives $E2\pi s\ell = \frac{1}{\epsilon_0}\rho\pi a^2\ell \Rightarrow \mathbf{E} = \frac{\rho a^2}{2\epsilon_0} \frac{\hat{\mathbf{s}}}{s}$, for *one* cylinder. For the combination, $\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = \frac{\rho a^2}{2\epsilon_0} \left(\frac{\hat{\mathbf{s}}_+}{s_+} - \frac{\hat{\mathbf{s}}_-}{s_-} \right)$, where

$$\mathbf{s}_{\pm} = \mathbf{s} \mp \frac{\mathbf{d}}{2};$$

$$\begin{aligned} \frac{\mathbf{s}_{\pm}}{s_{\pm}^2} &= \left(\mathbf{s} \mp \frac{\mathbf{d}}{2} \right) \left(s^2 + \frac{d^2}{4} \mp \mathbf{s} \cdot \mathbf{d} \right)^{-1} \cong \frac{1}{s^2} \left(\mathbf{s} \mp \frac{\mathbf{d}}{2} \right) \left(1 \mp \frac{\mathbf{s} \cdot \mathbf{d}}{s^2} \right)^{-1} \cong \frac{1}{s^2} \left(\mathbf{s} \mp \frac{\mathbf{d}}{2} \right) \left(1 \pm \frac{\mathbf{s} \cdot \mathbf{d}}{s^2} \right) \\ &= \frac{1}{s^2} \left(\mathbf{s} \pm \mathbf{s} \frac{(\mathbf{s} \cdot \mathbf{d})}{s^2} \mp \frac{\mathbf{d}}{2} \right) \quad (\text{keeping only 1st order terms in } \mathbf{d}). \end{aligned}$$

$$\left(\frac{\hat{\mathbf{s}}_+}{s_+} - \frac{\hat{\mathbf{s}}_-}{s_-} \right) = \frac{1}{s^2} \left[\left(\mathbf{s} + \mathbf{s} \frac{(\mathbf{s} \cdot \mathbf{d})}{s^2} - \frac{\mathbf{d}}{2} \right) - \left(\mathbf{s} - \mathbf{s} \frac{(\mathbf{s} \cdot \mathbf{d})}{s^2} + \frac{\mathbf{s}}{2} \right) \right] = \frac{1}{s^2} \left(2 \frac{\mathbf{s}(\mathbf{s} \cdot \mathbf{d})}{s^2} - \mathbf{d} \right).$$

$$\boxed{\mathbf{E}(\mathbf{s}) = \frac{a^2}{2\epsilon_0} \frac{1}{s^2} [2(\mathbf{P} \cdot \hat{\mathbf{s}}) \hat{\mathbf{s}} - \mathbf{P}], \quad \text{for } s > a.}$$

Problem 4.14

Total charge on the dielectric is $Q_{\text{tot}} = \oint_S \sigma_b da + \int_V \rho_b d\tau = \oint_S \mathbf{P} \cdot d\mathbf{a} - \int_V \nabla \cdot \mathbf{P} d\tau$. But the divergence theorem says $\oint_S \mathbf{P} \cdot d\mathbf{a} = \int_V \nabla \cdot \mathbf{P} d\tau$, so $Q_{\text{enc}} = 0$. qed

Problem 4.15

$$(a) \quad \rho_b = -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{k}{r} \right) = -\frac{k}{r^2}; \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = \begin{cases} +\mathbf{P} \cdot \hat{\mathbf{r}} = k/b & (\text{at } r = b), \\ -\mathbf{P} \cdot \hat{\mathbf{r}} = -k/a & (\text{at } r = a). \end{cases}$$

Gauss's law $\Rightarrow \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q_{\text{enc}}}{r^2} \hat{\mathbf{r}}$. For $r < a$, $Q_{\text{enc}} = 0$, so $\boxed{\mathbf{E} = \mathbf{0}}$. For $r > b$, $Q_{\text{enc}} = 0$ (Prob. 4.14), so $\boxed{\mathbf{E} = \mathbf{0}}$.

For $a < r < b$, $Q_{\text{enc}} = \left(\frac{-k}{a} \right) (4\pi a^2) + \int_a^r \left(\frac{-k}{\bar{r}^2} \right) 4\pi \bar{r}^2 d\bar{r} = -4\pi k a - 4\pi k(r - a) = -4\pi k r$; so $\boxed{\mathbf{E} = -(k/\epsilon_0 r) \hat{\mathbf{r}}}$.

$$(b) \quad \oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}} = 0 \Rightarrow \mathbf{D} = \mathbf{0} \text{ everywhere. } \mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \mathbf{0} \Rightarrow \mathbf{E} = (-1/\epsilon_0) \mathbf{P}, \text{ so}$$

$$\boxed{\mathbf{E} = \mathbf{0} \text{ (for } r < a \text{ and } r > b);} \quad \boxed{\mathbf{E} = -(k/\epsilon_0 r) \hat{\mathbf{r}} \text{ (for } a < r < b).}$$

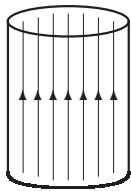
Problem 4.16

(a) Same as $\mathbf{E}_0$ minus the field at the center of a sphere with uniform polarization $\mathbf{P}$. The latter (Eq. 4.14) is $-\mathbf{P}/3\epsilon_0$. So $\boxed{\mathbf{E} = \mathbf{E}_0 + \frac{1}{3\epsilon_0} \mathbf{P}}$. $\mathbf{D} = \epsilon_0 \mathbf{E} = \epsilon_0 \mathbf{E}_0 + \frac{1}{3} \mathbf{P} = \mathbf{D}_0 - \mathbf{P} + \frac{1}{3} \mathbf{P}$, so $\boxed{\mathbf{D} = \mathbf{D}_0 - \frac{2}{3} \mathbf{P}}$.

(b) Same as $\mathbf{E}_0$ minus the field of $\pm$ charges at the two ends of the “needle”—but these are small, and far away, so $\boxed{\mathbf{E} = \mathbf{E}_0}$. $\mathbf{D} = \epsilon_0 \mathbf{E} = \epsilon_0 \mathbf{E}_0 = \mathbf{D}_0 - \mathbf{P}$, so $\boxed{\mathbf{D} = \mathbf{D}_0 - \mathbf{P}}$.

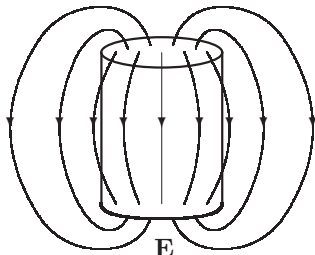
(c) Same as $\mathbf{E}_0$ minus the field of a parallel-plate capacitor with upper plate at $\sigma = P$. The latter is $-(1/\epsilon_0)P$, so $\boxed{\mathbf{E} = \mathbf{E}_0 + \frac{1}{\epsilon_0} \mathbf{P}}$. $\mathbf{D} = \epsilon_0 \mathbf{E} = \epsilon_0 \mathbf{E}_0 + \mathbf{P}$, so $\boxed{\mathbf{D} = \mathbf{D}_0}$.

Problem 4.17



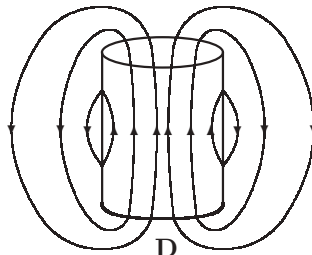
P

(uniform)



E

(field of two circular plates)



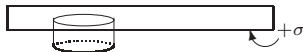
D

(same as **E** *outside*, but lines *continuous*, since $\nabla \cdot \mathbf{D} = 0$)

For more detailed figures see the solution to Problem 6.14, reading **P** for **M**, **E** for **H**, and **D** for **B**.

Problem 4.18

(a) Apply $\int \mathbf{D} \cdot d\mathbf{a} = Q_{f_{enc}}$ to the gaussian surface shown. $DA = \sigma A \Rightarrow \boxed{D = \sigma}$. (Note: $\mathbf{D} = \mathbf{0}$ inside the metal plate.) This is true in both slabs; **D** points *down*.



(b) $\mathbf{D} = \epsilon \mathbf{E} \Rightarrow E = \sigma/\epsilon_1$ in slab 1, $E = \sigma/\epsilon_2$ in slab 2. But $\epsilon = \epsilon_0 \epsilon_r$, so $\epsilon_1 = 2\epsilon_0$; $\epsilon_2 = \frac{3}{2}\epsilon_0$. $\boxed{E_1 = \sigma/2\epsilon_0}$,

$$\boxed{E_2 = 2\sigma/3\epsilon_0}.$$

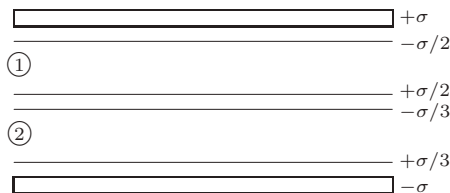
(c) $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, so $P = \epsilon_0 \chi_e d / (\epsilon_0 \epsilon_r) = (\chi_e / \epsilon_r) \sigma$; $\chi_e = \epsilon_r - 1 \Rightarrow P = (1 - \epsilon_r^{-1}) \sigma$. $\boxed{P_1 = \sigma/2}$, $\boxed{P_2 = \sigma/3}$.

(d) $V = E_1 a + E_2 a = (\sigma a / 6\epsilon_0)(3 + 4) = \boxed{7\sigma a / 6\epsilon_0}$.

(e) $\rho_b = 0$; $\boxed{\sigma_b = +P_1 \text{ at bottom of slab (1)} = \sigma/2, \sigma_b = -P_1 \text{ at top of slab (1)} = -\sigma/2;}$ $\boxed{\sigma_b = +P_2 \text{ at bottom of slab (2)} = \sigma/3, \sigma_b = -P_2 \text{ at top of slab (2)} = -\sigma/3.}$

(f) In slab 1: $\left\{ \begin{array}{l} \text{total surface charge above: } \sigma - (\sigma/2) = \sigma/2, \\ \text{total surface charge below: } (\sigma/2) - (\sigma/3) + (\sigma/3) - \sigma = -\sigma/2, \end{array} \right\} \Rightarrow E_1 = \frac{\sigma}{2\epsilon_0} \checkmark$

In slab 2: $\left\{ \begin{array}{l} \text{total surface charge above: } \sigma - (\sigma/2) + (\sigma/2) - (\sigma/3) = 2\sigma/3, \\ \text{total surface charge below: } (\sigma/3) - \sigma = -2\sigma/3, \end{array} \right\} \Rightarrow E_2 = \frac{2\sigma}{3\epsilon_0} \checkmark$



Problem 4.19

With no dielectric, $C_0 = A\epsilon_0/d$ (Eq. 2.54).

In configuration (a), with $+\sigma$ on upper plate, $-\sigma$ on lower, $D = \sigma$ between the plates. $E = \sigma/\epsilon_0$ (in air) and $E = \sigma/\epsilon$ (in dielectric). So $V = \frac{\sigma}{\epsilon_0} \frac{d}{2} + \frac{\sigma}{\epsilon} \frac{d}{2} = \frac{Qd}{2\epsilon_0 A} \left(1 + \frac{\epsilon_0}{\epsilon}\right)$.

$$C_a = \frac{Q}{V} = \frac{\epsilon_0 A}{d} \left(\frac{2}{1+1/\epsilon_r} \right) \Rightarrow \boxed{\frac{C_a}{C_0} = \frac{2\epsilon_r}{1+\epsilon_r}}.$$

In configuration (b), with potential difference V : $E = V/d$, so $\sigma = \epsilon_0 E = \epsilon_0 V/d$ (in air).
 $P = \epsilon_0 \chi_e E = \epsilon_0 \chi_e V/d$ (in dielectric), so $\sigma_b = -\epsilon_0 \chi_e V/d$ (at top surface of dielectric).
 $\sigma_{\text{tot}} = \epsilon_0 V/d = \sigma_f + \sigma_b = \sigma_f - \epsilon_0 \chi_e V/d$, so $\sigma_f = \epsilon_0 V(1 + \chi_e)/d = \epsilon_0 \epsilon_r V/d$ (on top plate above dielectric).
 $\Rightarrow C_b = \frac{Q}{V} = \frac{1}{V} \left(\sigma \frac{A}{2} + \sigma_f \frac{A}{2} \right) = \frac{A}{2V} \left(\epsilon_0 \frac{V}{d} + \epsilon_0 \frac{V}{d} \epsilon_r \right) = \frac{A\epsilon_0}{d} \left(\frac{1+\epsilon_r}{2} \right)$. $\boxed{\frac{C_b}{C_0} = \frac{1+\epsilon_r}{2}}.$

[Which is greater? $\frac{C_b}{C_0} - \frac{C_a}{C_0} = \frac{1+\epsilon_r}{2} - \frac{2\epsilon_r}{1+\epsilon_r} = \frac{(1+\epsilon_r)^2 - 4\epsilon_r}{2(1+\epsilon_r)} = \frac{1+2\epsilon_r+4\epsilon_r^2-4\epsilon_r}{2(1+\epsilon_r)} = \frac{(1-\epsilon_r)^2}{2(1+\epsilon_r)} > 0$. So $C_b > C_a$.]
 If the x axis points *down*:

	E	D	P
(a) air	$\frac{2\epsilon_r}{(\epsilon_r+1)} \frac{V}{d} \hat{\mathbf{x}}$	$\frac{2\epsilon_r}{(\epsilon_r+1)} \frac{\epsilon_0 V}{d} \hat{\mathbf{x}}$	0
(a) dielectric	$\frac{2}{(\epsilon_r+1)} \frac{V}{d} \hat{\mathbf{x}}$	$\frac{2\epsilon_r}{(\epsilon_r+1)} \frac{\epsilon_0 V}{d} \hat{\mathbf{x}}$	$\frac{2(\epsilon_r-1)}{(\epsilon_r+1)} \frac{\epsilon_0 V}{d} \hat{\mathbf{x}}$
(b) air	$\frac{V}{d} \hat{\mathbf{x}}$	$\frac{\epsilon_0 V}{d} \hat{\mathbf{x}}$	0
(b) dielectric	$\frac{V}{d} \hat{\mathbf{x}}$	$\epsilon_r \frac{\epsilon_0 V}{d} \hat{\mathbf{x}}$	$(\epsilon_r - 1) \frac{\epsilon_0 V}{d} \hat{\mathbf{x}}$

	σ_b (top surface)	σ_f (top plate)
(a)	$-\frac{2(\epsilon_r-1)}{(\epsilon_r+1)} \frac{\epsilon_0 V}{d}$	$\frac{2\epsilon_r}{(\epsilon_r+1)} \frac{\epsilon_0 V}{d}$
(b)	$-(\epsilon_r - 1) \frac{\epsilon_0 V}{d}$	$\epsilon_r \frac{\epsilon_0 V}{d}$ (left); $\frac{\epsilon_0 V}{d}$ (right)

Problem 4.20

$\int \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}} \Rightarrow D4\pi r^2 = \rho \frac{4}{3}\pi r^3 \Rightarrow D = \frac{1}{3}\rho r \Rightarrow \mathbf{E} = (\rho r/3\epsilon) \hat{\mathbf{r}}$, for $r < R$; $D4\pi r^2 = \rho \frac{4}{3}\pi R^3 \Rightarrow D = \rho R^3/3r^2 \Rightarrow \mathbf{E} = (\rho R^3/3\epsilon_0 r^2) \hat{\mathbf{r}}$, for $r > R$.

$$V = -\int_{\infty}^0 \mathbf{E} \cdot d\mathbf{l} = \frac{\rho R^3}{3\epsilon_0} \frac{1}{r} \Big|_{\infty}^R - \frac{\rho}{3\epsilon} \int_R^0 r dr = \frac{\rho R^2}{3\epsilon_0} + \frac{\rho}{3\epsilon} \frac{R^2}{2} = \boxed{\frac{\rho R^2}{3\epsilon_0} \left(1 + \frac{1}{2\epsilon_r} \right)}.$$

Problem 4.21

Let Q be the charge on a length ℓ of the inner conductor.

$$\oint \mathbf{D} \cdot d\mathbf{a} = D2\pi s\ell = Q \Rightarrow D = \frac{Q}{2\pi s\ell}; \quad E = \frac{Q}{2\pi\epsilon_0 s\ell} \quad (a < s < b), \quad E = \frac{Q}{2\pi\epsilon s\ell} \quad (b < r < c).$$

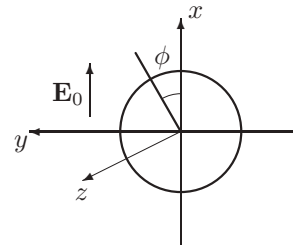
$$V = -\int_c^a \mathbf{E} \cdot d\mathbf{l} = \int_a^b \left(\frac{Q}{2\pi\epsilon_0\ell} \right) \frac{ds}{s} + \int_b^c \left(\frac{Q}{2\pi\epsilon\ell} \right) \frac{ds}{s} = \frac{Q}{2\pi\epsilon_0\ell} \left[\ln\left(\frac{b}{a}\right) + \frac{\epsilon_0}{\epsilon} \ln\left(\frac{c}{b}\right) \right].$$

$$\frac{C}{\ell} = \frac{Q}{V\ell} = \boxed{\frac{2\pi\epsilon_0}{\ln(b/a) + (1/\epsilon_r)\ln(c/b)}}.$$

Problem 4.22

Same method as Ex. 4.7: solve Laplace's equation for $V_{\text{in}}(s, \phi)$ ($s < a$) and $V_{\text{out}}(s, \phi)$ ($s > a$), subject to the boundary conditions

$$\begin{cases} \text{(i)} & V_{\text{in}} = V_{\text{out}} & \text{at } s = a, \\ \text{(ii)} & \epsilon \frac{\partial V_{\text{in}}}{\partial s} = \epsilon_0 \frac{\partial V_{\text{out}}}{\partial s} & \text{at } s = a, \\ \text{(iii)} & V_{\text{out}} \rightarrow -E_0 s \cos \phi & \text{for } s \gg a. \end{cases}$$



From Prob. 3.24 (invoking boundary condition (iii)):

$$V_{\text{in}}(s, \phi) = \sum_{k=1}^{\infty} s^k (a_k \cos k\phi + b_k \sin k\phi), \quad V_{\text{out}}(s, \phi) = -E_0 s \cos \phi + \sum_{k=1}^{\infty} s^{-k} (c_k \cos k\phi + d_k \sin k\phi).$$

(I eliminated the constant terms by setting $V = 0$ on the yz plane.) Condition (i) says

$$\sum a^k (a_k \cos k\phi + b_k \sin k\phi) = -E_0 a \cos \phi + \sum a^{-k} (c_k \cos k\phi + d_k \sin k\phi),$$

while (ii) says

$$\epsilon_r \sum k a^{k-1} (a_k \cos k\phi + b_k \sin k\phi) = -E_0 \cos \phi - \sum k a^{-k-1} (c_k \cos k\phi + d_k \sin k\phi).$$

Evidently $b_k = d_k = 0$ for all k , $a_k = c_k = 0$ unless $k = 1$, whereas for $k = 1$,

$$a a_1 = -E_0 a + a^{-1} c_1, \quad \epsilon_r a_1 = -E_0 - a^{-2} c_1.$$

Solving for a_1 ,

$$a_1 = -\frac{E_0}{(1 + \chi_e/2)}, \quad \text{so } V_{\text{in}}(s, \phi) = -\frac{E_0}{(1 + \chi_e/2)} s \cos \phi = -\frac{E_0}{(1 + \chi_e/2)} x,$$

and hence $\mathbf{E}_{\text{in}}(s, \phi) = -\frac{\partial V_{\text{in}}}{\partial x} \hat{\mathbf{x}} = \boxed{\frac{\mathbf{E}_0}{(1 + \chi_e/2)}}$. As in the spherical case (Ex. 4.7), the field inside is *uniform*.

Problem 4.23

$\mathbf{P}_0 = \epsilon_0 \chi_e \mathbf{E}_0$; $\mathbf{E}_1 = -\frac{1}{3\epsilon_0} \mathbf{P}_0 = -\frac{\chi_e}{3} \mathbf{E}_0$; $\mathbf{P}_1 = \epsilon_0 \chi_e \mathbf{E}_1 = -\frac{\epsilon_0 \chi_e^2}{3} \mathbf{E}_0$; $\mathbf{E}_2 = -\frac{1}{3\epsilon_0} \mathbf{P}_1 = \frac{\chi_e^2}{9} \mathbf{E}_0$; ... Evidently $\mathbf{E}_n = \left(-\frac{\chi_e}{3}\right)^n \mathbf{E}_0$, so

$$\mathbf{E} = \mathbf{E}_0 + \mathbf{E}_1 + \mathbf{E}_2 + \cdots = \left[\sum_{n=0}^{\infty} \left(-\frac{\chi_e}{3}\right)^n \right] \mathbf{E}_0.$$

The geometric series can be summed explicitly:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad \text{so } \boxed{\mathbf{E} = \frac{1}{(1 + \chi_e/3)} \mathbf{E}_0},$$

which agrees with Eq. 4.49. [Curiously, this method formally requires that $\chi_e < 3$ (else the infinite series diverges), yet the *result* is subject to no such restriction, since we can also get it by the method of Ex. 4.7.]

Problem 4.24

Potentials:

$$\begin{cases} V_{\text{out}}(r, \theta) = -E_0 r \cos \theta + \sum \frac{B_l}{r^{l+1}} P_l(\cos \theta), & (r > b); \\ V_{\text{med}}(r, \theta) = \sum \left(A_l r^l + \frac{\bar{B}_l}{r^{l+1}} \right) P_l(\cos \theta), & (a < r < b); \\ V_{\text{in}}(r, \theta) = 0, & (r < a). \end{cases}$$

Boundary Conditions:

$$\begin{cases} \text{(i)} & V_{\text{out}} = V_{\text{med}}, & (r = b); \\ \text{(ii)} & \epsilon \frac{\partial V_{\text{med}}}{\partial r} = \epsilon_0 \frac{\partial V_{\text{out}}}{\partial r}, & (r = b); \\ \text{(iii)} & V_{\text{med}} = 0, & (r = a). \end{cases}$$

$$\text{(i)} \Rightarrow -E_0 b \cos \theta + \sum \frac{B_l}{b^{l+1}} P_l(\cos \theta) = \sum \left(A_l b^l + \frac{\bar{B}_l}{b^{l+1}} \right) P_l(\cos \theta);$$

$$\text{(ii)} \Rightarrow \epsilon_r \sum \left[l A_l b^{l-1} - (l+1) \frac{\bar{B}_l}{b^{l+2}} \right] P_l(\cos \theta) = -E_0 \cos \theta - \sum (l+1) \frac{B_l}{b^{l+2}} P_l(\cos \theta);$$

$$\text{(iii)} \Rightarrow A_l a^l + \frac{\bar{B}_l}{a^{l+1}} = 0 \Rightarrow \bar{B}_l = -a^{2l+1} A_l.$$

For $l \neq 1$:

$$\text{(i)} \quad \frac{B_l}{b^{l+1}} = \left(A_l b^l - \frac{a^{2l+1} A_l}{b^{l+1}} \right) \Rightarrow B_l = A_l (b^{2l+1} - a^{2l+1});$$

$$\text{(ii)} \quad \epsilon_r \left[l A_l b^{l-1} + (l+1) \frac{a^{2l+1} A_l}{b^{l+2}} \right] = -(l+1) \frac{B_l}{b^{l+2}} \Rightarrow B_l = -\epsilon_r A_l \left[\left(\frac{l}{l+1} \right) b^{2l+1} + a^{2l+1} \right] \Rightarrow A_l = B_l = 0.$$

For $l = 1$:

$$\text{(i)} \quad -E_0 b + \frac{B_1}{b^2} = A_1 b - \frac{a^3 A_1}{b^2} \Rightarrow B_1 - E_0 b^3 = A_1 (b^3 - a^3);$$

$$\text{(ii)} \quad \epsilon_r \left(A_1 + 2 \frac{a^3 A_1}{b^3} \right) = -E_0 - 2 \frac{B_1}{b^3} \Rightarrow -2B_1 - E_0 b^3 = \epsilon_r A_1 (b^3 + 2a^3).$$

$$\text{So} \quad -3E_0 b^3 = A_1 [2(b^3 - a^3) + \epsilon_r (b^3 + 2a^3)]; \quad A_1 = \frac{-3E_0}{2[1 - (a/b)^3] + \epsilon_r [1 + 2(a/b)^3]}.$$

$$V_{\text{med}}(r, \theta) = \frac{-3E_0}{2[1 - (a/b)^3] + \epsilon_r [1 + 2(a/b)^3]} \left(r - \frac{a^3}{r^2} \right) \cos \theta,$$

$$\mathbf{E}(r, \theta) = -\nabla V_{\text{med}} = \frac{3E_0}{2[1 - (a/b)^3] + \epsilon_r [1 + 2(a/b)^3]} \left\{ \left(1 + \frac{2a^3}{r^3} \right) \cos \theta \hat{\mathbf{r}} - \left(1 - \frac{a^3}{r^3} \right) \sin \theta \hat{\boldsymbol{\theta}} \right\}.$$

Problem 4.25

There are four charges involved: (i) q , (ii) polarization charge surrounding q , (iii) surface charge (σ_b) on the top surface of the lower dielectric, (iv) surface charge (σ'_b) on the lower surface of the upper dielectric.

In view of Eq. 4.39, the bound charge (ii) is $q_p = -q(\chi'_e/(1 + \chi'_e))$, so the *total* (point) charge at $(0, 0, d)$ is $q_t = q + q_p = q/(1 + \chi'_e) = q/\epsilon'_r$. As in Ex. 4.8,

$$(a) \sigma_b = \epsilon_0 \chi_e \left[\frac{-1}{4\pi\epsilon_0} \frac{qd/\epsilon'_r}{(r^2 + d^2)^{\frac{3}{2}}} - \frac{\sigma_b}{2\epsilon_0} - \frac{\sigma'_b}{2\epsilon_0} \right] \quad (\text{here } \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = +P_z = \epsilon_0 \chi_e E_z);$$

$$(b) \sigma'_b = \epsilon_0 \chi'_e \left[\frac{1}{4\pi\epsilon_0} \frac{qd/\epsilon'_r}{(r^2 + d^2)^{\frac{3}{2}}} - \frac{\sigma_b}{2\epsilon_0} - \frac{\sigma'_b}{2\epsilon_0} \right] \quad (\text{here } \sigma_b = -P_z = -\epsilon_0 \chi'_e E_z).$$

Solve for σ_b, σ'_b : first divide by χ_e and χ'_e (respectively) and subtract:

$$\frac{\sigma'_b}{\chi'_e} - \frac{\sigma_b}{\chi_e} = \frac{1}{2\pi} \frac{qd/\epsilon'_r}{(r^2 + d^2)^{\frac{3}{2}}} \Rightarrow \sigma'_b = \chi'_e \left[\frac{\sigma_b}{\chi_e} + \frac{1}{2\pi} \frac{qd/\epsilon'_r}{(r^2 + d^2)^{\frac{3}{2}}} \right].$$

Plug this into (a) and solve for σ_b , using $\epsilon'_r = 1 + \chi'_e$:

$$\sigma_b = \frac{-1}{4\pi} \frac{qd/\epsilon'_r}{(r^2 + d^2)^{\frac{3}{2}}} \chi_e (1 + \chi'_e) - \frac{\sigma_b}{2} (\chi_e + \chi'_e), \text{ so } \boxed{\sigma_b = \frac{-1}{4\pi} \frac{qd}{(r^2 + d^2)^{\frac{3}{2}}} \frac{\chi_e}{[1 + (\chi_e + \chi'_e)/2]};}$$

$$\sigma'_b = \chi'_e \left\{ \frac{-1}{4\pi} \frac{qd}{(r^2 + d^2)^{\frac{3}{2}}} \frac{1}{[1 + (\chi_e + \chi'_e)/2]} + \frac{1}{2\pi} \frac{qd/\epsilon'_r}{(r^2 + d^2)^{\frac{3}{2}}} \right\}, \text{ so } \boxed{\sigma'_b = \frac{1}{4\pi} \frac{qd}{(r^2 + d^2)^{\frac{3}{2}}} \frac{\epsilon_r \chi'_e / \epsilon'_r}{[1 + (\chi_e + \chi'_e)/2]}.$$

The *total* bound surface charge is $\sigma_t = \sigma_b + \sigma'_b = \frac{1}{4\pi} \frac{qd}{(r^2 + d^2)^{\frac{3}{2}}} \frac{(\chi'_e - \chi_e)}{\epsilon'_r [1 + (\chi_e + \chi'_e)/2]}$ (which vanishes, as it should, when $\chi'_e = \chi_e$). The total bound charge is (compare Eq. 4.51):

$$q_t = \frac{(\chi'_e - \chi_e)q}{2\epsilon'_r [1 + (\chi_e + \chi'_e)/2]} = \boxed{\left(\frac{\epsilon'_r - \epsilon_r}{\epsilon'_r + \epsilon_r} \right) \frac{q}{\epsilon'_r}}, \text{ and hence}$$

$$\boxed{V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q/\epsilon'_r}{\sqrt{x^2 + y^2 + (z - d)^2}} + \frac{q_t}{\sqrt{x^2 + y^2 + (z + d)^2}} \right\}} \quad (\text{for } z > 0).$$

$$\text{Meanwhile, since } \frac{q}{\epsilon'_r} + q_t = \frac{q}{\epsilon'_r} \left[1 + \frac{\epsilon'_r - \epsilon_r}{\epsilon'_r + \epsilon_r} \right] = \frac{2q}{\epsilon'_r + \epsilon_r}, \quad \boxed{V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{[2q/(\epsilon'_r + \epsilon_r)]}{\sqrt{x^2 + y^2 + (z - d)^2}}} \quad (\text{for } z < 0).$$

Problem 4.26

From Ex. 4.5:

$$\mathbf{D} = \left\{ \begin{array}{ll} 0, & (r < a) \\ \frac{Q}{4\pi r^2} \hat{\mathbf{r}}, & (r > a) \end{array} \right\}, \quad \mathbf{E} = \left\{ \begin{array}{ll} 0, & (r < a) \\ \frac{Q}{4\pi\epsilon r^2} \hat{\mathbf{r}}, & (a < r < b) \\ \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}, & (r > b) \end{array} \right\}.$$

$$\begin{aligned} W &= \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau = \frac{1}{2} \frac{Q^2}{(4\pi)^2} 4\pi \left\{ \frac{1}{\epsilon} \int_a^b \frac{1}{r^2} \frac{1}{r^2} r^2 dr + \frac{1}{\epsilon_0} \int_b^\infty \frac{1}{r^2} dr \right\} = \frac{Q^2}{8\pi} \left\{ \frac{1}{\epsilon} \left(\frac{-1}{r} \right) \Big|_a^b + \frac{1}{\epsilon_0} \left(\frac{-1}{r} \right) \Big|_b^\infty \right\} \\ &= \frac{Q^2}{8\pi\epsilon_0} \left\{ \frac{1}{(1 + \chi_e)} \left(\frac{1}{a} - \frac{1}{b} \right) + \frac{1}{b} \right\} = \boxed{\frac{Q^2}{8\pi\epsilon_0(1 + \chi_e)} \left(\frac{1}{a} + \frac{\chi_e}{b} \right)}. \end{aligned}$$

Problem 4.27

Using Eq. 4.55: $W = \frac{\epsilon_0}{2} \int E^2 d\tau$. From Ex. 4.2 and Eq. 3.103,

$$\mathbf{E} = \left\{ \begin{array}{ll} \frac{-1}{3\epsilon_0} P \hat{\mathbf{z}}, & (r < R) \\ \frac{R^3 P}{3\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}), & (r > R) \end{array} \right\}, \quad \text{so}$$

$$W_{r < R} = \frac{\epsilon_0}{2} \left(\frac{P}{3\epsilon_0} \right)^2 \frac{4}{3} \pi R^3 = \frac{2\pi}{27} \frac{P^2 R^3}{\epsilon_0}.$$

$$\begin{aligned} W_{r > R} &= \frac{\epsilon_0}{2} \left(\frac{R^3 P}{3\epsilon_0} \right)^2 \int \frac{1}{r^6} (4 \cos^2 \theta + \sin^2 \theta) r^2 \sin \theta dr d\theta d\phi \\ &= \frac{(R^3 P)^2}{18\epsilon_0} 2\pi \int_0^\pi (1 + 3 \cos^2 \theta) \sin \theta d\theta \int_R^\infty \frac{1}{r^4} dr = \frac{\pi (R^3 P)^2}{9\epsilon_0} (-\cos \theta - \cos^3 \theta) \Big|_0^\pi \left(-\frac{1}{3r^3} \right) \Big|_R^\infty \\ &= \frac{\pi (R^3 P)^2}{9\epsilon_0} \left(\frac{4}{3R^3} \right) = \frac{4\pi R^3 P^2}{27\epsilon_0}. \end{aligned}$$

$$W_{\text{tot}} = \boxed{\frac{2\pi R^3 P^2}{9\epsilon_0}}.$$

This is the correct electrostatic energy of the configuration, but it is not the “total work necessary to assemble the system,” because it leaves out the mechanical energy involved in polarizing the molecules.

Using Eq. 4.58: $W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$. For $r > R$, $\mathbf{D} = \epsilon_0 \mathbf{E}$, so this contribution is the same as before. For $r < R$, $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = -\frac{1}{3} \mathbf{P} + \mathbf{P} = \frac{2}{3} \mathbf{P} = -2\epsilon_0 \mathbf{E}$, so $\frac{1}{2} \mathbf{D} \cdot \mathbf{E} = -2\frac{\epsilon_0}{2} E^2$, and this contribution is now $(-2) \left(\frac{2\pi}{27} \frac{P^2 R^3}{\epsilon_0} \right) = -\frac{4\pi}{27} \frac{R^3 P^2}{\epsilon_0}$, exactly cancelling the exterior term. *Conclusion:* $\boxed{W_{\text{tot}} = 0}$. This is not surprising, since the derivation in Sect. 4.4.3 calculates the work done on the *free* charge, and in this problem there is no free charge in sight. Since this is a nonlinear dielectric, however, the result cannot be interpreted as the “work necessary to assemble the configuration”—the latter would depend entirely on *how* you assemble it.

Problem 4.28

First find the capacitance, as a function of h :

$$\left. \begin{array}{l} \text{Air part: } E = \frac{2\lambda}{4\pi\epsilon_0 s} \implies V = \frac{2\lambda}{4\pi\epsilon_0} \ln(b/a), \\ \text{Oil part: } D = \frac{2\lambda'}{4\pi s} \implies E = \frac{2\lambda'}{4\pi\epsilon_s} \implies V = \frac{2\lambda'}{4\pi\epsilon} \ln(b/a), \end{array} \right\} \implies \frac{\lambda}{\epsilon_0} = \frac{\lambda'}{\epsilon}; \quad \lambda' = \frac{\epsilon}{\epsilon_0} \lambda = \epsilon_r \lambda.$$

$$Q = \lambda' h + \lambda(\ell - h) = \epsilon_r \lambda h - \lambda h + \lambda \ell = \lambda[(\epsilon_r - 1)h + \ell] = \lambda(\chi_e h + \ell), \text{ where } \ell \text{ is the total height.}$$

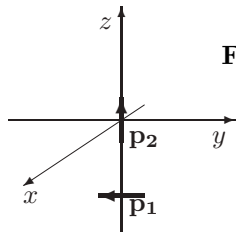
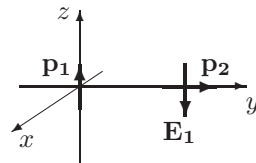
$$C = \frac{Q}{V} = \frac{\lambda(\chi_e h + \ell)}{2\lambda \ln(b/a)} 4\pi\epsilon_0 = 2\pi\epsilon_0 \frac{(\chi_e h + \ell)}{\ln(b/a)}.$$

$$\left. \begin{array}{l} \text{The net upward force is given by Eq. 4.64: } F = \frac{1}{2} V^2 \frac{dC}{dh} = \frac{1}{2} V^2 \frac{2\pi\epsilon_0 \chi_e}{\ln(b/a)}. \\ \text{The gravitational force down is } F = mg = \rho\pi(b^2 - a^2)gh. \end{array} \right\} \quad h = \frac{\epsilon_0 \chi_e V^2}{\rho(b^2 - a^2)g \ln(b/a)}.$$

Problem 4.29

$$(a) \text{ Eq. 4.5} \Rightarrow \mathbf{F}_2 = (\mathbf{p}_2 \cdot \nabla) \mathbf{E}_1 = p_2 \frac{\partial}{\partial y} (\mathbf{E}_1);$$

$$\text{Eq. 3.103} \Rightarrow \mathbf{E}_1 = \frac{p_1}{4\pi\epsilon_0 r^3} \hat{\boldsymbol{\theta}} = -\frac{p_1}{4\pi\epsilon_0 y^3} \hat{\mathbf{z}}. \text{ Therefore}$$



$$\mathbf{F}_2 = -\frac{p_1 p_2}{4\pi\epsilon_0} \left[\frac{d}{dy} \left(\frac{1}{y^3} \right) \right] \hat{\mathbf{z}} = \frac{3p_1 p_2}{4\pi\epsilon_0 y^4} \hat{\mathbf{z}}, \text{ or } \boxed{\mathbf{F}_2 = \frac{3p_1 p_2}{4\pi\epsilon_0 r^4} \hat{\mathbf{z}}} \text{ (upward).}$$

To calculate $\mathbf{F}_1$, put $\mathbf{p}_2$ at the origin, pointing in the z direction; then $\mathbf{p}_1$ is at $-r \hat{\mathbf{z}}$, and it points in the $-\hat{\mathbf{y}}$ direction. So $\mathbf{F}_1 = (\mathbf{p}_1 \cdot \nabla) \mathbf{E}_2 = -p_1 \frac{\partial \mathbf{E}_2}{\partial y} \Big|_{x=y=0, z=-r}$; we need $\mathbf{E}_2$ as a function of x , y , and z .

$$\text{From Eq. 3.104: } \mathbf{E}_2 = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \left[\frac{3(\mathbf{p}_2 \cdot \mathbf{r})\mathbf{r}}{r^2} - \mathbf{p}_2 \right], \text{ where } \mathbf{r} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}, \mathbf{p}_2 = p_2 \hat{\mathbf{z}}, \text{ and hence } \mathbf{p}_2 \cdot \mathbf{r} = p_2 z.$$

$$\mathbf{E}_2 = \frac{p_2}{4\pi\epsilon_0} \left[\frac{3z(x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) - (x^2 + y^2 + z^2) \hat{\mathbf{z}}}{(x^2 + y^2 + z^2)^{5/2}} \right] = \frac{p_2}{4\pi\epsilon_0} \left[\frac{3xz \hat{\mathbf{x}} + 3yz \hat{\mathbf{y}} - (x^2 + y^2 - 2z^2) \hat{\mathbf{z}}}{(x^2 + y^2 + z^2)^{5/2}} \right]$$

$$\frac{\partial \mathbf{E}_2}{\partial y} = \frac{p_2}{4\pi\epsilon_0} \left\{ -\frac{5}{2} \frac{2y}{r^7} [3xz \hat{\mathbf{x}} + 3yz \hat{\mathbf{y}} - (x^2 + y^2 - 2z^2) \hat{\mathbf{z}}] + \frac{1}{r^5} (3z \hat{\mathbf{y}} - 2y \hat{\mathbf{z}}) \right\};$$

$$\frac{\partial \mathbf{E}_2}{\partial y} \Big|_{(0,0)} = \frac{p_2}{4\pi\epsilon_0} \frac{3z}{r^5} \hat{\mathbf{y}}; \quad \mathbf{F}_1 = -p_1 \left(\frac{p_2}{4\pi\epsilon_0} \frac{-3r}{r^5} \hat{\mathbf{y}} \right) = \boxed{\frac{3p_1 p_2}{4\pi\epsilon_0 r^4} \hat{\mathbf{y}}}.$$

But $\hat{\mathbf{y}}$ in these coordinates corresponds to $-\hat{\mathbf{z}}$ in the original system, so these results *are* consistent with Newton's third law: $\mathbf{F}_1 = -\mathbf{F}_2$.

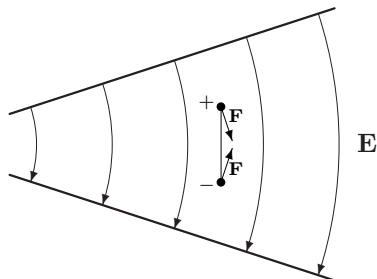
(b) From the remark following Eq. 4.5, $\mathbf{N}_2 = (\mathbf{p}_2 \times \mathbf{E}_1) + (\mathbf{r} \times \mathbf{F}_2)$. The first term was calculated in Prob. 4.5; the second we get from (a), using $\mathbf{r} = r \hat{\mathbf{y}}$:

$$\mathbf{p}_2 \times \mathbf{E}_1 = \frac{p_1 p_2}{4\pi\epsilon_0 r^3} (-\hat{\mathbf{x}}); \quad \mathbf{r} \times \mathbf{F}_2 = (r \hat{\mathbf{y}}) \times \left(\frac{3p_1 p_2}{4\pi\epsilon_0 r^4} \hat{\mathbf{z}} \right) = \frac{3p_1 p_2}{4\pi\epsilon_0 r^3} \hat{\mathbf{x}}; \text{ so } \boxed{\mathbf{N}_2 = \frac{2p_1 p_2}{4\pi\epsilon_0 r^3} \hat{\mathbf{x}}}.$$

This is equal and opposite to the torque on $\mathbf{p}_1$ due to $\mathbf{p}_2$, with respect to the center of $\mathbf{p}_1$ (see Prob. 4.5).

Problem 4.30

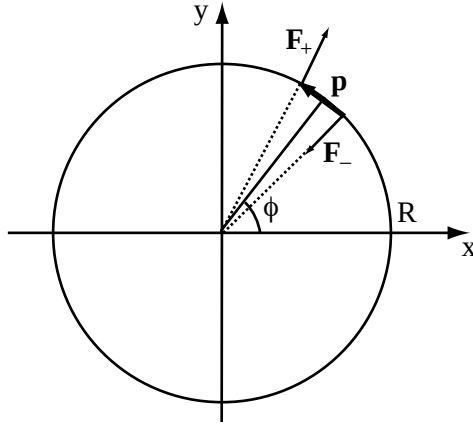
Net force is to the right (see diagram). Note that the field lines must bulge to the right, as shown, because $\mathbf{E}$ is perpendicular to the surface of each conductor.



Problem 4.31 In cylindrical coordinates (in the $z = 0$ plane), $\mathbf{p} = p \hat{\phi}$, $\mathbf{p} \cdot \nabla = p \frac{1}{s} \frac{\partial}{\partial \phi}$, and $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{s^2} \hat{\mathbf{s}}$, so

$$\mathbf{F} = (\mathbf{p} \cdot \nabla) \mathbf{E} = \left(\frac{p}{s} \frac{\partial}{\partial \phi} \right) \frac{1}{4\pi\epsilon_0} \frac{Q}{s^2} \hat{\mathbf{s}} = \frac{pQ}{4\pi\epsilon_0 s^3} \frac{\partial \hat{\mathbf{s}}}{\partial \phi} = \frac{pQ}{4\pi\epsilon_0 s^3} \hat{\phi} = \frac{Q}{4\pi\epsilon_0 R^3} \mathbf{p}. \quad \checkmark$$

Qualitatively, the forces on the negative and positive ends, though equal in magnitude, point in slightly different directions, and they combine to make a net force in the “forward” direction:



To keep the dipole going in a circle, there must be a centripetal force exerted by the track (we may as well take it to act at the center of the dipole, and it is irrelevant to the problem), and to keep it aiming in the tangential direction there must be a torque (which we could model by radial forces of equal magnitude acting at the two ends). Indeed, if the dipole has the orientation indicated in the figure, and is moving in the $\hat{\phi}$ direction, the torque exerted by Q is clockwise, whereas the rotation is counterclockwise, so these constraint forces must actually be *larger* than the forces exerted by Q , and the *net* force will be in the “backward” direction—tending to slow the dipole down. [If the motion is in the $-\hat{\phi}$ direction, then the *electrical* forces will dominate, and the net force will be in the direction of $\mathbf{p}$, but this again will tend to slow it down.]

Problem 4.32

(a) According to Eqs. 4.1 and 4.5, $\mathbf{F} = \alpha(\mathbf{E} \cdot \nabla) \mathbf{E}$. From the product rule,

$$\nabla E^2 = \nabla(\mathbf{E} \cdot \mathbf{E}) = 2\mathbf{E} \times (\nabla \times \mathbf{E}) + 2(\mathbf{E} \cdot \nabla) \mathbf{E}.$$

But in electrostatics $\nabla \times \mathbf{E} = \mathbf{0}$, so $(\mathbf{E} \cdot \nabla) \mathbf{E} = \frac{1}{2} \nabla(E^2)$, and hence

$$\mathbf{F} = \frac{1}{2} \alpha \nabla(E^2). \quad \checkmark$$

[It is tempting to start with Eq. 4.6, and write $\mathbf{F} = -\nabla U = \nabla(\mathbf{p} \cdot \mathbf{E}) = \alpha \nabla(\mathbf{E} \cdot \mathbf{E}) = \alpha \nabla(E^2)$. The error occurs in the third step: $\mathbf{p}$ should not have been differentiated, but after it is replaced by $\alpha \mathbf{E}$ we are differentiating *both* $\mathbf{E}$'s.]

(b) Suppose E^2 has a local maximum at point P . Then there is a sphere (of radius R) about P such that $E^2(P') < E^2(P)$, and hence $|\mathbf{E}(P')| < |\mathbf{E}(P)|$, for all points on the surface. But if there is no charge inside the

sphere, then Problem 3.4a says the average field over the spherical surface is equal to the value at the center:

$$\frac{1}{4\pi R^2} \int \mathbf{E} da = \mathbf{E}(P),$$

or, choosing the z axis to lie along $\mathbf{E}(P)$,

$$\frac{1}{4\pi R^2} \int E_z da = E(P).$$

But if E^2 has a maximum at P , then

$$\int E_z da \leq \int |\mathbf{E}| da < \int |\mathbf{E}(P)| da = 4\pi R^2 E(P),$$

and it follows that $E(P) < E(P)$, a contradiction. Therefore, E^2 cannot have a maximum in a charge-free region. [It can have a minimum, however; at the midpoint between two equal charges the field is zero, and this is obviously a minimum.]

Problem 4.33

$$\mathbf{P} = k\mathbf{r} = k(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}) \implies \rho_b = -\nabla \cdot \mathbf{P} = -k(1 + 1 + 1) = \boxed{-3k}.$$

$$\text{Total volume bound charge: } \boxed{Q_{\text{vol}} = -3ka^3}.$$

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}. \text{ At top surface, } \hat{\mathbf{n}} = \hat{\mathbf{z}}, z = a/2; \text{ so } \sigma_b = ka/2. \text{ Clearly, } \boxed{\sigma_b = ka/2} \text{ on all six surfaces.}$$

$$\text{Total surface bound charge: } \boxed{Q_{\text{surf}} = 6(ka/2)a^2 = 3ka^3}. \text{ Total bound charge is zero. } \checkmark$$

Problem 4.34

Say the high voltage is connected to the bottom plate, so the electric field points in the x direction, while the free charge density (σ_f) is positive on the lower plate and negative on the upper plate. (If you connect the battery the other way, all the signs will switch.) The susceptibility is $\chi_e = \frac{x}{d}$, and the permittivity is

$$\epsilon = \epsilon_0 \left(1 + \frac{x}{d}\right). \text{ Between the plates}$$

$$\mathbf{D} = \sigma_f \hat{\mathbf{x}}, \quad \mathbf{E} = \frac{1}{\epsilon} \mathbf{D} = \frac{\sigma_f}{\epsilon_0(1 + x/d)} \hat{\mathbf{x}}; \quad V = - \int_d^0 \mathbf{E} \cdot d\mathbf{l} = \frac{\sigma_f}{\epsilon_0} \int_0^d \frac{1}{(1 + x/d)} dx = \frac{\sigma_f}{\epsilon_0} d \ln \left(1 + \frac{x}{d}\right) \Big|_0^d = \frac{\sigma_f d}{\epsilon_0} \ln 2.$$

So

$$\sigma_f = \frac{\epsilon_0 V}{d \ln 2}, \quad \mathbf{E} = \frac{V}{d \ln 2} \frac{1}{(1 + x/d)} \hat{\mathbf{x}}, \quad \mathbf{P} = \epsilon_0 \chi_e \mathbf{E} = \frac{\epsilon_0 V}{d^2 \ln 2} \frac{x}{(1 + x/d)} \hat{\mathbf{x}}.$$

The bound charges are therefore

$$\rho_b = -\nabla \cdot \mathbf{P} = -\frac{\epsilon_0 V}{d^2 \ln 2} \left[\frac{1}{(1 + x/d)} - \frac{x/d}{(1 + x/d)^2} \right] = \boxed{-\frac{\epsilon_0 V}{d^2 \ln 2} \frac{1}{(1 + x/d)^2}}; \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = \begin{cases} \boxed{0} & (x = 0), \\ \boxed{\frac{\epsilon_0 V}{2d \ln 2}} & (x = d). \end{cases}$$

The total bound charge is

$$\begin{aligned} Q_b &= \int \rho_b d\tau + \int \sigma_b da = -\frac{\epsilon_0 V}{d^2 \ln 2} \int_0^d \frac{1}{(1 + x/d)^2} A dx + \frac{\epsilon_0 V}{2d \ln 2} A = \frac{\epsilon_0 V A}{d \ln 2} \left[-\frac{1}{d} \frac{-d}{(1 + x/d)} \Big|_0^d + \frac{1}{2} \right] \\ &= \frac{\epsilon_0 V A}{d \ln 2} \left(\frac{1}{2} - 1 + \frac{1}{2} \right) = 0, \quad \checkmark \end{aligned}$$

where A is the area of the plates.

Problem 4.35

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}} \Rightarrow \mathbf{D} = \frac{q}{4\pi r^2} \hat{\mathbf{r}}; \quad \mathbf{E} = \frac{1}{\epsilon} \mathbf{D} = \boxed{\frac{q}{4\pi\epsilon_0(1+\chi_e)} \frac{\hat{\mathbf{r}}}{r^2}}; \quad \mathbf{P} = \epsilon_0\chi_e\mathbf{E} = \boxed{\frac{q\chi_e}{4\pi(1+\chi_e)} \frac{\hat{\mathbf{r}}}{r^2}}.$$

$$\rho_b = -\nabla \cdot \mathbf{P} = -\frac{q\chi_e}{4\pi(1+\chi_e)} \left(\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} \right) = \boxed{-q \frac{\chi_e}{1+\chi_e} \delta^3(\mathbf{r})} \quad (\text{Eq. 1.99}); \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{r}} = \boxed{\frac{q\chi_e}{4\pi(1+\chi_e)R^2}};$$

$$Q_{\text{surf}} = \sigma_b(4\pi R^2) = \boxed{q \frac{\chi_e}{1+\chi_e}}. \quad \text{The compensating negative charge is at the center:}$$

$$\int \rho_b d\tau = -\frac{q\chi_e}{1+\chi_e} \int \delta^3(\mathbf{r}) d\tau = -q \frac{\chi_e}{1+\chi_e}.$$

Problem 4.36

$\mathbf{E}^{\parallel}$ is continuous (Eq. 4.29); $D_{\perp}$ is continuous (Eq. 4.26, with $\sigma_f = 0$). So $E_{x_1} = E_{x_2}$, $D_{y_1} = D_{y_2} \Rightarrow \epsilon_1 E_{y_1} = \epsilon_2 E_{y_2}$, and hence

$$\frac{\tan \theta_2}{\tan \theta_1} = \frac{E_{x_2}/E_{y_2}}{E_{x_1}/E_{y_1}} = \frac{E_{y_1}}{E_{y_2}} = \frac{\epsilon_2}{\epsilon_1}. \quad \text{qed}$$

If 1 is air and 2 is dielectric, $\tan \theta_2 / \tan \theta_1 = \epsilon_2 / \epsilon_0 > 1$, and the field lines bend *away* from the normal. This is the opposite of light rays, so a convex “lens” would *defocus* the field lines.

Problem 4.37

In view of Eq. 4.39, the *net* dipole moment at the center is $\mathbf{p}' = \mathbf{p} - \frac{\chi_e}{1+\chi_e} \mathbf{p} = \frac{1}{1+\chi_e} \mathbf{p} = \frac{1}{\epsilon_r} \mathbf{p}$. We want the potential produced by $\mathbf{p}'$ (at the center) and σ_b (at R). Use separation of variables:

$$\left\{ \begin{array}{l} \text{Outside: } V(r, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos \theta) \\ \text{Inside: } V(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{\epsilon_r r^2} + \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta) \quad (\text{Eqs. 3.66, 3.102}) \end{array} \right. \quad (\text{Eq. 3.72})$$

$$V \text{ continuous at } R \Rightarrow \left\{ \begin{array}{l} \frac{B_l}{R^{l+1}} = A_l R^l, \quad \text{or } B_l = R^{2l+1} A_l \quad (l \neq 1) \\ \frac{B_1}{R^2} = \frac{1}{4\pi\epsilon_0} \frac{p}{\epsilon_r R^2} + A_1 R, \text{ or } B_1 = \frac{p}{4\pi\epsilon_0 \epsilon_r} + A_1 R^3 \end{array} \right.$$

$$\begin{aligned} \left. \frac{\partial V}{\partial r} \right|_{R+} - \left. \frac{\partial V}{\partial r} \right|_{R-} &= -\sum (l+1) \frac{B_l}{R^{l+2}} P_l(\cos \theta) + \frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{\epsilon_r R^3} - \sum l A_l R^{l-1} P_l(\cos \theta) = -\frac{1}{\epsilon_0} \sigma_b \\ &= -\frac{1}{\epsilon_0} \mathbf{P} \cdot \hat{\mathbf{r}} = -\frac{1}{\epsilon_0} (\epsilon_0 \chi_e \mathbf{E} \cdot \hat{\mathbf{r}}) = \chi_e \left. \frac{\partial V}{\partial r} \right|_{R-} = \chi_e \left\{ -\frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{\epsilon_r R^3} + \sum l A_l R^{l-1} P_l(\cos \theta) \right\}. \end{aligned}$$

$$-(l+1) \frac{B_l}{R^{l+2}} - l A_l R^{l-1} = \chi_e l A_l R^{l-1} \quad (l \neq 1); \quad \text{or } -(2l+1) A_l R^{l-1} = \chi_e l A_l R^{l-1} \Rightarrow A_l = 0 \quad (l \neq 1).$$

$$\text{For } l = 1: -2 \frac{B_1}{R^3} + \frac{1}{4\pi\epsilon_0} \frac{2p}{\epsilon_r R^3} - A_1 = \chi_e \left(-\frac{1}{4\pi\epsilon_0} \frac{2p}{\epsilon_r R^3} + A_1 \right) - B_1 + \frac{p}{4\pi\epsilon_0 \epsilon_r} - \frac{A_1 R^3}{2} = -\frac{1}{4\pi\epsilon_0} \frac{\chi_e p}{\epsilon_r} + \chi_e \frac{A_1 R^3}{2};$$

$$-\frac{p}{4\pi\epsilon_0\epsilon_r} - A_1 R^3 + \frac{p}{4\pi\epsilon_0\epsilon_r} - \frac{A_1 R^3}{2} = -\frac{1}{4\pi\epsilon_0} \frac{\chi_e p}{\epsilon_r} + \chi_e \frac{A_1 R^3}{2} \Rightarrow \frac{A_1 R^3}{2} (3 + \chi_e) = \frac{1}{4\pi\epsilon_0} \frac{\chi_e p}{\epsilon_r}.$$

$$\Rightarrow A_1 = \frac{1}{4\pi\epsilon_0} \frac{2\chi_e p}{R^3 \epsilon_r (3 + \chi_e)} = \frac{1}{4\pi\epsilon_0} \frac{2(\epsilon_r - 1)p}{R^3 \epsilon_r (\epsilon_r + 2)}; \quad B_1 = \frac{p}{4\pi\epsilon_0\epsilon_r} \left[1 + \frac{2(\epsilon_r - 1)}{(\epsilon_r + 2)} \right] = \frac{p}{4\pi\epsilon_0\epsilon_r} \frac{3\epsilon_r}{\epsilon_r + 2}.$$

$$V(r, \theta) = \left(\frac{p \cos \theta}{4\pi\epsilon_0 r^2} \right) \left(\frac{3}{\epsilon_r + 2} \right) (r \geq R).$$

Meanwhile, for $r \leq R$, $V(r, \theta) = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{\epsilon_r r^2} + \frac{1}{4\pi\epsilon_0} \frac{pr \cos \theta}{R^3} \frac{2(\epsilon_r - 1)}{\epsilon_r (\epsilon_r + 2)}$

$$= \left[\frac{p \cos \theta}{4\pi\epsilon_0 r^2 \epsilon_r} \left[1 + 2 \left(\frac{\epsilon_r - 1}{\epsilon_r + 2} \right) \frac{r^3}{R^3} \right] \right] (r \leq R).$$

Problem 4.38

Given two solutions, V_1 (and $\mathbf{E}_1 = -\nabla V_1$, $\mathbf{D}_1 = \epsilon \mathbf{E}_1$) and V_2 ($\mathbf{E}_2 = -\nabla V_2$, $\mathbf{D}_2 = \epsilon \mathbf{E}_2$), define $V_3 \equiv V_2 - V_1$ ($\mathbf{E}_3 = \mathbf{E}_2 - \mathbf{E}_1$, $\mathbf{D}_3 = \mathbf{D}_2 - \mathbf{D}_1$).

$$\int_V \nabla \cdot (V_3 \mathbf{D}_3) d\tau = \int_S V_3 \mathbf{D}_3 \cdot d\mathbf{a} = 0, \quad (V_3 = 0 \text{ on } S), \text{ so } \int (\nabla V_3) \cdot \mathbf{D}_3 d\tau + \int V_3 (\nabla \cdot \mathbf{D}_3) d\tau = 0.$$

But $\nabla \cdot \mathbf{D}_3 = \nabla \cdot \mathbf{D}_2 - \nabla \cdot \mathbf{D}_1 = \rho_f - \rho_f = 0$, and $\nabla V_3 = \nabla V_2 - \nabla V_1 = -\mathbf{E}_2 + \mathbf{E}_1 = -\mathbf{E}_3$, so $\int \mathbf{E}_3 \cdot \mathbf{D}_3 d\tau = 0$. But $\mathbf{D}_3 = \mathbf{D}_2 - \mathbf{D}_1 = \epsilon \mathbf{E}_2 - \epsilon \mathbf{E}_1 = \epsilon \mathbf{E}_3$, so $\int \epsilon (E_3)^2 d\tau = 0$. But $\epsilon > 0$, so $\mathbf{E}_3 = 0$, so $V_2 - V_1 = \text{constant}$. But at surface, $V_2 = V_1$, so $V_2 = V_1$ everywhere. qed

Problem 4.39

(a) Proposed potential: $V(r) = V_0 \frac{R}{r}$. If so, then $\mathbf{E} = -\nabla V = V_0 \frac{R}{r^2} \hat{\mathbf{r}}$, in which case $\mathbf{P} = \epsilon_0 \chi_e V_0 \frac{R}{r^2} \hat{\mathbf{r}}$,

in the region $z < 0$. ($\mathbf{P} = 0$ for $z > 0$, of course.) Then $\sigma_b = \epsilon_0 \chi_e V_0 \frac{R}{R^2} (\hat{\mathbf{r}} \cdot \hat{\mathbf{n}}) = \left[-\frac{\epsilon_0 \chi_e V_0}{R} \right]$. (Note: $\hat{\mathbf{n}}$ points out of dielectric $\Rightarrow \hat{\mathbf{n}} = -\hat{\mathbf{r}}$.) This σ_b is on the surface at $r = R$. The flat surface $z = 0$ carries no bound charge, since $\hat{\mathbf{n}} = \hat{\mathbf{z}} \perp \hat{\mathbf{r}}$. Nor is there any volume bound charge (Eq. 4.39). If V is to have the required spherical symmetry, the *net* charge must be uniform:

$$\sigma_{\text{tot}} 4\pi R^2 = Q_{\text{tot}} = 4\pi\epsilon_0 R V_0 \quad (\text{since } V_0 = Q_{\text{tot}}/4\pi\epsilon_0 R), \text{ so } \sigma_{\text{tot}} = \epsilon_0 V_0/R. \text{ Therefore}$$

$$\sigma_f = \left\{ \begin{array}{l} (\epsilon_0 V_0/R), \text{ on northern hemisphere} \\ (\epsilon_0 V_0/R)(1 + \chi_e), \text{ on southern hemisphere} \end{array} \right\}.$$

(b) By construction, $\sigma_{\text{tot}} = \sigma_b + \sigma_f = \epsilon_0 V_0/R$ is uniform (on the northern hemisphere $\sigma_b = 0$, $\sigma_f = \epsilon_0 V_0/R$; on the southern hemisphere $\sigma_b = -\epsilon_0 \chi_e V_0/R$, so $\sigma_f = \epsilon V_0/R$). The potential of a uniformly charged sphere is

$$V_0 = \frac{Q_{\text{tot}}}{4\pi\epsilon_0 r} = \frac{\sigma_{\text{tot}}(4\pi R^2)}{4\pi\epsilon_0 r} = \frac{\epsilon_0 V_0}{R} \frac{R^2}{\epsilon_0 r} = V_0 \frac{R}{r}. \quad \checkmark$$

(c) Since everything is consistent, and the boundary conditions ($V = V_0$ at $r = R$, $V \rightarrow 0$ at ∞) are met, Prob. 4.38 guarantees that this is *the* solution.

(d) Figure (b) works the same way, but Fig. (a) does *not*: on the flat surface, $\mathbf{P}$ is *not* perpendicular to $\hat{\mathbf{n}}$, so we'd get bound charge on this surface, spoiling the symmetry.

Problem 4.40

$\mathbf{E}_{\text{ext}} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}$. Since the sphere is tiny, this is essentially constant, and hence $\mathbf{P} = \frac{\epsilon_0\chi_e}{1+\chi_e/3} \mathbf{E}_{\text{ext}}$ (Ex. 4.7).

$$\begin{aligned} \mathbf{F} &= \int \left(\frac{\epsilon_0\chi_e}{1+\chi_e/3} \right) \left(\frac{\lambda}{2\pi\epsilon_0 s} \right) \frac{d}{ds} \left(\frac{\lambda}{2\pi\epsilon_0 s} \right) \hat{\mathbf{s}} d\tau = \left(\frac{\epsilon_0\chi_e}{1+\chi_e/3} \right) \left(\frac{\lambda}{2\pi\epsilon_0} \right)^2 \left(\frac{1}{s} \right) \left(\frac{-1}{s^2} \right) \hat{\mathbf{s}} \int d\tau \\ &= \frac{-\chi_e}{1+\chi_e/3} \left(\frac{\lambda^2}{4\pi^2\epsilon_0} \right) \frac{1}{s^3} \frac{4}{3} \pi R^3 \hat{\mathbf{s}} = \boxed{- \left(\frac{\chi_e}{3+\chi_e} \right) \frac{\lambda^2 R^3}{\pi\epsilon_0 s^3} \hat{\mathbf{s}}}. \end{aligned}$$

Problem 4.41

The density of atoms is $N = \frac{1}{(4/3)\pi R^3}$. The macroscopic field $\mathbf{E}$ is $\mathbf{E}_{\text{self}} + \mathbf{E}_{\text{else}}$, where $\mathbf{E}_{\text{self}}$ is the average field over the sphere due to the atom itself.

$$\mathbf{p} = \alpha \mathbf{E}_{\text{else}} \Rightarrow \mathbf{P} = N\alpha \mathbf{E}_{\text{else}}.$$

[Actually, it is the field at the *center*, not the average over the sphere, that belongs here, but the two are in fact equal, as we found in Prob. 3.47d.] Now

$$\mathbf{E}_{\text{self}} = -\frac{1}{4\pi\epsilon_0} \frac{\mathbf{p}}{R^3}$$

(Eq. 3.105), so

$$\mathbf{E} = -\frac{1}{4\pi\epsilon_0} \frac{\alpha}{R^3} \mathbf{E}_{\text{else}} + \mathbf{E}_{\text{else}} = \left(1 - \frac{\alpha}{4\pi\epsilon_0 R^3} \right) \mathbf{E}_{\text{else}} = \left(1 - \frac{N\alpha}{3\epsilon_0} \right) \mathbf{E}_{\text{else}}.$$

So

$$\mathbf{P} = \frac{N\alpha}{(1 - N\alpha/3\epsilon_0)} \mathbf{E} = \epsilon_0\chi_e \mathbf{E},$$

and hence

$$\chi_e = \frac{N\alpha/\epsilon_0}{(1 - N\alpha/3\epsilon_0)}.$$

Solving for α :

$$\chi_e - \frac{N\alpha}{3\epsilon_0} \chi_e = \frac{N\alpha}{\epsilon_0} \Rightarrow \frac{N\alpha}{\epsilon_0} \left(1 + \frac{\chi_e}{3} \right) = \chi_e,$$

or

$$\alpha = \frac{\epsilon_0}{N} \frac{\chi_e}{(1 + \chi_e/3)} = \frac{3\epsilon_0}{N} \frac{\chi_e}{(3 + \chi_e)}. \quad \text{But } \chi_e = \epsilon_r - 1, \text{ so } \alpha = \frac{3\epsilon_0}{N} \left(\frac{\epsilon_r - 1}{\epsilon_r + 2} \right). \quad \text{qed}$$

Problem 4.42

For an ideal gas, $N = \text{Avagadro's number}/22.4 \text{ liters} = (6.02 \times 10^{23})/(22.4 \times 10^{-3}) = 2.7 \times 10^{25}$. $N\alpha/\epsilon_0 = (2.7 \times 10^{25})(4\pi\epsilon_0 \times 10^{-30})\beta/\epsilon_0 = 3.4 \times 10^{-4}\beta$, where β is the number listed in Table 4.1.

$$\left. \begin{array}{l} \text{H: } \beta = 0.667, N\alpha/\epsilon_0 = (3.4 \times 10^{-4})(0.67) = 2.3 \times 10^{-4}, \chi_e = 2.5 \times 10^{-4} \\ \text{He: } \beta = 0.205, N\alpha/\epsilon_0 = (3.4 \times 10^{-4})(0.21) = 7.1 \times 10^{-5}, \chi_e = 6.5 \times 10^{-5} \\ \text{Ne: } \beta = 0.396, N\alpha/\epsilon_0 = (3.4 \times 10^{-4})(0.40) = 1.4 \times 10^{-4}, \chi_e = 1.3 \times 10^{-4} \\ \text{Ar: } \beta = 1.64, N\alpha/\epsilon_0 = (3.4 \times 10^{-4})(1.64) = 5.6 \times 10^{-4}, \chi_e = 5.2 \times 10^{-4} \end{array} \right\} \text{agreement is quite good.}$$

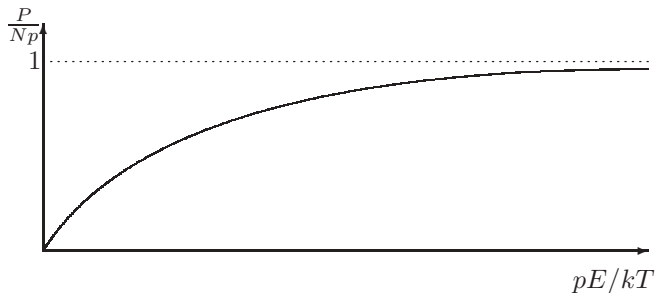
Problem 4.43

(a) Doing the (trivial) ϕ integral, and changing the remaining integration variable from θ to u ($du = pE \sin \theta d\theta$),

$$\begin{aligned}\langle u \rangle &= \frac{\int_{-pE}^{pE} u e^{-u/kT} du}{\int_{-pE}^{pE} e^{-u/kT} du} = \frac{(kT)^2 e^{-u/kT} [-(u/kT) - 1] \Big|_{-pE}^{pE}}{-kT e^{-u/kT} \Big|_{-pE}^{pE}} \\ &= kT \left\{ \frac{[e^{-pE/kT} - e^{pE/kT}] + [(pE/kT)e^{-pE/kT} + (pE/kT)e^{pE/kT}]}{e^{-pE/kT} - e^{pE/kT}} \right\} \\ &= kT - pE \left[\frac{e^{pE/kT} + e^{-pE/kT}}{e^{pE/kT} - e^{-pE/kT}} \right] = kT - pE \coth \left(\frac{pE}{kT} \right).\end{aligned}$$

$$\mathbf{P} = N \langle \mathbf{p} \rangle; \quad \mathbf{p} = \langle p \cos \theta \rangle \hat{\mathbf{E}} = \langle \mathbf{p} \cdot \mathbf{E} \rangle (\hat{\mathbf{E}}/E) = -\langle u \rangle (\hat{\mathbf{E}}/E); \quad P = Np \frac{-\langle u \rangle}{pE} = \boxed{Np \left\{ \coth \left(\frac{pE}{kT} \right) - \frac{kT}{pE} \right\}}.$$

Let $y \equiv P/Np$, $x \equiv pE/kT$. Then $y = \coth x - 1/x$. As $x \rightarrow 0$, $y = \left(\frac{1}{x} + \frac{x}{3} - \frac{x^3}{45} + \dots \right) - \frac{1}{x} = \frac{x}{3} - \frac{x^3}{45} + \dots \rightarrow 0$, so the graph starts at the origin, with an initial slope of $1/3$. As $x \rightarrow \infty$, $y \rightarrow \coth(\infty) = 1$, so the graph approaches asymptotically to $y = 1$ (see Figure).



(b) For small x , $y \approx \frac{1}{3}x$, so $\frac{P}{Np} \approx \frac{pE}{3kT}$, or $P \approx \frac{Np^2}{3kT} E = \epsilon_0 \chi_e E \Rightarrow P$ is proportional to E , and $\chi_e = \frac{Np^2}{3\epsilon_0 kT}$.

For water at $20^\circ = 293 \text{ K}$, $p = 6.1 \times 10^{-30} \text{ C m}$; $N = \frac{\text{molecules}}{\text{volume}} = \frac{\text{molecules}}{\text{mole}} \times \frac{\text{moles}}{\text{gram}} \times \frac{\text{grams}}{\text{volume}}$.
 $N = (6.0 \times 10^{23}) \times \left(\frac{1}{18} \right) \times (10^6) = 0.33 \times 10^{29}$; $\chi_e = \frac{(0.33 \times 10^{29})(6.1 \times 10^{-30})^2}{(3)(8.85 \times 10^{-12})(1.38 \times 10^{-23})(293)} = \boxed{12.}$ Table 4.2 gives an experimental value of 79, so it's pretty far off.

For water vapor at $100^\circ = 373 \text{ K}$, treated as an ideal gas, $\frac{\text{volume}}{\text{mole}} = (22.4 \times 10^{-3}) \times \left(\frac{373}{293} \right) = 2.85 \times 10^{-2} \text{ m}^3$.

$$N = \frac{6.0 \times 10^{23}}{2.85 \times 10^{-2}} = 2.11 \times 10^{25}; \quad \chi_e = \frac{(2.11 \times 10^{25})(6.1 \times 10^{-30})^2}{(3)(8.85 \times 10^{-12})(1.38 \times 10^{-23})(373)} = \boxed{5.7 \times 10^{-3}}.$$

Table 4.2 gives 5.9×10^{-3} , so this time the agreement is quite good.

Chapter 5

Magnetostatics

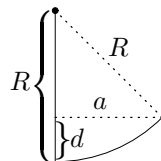
Problem 5.1

Since $\mathbf{v} \times \mathbf{B}$ points upward, and that is also the direction of the force, q must be positive. To find R , in terms of a and d , use the pythagorean theorem:

$$(R - d)^2 + a^2 = R^2 \Rightarrow R^2 - 2Rd + d^2 + a^2 = R^2 \Rightarrow R = \frac{a^2 + d^2}{2d}.$$

The cyclotron formula then gives

$$p = qBR = \boxed{qB \frac{a^2 + d^2}{2d}}.$$



Problem 5.2

The general solution is (Eq. 5.6):

$$y(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t) + \frac{E}{B}t + C_3; \quad z(t) = C_2 \cos(\omega t) - C_1 \sin(\omega t) + C_4.$$

(a) $y(0) = z(0) = 0$; $\dot{y}(0) = E/B$; $\dot{z}(0) = 0$. Use these to determine C_1 , C_2 , C_3 , and C_4 .

$y(0) = 0 \Rightarrow C_1 + C_3 = 0$; $\dot{y}(0) = \omega C_2 + E/B = E/B \Rightarrow C_2 = 0$; $z(0) = 0 \Rightarrow C_2 + C_4 = 0 \Rightarrow C_4 = 0$;

$\dot{z}(0) = 0 \Rightarrow C_1 = 0$, and hence also $C_3 = 0$. So $y(t) = Et/B$; $z(t) = 0$. Does this make sense? The magnetic force is $q(\mathbf{v} \times \mathbf{B}) = -q(E/B)\mathbf{B} \hat{\mathbf{z}} = -q\mathbf{E}$, which exactly cancels the electric force; since there is no net force, the particle moves in a straight line at constant speed. ✓

(b) Assuming it starts from the origin, so $C_3 = -C_1$, $C_4 = -C_2$, we have $\dot{z}(0) = 0 \Rightarrow C_1 = 0 \Rightarrow C_3 = 0$;

$$\dot{y}(0) = \frac{E}{2B} \Rightarrow C_2\omega + \frac{E}{B} = \frac{E}{2B} \Rightarrow C_2 = -\frac{E}{2\omega B} = -C_4; \quad y(t) = -\frac{E}{2\omega B} \sin(\omega t) + \frac{E}{B}t;$$

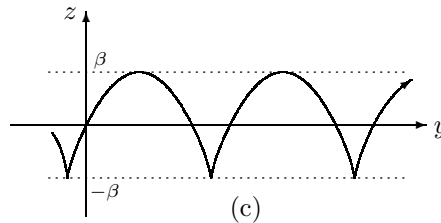
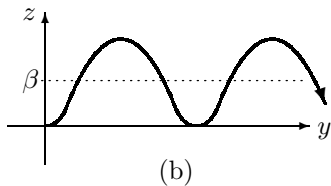
$$z(t) = -\frac{E}{2\omega B} \cos(\omega t) + \frac{E}{2\omega B}, \text{ or } \boxed{y(t) = \frac{E}{2\omega B} [2\omega t - \sin(\omega t)]; \quad z(t) = \frac{E}{2\omega B} [1 - \cos(\omega t)]}. \text{ Let } \beta \equiv E/2\omega B.$$

Then $y(t) = \beta [2\omega t - \sin(\omega t)]$; $z(t) = \beta [1 - \cos(\omega t)]$; $(y - 2\beta\omega t) = -\beta \sin(\omega t)$, $(z - \beta) = -\beta \cos(\omega t) \Rightarrow (y - 2\beta\omega t)^2 + (z - \beta)^2 = \beta^2$. This is a circle of radius β whose center moves to the right at constant speed: $y_0 = 2\beta\omega t$; $z_0 = \beta$.

$$(c) \dot{z}(0) = \dot{y}(0) = \frac{E}{B} \Rightarrow -C_1\omega = \frac{E}{B} \Rightarrow C_1 = -C_3 = -\frac{E}{\omega B}; \quad C_2\omega + \frac{E}{B} = \frac{E}{B} \Rightarrow C_2 = C_4 = 0.$$

$$y(t) = -\frac{E}{\omega B} \cos(\omega t) + \frac{E}{B}t + \frac{E}{\omega B}; \quad z(t) = \frac{E}{\omega B} \sin(\omega t). \quad \boxed{y(t) = \frac{E}{\omega B} [1 + \omega t - \cos(\omega t)]; \quad z(t) = \frac{E}{\omega B} \sin(\omega t).}$$

Let $\beta \equiv E/\omega B$; then $[y - \beta(1 + \omega t)] = -\beta \cos(\omega t)$, $z = \beta \sin(\omega t)$; $[y - \beta(1 + \omega t)]^2 + z^2 = \beta^2$. This is a circle of radius β whose center is at $y_0 = \beta(1 + \omega t)$, $z_0 = 0$.



Problem 5.3

(a) From Eq. 5.2, $\mathbf{F} = q[\mathbf{E} + (\mathbf{v} \times \mathbf{B})] = 0 \Rightarrow E = vB \Rightarrow \boxed{v = \frac{E}{B}}.$

(b) From Eq. 5.3, $mv = qBR \Rightarrow \frac{q}{m} = \frac{v}{BR} = \boxed{\frac{E}{B^2 R}}.$

Problem 5.4

Suppose I flows counterclockwise (if not, change the sign of the answer). The force on the left side (toward the left) cancels the force on the right side (toward the right); the force on the top is $IaB = Iak(a/2) = Ika^2/2$, (pointing upward), and the force on the bottom is $IaB = -Ika^2/2$ (also upward). So the net force is $\mathbf{F} = \boxed{Ika^2 \hat{\mathbf{z}}}$.

Problem 5.5

(a) $\boxed{K = \frac{I}{2\pi a}}$, because the length-perpendicular-to-flow is the circumference.

(b) $J = \frac{\alpha}{s} \Rightarrow I = \int J da = \alpha \int \frac{1}{s} ds d\phi = 2\pi\alpha \int ds = 2\pi\alpha a \Rightarrow \alpha = \frac{I}{2\pi a}; J = \boxed{\frac{I}{2\pi as}}.$

Problem 5.6

(a) $v = \omega r$, so $\boxed{K = \sigma\omega r}.$ (b) $\mathbf{v} = \omega r \sin \theta \hat{\phi} \Rightarrow \boxed{\mathbf{J} = \rho\omega r \sin \theta \hat{\phi}},$ where $\rho \equiv Q/(4/3)\pi R^3$.

Problem 5.7

$\frac{d\mathbf{p}}{dt} = \frac{d}{dt} \int_V \rho \mathbf{r} d\tau = \int \left(\frac{\partial \rho}{\partial t} \right) \mathbf{r} d\tau = - \int (\nabla \cdot \mathbf{J}) \mathbf{r} d\tau$ (by the continuity equation). Now product rule #5 says $\nabla \cdot (x\mathbf{J}) = x(\nabla \cdot \mathbf{J}) + \mathbf{J} \cdot (\nabla x)$. But $\nabla x = \hat{\mathbf{x}}$, so $\nabla \cdot (x\mathbf{J}) = x(\nabla \cdot \mathbf{J}) + J_x$. Thus $\int_V (\nabla \cdot \mathbf{J}) x d\tau = \int_V \nabla \cdot (x\mathbf{J}) d\tau - \int_V J_x d\tau$. The first term is $\int_S x\mathbf{J} \cdot d\mathbf{a}$ (by the divergence theorem), and since $\mathbf{J}$ is entirely inside V , it is zero on the surface S . Therefore $\int_V (\nabla \cdot \mathbf{J}) x d\tau = - \int_V J_x d\tau$, or, combining this with the y and z components, $\int_V (\nabla \cdot \mathbf{J}) \mathbf{r} d\tau = - \int_V \mathbf{J} d\tau$. Or, referring back to the first line, $\frac{d\mathbf{p}}{dt} = \int \mathbf{J} d\tau$. qed

Here's a quicker method, if the distribution consists of a collection of point charges. Use Eqs. 5.30 and 3.100:

$$\int \mathbf{J} d\tau = \sum q_i \mathbf{v}_i = \frac{d}{dt} \sum q_i \mathbf{r}_i = \frac{d\mathbf{p}}{dt}.$$

Problem 5.8

(a) Use Eq. 5.37, with $z = R$, $\theta_2 = -\theta_1 = 45^\circ$, and four sides: $B = \boxed{\frac{\sqrt{2}\mu_0 I}{\pi R}}$.

(b) $z = R$, $\theta_2 = -\theta_1 = \frac{\pi}{n}$, and n sides: $B = \boxed{\frac{n\mu_0 I}{2\pi R} \sin(\pi/n)}$.

(c) For small θ , $\sin \theta \approx \theta$. So as $n \rightarrow \infty$, $B \rightarrow \frac{n\mu_0 I}{2\pi R} \left(\frac{\pi}{n}\right) = \boxed{\frac{\mu_0 I}{2R}}$ (same as Eq. 5.41, with $z = 0$).

Problem 5.9

(a) The straight segments produce no field at P . The two quarter-circles give $B = \boxed{\frac{\mu_0 I}{8} \left(\frac{1}{a} - \frac{1}{b}\right)}$ (out).

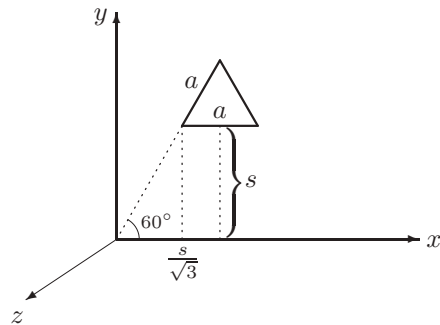
(b) The two half-lines are the same as one infinite line: $\frac{\mu_0 I}{2\pi R}$; the half-circle contributes $\frac{\mu_0 I}{4R}$.

So $B = \boxed{\frac{\mu_0 I}{4R} \left(1 + \frac{2}{\pi}\right)}$ (into the page).

Problem 5.10

(a) The forces on the two sides cancel. At the bottom, $B = \frac{\mu_0 I}{2\pi s} \Rightarrow F = \left(\frac{\mu_0 I}{2\pi s}\right) Ia = \frac{\mu_0 I^2 a}{2\pi s}$ (up). At the top, $B = \frac{\mu_0 I}{2\pi(s+a)} \Rightarrow F = \frac{\mu_0 I^2 a}{2\pi(s+a)}$ (down). The net force is $\boxed{\frac{\mu_0 I^2 a^2}{2\pi s(s+a)}}$ (up).

(b) The force on the bottom is the same as before, $\mu_0 I^2 a / 2\pi s$ (up). On the left side, $\mathbf{B} = \frac{\mu_0 I}{2\pi y} \hat{\mathbf{z}}$;
 $d\mathbf{F} = I(d\mathbf{l} \times \mathbf{B}) = I(dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}) \times \left(\frac{\mu_0 I}{2\pi y} \hat{\mathbf{z}}\right) = \frac{\mu_0 I^2}{2\pi y} (-dx \hat{\mathbf{y}} + dy \hat{\mathbf{x}})$. But the x component cancels the corresponding term from the right side, and $F_y = -\frac{\mu_0 I^2}{2\pi} \int_{s/\sqrt{3}}^{(s/\sqrt{3}+a/2)} \frac{1}{y} dx$. Here $y = \sqrt{3}x$, so
 $F_y = -\frac{\mu_0 I^2}{2\sqrt{3}\pi} \ln \left(\frac{s/\sqrt{3} + a/2}{s/\sqrt{3}}\right) = -\frac{\mu_0 I^2}{2\sqrt{3}\pi} \ln \left(1 + \frac{\sqrt{3}a}{2s}\right)$. The force on the right side is the same, so the net force on the triangle is $\boxed{\frac{\mu_0 I^2}{2\pi} \left[\frac{a}{s} - \frac{2}{\sqrt{3}} \ln \left(1 + \frac{\sqrt{3}a}{2s}\right)\right]}$.



Problem 5.11

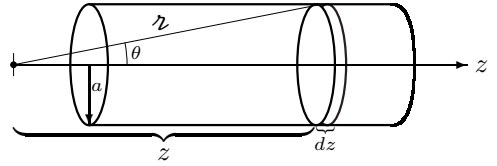
Use Eq. 5.41 for a ring of width dz , with $I \rightarrow nI dz$:

$$B = \frac{\mu_0 n I}{2} \int \frac{a^2}{(a^2 + z^2)^{3/2}} dz. \text{ But } z = a \cot \theta,$$

$$\text{so } dz = -\frac{a}{\sin^2 \theta} d\theta, \text{ and } \frac{1}{(a^2 + z^2)^{3/2}} = \frac{\sin^3 \theta}{a^3}.$$

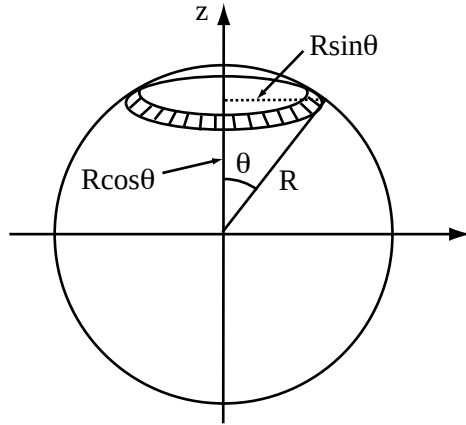
So

$$B = \frac{\mu_0 n I}{2} \int \frac{a^2 \sin^3 \theta}{a^3 \sin^2 \theta} (-a d\theta) = -\frac{\mu_0 n I}{2} \int \sin \theta d\theta = \frac{\mu_0 n I}{2} \cos \theta \Big|_{\theta_1}^{\theta_2} = \boxed{\frac{\mu_0 n I}{2} (\cos \theta_2 - \cos \theta_1)}.$$



For an infinite solenoid, $\theta_2 = 0$, $\theta_1 = \pi$, so $(\cos \theta_2 - \cos \theta_1) = 1 - (-1) = 2$, and $B = \boxed{\mu_0 n I}$. ✓

Problem 5.12



Field (at center of sphere) due to the ring at θ (see figure) is (Eq. 5.41):

$$dB = \frac{\mu_0 dI}{2} \frac{(R \sin \theta)^2}{[(R \sin \theta)^2 + (R \cos \theta)^2]^{3/2}} = \frac{\mu_0}{2R} \sin^2 \theta dI.$$

$$dI = K R d\theta, \quad K = \sigma v, \quad \sigma = \frac{Q}{4\pi R^2}, \quad v = \omega R \sin \theta,$$

so

$$dI = \frac{Q}{4\pi R^2} \omega R \sin \theta R d\theta = \frac{Q\omega}{4\pi} \sin \theta d\theta.$$

$$B = \frac{\mu_0}{2R} \frac{Q\omega}{4\pi} \int_0^\pi \sin^3 \theta d\theta = \frac{\mu_0 Q\omega}{8\pi R} \left(\frac{4}{3}\right). \quad \mathbf{B} = \boxed{\frac{\mu_0 Q\omega}{6\pi R} \hat{\mathbf{z}}}.$$

Problem 5.13

Magnetic attraction per unit length (Eqs. 5.40 and 5.13): $f_m = \frac{\mu_0}{2\pi} \frac{\lambda^2 v^2}{d}$.

Electric field of one wire (Eq. 2.9): $E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{s}$. Electric repulsion per unit length on the other wire:

$$f_e = \frac{1}{2\pi\epsilon_0} \frac{\lambda^2}{d}. \text{ They balance when } \mu_0 v^2 = \frac{1}{\epsilon_0}, \text{ or } \boxed{v = \frac{1}{\sqrt{\epsilon_0 \mu_0}}}. \text{ Putting in the numbers,}$$

$v = \frac{1}{\sqrt{(8.85 \times 10^{-12})(4\pi \times 10^{-7})}} = \boxed{3.00 \times 10^8 \text{ m/s.}}$ This is precisely the *speed of light(!)*, so in fact you could *never* get the wires going fast enough; the electric force always dominates.

Problem 5.14

$$(a) \oint \mathbf{B} \cdot d\mathbf{l} = B 2\pi s = \mu_0 I_{\text{enc}} \Rightarrow \boxed{\mathbf{B} = \begin{cases} \mathbf{0}, & \text{for } s < a; \\ \frac{\mu_0 I}{2\pi s} \hat{\phi}, & \text{for } s > a. \end{cases}}$$

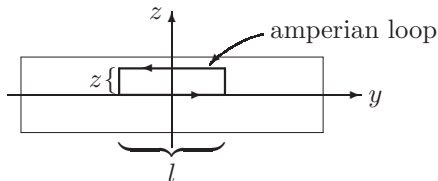
$$(b) J = ks; \quad I = \int_0^a J da = \int_0^a ks(2\pi s) ds = \frac{2\pi ka^3}{3} \Rightarrow k = \frac{3I}{2\pi a^3}. \quad I_{\text{enc}} = \int_0^s J da = \int_0^s k\bar{s}(2\pi\bar{s}) d\bar{s} = \frac{2\pi ks^3}{3} = I \frac{s^3}{a^3}, \text{ for } s < a; \quad I_{\text{enc}} = I, \text{ for } s > a. \text{ So } \boxed{\mathbf{B} = \begin{cases} \frac{\mu_0 I s^2}{2\pi a^3} \hat{\phi}, & \text{for } s < a; \\ \frac{\mu_0 I}{2\pi s} \hat{\phi}, & \text{for } s > a. \end{cases}}$$

Problem 5.15

By the right-hand-rule, the field points in the $-\hat{\mathbf{y}}$ direction for $z > 0$, and in the $+\hat{\mathbf{y}}$ direction for $z < 0$. At $z = 0$, $B = 0$. Use the amperian loop shown:

$$\oint \mathbf{B} \cdot d\mathbf{l} = Bl = \mu_0 I_{\text{enc}} = \mu_0 lzJ \Rightarrow \boxed{\mathbf{B} = -\mu_0 Jz \hat{\mathbf{y}}} \quad (-a < z < a). \quad \text{If } z > a, I_{\text{enc}} = \mu_0 laJ,$$

$$\text{so } \boxed{\mathbf{B} = \begin{cases} -\mu_0 Ja \hat{\mathbf{y}}, & \text{for } z > +a; \\ +\mu_0 Ja \hat{\mathbf{y}}, & \text{for } z < -a. \end{cases}}$$



Problem 5.16

The field inside a solenoid is $\mu_0 nI$, and outside it is zero. The outer solenoid's field points to the right ($\hat{\mathbf{z}}$), whereas the inner one points to the left ($-\hat{\mathbf{z}}$). So: (i) $\boxed{\mathbf{B} = \mu_0 I(n_2 - n_1) \hat{\mathbf{z}}}$, (ii) $\boxed{\mathbf{B} = \mu_0 I n_2 \hat{\mathbf{z}}}$, (iii) $\boxed{\mathbf{B} = \mathbf{0}}$.

Problem 5.17

From Ex. 5.8, the top plate produces a field $\mu_0 K/2$ (aiming *out of the page*, for points above it, and *into the page*, for points below). The bottom plate produces a field $\mu_0 K/2$ (aiming *into the page*, for points above it, and *out of the page*, for points below). Above and below *both* plates the two fields cancel; *between* the plates they add up to $\mu_0 K$, pointing *in*.

(a) $\boxed{B = \mu_0 \sigma v \text{ (in)}}$ between the plates, $\boxed{B = 0}$ elsewhere.

(b) The Lorentz force law says $\mathbf{F} = \int (\mathbf{K} \times \mathbf{B}) da$, so the force *per unit area* is $\mathbf{f} = \mathbf{K} \times \mathbf{B}$. Here $K = \sigma v$, to the right, and $\mathbf{B}$ (the field of the lower plate) is $\mu_0 \sigma v/2$, into the page. So $\boxed{f_m = \mu_0 \sigma^2 v^2/2 \text{ (up)}}$.

(c) The electric field of the lower plate is $\sigma/2\epsilon_0$; the electric force per unit area on the upper plate is $\boxed{f_e = \sigma^2/2\epsilon_0 \text{ (down)}}$. They balance if $\mu_0 v^2 = 1/\epsilon_0$, or $\boxed{v = 1/\sqrt{\epsilon_0 \mu_0} = c}$ (the speed of light), as in Prob. 5.13.

Problem 5.18

We might as well orient the axes so the field point $\mathbf{r}$ lies on the y axis: $\mathbf{r} = (0, y, 0)$. Consider a source point at (x', y', z') on loop #1:

$$\boldsymbol{r} = -x' \hat{\mathbf{x}} + (y - y') \hat{\mathbf{y}} - z' \hat{\mathbf{z}}; \quad d\mathbf{l}' = dx' \hat{\mathbf{x}} + dy' \hat{\mathbf{y}};$$

$$d\mathbf{l}' \times \mathbf{z} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ dx' & dy' & 0 \\ -x' & (y-y') & -z' \end{vmatrix} = (-z' dy') \hat{\mathbf{x}} + (z' dx') \hat{\mathbf{y}} + [(y-y') dx' + x' dy'] \hat{\mathbf{z}}.$$

$$d\mathbf{B}_1 = \frac{\mu_0 I}{4\pi} \frac{d\mathbf{l}' \times \mathbf{z}}{r^3} = \frac{\mu_0 I}{4\pi} \frac{(-z' dy') \hat{\mathbf{x}} + (z' dx') \hat{\mathbf{y}} + [(y-y') dx' + x' dy'] \hat{\mathbf{z}}}{[(x')^2 + (y-y')^2 + (z')^2]^{3/2}}.$$

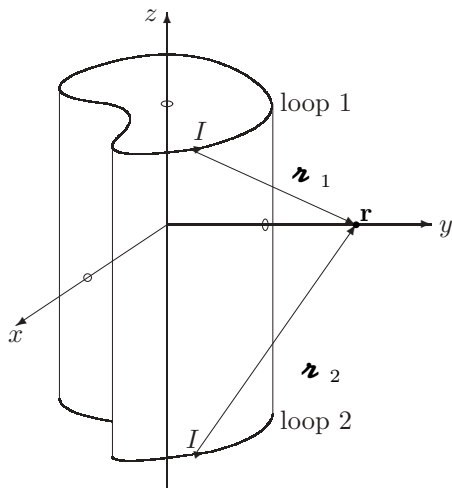
Now consider the symmetrically placed source element on loop #2, at $(x', y', -z')$. Since z' changes sign, while everything else is the same, the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components from $d\mathbf{B}_1$ and $d\mathbf{B}_2$ cancel, leaving only a $\hat{\mathbf{z}}$ component. $\square$

With this, Ampère's law yields immediately:

$$\mathbf{B} = \begin{cases} \mu_0 n I \hat{\mathbf{z}}, & \text{inside the solenoid;} \\ 0, & \text{outside} \end{cases}$$

(the same as for a circular solenoid—Ex. 5.9).

For the toroid, $N/2\pi s = n$ (the number of turns per unit length), so Eq. 5.60 yields $B = \mu_0 n I$ inside, and zero outside, consistent with the solenoid. [Note: $N/2\pi s = n$ applies only if the toroid is large in circumference, so that s is essentially constant over the cross-section.]



Problem 5.19

[It doesn't matter.] According to Theorem 2, in Sect. 1.6.2, $\int \mathbf{J} \cdot d\mathbf{a}$ is independent of surface, for any given boundary line, provided that $\mathbf{J}$ is divergenceless, which it *is*, for steady currents (Eq. 5.33).

Problem 5.20

$$(a) \rho = \frac{\text{charge}}{\text{volume}} = \frac{\text{charge}}{\text{atom}} \cdot \frac{\text{atoms}}{\text{mole}} \cdot \frac{\text{moles}}{\text{gram}} \cdot \frac{\text{grams}}{\text{volume}} = (e)(N) \left(\frac{1}{M} \right) (d), \text{ where}$$

$$\begin{aligned} e &= \text{charge of electron} &= 1.6 \times 10^{-19} \text{ C}, \\ N &= \text{Avogadro's number} &= 6.0 \times 10^{23} \text{ mole}, \\ M &= \text{atomic mass of copper} &= 64 \text{ gm/mole}, \\ d &= \text{density of copper} &= 9.0 \text{ gm/cm}^3. \end{aligned}$$

$$\rho = (1.6 \times 10^{-19})(6.0 \times 10^{23}) \left(\frac{9.0}{64} \right) = \boxed{1.4 \times 10^4 \text{ C/cm}^3}.$$

$$(b) J = \frac{I}{\pi s^2} = \rho v \Rightarrow v = \frac{I}{\pi s^2 \rho} = \frac{1}{\pi (2.5 \times 10^{-3})^2 (1.4 \times 10^4)} = \boxed{9.1 \times 10^{-3} \text{ cm/s}}, \text{ or about 33 cm/hr. This is astonishingly small—literally slower than a snail's pace.}$$

$$(c) \text{ From Eq. 5.40, } f_m = \frac{\mu_0}{2\pi} \left(\frac{I_1 I_2}{d} \right) = \frac{(4\pi \times 10^{-7})}{2\pi} = \boxed{2 \times 10^{-7} \text{ N/cm.}}$$

$$(d) E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{d}; \quad f_e = \frac{1}{2\pi\epsilon_0} \left(\frac{\lambda_1 \lambda_2}{d} \right) = \frac{1}{v^2} \frac{1}{2\pi\epsilon_0} \left(\frac{I_1 I_2}{d} \right) = \left(\frac{c^2}{v^2} \right) \frac{\mu_0}{2\pi} \left(\frac{I_1 I_2}{d} \right) = \frac{c^2}{v^2} f_m, \text{ where}$$

$$c \equiv 1/\sqrt{\epsilon_0 \mu_0} = 3.00 \times 10^8 \text{ m/s. Here } \frac{f_e}{f_m} = \frac{c^2}{v^2} = \left(\frac{3.0 \times 10^{10}}{9.1 \times 10^{-3}} \right)^2 = \boxed{1.1 \times 10^{25}}.$$

$$f_e = (1.1 \times 10^{25})(2 \times 10^{-7}) = \boxed{2 \times 10^{18} \text{ N/cm.}}$$

Problem 5.21

Ampère's law says $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$. Together with the continuity equation (5.29) this gives $\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 \nabla \cdot \mathbf{J} = -\mu_0 \partial \rho / \partial t$, which is *inconsistent* with $\text{div}(\text{curl})=0$ unless ρ is constant (magnetostatics). The other Maxwell equations are OK: $\nabla \times \mathbf{E} = \mathbf{0} \Rightarrow \nabla \cdot (\nabla \times \mathbf{E}) = 0$ ($\checkmark$), and as for the two divergence equations, there is no relevant vanishing second derivative (the other one is $\text{curl}(\text{grad})$, which doesn't involve the divergence).

Problem 5.22

At this stage I'd expect no changes in Gauss's law or Ampère's law. The divergence of $\mathbf{B}$ would take the form $\boxed{\nabla \cdot \mathbf{B} = \alpha_0 \rho_m}$, where ρ_m is the density of magnetic charge, and α_0 is some constant (analogous to ϵ_0 and μ_0). The curl of $\mathbf{E}$ becomes $\boxed{\nabla \times \mathbf{E} = \beta_0 \mathbf{J}_m}$, where $\mathbf{J}_m$ is the magnetic current density (representing the flow of magnetic charge), and β_0 is another constant. Presumably magnetic charge is conserved, so ρ_m and $\mathbf{J}_m$ satisfy a continuity equation: $\nabla \cdot \mathbf{J}_m = -\partial \rho_m / \partial t$.

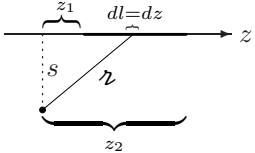
As for the Lorentz force law, one might guess something of the form $q_m[\mathbf{B} + (\mathbf{v} \times \mathbf{E})]$ (where q_m is the magnetic charge). But this is dimensionally impossible, since E has the same units as vB . Evidently we need to divide $(\mathbf{v} \times \mathbf{E})$ by something with the dimensions of velocity-squared. The natural candidate is

$$c^2 = 1/\epsilon_0 \mu_0: \quad \boxed{\mathbf{F} = q_e [\mathbf{E} + (\mathbf{v} \times \mathbf{B})] + q_m \left[\mathbf{B} - \frac{1}{c^2} (\mathbf{v} \times \mathbf{E}) \right]}.$$

In this form the magnetic analog to Coulomb's law reads $\mathbf{F} = \frac{\alpha_0}{4\pi} \frac{q_{m1} q_{m2}}{r^2} \hat{\mathbf{r}}$, so to determine α_0 we would first introduce (arbitrarily) a unit of magnetic charge, then measure the force between unit charges at a given separation. [For further details, and an explanation of the minus sign in the force law, see Prob. 7.38.]

Problem 5.23

$$\begin{aligned} \mathbf{A} &= \frac{\mu_0}{4\pi} \int \frac{I \hat{\mathbf{z}}}{z} dz = \frac{\mu_0 I}{4\pi} \hat{\mathbf{z}} \int_{z_1}^{z_2} \frac{dz}{\sqrt{z^2 + s^2}} \\ &= \frac{\mu_0 I}{4\pi} \hat{\mathbf{z}} \left[\ln \left(z + \sqrt{z^2 + s^2} \right) \right] \Big|_{z_1}^{z_2} = \boxed{\frac{\mu_0 I}{4\pi} \ln \left[\frac{z_2 + \sqrt{(z_2)^2 + s^2}}{z_1 + \sqrt{(z_1)^2 + s^2}} \right] \hat{\mathbf{z}}} \end{aligned}$$



$$\begin{aligned} \mathbf{B} &= \nabla \times \mathbf{A} = -\frac{\partial A}{\partial s} \hat{\phi} = -\frac{\mu_0 I}{4\pi} \left[\frac{1}{z_2 + \sqrt{(z_2)^2 + s^2}} \frac{s}{\sqrt{(z_2)^2 + s^2}} - \frac{1}{z_1 + \sqrt{(z_1)^2 + s^2}} \frac{s}{\sqrt{(z_1)^2 + s^2}} \right] \hat{\phi} \\ &= -\frac{\mu_0 I s}{4\pi} \left[\frac{z_2 - \sqrt{(z_2)^2 + s^2}}{(z_2)^2 - [(z_2)^2 + s^2]} \frac{1}{\sqrt{(z_2)^2 + s^2}} - \frac{z_1 - \sqrt{(z_1)^2 + s^2}}{z_1^2 - [(z_1)^2 + s^2]} \frac{1}{\sqrt{(z_1)^2 + s^2}} \right] \hat{\phi} \\ &= -\frac{\mu_0 I s}{4\pi} \left(-\frac{1}{s^2} \right) \left[\frac{z_2}{\sqrt{(z_2)^2 + s^2}} - 1 - \frac{z_1}{\sqrt{(z_1)^2 + s^2}} + 1 \right] \hat{\phi} = \frac{\mu_0 I}{4\pi s} \left[\frac{z_2}{\sqrt{(z_2)^2 + s^2}} - \frac{z_1}{\sqrt{(z_1)^2 + s^2}} \right] \hat{\phi}, \\ &\text{or, since } \sin \theta_1 = \frac{z_1}{\sqrt{(z_1)^2 + s^2}} \text{ and } \sin \theta_2 = \frac{z_2}{\sqrt{(z_2)^2 + s^2}}, \\ &= \boxed{\frac{\mu_0 I}{4\pi s} (\sin \theta_2 - \sin \theta_1) \hat{\phi}} \quad (\text{as in Eq. 5.37}). \end{aligned}$$

Problem 5.24

$$A_\phi = k \Rightarrow \mathbf{B} = \nabla \times \mathbf{A} = \frac{1}{s} \frac{\partial}{\partial s} (sk) \hat{\mathbf{z}} = \frac{k}{s} \hat{\mathbf{z}}; \quad \mathbf{J} = \frac{1}{\mu_0} (\nabla \times \mathbf{B}) = \frac{1}{\mu_0} \left[-\frac{\partial}{\partial s} \left(\frac{k}{s} \right) \right] \hat{\phi} = \boxed{\frac{k}{\mu_0 s^2} \hat{\phi}}.$$

Problem 5.25

$\nabla \cdot \mathbf{A} = -\frac{1}{2} \nabla \cdot (\mathbf{r} \times \mathbf{B}) = -\frac{1}{2} [\mathbf{B} \cdot (\nabla \times \mathbf{r}) - \mathbf{r} \cdot (\nabla \times \mathbf{B})] = 0$, since $\nabla \times \mathbf{B} = \mathbf{0}$ ($\mathbf{B}$ is uniform) and $\nabla \times \mathbf{r} = \mathbf{0}$ (Prob. 1.63). $\nabla \times \mathbf{A} = -\frac{1}{2} \nabla \times (\mathbf{r} \times \mathbf{B}) = -\frac{1}{2} [(\mathbf{B} \cdot \nabla) \mathbf{r} - (\mathbf{r} \cdot \nabla) \mathbf{B} + \mathbf{r}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{r})]$. But $(\mathbf{r} \cdot \nabla) \mathbf{B} = \mathbf{0}$ and $\nabla \cdot \mathbf{B} = 0$ (since $\mathbf{B}$ is uniform), and $\nabla \cdot \mathbf{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$. Finally, $(\mathbf{B} \cdot \nabla) \mathbf{r} = \left(B_x \frac{\partial}{\partial x} + B_y \frac{\partial}{\partial y} + B_z \frac{\partial}{\partial z} \right) (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}} = \mathbf{B}$. So $\nabla \times \mathbf{A} = -\frac{1}{2} (\mathbf{B} - 3\mathbf{B}) = \mathbf{B}$. qed

Problem 5.26

(a) $\mathbf{A}$ is parallel (or antiparallel) to $\mathbf{I}$, and is a function only of s (the distance from the wire). In cylindrical coordinates, then, $\mathbf{A} = A(s) \hat{\mathbf{z}}$, so $\mathbf{B} = \nabla \times \mathbf{A} = -\frac{\partial A}{\partial s} \hat{\phi} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$ (the field of an infinite wire). Therefore $\frac{\partial A}{\partial s} = -\frac{\mu_0 I}{2\pi s}$, and $\mathbf{A}(\mathbf{r}) = -\frac{\mu_0 I}{2\pi} \ln(s/a) \hat{\mathbf{z}}$ (the constant a is arbitrary; you could use 1, but then the units look fishy). $\nabla \cdot \mathbf{A} = \frac{\partial A_z}{\partial z} = 0$. ✓ $\nabla \times \mathbf{A} = -\frac{\partial A_z}{\partial s} \hat{\phi} = \frac{\mu_0 I}{2\pi s} \hat{\phi} = \mathbf{B}$. ✓

(b) Here Ampère's law gives $\oint \mathbf{B} \cdot d\mathbf{l} = B 2\pi s = \mu_0 I_{\text{enc}} = \mu_0 J \pi s^2 = \mu_0 \frac{I}{\pi R^2} \pi s^2 = \frac{\mu_0 I s^2}{R^2}$. $\mathbf{B} = \frac{\mu_0 I s}{2\pi R^2} \hat{\phi}$. $\frac{\partial A}{\partial s} = -\frac{\mu_0 I}{2\pi} \frac{s}{R^2} \Rightarrow \mathbf{A} = -\frac{\mu_0 I}{4\pi R^2} (s^2 - b^2) \hat{\mathbf{z}}$. Here b is again arbitrary, except that since $\mathbf{A}$ must be continuous at R , $-\frac{\mu_0 I}{2\pi} \ln(R/a) = -\frac{\mu_0 I}{4\pi R^2} (R^2 - b^2)$, which means that we must pick a and b such that

$$2 \ln(R/b) = 1 - (b/R)^2. \text{ I'll use } a = b = R. \text{ Then } \mathbf{A} = \begin{cases} -\frac{\mu_0 I}{4\pi R^2} (s^2 - R^2) \hat{\mathbf{z}}, & \text{for } s \leq R; \\ -\frac{\mu_0 I}{2\pi} \ln(s/R) \hat{\mathbf{z}}, & \text{for } s \geq R. \end{cases}$$

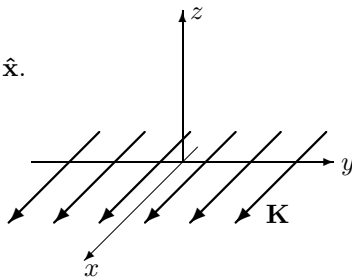
Problem 5.27

$\mathbf{K} = K \hat{\mathbf{x}} \Rightarrow \mathbf{B} = \pm \frac{\mu_0 K}{2} \hat{\mathbf{y}}$ (plus for $z < 0$, minus for $z > 0$).

$\mathbf{A}$ is parallel (or antiparallel) to $\mathbf{K}$, and depends only on z , so $\mathbf{A} = A(z) \hat{\mathbf{x}}$.

$$\mathbf{B} = \nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A(z) & 0 & 0 \end{vmatrix} = \frac{\partial A}{\partial z} \hat{\mathbf{y}} = \pm \frac{\mu_0 K}{2} \hat{\mathbf{y}}.$$

$$\mathbf{A} = -\frac{\mu_0 K}{2} |z| \hat{\mathbf{x}} \text{ will do the job—or this plus any constant.}$$



Problem 5.28

(a) $\nabla \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \nabla \cdot \left(\frac{\mathbf{J}}{r} \right) d\tau'$; $\nabla \cdot \left(\frac{\mathbf{J}}{r} \right) = \frac{1}{r} (\nabla \cdot \mathbf{J}) + \mathbf{J} \cdot \nabla \left(\frac{1}{r} \right)$. But the first term is zero, because $\mathbf{J}(\mathbf{r}')$ is a function of the *source* coordinates, not the *field* coordinates. And since $\mathbf{r} = \mathbf{r} - \mathbf{r}'$, $\nabla \left(\frac{1}{r} \right) = -\nabla' \left(\frac{1}{r} \right)$. So $\nabla \cdot \left(\frac{\mathbf{J}}{r} \right) = -\mathbf{J} \cdot \nabla' \left(\frac{1}{r} \right)$. But $\nabla' \cdot \left(\frac{\mathbf{J}}{r} \right) = \frac{1}{r} (\nabla' \cdot \mathbf{J}) + \mathbf{J} \cdot \nabla' \left(\frac{1}{r} \right)$, and $\nabla' \cdot \mathbf{J} = 0$ in magnetostatics (Eq. 5.33). So $\nabla \cdot \left(\frac{\mathbf{J}}{r} \right) = -\nabla' \cdot \left(\frac{\mathbf{J}}{r} \right)$, and hence, by the divergence theorem, $\nabla \cdot \mathbf{A} = -\frac{\mu_0}{4\pi} \int \nabla' \cdot \left(\frac{\mathbf{J}}{r} \right) d\tau' =$

$-\frac{\mu_0}{4\pi} \oint \frac{\mathbf{J}}{r} \cdot d\mathbf{a}'$, where the integral is now over the *surface* surrounding all the currents. But $\mathbf{J} = \mathbf{0}$ on this surface, so $\nabla \cdot \mathbf{A} = 0$. ✓

(b) $\nabla \times \mathbf{A} = \frac{\mu_0}{4\pi} \int \nabla \times \left(\frac{\mathbf{J}}{r} \right) d\tau' = \frac{\mu_0}{4\pi} \int \left[\frac{1}{r} (\nabla \times \mathbf{J}) - \mathbf{J} \times \nabla \left(\frac{1}{r} \right) \right] d\tau'$. But $\nabla \times \mathbf{J} = \mathbf{0}$ (since $\mathbf{J}$ is not a function of $\mathbf{r}$), and $\nabla \left(\frac{1}{r} \right) = -\frac{\hat{\mathbf{r}}}{r^2}$ (Eq. 1.101), so $\nabla \times \mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} \times \hat{\mathbf{r}}}{r^2} d\tau' = \mathbf{B}$. ✓

(c) $\nabla^2 \mathbf{A} = \frac{\mu_0}{4\pi} \int \nabla^2 \left(\frac{\mathbf{J}}{r} \right) d\tau'$. But $\nabla^2 \left(\frac{\mathbf{J}}{r} \right) = \mathbf{J} \nabla^2 \left(\frac{1}{r} \right)$ (once again, $\mathbf{J}$ is a *constant*, as far as differentiation with respect to $\mathbf{r}$ is concerned), and $\nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^3(\mathbf{r})$ (Eq. 1.102). So

$$\nabla^2 \mathbf{A} = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') [-4\pi \delta^3(\mathbf{r})] d\tau' = -\mu_0 \mathbf{J}(\mathbf{r}). \quad \checkmark$$

Problem 5.29

$$\mu_0 I = \oint \mathbf{B} \cdot d\mathbf{l} = - \int_{\mathbf{a}}^{\mathbf{b}} \nabla U \cdot d\mathbf{l} = -[U(\mathbf{b}) - U(\mathbf{a})] \text{ (by the gradient theorem), so } U(\mathbf{b}) \neq U(\mathbf{a}). \quad \text{qed}$$

For an infinite straight wire, $\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$. $U = -\frac{\mu_0 I \phi}{2\pi}$ would do the job, in the sense that

$-\nabla U = \frac{\mu_0 I}{2\pi} \nabla(\phi) = \frac{\mu_0 I}{2\pi} \frac{1}{s} \frac{\partial \phi}{\partial \phi} \hat{\phi} = \mathbf{B}$. But when ϕ advances by 2π , this function does *not* return to its initial value; it works (say) for $0 \leq \phi < 2\pi$, but at 2π it “jumps” back to zero.

Problem 5.30

Use Eq. 5.69, with $R \rightarrow \bar{r}$ and $\sigma \rightarrow \rho d\bar{r}$:

$$\begin{aligned} \mathbf{A} &= \frac{\mu_0 \omega \rho}{3} \frac{\sin \theta}{r^2} \hat{\phi} \int_0^r \bar{r}^4 d\bar{r} + \frac{\mu_0 \omega \rho}{3} r \sin \theta \hat{\phi} \int_r^R \bar{r} d\bar{r} \\ &= \left(\frac{\mu_0 \omega \rho}{3} \right) \sin \theta \left[\frac{1}{r^2} \left(\frac{r^5}{5} \right) + \frac{r}{2} (R^2 - r^2) \right] \hat{\phi} = \frac{\mu_0 \omega \rho}{2} r \sin \theta \left(\frac{R^2}{3} - \frac{r^2}{5} \right) \hat{\phi}. \\ \mathbf{B} &= \nabla \times \mathbf{A} = \frac{\mu_0 \omega \rho}{2} \left\{ \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta r \sin \theta \left(\frac{R^2}{3} - \frac{r^2}{5} \right) \right] \hat{\mathbf{r}} - \frac{1}{r} \frac{\partial}{\partial r} \left[r^2 \sin \theta \left(\frac{R^2}{3} - \frac{r^2}{5} \right) \right] \hat{\theta} \right\} \\ &= \boxed{\mu_0 \omega \rho \left[\left(\frac{R^2}{3} - \frac{r^2}{5} \right) \cos \theta \hat{\mathbf{r}} - \left(\frac{R^2}{3} - \frac{2r^2}{5} \right) \sin \theta \hat{\theta} \right]}. \end{aligned}$$

Problem 5.31

$$(a) \left\{ \begin{array}{l} -\frac{\partial W_z}{\partial x} = F_y \Rightarrow W_z(x, y, z) = -\int_0^x F_y(x', y, z) dx' + C_1(y, z). \\ \frac{\partial W_y}{\partial x} = F_z \Rightarrow W_y(x, y, z) = +\int_0^x F_z(x', y, z) dx' + C_2(y, z). \end{array} \right\}$$

These satisfy (ii) and (iii), for *any* C_1 and C_2 ; it remains to choose these functions so as to satisfy (i):

$$\begin{aligned} -\int_0^x \frac{\partial F_y(x', y, z)}{\partial y} dx' + \frac{\partial C_1}{\partial y} - \int_0^x \frac{\partial F_z(x', y, z)}{\partial z} dx' - \frac{\partial C_2}{\partial z} &= F_x(x, y, z). \quad \text{But } \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = 0, \text{ so} \\ \int_0^x \frac{\partial F_x(x', y, z)}{\partial x'} dx' + \frac{\partial C_1}{\partial y} - \frac{\partial C_2}{\partial z} &= F_x(x, y, z). \quad \text{Now } \int_0^x \frac{\partial F_x(x', y, z)}{\partial x'} dx' = F_x(x, y, z) - F_x(0, y, z), \text{ so} \end{aligned}$$

$\frac{\partial C_1}{\partial y} - \frac{\partial C_2}{\partial z} = F_x(0, y, z)$. We may as well pick $C_2 = 0$, $C_1(y, z) = \int_0^y F_x(0, y', z) dy'$, and we're done, with

$$W_x = 0; \quad W_y = \int_0^x F_z(x', y, z) dx'; \quad W_z = \int_0^y F_x(0, y', z) dy' - \int_0^x F_y(x', y, z) dx'.$$

$$\begin{aligned} \text{(b) } \nabla \times \mathbf{W} &= \left(\frac{\partial W_z}{\partial y} - \frac{\partial W_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial W_x}{\partial z} - \frac{\partial W_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial W_y}{\partial x} - \frac{\partial W_x}{\partial y} \right) \hat{\mathbf{z}} \\ &= \left[F_x(0, y, z) - \int_0^x \frac{\partial F_y(x', y, z)}{\partial y} dx' - \int_0^x \frac{\partial F_z(x', y, z)}{\partial z} dx' \right] \hat{\mathbf{x}} + [0 + F_y(x, y, z)] \hat{\mathbf{y}} + [F_z(x, y, z) - 0] \hat{\mathbf{z}}. \end{aligned}$$

But $\nabla \cdot \mathbf{F} = 0$, so the $\hat{\mathbf{x}}$ term is $\left[F_x(0, y, z) + \int_0^x \frac{\partial F_x(x', y, z)}{\partial x'} dx' \right] = F_x(0, y, z) + F_x(x, y, z) - F_x(0, y, z)$,

so $\nabla \times \mathbf{W} = \mathbf{F}$. ✓

$\nabla \cdot \mathbf{W} = \frac{\partial W_x}{\partial x} + \frac{\partial W_y}{\partial y} + \frac{\partial W_z}{\partial z} = 0 + \int_0^x \frac{\partial F_z(x', y, z)}{\partial y} dx' + \int_0^y \frac{\partial F_x(0, y', z)}{\partial z} dy' - \int_0^x \frac{\partial F_y(x', y, z)}{\partial z} dx' \neq 0$,
in general.

$$\text{(c) } W_y = \int_0^x x' dx' = \frac{x^2}{2}; \quad W_z = \int_0^y y' dy' - \int_0^x z dx' = \frac{y^2}{2} - zx.$$

$$\boxed{\mathbf{W} = \frac{x^2}{2} \hat{\mathbf{y}} + \left(\frac{y^2}{2} - zx \right) \hat{\mathbf{z}}.} \quad \nabla \times \mathbf{W} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & x^2/2 & (y^2/2 - zx) \end{vmatrix} = y \hat{\mathbf{x}} + z \hat{\mathbf{y}} + x \hat{\mathbf{z}} = \mathbf{F}. \quad \checkmark$$

Problem 5.32

(a) At the surface of the solenoid, $\mathbf{B}_{\text{above}} = 0$, $\mathbf{B}_{\text{below}} = \mu_0 n I \hat{\mathbf{z}} = \mu_0 K \hat{\mathbf{z}}$; $\hat{\mathbf{n}} = \hat{\mathbf{s}}$; so $\mathbf{K} \times \hat{\mathbf{n}} = -K \hat{\mathbf{z}}$. Evidently Eq. 5.76 holds. ✓

(b) In Eq. 5.69, both expressions reduce to $(\mu_0 R^2 \omega \sigma / 3) \sin \theta \hat{\phi}$ at the surface, so Eq. 5.77 is satisfied.
 $\left. \frac{\partial \mathbf{A}}{\partial r} \right|_{R^+} = \frac{\mu_0 R^4 \omega \sigma}{3} \left(-\frac{2 \sin \theta}{r^3} \right) \hat{\phi} \Big|_R = -\frac{2 \mu_0 R \omega \sigma}{3} \sin \theta \hat{\phi}; \quad \left. \frac{\partial \mathbf{A}}{\partial r} \right|_{R^-} = \frac{\mu_0 R \omega \sigma}{3} \sin \theta \hat{\phi}$. So the left side of Eq. 5.78 is $-\mu_0 R \omega \sigma \sin \theta \hat{\phi}$. Meanwhile $\mathbf{K} = \sigma \mathbf{v} = \sigma (\boldsymbol{\omega} \times \mathbf{r}) = \sigma \omega R \sin \theta \hat{\phi}$, so the right side of Eq. 5.78 is $-\mu_0 \sigma \omega R \sin \theta \hat{\phi}$, and the equation is satisfied.

Problem 5.33

Because $\mathbf{A}_{\text{above}} = \mathbf{A}_{\text{below}}$ at every point on the surface, it follows that $\frac{\partial \mathbf{A}}{\partial x}$ and $\frac{\partial \mathbf{A}}{\partial y}$ are the same above and below; any discontinuity is confined to the normal derivative.

$\mathbf{B}_{\text{above}} - \mathbf{B}_{\text{below}} = \left(-\frac{\partial A_{y\text{above}}}{\partial z} + \frac{\partial A_{y\text{below}}}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial A_{x\text{above}}}{\partial z} - \frac{\partial A_{x\text{below}}}{\partial z} \right) \hat{\mathbf{y}}$. But Eq. 5.76 says this equals $\mu_0 K (-\hat{\mathbf{y}})$. So $\frac{\partial A_{y\text{above}}}{\partial z} = \frac{\partial A_{y\text{below}}}{\partial z}$, and $\frac{\partial A_{x\text{above}}}{\partial z} - \frac{\partial A_{x\text{below}}}{\partial z} = -\mu_0 K$. Thus the *normal* derivative of the *component of A parallel to K* suffers a discontinuity $-\mu_0 K$, or, more compactly:

$$\boxed{\frac{\partial \mathbf{A}_{\text{above}}}{\partial n} - \frac{\partial \mathbf{A}_{\text{below}}}{\partial n} = -\mu_0 \mathbf{K}.}$$

Problem 5.34

(Same idea as Prob. 3.36.) Write $\mathbf{m} = (\mathbf{m} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} + (\mathbf{m} \cdot \hat{\boldsymbol{\theta}}) \hat{\boldsymbol{\theta}} = m \cos \theta \hat{\mathbf{r}} - m \sin \theta \hat{\boldsymbol{\theta}}$ (Fig. 5.54). Then $3(\mathbf{m} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{m} = 3m \cos \theta \hat{\mathbf{r}} - m \cos \theta \hat{\mathbf{r}} + m \sin \theta \hat{\boldsymbol{\theta}} = 2m \cos \theta \hat{\mathbf{r}} + m \sin \theta \hat{\boldsymbol{\theta}}$, and Eq. 5.89 $\Leftrightarrow$ Eq. 5.88. qed

Problem 5.35

$$\text{(a) } \mathbf{m} = I \mathbf{a} = \boxed{I \pi R^2 \hat{\mathbf{z}}.}$$

$$\text{(b) } \mathbf{B} \approx \boxed{\frac{\mu_0}{4\pi} \frac{I \pi R^2}{r^3} \left(2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}} \right).}$$

(c) On the z axis, $\theta = 0$, $r = z$, $\hat{\mathbf{r}} = \hat{\mathbf{z}}$ (for $z > 0$), so $\mathbf{B} \approx \frac{\mu_0 I R^2}{2z^3} \hat{\mathbf{z}}$ (for $z < 0$, $\theta = \pi$, $\hat{\mathbf{r}} = -\hat{\mathbf{z}}$, so the field is the same, with $|z|^3$ in place of z^3). The exact answer (Eq. 5.41) reduces (for $z \gg R$) to $B \approx \mu_0 I R^2 / 2|z|^3$, so they agree.

Problem 5.36

The field of one side is given by Eq. 5.37, with $s \rightarrow \sqrt{z^2 + (w/2)^2}$ and $\sin \theta_2 = -\sin \theta_1 = \frac{(w/2)}{\sqrt{z^2 + w^2/2}}$;

$B = \frac{\mu_0 I}{4\pi} \frac{w}{\sqrt{z^2 + (w^2/4)} \sqrt{z^2 + (w^2/2)}}$. To pick off the vertical component, multiply by $\sin \phi = \frac{(w/2)}{\sqrt{z^2 + (w^2/2)^2}}$; for all four

sides, multiply by 4: $\mathbf{B} = \frac{\mu_0 I}{2\pi} \frac{w^2}{(z^2 + w^2/4) \sqrt{z^2 + w^2/2}} \hat{\mathbf{z}}$.

For $z \gg w$, $\mathbf{B} \approx \frac{\mu_0 I w^2}{2\pi z^3} \hat{\mathbf{z}}$. The field of a dipole $\boxed{m = I w^2}$,

for points on the z axis (Eq. 5.88, with $r \rightarrow z$, $\hat{\mathbf{r}} \rightarrow \hat{\mathbf{z}}$, $\theta = 0$) is $\mathbf{B} = \frac{\mu_0 m}{2\pi z^3} \hat{\mathbf{z}}$. ✓

Problem 5.37

(a) For a ring, $m = I \pi r^2$. Here $I \rightarrow \sigma v dr = \sigma \omega r dr$, so $m = \int_0^R \pi r^2 \sigma \omega r dr = \boxed{\pi \sigma \omega R^4 / 4}$.

(b) The total charge on the shaded ring is $dq = \sigma(2\pi R \sin \theta) R d\theta$. The time for one revolution is $dt = 2\pi / \omega$.

So the current in the ring is $I = \frac{dq}{dt} = \sigma \omega R^2 \sin \theta d\theta$. The area of the ring is $\pi(R \sin \theta)^2$, so the magnetic moment of the ring is $dm = (\sigma \omega R^2 \sin \theta d\theta) \pi R^2 \sin^2 \theta$, and the total dipole moment of the shell is

$$m = \sigma \omega \pi R^4 \int_0^\pi \sin^3 \theta d\theta = (4/3) \sigma \omega \pi R^4, \text{ or } \boxed{\mathbf{m} = \frac{4\pi}{3} \sigma \omega R^4 \hat{\mathbf{z}}}.$$

The dipole term in the multipole expansion for $\mathbf{A}$ is therefore $\mathbf{A}_{\text{dip}} = \frac{\mu_0}{4\pi} \frac{4\pi}{3} \sigma \omega R^4 \frac{\sin \theta}{r^2} \hat{\phi} = \frac{\mu_0 \sigma \omega R^4}{3} \frac{\sin \theta}{r^2} \hat{\phi}$, which is also the *exact* potential (Eq. 5.69); evidently a spinning sphere produces a perfect dipole field, with no higher multipole contributions.

Problem 5.38

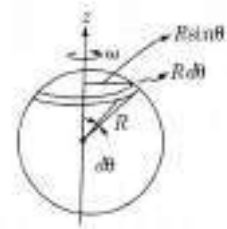
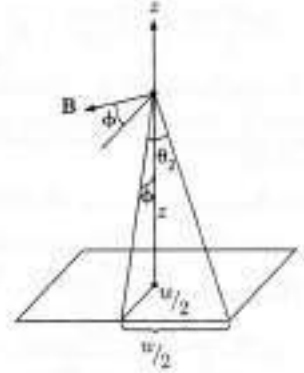
(a) $I d\mathbf{l} \rightarrow \mathbf{J} d\tau$, so $\boxed{\mathbf{A} = \frac{\mu_0}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \theta) \mathbf{J} d\tau}$.

(b) $\mathbf{A}_{\text{mon}} = \frac{\mu_0}{4\pi r} \int \mathbf{J} d\tau = \frac{\mu_0}{4\pi r} \frac{d\mathbf{p}}{dt}$ (Prob. 5.7), where $\mathbf{p}$ is the total electric dipole moment. In magnetostatics, $\mathbf{p}$ is constant, so $d\mathbf{p}/dt = \mathbf{0}$, and hence $\mathbf{A}_{\text{mon}} = \mathbf{0}$. qed

(c) $\mathbf{m} = I \mathbf{a} = \frac{1}{2} I \oint (\mathbf{r} \times d\mathbf{l}) \rightarrow \mathbf{m} = \frac{1}{2} \int (\mathbf{r} \times \mathbf{J}) d\tau$. qed

Problem 5.39

(a) Yes. (Magnetic forces do no work.)



$$(b) \mathbf{B} = \frac{\mu_0}{2\pi s} \hat{\phi}; \quad \mathbf{v} = \dot{s} \hat{\mathbf{s}} + s \dot{\phi} \hat{\phi} + \dot{z} \hat{\mathbf{z}}; \quad (\mathbf{v} \times \mathbf{B}) = \begin{vmatrix} \hat{\mathbf{s}} & \hat{\phi} & \hat{\mathbf{z}} \\ \dot{s} & s\dot{\phi} & \dot{z} \\ 0 & \frac{\mu_0}{2\pi s} & 0 \end{vmatrix} = \left(\frac{\mu_0}{2\pi s} \right) (-\dot{z} \hat{\mathbf{s}} + \dot{s} \hat{\mathbf{z}}).$$

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}) = \left(\frac{q\mu_0}{2\pi s} \right) (-\dot{z} \hat{\mathbf{r}} + \dot{s} \hat{\mathbf{z}}).$$

$$(c) \mathbf{a} = (\ddot{s} - s\dot{\phi}^2) \hat{\mathbf{s}} + (s\ddot{\phi} + 2\dot{s}\dot{\phi}) \hat{\phi} + \ddot{z} \hat{\mathbf{z}} \text{ (see, for example J. R. Taylor, } \textit{Classical Mechanics}, \text{ Eq. 1.47).}$$

$$\mathbf{F} = m\mathbf{a} \Rightarrow (\ddot{s} - s\dot{\phi}^2) \hat{\mathbf{s}} + (s\ddot{\phi} + 2\dot{s}\dot{\phi}) \hat{\phi} + \ddot{z} \hat{\mathbf{z}} = \frac{\alpha}{s} (-\dot{z} \hat{\mathbf{s}} + \dot{s} \hat{\mathbf{z}}),$$

where $\alpha \equiv \left(\frac{q\mu_0}{2\pi m} \right)$. Thus

$$\boxed{\ddot{s} - s\dot{\phi}^2 = -\alpha \frac{\dot{z}}{s}}, \quad \boxed{s\ddot{\phi} + 2\dot{s}\dot{\phi} = 0}, \quad \boxed{\ddot{z} = \alpha \frac{\dot{s}}{s}}.$$

(d) If $\dot{z}$ is constant, then $\ddot{z} = 0$, and it follows from the third equation that $\dot{s} = 0$, and hence s is constant.

Then the first equation says $\dot{\phi}^2 = \alpha(\dot{z}/s^2)$, so $\boxed{\dot{\phi} = \pm \frac{1}{s} \sqrt{\alpha \dot{z}}}$ is also constant (the second equation holds automatically). The charge moves in a helix around the wire.

Problem 5.40

The mobile charges *do* pull in toward the axis, but the resulting concentration of (negative) charge sets up an *electric* field that repels away further accumulation. Equilibrium is reached when the electric repulsion on a mobile charge q balances the magnetic attraction: $\mathbf{F} = q[\mathbf{E} + (\mathbf{v} \times \mathbf{B})] = \mathbf{0} \Rightarrow \mathbf{E} = -(\mathbf{v} \times \mathbf{B})$. Say the current is in the z direction: $\mathbf{J} = \rho_- v \hat{\mathbf{z}}$ (where ρ_- and v are both negative).

$$\oint \mathbf{B} \cdot d\mathbf{l} = B 2\pi s = \mu_0 J \pi s^2 \Rightarrow \mathbf{B} = \frac{\mu_0 \rho_- v s}{2} \hat{\phi};$$

$$\int \mathbf{E} \cdot d\mathbf{a} = E 2\pi s l = \frac{1}{\epsilon_0} (\rho_+ + \rho_-) \pi s^2 l \Rightarrow \mathbf{E} = \frac{1}{2\epsilon_0} (\rho_+ + \rho_-) s \hat{\mathbf{s}}.$$

$$\frac{1}{2\epsilon_0} (\rho_+ + \rho_-) s \hat{\mathbf{s}} = - \left[(v \hat{\mathbf{z}}) \times \left(\frac{\mu_0 \rho_- v s}{2} \hat{\phi} \right) \right] = \frac{\mu_0}{2} \rho_- v^2 s \hat{\mathbf{s}} \Rightarrow \rho_+ + \rho_- = \rho_- (\epsilon_0 \mu_0 v^2) = \rho_- \left(\frac{v^2}{c^2} \right).$$

Evidently $\rho_+ = -\rho_- \left(1 - \frac{v^2}{c^2} \right) = \frac{\rho_-}{\gamma^2}$, or $\rho_- = -\gamma^2 \rho_+$. In this naive model the mobile negative charges fill a smaller inner cylinder, leaving a shell of positive (stationary) charge at the outside. But since $v \ll c$, the effect is extremely small.

Problem 5.41

(a) If *positive* charges flow to the *right*, they are deflected down, and the bottom plate acquires a *positive* charge.

(b) $qvB = qE \Rightarrow E = vB \Rightarrow V = Et = \boxed{vBt}$, with the *bottom* at higher potential.

(c) If *negative* charges flow to the *left*, they are *also* deflected down, and the bottom plate acquires a *negative* charge. The potential difference is still the same, but this time the *top* plate is at the higher potential.

Problem 5.42

From Eq. 5.17, $\mathbf{F} = I \int (d\mathbf{l} \times \mathbf{B})$. But $\mathbf{B}$ is constant, in this case, so it comes outside the integral: $\mathbf{F} = I (\int d\mathbf{l}) \times \mathbf{B}$, and $\int d\mathbf{l} = \mathbf{w}$, the vector displacement from the point at which the wire first enters the field to the point where it leaves. Since $\mathbf{w}$ and $\mathbf{B}$ are perpendicular, $F = IBw$, and $\mathbf{F}$ is perpendicular to $\mathbf{w}$.

Problem 5.43

The angular momentum acquired by the particle as it moves out from the center to the edge is

$$\mathbf{L} = \int \frac{d\mathbf{L}}{dt} dt = \int \mathbf{N} dt = \int (\mathbf{r} \times \mathbf{F}) dt = \int \mathbf{r} \times q(\mathbf{v} \times \mathbf{B}) dt = q \int \mathbf{r} \times (d\mathbf{l} \times \mathbf{B}) = q \left[\int (\mathbf{r} \cdot \mathbf{B}) d\mathbf{l} - \int \mathbf{B}(\mathbf{r} \cdot d\mathbf{l}) \right].$$

But $\mathbf{r}$ is perpendicular to $\mathbf{B}$, so $\mathbf{r} \cdot \mathbf{B} = 0$, and $\mathbf{r} \cdot d\mathbf{l} = \mathbf{r} \cdot d\mathbf{r} = \frac{1}{2}d(\mathbf{r} \cdot \mathbf{r}) = \frac{1}{2}d(r^2) = r dr = (1/2\pi)(2\pi r dr)$.

So $\mathbf{L} = -\frac{q}{2\pi} \int_0^R \mathbf{B} 2\pi r dr = -\frac{q}{2\pi} \int \mathbf{B} da$. It follows that $L = -\frac{q}{2\pi} \Phi$, where $\Phi = \int B da$ is the total flux.

In particular, if $\Phi = 0$, then $L = 0$, and the charge emerges with zero angular momentum, which means it is going along a radial line. qed

Problem 5.44

From Eq. 5.24, $\mathbf{F} = \int (\mathbf{K} \times \mathbf{B}_{\text{ave}}) da$. Here $\mathbf{K} = \sigma \mathbf{v}$, $\mathbf{v} = \omega R \sin \theta \hat{\phi}$, $da = R^2 \sin \theta d\theta d\phi$, and $\mathbf{B}_{\text{ave}} = \frac{1}{2}(\mathbf{B}_{\text{in}} + \mathbf{B}_{\text{out}})$. From Eq. 5.70,

$$\mathbf{B}_{\text{in}} = \frac{2}{3} \mu_0 \sigma R \omega \hat{\mathbf{z}} = \frac{2}{3} \mu_0 \sigma R \omega (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\theta}). \text{ From Eq. 5.69,}$$

$$\begin{aligned} \mathbf{B}_{\text{out}} &= \nabla \times \mathbf{A} = \nabla \times \left(\frac{\mu_0 R^4 \omega \sigma}{3} \frac{\sin \theta}{r^2} \hat{\phi} \right) = \frac{\mu_0 R^4 \omega \sigma}{3} \left[\frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{\sin^2 \theta}{r^2} \right) \hat{\mathbf{r}} - \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{\sin \theta}{r} \right) \hat{\theta} \right] \\ &= \frac{\mu_0 R^4 \omega \sigma}{3r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\theta}) = \frac{\mu_0 R \omega \sigma}{3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\theta}) \text{ (since } r = R). \end{aligned}$$

$$\mathbf{B}_{\text{ave}} = \frac{\mu_0 R \omega \sigma}{6} (4 \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\theta}).$$

$$\mathbf{K} \times \mathbf{B}_{\text{ave}} = (\sigma \omega R \sin \theta) \left(\frac{\mu_0 R \omega \sigma}{6} \right) [\hat{\phi} \times (4 \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\theta})] = \frac{\mu_0}{6} (\sigma \omega R)^2 (4 \cos \theta \hat{\theta} + \sin \theta \hat{\mathbf{r}}) \sin \theta.$$

Picking out the z component of $\hat{\theta}$ (namely, $-\sin \theta$) and of $\hat{\mathbf{r}}$ (namely, $\cos \theta$), we have

$(\mathbf{K} \times \mathbf{B}_{\text{ave}})_z = -\frac{\mu_0}{2} (\sigma \omega R)^2 \sin^2 \theta \cos \theta$, so

$$F_z = -\frac{\mu_0}{2} (\sigma \omega R)^2 R^2 \int \sin^3 \theta \cos \theta d\theta d\phi = -\frac{\mu_0}{2} (\sigma \omega R^2)^2 2\pi \left(\frac{\sin^4 \theta}{4} \right) \Big|_0^{\pi/2}, \text{ or } \boxed{\mathbf{F} = -\frac{\mu_0 \pi}{4} (\sigma \omega R^2)^2 \hat{\mathbf{z}}}.$$

Problem 5.45

$$(a) \mathbf{F} = m\mathbf{a} = q_e(\mathbf{v} \times \mathbf{B}) = \frac{\mu_0 q_e q_m}{4\pi r^2} (\mathbf{v} \times \hat{\mathbf{r}}); \quad \boxed{\mathbf{a} = \frac{\mu_0 q_e q_m}{4\pi m r^3} (\mathbf{v} \times \mathbf{r})}.$$

(b) Because $\mathbf{a} \perp \mathbf{v}$, $\mathbf{a} \cdot \mathbf{v} = 0$. But $\mathbf{a} \cdot \mathbf{v} = \frac{1}{2} \frac{d}{dt} (\mathbf{v} \cdot \mathbf{v}) = \frac{1}{2} \frac{d}{dt} (v^2) = v \frac{dv}{dt}$. So $\frac{dv}{dt} = 0$. qed

$$\begin{aligned} (c) \frac{d\mathbf{Q}}{dt} &= m(\mathbf{v} \times \mathbf{v}) + m(\mathbf{r} \times \mathbf{a}) - \frac{\mu_0 q_e q_m}{4\pi} \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right) = 0 + \frac{\mu_0 q_e q_m}{4\pi r^3} [\mathbf{r} \times (\mathbf{v} \times \mathbf{r})] - \frac{\mu_0 q_e q_m}{4\pi} \left(\frac{\mathbf{v}}{r} - \frac{\mathbf{r}}{r^2} \frac{dr}{dt} \right) \\ &= \frac{\mu_0 q_e q_m}{4\pi} \left\{ \frac{1}{r^3} [r^2 \mathbf{v} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{r}] - \frac{\mathbf{v}}{r} + \frac{\mathbf{r}}{r^2} \frac{d}{dt} (\sqrt{\mathbf{r} \cdot \mathbf{r}}) \right\} = \frac{\mu_0 q_e q_m}{4\pi} \left[\frac{\mathbf{v}}{r} - \frac{(\hat{\mathbf{r}} \cdot \mathbf{v})}{r} \hat{\mathbf{r}} - \frac{\mathbf{v}}{r} + \frac{\hat{\mathbf{r}}}{2r} \frac{2(\mathbf{r} \cdot \mathbf{v})}{r} \right] = \mathbf{0}. \quad \checkmark \end{aligned}$$

(d) (i) $\mathbf{Q} \cdot \hat{\phi} = Q(\hat{\mathbf{z}} \cdot \hat{\phi}) = m(\mathbf{r} \times \mathbf{v}) \cdot \hat{\phi} - \frac{\mu_0 q_e q_m}{4\pi} (\hat{\mathbf{r}} \cdot \hat{\phi})$. But $\hat{\mathbf{z}} \cdot \hat{\phi} = \hat{\mathbf{r}} \cdot \hat{\phi} = 0$, so $(\mathbf{r} \times \mathbf{v}) \cdot \hat{\phi} = 0$. But

$\mathbf{r} = r \hat{\mathbf{r}}$, and $\mathbf{v} = \frac{dl}{dt} = \dot{r} \hat{\mathbf{r}} + r \dot{\theta} \hat{\boldsymbol{\theta}} + r \sin \theta \dot{\phi} \hat{\boldsymbol{\phi}}$ (where dots denote differentiation with respect to time), so

$$\mathbf{r} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ r & 0 & 0 \\ \dot{r} & r \dot{\theta} & r \sin \theta \dot{\phi} \end{vmatrix} = (-r^2 \sin \theta \dot{\phi}) \hat{\boldsymbol{\theta}} + (r^2 \dot{\theta}) \hat{\boldsymbol{\phi}}.$$

Therefore $(\mathbf{r} \times \mathbf{v}) \cdot \hat{\boldsymbol{\phi}} = r^2 \dot{\theta} = 0$, so θ is constant. qed

(ii) $\mathbf{Q} \cdot \hat{\mathbf{r}} = Q(\hat{\mathbf{z}} \cdot \hat{\mathbf{r}}) = m(\mathbf{r} \times \mathbf{v}) \cdot \hat{\mathbf{r}} - \frac{\mu_0 q_e q_m}{4\pi} (\hat{\mathbf{z}} \cdot \hat{\mathbf{r}})$. But $\hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = \cos \theta$, and $(\mathbf{r} \times \mathbf{v}) \perp \mathbf{r} \Rightarrow (\mathbf{r} \times \mathbf{v}) \cdot \hat{\mathbf{r}} = 0$, so $Q \cos \theta = -\frac{\mu_0 q_e q_m}{4\pi}$, or $Q = -\frac{\mu_0 q_e q_m}{4\pi \cos \theta}$. And since θ is constant, so too is Q . qed

(iii) $\mathbf{Q} \cdot \hat{\boldsymbol{\theta}} = Q(\hat{\mathbf{z}} \cdot \hat{\boldsymbol{\theta}}) = m(\mathbf{r} \times \mathbf{v}) \cdot \hat{\boldsymbol{\theta}} - \frac{\mu_0 q_e q_m}{4\pi} (\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\theta}})$. But $\hat{\mathbf{z}} \cdot \hat{\boldsymbol{\theta}} = -\sin \theta$, $\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\theta}} = 0$, and $(\mathbf{r} \times \mathbf{v}) \cdot \hat{\boldsymbol{\theta}} = -r^2 \sin \theta \dot{\phi}$ (from (i)), so $-Q \sin \theta = -mr^2 \sin \theta \dot{\phi} \Rightarrow \dot{\phi} = \frac{Q}{mr^2} = \frac{k}{r^2}$, with $k \equiv \frac{Q}{m} = -\frac{\mu_0 q_e q_m}{4\pi m \cos \theta}$.

(e) $v^2 = \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$, but $\dot{\theta} = 0$ and $\dot{\phi} = \frac{k}{r^2}$, so $\dot{r}^2 = v^2 - r^2 \sin^2 \theta \frac{k^2}{r^4} = v^2 - \frac{k^2 \sin^2 \theta}{r^2}$.

$$\left(\frac{dr}{d\phi} \right)^2 = \frac{\dot{r}^2}{\dot{\phi}^2} = \frac{v^2 - (k \sin \theta / r)^2}{(k^2 / r^4)} = r^2 \left[\left(\frac{vr}{k} \right)^2 - \sin^2 \theta \right]; \quad \frac{dr}{d\phi} = r \sqrt{\left(\frac{vr}{k} \right)^2 - \sin^2 \theta}.$$

(f) $\int \frac{dr}{r \sqrt{(vr/k)^2 - \sin^2 \theta}} = \int d\phi \Rightarrow \phi - \phi_0 = \frac{1}{\sin \theta} \sec^{-1} \left(\frac{vr}{k \sin \theta} \right); \sec[(\phi - \phi_0) \sin \theta] = \frac{vr}{k \sin \theta}$, or

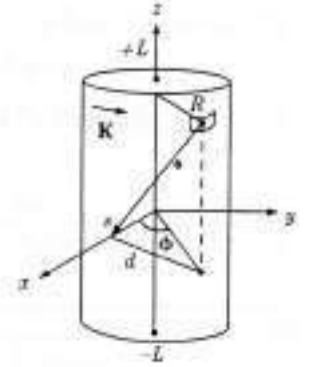
$$r(\phi) = \frac{A}{\cos[(\phi - \phi_0) \sin \theta]}, \quad \text{where } A \equiv -\frac{\mu_0 q_e q_m \tan \theta}{4\pi m v}.$$

Problem 5.46

Put the field point on the x axis, so $\mathbf{r} = (s, 0, 0)$. Then $\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{(\mathbf{K} \times \hat{\boldsymbol{\nu}})}{\boldsymbol{\nu}^2} da$; $da = R d\phi dz$; $\mathbf{K} = K \hat{\boldsymbol{\phi}} = K(-\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}})$; $\hat{\boldsymbol{\nu}} = (s - R \cos \phi) \hat{\mathbf{x}} - R \sin \phi \hat{\mathbf{y}} - z \hat{\mathbf{z}}$.

$$\mathbf{K} \times \hat{\boldsymbol{\nu}} = K \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ -\sin \phi & \cos \phi & 0 \\ (s - R \cos \phi) & (-R \sin \phi) & (-z) \end{vmatrix} =$$

$K [(-z \cos \phi) \hat{\mathbf{x}} + (-z \sin \phi) \hat{\mathbf{y}} + (R - s \cos \phi) \hat{\mathbf{z}}];$
 $\boldsymbol{\nu}^2 = z^2 + R^2 + s^2 - 2Rs \cos \phi$. The x and y components integrate to zero (z integrand is odd, as in Prob. 5.18).



$$B_z = \frac{\mu_0}{4\pi} KR \int \frac{(R - s \cos \phi)}{(z^2 + R^2 + s^2 - 2Rs \cos \phi)^{3/2}} d\phi dz = \frac{\mu_0 KR}{4\pi} \int_0^{2\pi} (R - s \cos \phi) \left\{ \int_{-\infty}^{\infty} \frac{dz}{(z^2 + d^2)^{3/2}} \right\} d\phi,$$

$$\text{where } d^2 \equiv R^2 + s^2 - 2Rs \cos \phi. \quad \text{Now } \int_{-\infty}^{\infty} \frac{dz}{(z^2 + d^2)^{3/2}} = \frac{2z}{d^2 \sqrt{z^2 + d^2}} \Big|_0^{\infty} = \frac{2}{d^2}.$$

$$= \frac{\mu_0 KR}{2\pi} \int_0^{2\pi} \frac{(R - s \cos \phi)}{(R^2 + s^2 - 2Rs \cos \phi)} d\phi; \quad (R - s \cos \phi) = \frac{1}{2R} [(R^2 - s^2) + (R^2 + s^2 - 2Rs \cos \phi)].$$

$$= \frac{\mu_0 K}{4\pi} \left[(R^2 - s^2) \int_0^{2\pi} \frac{d\phi}{(R^2 + s^2 - 2Rs \cos \phi)} + \int_0^{2\pi} d\phi \right].$$

$$\int_0^{2\pi} \frac{d\phi}{a + b \cos \phi} = 2 \int_0^\pi \frac{d\phi}{a + b \cos \phi} = \frac{4}{\sqrt{a^2 - b^2}} \tan^{-1} \left[\frac{\sqrt{a^2 - b^2} \tan(\phi/2)}{a + b} \right] \Big|_0^\pi$$

$$= \frac{4}{\sqrt{a^2 - b^2}} \tan^{-1} \left[\frac{\sqrt{a^2 - b^2} \tan(\pi/2)}{a + b} \right] = \frac{4}{\sqrt{a^2 - b^2}} \left(\frac{\pi}{2} \right) = \frac{2\pi}{\sqrt{a^2 - b^2}}. \text{ Here } a = R^2 + s^2,$$

$b = -2Rs$, so $a^2 - b^2 = R^4 + 2R^2s^2 + s^4 - 4R^2s^2 = R^4 - 2R^2s^2 + s^4 = (R^2 - s^2)^2$; $\sqrt{a^2 - b^2} = |R^2 - s^2|$.

$$B_z = \frac{\mu_0 K}{4\pi} \left[\frac{(R^2 - s^2)}{|R^2 - s^2|} 2\pi + 2\pi \right] = \frac{\mu_0 K}{2} \left(\frac{R^2 - s^2}{|R^2 - s^2|} + 1 \right).$$

Inside the solenoid, $s < R$, so $B_z = \frac{\mu_0 K}{2} (1+1) = \mu_0 K$. *Outside* the solenoid, $s > R$, so $B_z = \frac{\mu_0 K}{2} (-1+1) = 0$.

Here $K = nI$, so $\boxed{\mathbf{B} = \mu_0 nI \hat{\mathbf{z}}(\text{inside}), \text{ and } \mathbf{0}(\text{outside})}$ (as we found more easily using Ampère's law, in Ex. 5.9).

Problem 5.47

(a) From Eq. 5.41, $\boxed{\mathbf{B} = \frac{\mu_0 I R^2}{2} \left\{ \frac{1}{[R^2 + (d/2 + z)^2]^{3/2}} + \frac{1}{[R^2 + (d/2 - z)^2]^{3/2}} \right\}}.$

$$\frac{\partial B}{\partial z} = \frac{\mu_0 I R^2}{2} \left\{ \frac{(-3/2)2(d/2 + z)}{[R^2 + (d/2 + z)^2]^{5/2}} + \frac{(-3/2)2(d/2 - z)(-1)}{[R^2 + (d/2 - z)^2]^{5/2}} \right\}$$

$$= \frac{3\mu_0 I R^2}{2} \left\{ \frac{-(d/2 + z)}{[R^2 + (d/2 + z)^2]^{5/2}} + \frac{(d/2 - z)}{[R^2 + (d/2 - z)^2]^{5/2}} \right\}.$$

$$\left. \frac{\partial B}{\partial z} \right|_{z=0} = \frac{3\mu_0 I R^2}{2} \left\{ \frac{-d/2}{[R^2 + (d/2)^2]^{5/2}} + \frac{d/2}{[R^2 + (d/2)^2]^{5/2}} \right\} = 0. \quad \checkmark$$

(b) Differentiating again:

$$\frac{\partial^2 B}{\partial z^2} = \frac{3\mu_0 I R^2}{2} \left\{ \frac{-1}{[R^2 + (d/2 + z)^2]^{5/2}} + \frac{-(d/2 + z)(-5/2)2(d/2 + z)}{[R^2 + (d/2 + z)^2]^{7/2}} \right.$$

$$\left. + \frac{-1}{[R^2 + (d/2 - z)^2]^{5/2}} + \frac{(d/2 - z)(-5/2)2(d/2 - z)(-1)}{[R^2 + (d/2 - z)^2]^{7/2}} \right\}.$$

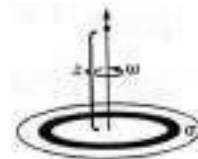
$$\left. \frac{\partial^2 B}{\partial z^2} \right|_{z=0} = \frac{3\mu_0 I R^2}{2} \left\{ \frac{-2}{[R^2 + (d/2)^2]^{5/2}} + \frac{2(5/2)2(d/2)^2}{[R^2 + (d/2)^2]^{7/2}} \right\} = \frac{3\mu_0 I R^2}{[R^2 + (d/2)^2]^{7/2}} \left(-R^2 - \frac{d^2}{4} + \frac{5d^2}{4} \right)$$

$$= \frac{3\mu_0 I R^2}{[R^2 + (d/2)^2]^{7/2}} (d^2 - R^2). \quad \text{Zero if } \boxed{d = R}, \text{ in which case}$$

$$B(0) = \frac{\mu_0 I R^2}{2} \left\{ \frac{1}{[R^2 + (R/2)^2]^{3/2}} + \frac{1}{[R^2 + (R/2)^2]^{3/2}} \right\} = \mu_0 I R^2 \frac{1}{(5R^2/4)^{3/2}} = \boxed{\frac{8\mu_0 I}{5^{3/2} R}}.$$

Problem 5.48

The total charge on the shaded ring is $dq = \sigma(2\pi r) dr$. The time for one revolution is $dt = 2\pi/\omega$. So the current in the ring is $I = \frac{dq}{dt} = \sigma\omega r dr$. From Eq. 5.41, the magnetic field of



this ring (for points on the axis) is $d\mathbf{B} = \frac{\mu_0}{2} \sigma \omega r \frac{r^2}{(r^2 + z^2)^{3/2}} dr \hat{\mathbf{z}}$, and the total field of the disk is

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0 \sigma \omega}{2} \int_0^R \frac{r^3 dr}{(r^2 + z^2)^{3/2}} \hat{\mathbf{z}}. \quad \text{Let } u \equiv r^2, \text{ so } du = 2r dr. \quad \text{Then} \\ &= \frac{\mu_0 \sigma \omega}{4} \int_0^{R^2} \frac{u du}{(u + z^2)^{3/2}} = \frac{\mu_0 \sigma \omega}{4} \left[2 \left(\frac{u + 2z^2}{\sqrt{u + z^2}} \right) \right] \Big|_0^{R^2} = \boxed{\frac{\mu_0 \sigma \omega}{2} \left[\frac{(R^2 + 2z^2)}{\sqrt{R^2 + z^2}} - 2z \right] \hat{\mathbf{z}}}. \end{aligned}$$

When $z \gg R$, the term in square brackets is

$$\begin{aligned} [\] &= \frac{2z^2(1 + R^2/2z^2)}{z[1 + (R/z)^2]^{-1/2}} - 2z \approx 2z \left[\left(1 + \frac{R^2}{2z^2} \right) \left(1 - \frac{1}{2} \frac{R^2}{z^2} + \frac{3}{8} \frac{R^4}{z^4} \right) - 1 \right] \\ &\approx 2z \left(1 - \frac{R^2}{2z^2} + \frac{3}{8} \frac{R^4}{z^4} + \frac{R^2}{2z^2} - \frac{R^4}{4z^4} - 1 \right) = 2z \left(\frac{1}{8} \frac{R^4}{z^4} \right) = \frac{R^4}{4z^4}, \end{aligned}$$

so

$$\mathbf{B} \approx \frac{\mu_0 \sigma \omega}{2} \frac{R^4}{4z^4} \hat{\mathbf{z}} = \frac{\mu_0 \sigma \omega R^4}{8z^3} \hat{\mathbf{z}}.$$

Meanwhile, from Eq. 5.88, the dipole field is

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0 m}{4\pi r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}),$$

and for points on the $+z$ axis $\theta = 0$, $r = z$, $\hat{\mathbf{r}} = \hat{\mathbf{z}}$, so $\mathbf{B}_{\text{dip}} = \frac{\mu_0 m}{2\pi z^3} \hat{\mathbf{z}}$. In this case (Problem 5.37a) $m = \pi \sigma \omega R^4/4$, so $\mathbf{B}_{\text{dip}} = \frac{\mu_0 \sigma \omega R^4}{8z^3} \hat{\mathbf{z}}$, in agreement with the approximation. $\checkmark$

Problem 5.49

$\mathbf{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times \boldsymbol{\mathcal{R}}}{\mathcal{R}^3}$; $\boldsymbol{\mathcal{R}} = -R \cos \phi \hat{\mathbf{x}} + (y - R \sin \phi) \hat{\mathbf{y}} + z \hat{\mathbf{z}}$. (For simplicity I'll drop the prime on ϕ .) $\mathcal{R}^2 = R^2 \cos^2 \phi + y^2 - 2Ry \sin \phi + R^2 \sin^2 \phi + z^2 = R^2 + y^2 + z^2 - 2Ry \sin \phi$. The source coordinates (x', y', z') satisfy $x' = R \cos \phi \Rightarrow dx' = -R \sin \phi d\phi$; $y' = R \sin \phi \Rightarrow dy' = R \cos \phi d\phi$; $z' = 0 \Rightarrow dz' = 0$. So $d\mathbf{l}' = -R \sin \phi d\phi \hat{\mathbf{x}} + R \cos \phi d\phi \hat{\mathbf{y}}$.

$$d\mathbf{l}' \times \boldsymbol{\mathcal{R}} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ -R \sin \phi d\phi & R \cos \phi d\phi & 0 \\ -R \cos \phi & (y - R \sin \phi) & z \end{vmatrix} = (Rz \cos \phi d\phi) \hat{\mathbf{x}} + (Rz \sin \phi d\phi) \hat{\mathbf{y}} + (-Ry \sin \phi d\phi + R^2 d\phi) \hat{\mathbf{z}}.$$

$$B_x = \frac{\mu_0 I R z}{4\pi} \int_0^{2\pi} \frac{\cos \phi d\phi}{(R^2 + y^2 + z^2 - 2Ry \sin \phi)^{3/2}} = \frac{\mu_0 I R z}{4\pi} \frac{1}{Ry} \frac{1}{\sqrt{R^2 + y^2 + z^2 - 2Ry \sin \phi}} \Big|_0^{2\pi} = 0,$$

since $\sin \phi = 0$ at both limits. The y and z components are elliptic integrals, and cannot be expressed in terms of elementary functions.

$$\boxed{B_x = 0; \quad B_y = \frac{\mu_0 I R z}{4\pi} \int_0^{2\pi} \frac{\sin \phi d\phi}{(R^2 + y^2 + z^2 - 2Ry \sin \phi)^{3/2}}; \quad B_z = \frac{\mu_0 I R}{4\pi} \int_0^{2\pi} \frac{(R - y \sin \phi) d\phi}{(R^2 + y^2 + z^2 - 2Ry \sin \phi)^{3/2}}.}$$

Problem 5.50

From the Biot-Savart law, the field of loop #1 is $\mathbf{B} = \frac{\mu_0 I_1}{4\pi} \oint_1 \frac{d\mathbf{l}_1 \times \hat{\mathbf{r}}}{r^2}$; the force on loop #2 is

$$\mathbf{F} = I_2 \oint_2 d\mathbf{l}_2 \times \mathbf{B} = \frac{\mu_0}{4\pi} I_1 I_2 \oint_1 \oint_2 \frac{d\mathbf{l}_2 \times (d\mathbf{l}_1 \times \hat{\mathbf{r}})}{r^2}. \quad \text{Now } d\mathbf{l}_2 \times (d\mathbf{l}_1 \times \hat{\mathbf{r}}) = d\mathbf{l}_1 (d\mathbf{l}_2 \cdot \hat{\mathbf{r}}) - \hat{\mathbf{r}} (d\mathbf{l}_1 \cdot d\mathbf{l}_2), \text{ so}$$

$$\mathbf{F} = -\frac{\mu_0}{4\pi} I_1 I_2 \left\{ \oint \oint \frac{\hat{\mathbf{r}}}{r^2} (d\mathbf{l}_1 \cdot d\mathbf{l}_2) - \oint d\mathbf{l}_1 \oint \frac{(d\mathbf{l}_2 \cdot \hat{\mathbf{r}})}{r^2} \right\}$$

The first term is what we want. It remains to show that the second term is zero:

$$\begin{aligned} \hat{\mathbf{r}} &= (x_2 - x_1) \hat{\mathbf{x}} + (y_2 - y_1) \hat{\mathbf{y}} + (z_2 - z_1) \hat{\mathbf{z}}, \text{ so } \nabla_2 (1/r) = \frac{\partial}{\partial x_2} [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{-1/2} \hat{\mathbf{x}} \\ &+ \frac{\partial}{\partial y_2} [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{-1/2} \hat{\mathbf{y}} + \frac{\partial}{\partial z_2} [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{-1/2} \hat{\mathbf{z}} \\ &= -\frac{(x_2 - x_1)}{r^3} \hat{\mathbf{x}} - \frac{(y_2 - y_1)}{r^3} \hat{\mathbf{y}} - \frac{(z_2 - z_1)}{r^3} \hat{\mathbf{z}} = -\frac{\hat{\mathbf{r}}}{r^3} = -\frac{\hat{\mathbf{r}}}{r^2}. \text{ So } \oint \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{l}_2 = -\oint \nabla_2 \left(\frac{1}{r} \right) \cdot d\mathbf{l}_2 = 0 \text{ (by} \end{aligned}$$

Corollary 2 in Sect. 1.3.3). $\square$

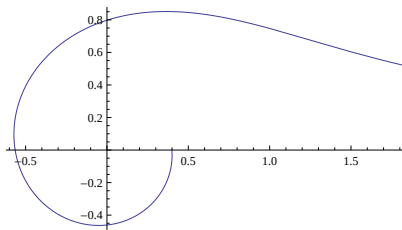
Problem 5.51

(a) $\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint \frac{d\mathbf{l} \times \hat{\mathbf{r}}}{r^2} = -\frac{\mu_0 I}{4\pi} \oint \frac{d\mathbf{l} \times \hat{\mathbf{r}}}{r^2}, \quad d\mathbf{l} \times \hat{\mathbf{r}} = -dl \sin \phi \hat{\mathbf{z}}, \quad dl \sin \phi = r d\theta$ (see diagram in the book),

so $B = \frac{\mu_0 I}{4\pi} \oint \frac{d\theta}{r}. \quad \checkmark$

(b) For a circular loop, $r = R$ is constant; $B = \frac{\mu_0 I}{4\pi R} \oint d\theta = \frac{\mu_0 I}{2R}$, which agrees with Eq. 5.41 ($z = 0$).

(c)



$$B = \frac{\mu_0 I}{4\pi a} \int_0^{2\pi} \sqrt{\theta} d\theta = \frac{\mu_0 I}{4\pi a} \left[\frac{2}{3} \theta^{3/2} \right] \bigg|_0^{2\pi} = \boxed{\frac{\mu_0 I \sqrt{2\pi}}{3a}}.$$

(d) $B = \frac{\mu_0 I}{4\pi p} \int_0^{2\pi} (1 + e \cos \theta) d\theta = \frac{\mu_0 I}{4\pi p} (2\pi) = \boxed{\frac{\mu_0 I}{2p}}. \quad \checkmark$

Problem 5.52

(a)

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} \times \hat{\mathbf{r}}}{r^2} d\tau \quad \Rightarrow \quad \mathbf{A} = \frac{1}{4\pi} \int \frac{\mathbf{B} \times \hat{\mathbf{r}}}{r^2} d\tau.$$

(b) Poisson's equation (Eq. 2.24) says $\nabla^2 V = -\frac{1}{\epsilon_0} \rho$. For *dielectrics* (with no *free* charge), $\rho_b = -\nabla \cdot \mathbf{P}$ (Eq. 4.12), and the resulting potential is $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{r}}}{r^2} d\tau'$. In *general*, $\rho = \epsilon_0 \nabla \cdot \mathbf{E}$ (Gauss's law),

so the analogy is $\mathbf{P} \rightarrow -\epsilon_0 \mathbf{E}$, and hence $V(\mathbf{r}) = -\frac{1}{4\pi} \int \frac{\mathbf{E}(\mathbf{r}') \cdot \hat{\mathbf{z}}}{r^2} d\tau'$. qed

[There are many other ways to obtain this result. For example, using Eq. 1.100:

$$\nabla \cdot \left(\frac{\hat{\mathbf{z}}}{r^2} \right) = -\nabla' \cdot \left(\frac{\hat{\mathbf{z}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r}) = 4\pi\delta^3(\mathbf{r} - \mathbf{r}'),$$

$$V(\mathbf{r}) = \int V(\mathbf{r}')\delta^3(\mathbf{r} - \mathbf{r}') d\tau' = -\frac{1}{4\pi} \int V(\mathbf{r}')\nabla' \cdot \left(\frac{\hat{\mathbf{z}}}{r^2} \right) d\tau' = \frac{1}{4\pi} \int \frac{\hat{\mathbf{z}}}{r^2} \cdot [\nabla' V(\mathbf{r}')] d\tau' - \frac{1}{4\pi} \oint V(\mathbf{r}') \frac{\hat{\mathbf{z}}}{r^2} \cdot d\mathbf{a}'$$

(Eq. 1.59). But $\nabla' V(\mathbf{r}') = -\mathbf{E}(\mathbf{r}')$, and the surface integral $\rightarrow 0$ at ∞ , so $V(\mathbf{r}) = -\frac{1}{4\pi} \int \frac{\mathbf{E}(\mathbf{r}') \cdot \hat{\mathbf{z}}}{r^2} d\tau'$, as before. You can also *check* the result, by computing its gradient—but it's not easy.]

Problem 5.53

(a) For uniform $\mathbf{B}$, $\int_0^{\mathbf{r}} (\mathbf{B} \times d\mathbf{l}) = \mathbf{B} \times \int_0^{\mathbf{r}} d\mathbf{l} = \boxed{\mathbf{B} \times \mathbf{r}} \neq \mathbf{A} = -\frac{1}{2}(\mathbf{B} \times \mathbf{r})$.

(b) $\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$, so $\oint \mathbf{B} \times d\mathbf{l} = \left(\frac{\mu_0 I}{2\pi a} \hat{s} - \frac{\mu_0 I}{2\pi b} \hat{s} \right) w = \boxed{\frac{\mu_0 I w}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right) \hat{s} \neq \mathbf{0}}$.

(c) $\mathbf{A} = -\mathbf{r} \times \mathbf{B} \int_0^1 \lambda d\lambda = \boxed{-\frac{1}{2}(\mathbf{r} \times \mathbf{B})}$.

(d) $\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$; $\mathbf{B}(\lambda \mathbf{r}) = \frac{\mu_0 I}{2\pi \lambda s} \hat{\phi}$; $\mathbf{A} = -\frac{\mu_0 I}{2\pi s} (\mathbf{r} \times \hat{\phi}) \int_0^1 \lambda \frac{1}{\lambda} d\lambda = -\frac{\mu_0 I}{2\pi s} (\mathbf{r} \times \hat{\phi})$. But $\mathbf{r}$ here is the vector from the origin—in cylindrical coordinates $\mathbf{r} = s\hat{s} + z\hat{z}$. So $\mathbf{A} = -\frac{\mu_0 I}{2\pi s} [s(\hat{s} \times \hat{\phi}) + z(\hat{z} \times \hat{\phi})]$, and $(\hat{s} \times \hat{\phi}) = \hat{z}$, $(\hat{z} \times \hat{\phi}) = -\hat{s}$. So $\boxed{\mathbf{A} = \frac{\mu_0 I}{2\pi s} (z\hat{s} - s\hat{z})}$.

The examples in (c) and (d) happen to be divergenceless, but this is not in general the case. For (letting $\mathbf{L} \equiv \int_0^1 \lambda \mathbf{B}(\lambda \mathbf{r}) d\lambda$, for short) $\nabla \cdot \mathbf{A} = -\nabla \cdot (\mathbf{r} \times \mathbf{L}) = -[\mathbf{L} \cdot (\nabla \times \mathbf{r}) - \mathbf{r} \cdot (\nabla \times \mathbf{L})] = \mathbf{r} \cdot (\nabla \times \mathbf{L})$, and $\nabla \times \mathbf{L} = \int_0^1 \lambda [\nabla \times \mathbf{B}(\lambda \mathbf{r})] d\lambda = \int_0^1 \lambda^2 [\nabla_\lambda \times \mathbf{B}(\lambda \mathbf{r})] d\lambda = \mu_0 \int_0^1 \lambda^2 \mathbf{J}(\lambda \mathbf{r}) d\lambda$, so $\nabla \cdot \mathbf{A} = \mu_0 \mathbf{r} \cdot \int_0^1 \lambda^2 \mathbf{J}(\lambda \mathbf{r}) d\lambda$, and it vanishes in regions where $\mathbf{J} = \mathbf{0}$ (which is why the examples in (c) and (d) were divergenceless). To construct an explicit counterexample, we need the field at a point where $\mathbf{J} \neq \mathbf{0}$ —say, *inside* a wire with uniform current.

Here Ampère's law gives $B 2\pi s = \mu_0 I_{\text{enc}} = \mu_0 J \pi s^2 \Rightarrow \mathbf{B} = \frac{\mu_0 J}{2} s \hat{\phi}$, so

$$\mathbf{A} = -\mathbf{r} \times \int_0^1 \lambda \left(\frac{\mu_0 J}{2} \right) \lambda s \hat{\phi} d\lambda = -\frac{\mu_0 J}{6} s (\mathbf{r} \times \hat{\phi}) = \frac{\mu_0 J s}{6} (z\hat{s} - s\hat{z}).$$

$$\nabla \cdot \mathbf{A} = \frac{\mu_0 J}{6} \left[\frac{1}{s} \frac{\partial}{\partial s} (s^2 z) + \frac{\partial}{\partial z} (-s^2) \right] = \frac{\mu_0 J}{6} \left(\frac{1}{s} 2sz \right) = \frac{\mu_0 J z}{3} \neq 0.$$

Conclusion: $\boxed{\text{(ii) does not automatically yield } \nabla \cdot \mathbf{A} = 0.}$

Problem 5.54

(a) Exploit the analogy with the electrical case:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}] \quad (\text{Eq. 3.104}) = -\nabla V, \quad \text{with } V = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} \quad (\text{Eq. 3.102}).$$

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{m}] \quad (\text{Eq. 5.89}) = -\nabla U, \quad (\text{Eq. 5.67}).$$

Evidently the prescription is $\mathbf{p}/\epsilon_0 \rightarrow \mu_0 \mathbf{m} : \boxed{U(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \cdot \hat{\mathbf{r}}}{r^2}}.$

(b) Comparing Eqs. 5.69 and 5.87, the dipole moment of the shell is $\mathbf{m} = (4\pi/3)\omega\sigma R^4 \hat{\mathbf{z}}$ (which we also got in Prob. 5.37). Using the result of (a), then, $\boxed{U(\mathbf{r}) = \frac{\mu_0\omega\sigma R^4}{3} \frac{\cos\theta}{r^2}}$ for $r > R$.

Inside the shell, the field is uniform (Eq. 5.70): $\mathbf{B} = \frac{2}{3}\mu_0\sigma\omega R \hat{\mathbf{z}}$, so $U(\mathbf{r}) = -\frac{2}{3}\mu_0\sigma\omega R z + \text{constant}$. We may as well pick the constant to be zero, so $\boxed{U(\mathbf{r}) = -\frac{2}{3}\mu_0\sigma\omega R r \cos\theta}$ for $r < R$.

[Notice that $U(\mathbf{r})$ is *not continuous* at the surface ($r = R$): $U_{\text{in}}(R) = -\frac{2}{3}\mu_0\sigma\omega R^2 \cos\theta \neq U_{\text{out}}(R) = \frac{1}{3}\mu_0\sigma\omega R^2 \cos\theta$. As I warned you on p. 245: if you insist on using magnetic scalar potentials, keep away from places where there is current!]

(c)

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0\omega Q}{4\pi R} \left[\left(1 - \frac{3r^2}{5R^2}\right) \cos\theta \hat{\mathbf{r}} - \left(1 - \frac{6r^2}{5R^2}\right) \sin\theta \hat{\boldsymbol{\theta}} \right] = -\nabla U = -\frac{\partial U}{\partial r} \hat{\mathbf{r}} - \frac{1}{r} \frac{\partial U}{\partial \theta} \hat{\boldsymbol{\theta}} - \frac{1}{r \sin\theta} \frac{\partial U}{\partial \phi} \hat{\boldsymbol{\phi}}. \\ \frac{\partial U}{\partial \phi} &= 0 \Rightarrow U(r, \theta, \phi) = U(r, \theta). \\ \frac{1}{r} \frac{\partial U}{\partial \theta} &= \left(\frac{\mu_0\omega Q}{4\pi R} \right) \left(1 - \frac{6r^2}{5R^2}\right) \sin\theta \Rightarrow U(r, \theta) = -\left(\frac{\mu_0\omega Q}{4\pi R} \right) \left(1 - \frac{6r^2}{5R^2}\right) r \cos\theta + f(r). \\ \frac{\partial U}{\partial r} &= -\left(\frac{\mu_0\omega Q}{4\pi R} \right) \left(1 - \frac{3r^2}{5R^2}\right) \cos\theta \Rightarrow U(r, \theta) = -\left(\frac{\mu_0\omega Q}{4\pi R} \right) \left(r - \frac{r^3}{5R^2}\right) \cos\theta + g(\theta). \end{aligned}$$

Equating the two expressions:

$$-\left(\frac{\mu_0\omega Q}{4\pi R} \right) \left(1 - \frac{6r^2}{5R^2}\right) r \cos\theta + f(r) = -\left(\frac{\mu_0\omega Q}{4\pi R} \right) \left(1 - \frac{r^2}{5R^2}\right) r \cos\theta + g(\theta),$$

or

$$\left(\frac{\mu_0\omega Q}{4\pi R^3} \right) r^3 \cos\theta + f(r) = g(\theta).$$

But there is no way to write $r^3 \cos\theta$ as the sum of a function of θ and a function of r , so we're stuck. The reason is that you can't *have* a scalar magnetic potential in a region where the current is nonzero.

Problem 5.55

(a) $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, and $\nabla \cdot \mathbf{A} = 0$, $\nabla \times \mathbf{A} = \mathbf{B} \Rightarrow \mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}}{z} d\tau'$, so

$$\nabla \cdot \mathbf{A} = 0, \nabla \times \mathbf{A} = \mathbf{B}, \text{ and } \nabla \cdot \mathbf{W} = 0 \text{ (we'll choose it so), } \nabla \times \mathbf{W} = \mathbf{A} \Rightarrow \boxed{\mathbf{W} = \frac{1}{4\pi} \int \frac{\mathbf{B}}{z} d\tau'}.$$

(b) $\mathbf{W}$ will be proportional to $\mathbf{B}$ and to two factors of $\mathbf{r}$ (since differentiating *twice* must recover $\mathbf{B}$), so I'll try something of the form $\mathbf{W} = \alpha \mathbf{r}(\mathbf{r} \cdot \mathbf{B}) + \beta r^2 \mathbf{B}$, and see if I can pick the constants α and β in such a way that $\nabla \cdot \mathbf{W} = 0$ and $\nabla \times \mathbf{W} = \mathbf{A}$.

$$\nabla \cdot \mathbf{W} = \alpha [(\mathbf{r} \cdot \mathbf{B})(\nabla \cdot \mathbf{r}) + \mathbf{r} \cdot \nabla(\mathbf{r} \cdot \mathbf{B})] + \beta [r^2(\nabla \cdot \mathbf{B}) + \mathbf{B} \cdot \nabla(r^2)]. \quad \nabla \mathbf{r} = \frac{\partial x}{\partial x} \hat{\mathbf{x}} + \frac{\partial y}{\partial y} \hat{\mathbf{y}} + \frac{\partial z}{\partial z} \hat{\mathbf{z}} = 1 + 1 + 1 = 3;$$

$\nabla(\mathbf{r} \cdot \mathbf{B}) = \mathbf{r} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{r}) + (\mathbf{r} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{r}$; but $\mathbf{B}$ is constant, so all derivatives of $\mathbf{B}$ vanish, and $\nabla \times \mathbf{r} = 0$ (Prob. 1.63), so

$$\nabla(\mathbf{r} \cdot \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{r} = \left(B_x \frac{\partial}{\partial x} + B_y \frac{\partial}{\partial y} + B_z \frac{\partial}{\partial z} \right) (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}} = \mathbf{B};$$

$$\nabla(r^2) = \left(\hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2) = 2x \hat{\mathbf{x}} + 2y \hat{\mathbf{y}} + 2z \hat{\mathbf{z}} = 2\mathbf{r}. \text{ So}$$

$\nabla \cdot \mathbf{W} = \alpha [3(\mathbf{r} \cdot \mathbf{B}) + (\mathbf{r} \cdot \mathbf{B})] + \beta [0 + 2(\mathbf{r} \cdot \mathbf{B})] = 2(\mathbf{r} \cdot \mathbf{B})(2\alpha + \beta)$, which is zero if $2\alpha + \beta = 0$.

$\nabla \times \mathbf{W} = \alpha [(\mathbf{r} \cdot \mathbf{B})(\nabla \times \mathbf{r}) - \mathbf{r} \times \nabla(\mathbf{r} \cdot \mathbf{B})] + \beta [r^2(\nabla \times \mathbf{B}) - \mathbf{B} \times \nabla(r^2)] = \alpha [0 - (\mathbf{r} \times \mathbf{B})] + \beta [0 - 2(\mathbf{B} \times \mathbf{r})]$
 $= -(\mathbf{r} \times \mathbf{B})(\alpha - 2\beta) = -\frac{1}{2}(\mathbf{r} \times \mathbf{B})$ (Prob. 5.25). So we want $\alpha - 2\beta = 1/2$. Evidently $\alpha - 2(-2\alpha) = 5\alpha = 1/2$,

or $\alpha = 1/10; \beta = -2\alpha = -1/5$. *Conclusion:* $\mathbf{W} = \frac{1}{10} [\mathbf{r}(\mathbf{r} \cdot \mathbf{B}) - 2r^2\mathbf{B}]$. (But this is certainly not unique.)

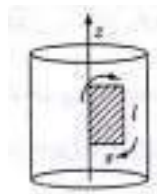
(c) $\nabla \times \mathbf{W} = \mathbf{A} \Rightarrow \int (\nabla \times \mathbf{W}) \cdot d\mathbf{a} = \int \mathbf{A} \cdot d\mathbf{a}$. Or $\oint \mathbf{W} \cdot d\mathbf{l} = \int \mathbf{A} \cdot d\mathbf{a}$. Integrate around the amperian loop shown, taking $\mathbf{W}$ to point parallel to the axis, and choosing $\mathbf{W} = 0$ on the axis:

$$-Wl = \int_0^s \left(\frac{\mu_0 n I}{2} \right) l \bar{s} d\bar{s} = \frac{\mu_0 n I}{2} \frac{s^2 l}{2} \quad (\text{using Eq. 5.72 for } \mathbf{A}).$$

$$\mathbf{W} = -\frac{\mu_0 n I s^2}{4} \hat{\mathbf{z}} \quad (s < R).$$

$$\text{For } s > R, -Wl = \frac{\mu_0 n I R^2 l}{4} + \int_R^s \left(\frac{\mu_0 n I}{2} \right) \frac{R^2}{\bar{s}} l d\bar{s} = \frac{\mu_0 n I R^2 l}{4} + \frac{\mu_0 n I R^2 l}{2} \ln(s/R);$$

$$\mathbf{W} = -\frac{\mu_0 n I R^2}{4} [1 + 2 \ln(s/R)] \hat{\mathbf{z}} \quad (s > R).$$



Problem 5.56

Apply the divergence theorem to the function $[\mathbf{U} \times (\nabla \times \mathbf{V})]$, noting (from the product rule) that $\nabla \cdot [\mathbf{U} \times (\nabla \times \mathbf{V})] = (\nabla \times \mathbf{V}) \cdot (\nabla \times \mathbf{U}) - \mathbf{U} \cdot [\nabla \times (\nabla \times \mathbf{V})]$:

$$\int \nabla \cdot [\mathbf{U} \times (\nabla \times \mathbf{V})] d\tau = \int \{(\nabla \times \mathbf{V}) \cdot (\nabla \times \mathbf{U}) - \mathbf{U} \cdot [\nabla \times (\nabla \times \mathbf{V})]\} d\tau = \oint [\mathbf{U} \times (\nabla \times \mathbf{V})] \cdot d\mathbf{a}.$$

As always, suppose we have *two* solutions, $\mathbf{B}_1$ (and $\mathbf{A}_1$) and $\mathbf{B}_2$ (and $\mathbf{A}_2$). Define $\mathbf{B}_3 \equiv \mathbf{B}_2 - \mathbf{B}_1$ (and $\mathbf{A}_3 \equiv \mathbf{A}_2 - \mathbf{A}_1$), so that $\nabla \times \mathbf{A}_3 = \mathbf{B}_3$ and $\nabla \times \mathbf{B}_3 = \nabla \times \mathbf{B}_1 - \nabla \times \mathbf{B}_2 = \mu_0 \mathbf{J} - \mu_0 \mathbf{J} = \mathbf{0}$. Set $\mathbf{U} = \mathbf{V} = \mathbf{A}_3$ in the above identity:

$$\int \{(\nabla \times \mathbf{A}_3) \cdot (\nabla \times \mathbf{A}_3) - \mathbf{A}_3 \cdot [\nabla \times (\nabla \times \mathbf{A}_3)]\} d\tau = \int \{(\mathbf{B}_3) \cdot (\mathbf{B}_3) - \mathbf{A}_3 \cdot [\nabla \times \mathbf{B}_3]\} d\tau = \int (B_3)^2 d\tau$$

$$= \oint [\mathbf{A}_3 \times (\nabla \times \mathbf{A}_3)] \cdot d\mathbf{a} = \oint (\mathbf{A}_3 \times \mathbf{B}_3) \cdot d\mathbf{a}. \text{ But either } \mathbf{A} \text{ is specified (in which case } \mathbf{A}_3 = \mathbf{0}), \text{ or else } \mathbf{B} \text{ is}$$

specified (in which case $\mathbf{B}_3 = \mathbf{0}$), at the surface. In either case $\oint (\mathbf{A}_3 \times \mathbf{B}_3) \cdot d\mathbf{a} = 0$. So $\int (B_3)^2 d\tau = 0$, and hence $\mathbf{B}_1 = \mathbf{B}_2$. *qed*

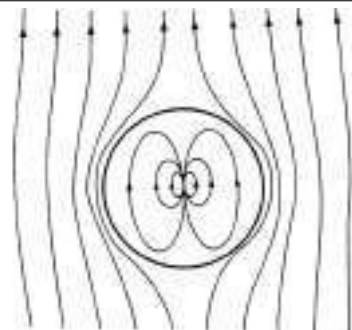
Problem 5.57

From Eq. 5.88, $\mathbf{B}_{\text{tot}} = B_0 \hat{\mathbf{z}} - \frac{\mu_0 m_0}{4\pi r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}})$. There-

fore $\mathbf{B} \cdot \hat{\mathbf{r}} = B_0 (\hat{\mathbf{z}} \cdot \hat{\mathbf{r}}) - \frac{\mu_0 m_0}{4\pi r^3} 2 \cos \theta = \left(B_0 - \frac{\mu_0 m_0}{2\pi r^3} \right) \cos \theta$.

This is zero, for all θ , when $r = R$, given by $B_0 = \frac{\mu_0 m_0}{2\pi R^3}$, or

$$R = \left(\frac{\mu_0 m_0}{2\pi B_0} \right)^{1/3}. \text{ Evidently no field lines cross this sphere.}$$



Problem 5.58

$$(a) \ I = \frac{Q}{(2\pi/\omega)} = \frac{Q\omega}{2\pi}; \ a = \pi R^2; \ \mathbf{m} = \frac{Q\omega}{2\pi} \pi R^2 \hat{\mathbf{z}} = \frac{Q}{2} \omega R^2 \hat{\mathbf{z}}. \quad L = RMv = M\omega R^2; \ \mathbf{L} = M\omega R^2 \hat{\mathbf{z}}.$$

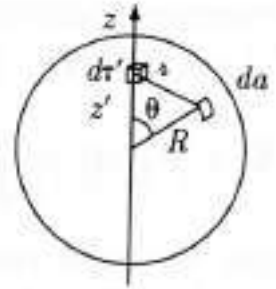
$$\frac{m}{L} = \frac{Q}{2} \frac{\omega R^2}{M\omega R^2} = \frac{Q}{2M}. \quad \boxed{\mathbf{m} = \left(\frac{Q}{2M}\right) \mathbf{L}}, \text{ and the gyromagnetic ratio is } \boxed{g = \frac{Q}{2M}}.$$

(b) Because g is *independent of R* , the same ratio applies to all “donuts”, and hence to the entire sphere (or any other figure of revolution): $\boxed{g = \frac{Q}{2M}}.$

$$(c) \ m = \frac{e}{2m} \frac{\hbar}{2} = \frac{e\hbar}{4m} = \frac{(1.60 \times 10^{-19})(1.05 \times 10^{-34})}{4(9.11 \times 10^{-31})} = \boxed{4.61 \times 10^{-24} \text{ A m}^2}.$$

Problem 5.59

(a) $\mathbf{B}_{\text{ave}} = \frac{1}{(3/4)\pi R^3} \int \mathbf{B} \, d\tau = \frac{3}{4\pi R^3} \int (\nabla \times \mathbf{A}) \, d\tau =$
 $-\frac{3}{4\pi R^3} \oint \mathbf{A} \times d\mathbf{a} = -\frac{3}{4\pi R^3} \frac{\mu_0}{4\pi} \oint \left\{ \int \frac{\mathbf{J}}{r} \, d\tau' \right\} \times d\mathbf{a} =$
 $-\frac{3\mu_0}{(4\pi)^2 R^3} \int \mathbf{J} \times \left\{ \oint \frac{1}{r} \, d\mathbf{a} \right\} \, d\tau'.$ Note that $\mathbf{J}$ depends on the source point $\mathbf{r}'$, not on the field point $\mathbf{r}$. To do the surface integral, choose the (x, y, z) coordinates so that $\mathbf{r}'$ lies on the z axis (see diagram). Then $r = \sqrt{R^2 + (z')^2 - 2Rz' \cos \theta}$, while $d\mathbf{a} = R^2 \sin \theta \, d\theta \, d\phi \, \hat{\mathbf{r}}$. By symmetry, the x and y components must integrate to zero; since the z component of $\hat{\mathbf{r}}$ is $\cos \theta$, we have



$$\oint \frac{1}{r} \, d\mathbf{a} = \hat{\mathbf{z}} \int \frac{\cos \theta}{\sqrt{R^2 + (z')^2 - 2Rz' \cos \theta}} R^2 \sin \theta \, d\theta \, d\phi = 2\pi R^2 \hat{\mathbf{z}} \int_0^\pi \frac{\cos \theta \sin \theta}{\sqrt{R^2 + (z')^2 - 2Rz' \cos \theta}} \, d\theta.$$

Let $u \equiv \cos \theta$, so $du = -\sin \theta \, d\theta$.

$$\begin{aligned} &= 2\pi R^2 \hat{\mathbf{z}} \int_{-1}^1 \frac{u}{\sqrt{R^2 + (z')^2 - 2Rz'u}} \, du \\ &= 2\pi R^2 \hat{\mathbf{z}} \left\{ -\frac{2[2(R^2 + (z')^2) + 2Rz'u]}{3(2Rz')^2} \sqrt{R^2 + (z')^2 - 2Rz'u} \right\} \Big|_{-1}^1 \\ &= -\frac{2\pi R^2 \hat{\mathbf{z}}}{3(Rz')^2} \left\{ [R^2 + (z')^2 + Rz'] \sqrt{R^2 + (z')^2 - 2Rz'} - [R^2 + (z')^2 - Rz'] \sqrt{R^2 + (z')^2 + 2Rz'} \right\} \\ &= -\left[\frac{2\pi}{3(z')^2} \hat{\mathbf{z}} \right] \left\{ [R^2 + (z')^2 + Rz'] |R - z'| - [R^2 + (z')^2 - Rz'] (R + z') \right\} \\ &= \begin{cases} \frac{4\pi}{3} z' \hat{\mathbf{z}} = \frac{4\pi}{3} \mathbf{r}', & (r' < R); \\ \frac{4\pi R^3}{3(z')^2} \hat{\mathbf{z}} = \frac{4\pi}{3} \frac{R^3}{(r')^3} \mathbf{r}', & (r' > R). \end{cases} \end{aligned}$$

For now we want $r' < R$, so $\mathbf{B}_{\text{ave}} = -\frac{3\mu_0}{(4\pi)^2 R^3} \frac{4\pi}{3} \int (\mathbf{J} \times \mathbf{r}') d\tau' = -\frac{\mu_0}{4\pi R^3} \int (\mathbf{J} \times \mathbf{r}') d\tau'$. Now $\mathbf{m} = \frac{1}{2} \int (\mathbf{r} \times \mathbf{J}) d\tau$ (Eq. 5.90), so $\mathbf{B}_{\text{ave}} = \frac{\mu_0}{4\pi} \frac{2\mathbf{m}}{R^3}$. qed

(b) This time $r' > R$, so $\mathbf{B}_{\text{ave}} = -\frac{3\mu_0}{(4\pi)^2 R^3} \frac{4\pi}{3} R^3 \int \left(\mathbf{J} \times \frac{\mathbf{r}'}{(r')^3} \right) d\tau' = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} \times \hat{\mathbf{z}}}{\mathbf{z}^2} d\tau'$, where $\mathbf{z}$ now goes from the source point to the *center* ($\mathbf{z} = -\mathbf{r}'$). Thus $\mathbf{B}_{\text{ave}} = \mathbf{B}_{\text{cen}}$. qed

Problem 5.60

(a) Problem 5.53 gives the dipole moment of a shell: $\mathbf{m} = \frac{4\pi}{3} \sigma \omega R^4 \hat{\mathbf{z}}$. Let $R \rightarrow r, \sigma \rightarrow \rho dr$, and integrate:

$$\mathbf{m} = \frac{4\pi}{3} \omega \rho \hat{\mathbf{z}} \int_0^R r^4 dr = \frac{4\pi}{3} \omega \rho \frac{R^5}{5} \hat{\mathbf{z}}. \quad \text{But } \rho = \frac{Q}{(4/3)\pi R^3}, \text{ so } \boxed{\mathbf{m} = \frac{1}{5} Q \omega R^2 \hat{\mathbf{z}}.}$$

$$(b) \mathbf{B}_{\text{ave}} = \frac{\mu_0}{4\pi} \frac{2\mathbf{m}}{R^3} = \boxed{\frac{\mu_0}{4\pi} \frac{2Q\omega}{5R} \hat{\mathbf{z}}.}$$

$$(c) \mathbf{A} \cong \frac{\mu_0}{4\pi} \frac{m \sin \theta}{r^2} \hat{\phi} = \boxed{\frac{\mu_0}{4\pi} \frac{Q\omega R^2}{5} \frac{\sin \theta}{r^2} \hat{\phi}.}$$

(d) Use Eq. 5.69, with $R \rightarrow \bar{r}, \sigma \rightarrow \rho d\bar{r}$, and integrate:

$$\mathbf{A} = \frac{\mu_0 \omega \rho}{3} \frac{\sin \theta}{r^2} \hat{\phi} \int_0^R \bar{r}^4 d\bar{r} = \frac{\mu_0 \omega}{3} \frac{3Q}{4\pi R^3} \frac{\sin \theta}{r^2} \frac{R^5}{5} \hat{\phi} = \boxed{\frac{\mu_0}{4\pi} \frac{Q\omega R^2}{5} \frac{\sin \theta}{r^2} \hat{\phi}.}$$

This is *identical* to (c); evidently the field is *pure* dipole, for points outside the sphere.

(e) According to Prob. 5.30, the field is $\mathbf{B} = \frac{\mu_0 \omega Q}{4\pi R} \left[\left(1 - \frac{3r^2}{5R^2} \right) \cos \theta \hat{\mathbf{r}} - \left(1 - \frac{6r^2}{5R^2} \right) \sin \theta \hat{\theta} \right]$. The average obviously points in the z direction, so take the z component of $\hat{\mathbf{r}} (\cos \theta)$ and $\hat{\theta} (-\sin \theta)$:

$$\begin{aligned} B_{\text{ave}} &= \frac{\mu_0 \omega Q}{4\pi R} \frac{1}{(4/3)\pi R^3} \int \left[\left(1 - \frac{3r^2}{5R^2} \right) \cos^2 \theta + \left(1 - \frac{6r^2}{5R^2} \right) \sin^2 \theta \right] r^2 \sin \theta dr d\theta d\phi \\ &= \frac{3\mu_0 \omega Q}{(4\pi R^2)^2} 2\pi \int_0^\pi \left[\left(\frac{r^3}{3} - \frac{3}{5} \frac{R^5}{5R^2} \right) \cos^2 \theta + \left(\frac{R^3}{3} - \frac{6}{5} \frac{R^5}{5R^2} \right) \sin^2 \theta \right] \sin \theta d\theta \\ &= \frac{3\mu_0 \omega Q}{8\pi R^4} R^3 \int_0^\pi \left(\frac{16}{75} \cos^2 \theta + \frac{7}{75} \sin^2 \theta \right) \sin \theta d\theta = \frac{3\mu_0 \omega Q}{8\pi R} \frac{1}{75} \int_0^\pi (7 + 9 \cos^2 \theta) \sin \theta d\theta \\ &= \frac{\mu_0 \omega Q}{200\pi R} (-7 \cos \theta - 3 \cos^3 \theta) \Big|_0^\pi = \frac{\mu_0 \omega Q}{200\pi R} (20) = \frac{\mu_0 \omega Q}{10\pi R} \text{ (same as (b)). } \checkmark \end{aligned}$$

Problem 5.61

The issue (and the integral) is identical to the one in Prob. 3.48. The resolution (as before) is to regard Eq. 5.89 as correct outside an infinitesimal sphere centered at the dipole. *Inside* this sphere the field is a delta-function, $\mathbf{A} \delta^3(\mathbf{r})$, with $\mathbf{A}$ selected so as to make the average field consistent with Prob. 5.59:

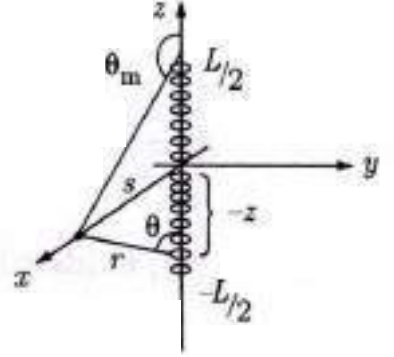
$$\mathbf{B}_{\text{ave}} = \frac{1}{(4/3)\pi R^3} \int \mathbf{A} \delta^3(\mathbf{r}) d\tau = \frac{3}{4\pi R^3} \mathbf{A} = \frac{\mu_0}{4\pi} \frac{2\mathbf{m}}{R^3} \Rightarrow \mathbf{A} = \frac{2\mu_0 \mathbf{m}}{3}. \quad \text{The added term is } \boxed{\frac{2\mu_0}{3} \mathbf{m} \delta^3(\mathbf{r}).}$$

Problem 5.62

For a dipole at the origin and a field point in the xz plane ($\phi = 0$), we have

$$\begin{aligned}\mathbf{B} &= \frac{\mu_0}{4\pi} \frac{m}{r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}) = \frac{\mu_0}{4\pi} \frac{m}{r^3} [2 \cos \theta (\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{z}}) + \sin \theta (\cos \theta \hat{\mathbf{x}} - \sin \theta \hat{\mathbf{z}})] \\ &= \frac{\mu_0}{4\pi} \frac{m}{r^3} [3 \sin \theta \cos \theta \hat{\mathbf{x}} + (2 \cos^2 \theta - \sin^2 \theta) \hat{\mathbf{z}}].\end{aligned}$$

Here we have a *stack* of such dipoles, running from $z = -L/2$ to $z = +L/2$. Put the field point at s on the x axis. The $\hat{\mathbf{x}}$ components cancel (because of symmetrically placed dipoles above and below $z = 0$), leaving $\mathbf{B} = \frac{\mu_0}{4\pi} 2\mathcal{M} \hat{\mathbf{z}} \int_0^{L/2} \frac{(3 \cos^2 \theta - 1)}{r^3} dz$, where $\mathcal{M}$ is the dipole moment per unit length: $m = I\pi R^2 = (\sigma v h)\pi R^2 = \sigma \omega R \pi R^2 h \Rightarrow \mathcal{M} = \frac{m}{h} = \pi \sigma \omega R^3$. Now $\sin \theta = \frac{s}{r}$, so $\frac{1}{r^3} = \frac{\sin^3 \theta}{s^3}$; $z = -s \cot \theta \Rightarrow dz = \frac{s}{\sin^2 \theta} d\theta$. Therefore



$$\begin{aligned}\mathbf{B} &= \frac{\mu_0}{2\pi} (\pi \sigma \omega R^3) \hat{\mathbf{z}} \int_{\pi/2}^{\theta_m} (3 \cos^2 \theta - 1) \frac{\sin^3 \theta}{s^3} \frac{s}{\sin^2 \theta} d\theta = \frac{\mu_0 \sigma \omega R^3}{2s^2} \hat{\mathbf{z}} \int_{\pi/2}^{\theta_m} (3 \cos^2 \theta - 1) \sin \theta d\theta \\ &= \frac{\mu_0 \sigma \omega R^3}{2s^2} \hat{\mathbf{z}} (-\cos^3 \theta + \cos \theta) \Big|_{\pi/2}^{\theta_m} = \frac{\mu_0 \sigma \omega R^3}{2s^2} \cos \theta_m (1 - \cos^2 \theta_m) \hat{\mathbf{z}} = \frac{\mu_0 \sigma \omega R^3}{2s^2} \cos \theta_m \sin^2 \theta_m \hat{\mathbf{z}}.\end{aligned}$$

But $\sin \theta_m = \frac{s}{\sqrt{s^2 + (L/2)^2}}$, and $\cos \theta_m = \frac{-(L/2)}{\sqrt{s^2 + (L/2)^2}}$, so $\mathbf{B} = -\frac{\mu_0 \sigma \omega R^3 L}{4[s^2 + (L/2)^2]^{3/2}} \hat{\mathbf{z}}.$

Chapter 6

Magnetic Fields in Matter

Problem 6.1

$$\mathbf{N} = \mathbf{m}_2 \times \mathbf{B}_1; \quad \mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m}_1 \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}_1]; \quad \hat{\mathbf{r}} = \hat{\mathbf{y}}; \quad \mathbf{m}_1 = m_1 \hat{\mathbf{z}}; \quad \mathbf{m}_2 = m_2 \hat{\mathbf{y}}. \quad \mathbf{B}_1 = -\frac{\mu_0}{4\pi} \frac{m_1}{r^3} \hat{\mathbf{z}}.$$

$$\mathbf{N} = -\frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^3} (\hat{\mathbf{y}} \times \hat{\mathbf{z}}) = -\frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^3} \hat{\mathbf{x}}. \quad \text{Here } m_1 = \pi a^2 I, m_2 = b^2 I. \quad \text{So } \boxed{\mathbf{N} = -\frac{\mu_0}{4} \frac{(abI)^2}{r^3} \hat{\mathbf{x}}.} \quad \text{Final orientation :}$$

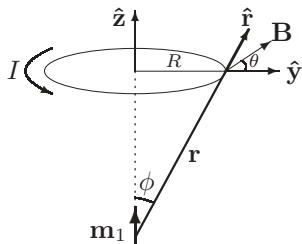
downward $(-\hat{\mathbf{z}})$.

Problem 6.2

$d\mathbf{F} = I d\mathbf{l} \times \mathbf{B}$; $d\mathbf{N} = \mathbf{r} \times d\mathbf{F} = I \mathbf{r} \times (d\mathbf{l} \times \mathbf{B})$. Now (Prob. 1.6): $\mathbf{r} \times (d\mathbf{l} \times \mathbf{B}) + d\mathbf{l} \times (\mathbf{B} \times \mathbf{r}) + \mathbf{B} \times (\mathbf{r} \times d\mathbf{l}) = \mathbf{0}$. But $d[\mathbf{r} \times (\mathbf{r} \times \mathbf{B})] = d\mathbf{r} \times (\mathbf{r} \times \mathbf{B}) + \mathbf{r} \times (d\mathbf{r} \times \mathbf{B})$ (since $\mathbf{B}$ is constant), and $d\mathbf{r} = d\mathbf{l}$, so $d\mathbf{l} \times (\mathbf{B} \times \mathbf{r}) = \mathbf{r} \times (d\mathbf{l} \times \mathbf{B}) - d[\mathbf{r} \times (\mathbf{r} \times \mathbf{B})]$. Hence $2\mathbf{r} \times (d\mathbf{l} \times \mathbf{B}) = d[\mathbf{r} \times (\mathbf{r} \times \mathbf{B})] - \mathbf{B} \times (\mathbf{r} \times d\mathbf{l})$. $d\mathbf{N} = \frac{1}{2} I \{d[\mathbf{r} \times (\mathbf{r} \times \mathbf{B})] - \mathbf{B} \times (\mathbf{r} \times d\mathbf{l})\}$. $\therefore \mathbf{N} = \frac{1}{2} I \{ \oint d[\mathbf{r} \times (\mathbf{r} \times \mathbf{B})] - \mathbf{B} \times \oint (\mathbf{r} \times d\mathbf{l}) \}$. But the first term is zero ($\oint d(\cdots) = \mathbf{0}$), and the second integral is $2\mathbf{a}$ (Eq. 1.107). So $\mathbf{N} = -I(\mathbf{B} \times \mathbf{a}) = \mathbf{m} \times \mathbf{B}$. $\quad \text{qed}$

Problem 6.3

(a)



According to Eq. 6.2, $F = 2\pi IRB \cos \theta$. But $\mathbf{B} = \frac{\mu_0}{4\pi} \frac{[3(\mathbf{m}_1 \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}_1]}{r^3}$, and $B \cos \theta = \mathbf{B} \cdot \hat{\mathbf{y}}$, so $B \cos \theta = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\mathbf{m}_1 \cdot \hat{\mathbf{r}})(\hat{\mathbf{r}} \cdot \hat{\mathbf{y}}) - (\mathbf{m}_1 \cdot \hat{\mathbf{y}})]$. But $\mathbf{m}_1 \cdot \hat{\mathbf{y}} = 0$ and $\hat{\mathbf{r}} \cdot \hat{\mathbf{y}} = \sin \phi$, while $\mathbf{m}_1 \cdot \hat{\mathbf{r}} = m_1 \cos \theta$. $\therefore B \cos \theta = \frac{\mu_0}{4\pi} \frac{1}{r^3} 3m_1 \sin \phi \cos \phi$.

$$F = 2\pi IR \frac{\mu_0}{4\pi} \frac{1}{r^3} 3m_1 \sin \phi \cos \phi. \quad \text{Now } \sin \phi = \frac{R}{r}, \quad \cos \phi = \sqrt{r^2 - R^2}/r, \quad \text{so } F = 3\frac{\mu_0}{2} m_1 IR^2 \frac{\sqrt{r^2 - R^2}}{r^5}.$$

$$\text{But } IR^2 \pi = m_2, \quad \text{so } F = \frac{3\mu_0}{2\pi} m_1 m_2 \frac{\sqrt{r^2 - R^2}}{r^5}, \quad \text{while for a dipole, } R \ll r, \quad \text{so } \boxed{F = \frac{3\mu_0}{2\pi} \frac{m_1 m_2}{r^4}}.$$

$$(b) \quad \mathbf{F} = \nabla(\mathbf{m}_2 \cdot \mathbf{B}) = (\mathbf{m}_2 \cdot \nabla) \mathbf{B} = (m_2 \frac{d}{dz}) \left[\underbrace{\frac{\mu_0}{4\pi} \frac{1}{z^3} (3(\mathbf{m}_1 \cdot \hat{\mathbf{z}})\hat{\mathbf{z}} - \mathbf{m}_1)}_{2\mathbf{m}_1} \right] = \frac{\mu_0}{2\pi} m_1 m_2 \hat{\mathbf{z}} \underbrace{\frac{d}{dz} \left(\frac{1}{z^3} \right)}_{-3 \frac{1}{z^4}},$$

$$\text{or, since } z = r: \quad \boxed{\mathbf{F} = -\frac{3\mu_0}{2\pi} \frac{m_1 m_2}{r^4} \hat{\mathbf{z}}.}$$

Problem 6.4

$$\begin{aligned}
 d\mathbf{F} &= I \{ (dy \hat{\mathbf{y}}) \times \mathbf{B}(0, y, 0) + (dz \hat{\mathbf{z}}) \times \mathbf{B}(0, \epsilon, z) - (dy \hat{\mathbf{y}}) \times \mathbf{B}(0, y, \epsilon) - (dz \hat{\mathbf{z}}) \times \mathbf{B}(0, 0, z) \} \\
 &= I \left\{ -(dy \hat{\mathbf{y}}) \times \underbrace{[\mathbf{B}(0, y, \epsilon) - \mathbf{B}(0, y, 0)]}_{\approx \epsilon \frac{\partial \mathbf{B}}{\partial z}} + (dz \hat{\mathbf{z}}) \times \underbrace{[\mathbf{B}(0, \epsilon, z) - \mathbf{B}(0, 0, z)]}_{\approx \epsilon \frac{\partial \mathbf{B}}{\partial y}} \right\} \\
 &\Rightarrow I \epsilon^2 \left\{ \hat{\mathbf{z}} \times \frac{\partial \mathbf{B}}{\partial y} - \hat{\mathbf{y}} \times \frac{\partial \mathbf{B}}{\partial z} \right\} \cdot \left[\text{Note that } \int dy \frac{\partial \mathbf{B}}{\partial z} \Big|_{0, y, 0} \approx \epsilon \frac{\partial \mathbf{B}}{\partial z} \Big|_{0, 0, 0} \text{ and } \int dz \frac{\partial \mathbf{B}}{\partial y} \Big|_{0, 0, z} \approx \epsilon \frac{\partial \mathbf{B}}{\partial y} \Big|_{0, 0, 0} \right]
 \end{aligned}$$

$$\begin{aligned}
 \mathbf{F} &= m \left\{ \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & 1 \\ \frac{\partial B_x}{\partial y} & \frac{\partial B_y}{\partial y} & \frac{\partial B_z}{\partial y} \end{vmatrix} - \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 1 & 0 \\ \frac{\partial B_x}{\partial z} & \frac{\partial B_y}{\partial z} & \frac{\partial B_z}{\partial z} \end{vmatrix} \right\} = m \left\{ \hat{\mathbf{y}} \frac{\partial B_x}{\partial y} - \hat{\mathbf{x}} \frac{\partial B_y}{\partial y} - \hat{\mathbf{x}} \frac{\partial B_z}{\partial z} + \hat{\mathbf{z}} \frac{\partial B_x}{\partial z} \right\} \\
 &= m \left[\hat{\mathbf{x}} \frac{\partial B_x}{\partial x} + \hat{\mathbf{y}} \frac{\partial B_x}{\partial y} + \hat{\mathbf{z}} \frac{\partial B_x}{\partial z} \right] \quad \left(\text{using } \nabla \cdot \mathbf{B} = 0 \text{ to write } \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = -\frac{\partial B_x}{\partial x} \right).
 \end{aligned}$$

But $\mathbf{m} \cdot \mathbf{B} = mB_x$ (since $\mathbf{m} = m\hat{\mathbf{x}}$, here), so $\nabla(\mathbf{m} \cdot \mathbf{B}) = m\nabla(B_x) = m \left(\frac{\partial B_x}{\partial x} \hat{\mathbf{x}} + \frac{\partial B_x}{\partial y} \hat{\mathbf{y}} + \frac{\partial B_x}{\partial z} \hat{\mathbf{z}} \right)$.

Therefore $\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$. qed

Problem 6.5

(a) $\mathbf{B} = \mu_0 J_0 x \hat{\mathbf{y}}$ (Prob. 5.15).

$\mathbf{m} \cdot \mathbf{B} = 0$, so Eq. 6.3 says $\boxed{\mathbf{F} = \mathbf{0}}$.

(b) $\mathbf{m} \cdot \mathbf{B} = m_0 \mu_0 J_0 x$, so $\boxed{\mathbf{F} = m_0 \mu_0 J_0 \hat{\mathbf{x}}}$.

(c) Use product rule #4: $\nabla(\mathbf{p} \cdot \mathbf{E})$

$$= \mathbf{p} \times (\nabla \times \mathbf{E}) + \mathbf{E} \times (\nabla \times \mathbf{p}) + (\mathbf{p} \cdot \nabla) \mathbf{E} + (\mathbf{E} \cdot \nabla) \mathbf{p}.$$

But $\mathbf{p}$ does not depend on (x, y, z) , so the second

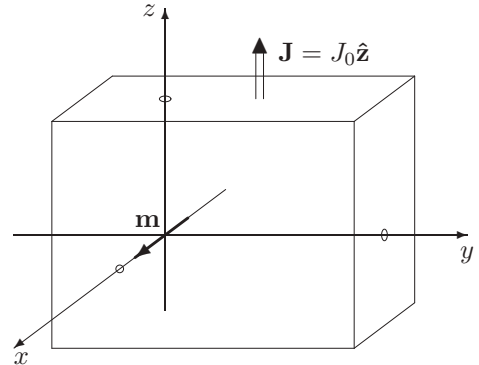
and fourth terms vanish, and $\nabla \times \mathbf{E} = 0$, so the

first term is zero. Hence $\nabla(\mathbf{p} \cdot \mathbf{E}) = (\mathbf{p} \cdot \nabla) \mathbf{E}$. qed

This argument does *not* apply to the magnetic analog,

since $\nabla \times \mathbf{B} \neq 0$. In fact, $\nabla(\mathbf{m} \cdot \mathbf{B}) = (\mathbf{m} \cdot \nabla) \mathbf{B} + \mu_0(\mathbf{m} \times \mathbf{J})$.

$$(\mathbf{m} \cdot \nabla) \mathbf{B}_a = m_0 \frac{\partial}{\partial x} (\mathbf{B}) = m_0 \mu_0 J_0 \hat{\mathbf{y}}, \quad (\mathbf{m} \cdot \nabla) \mathbf{B}_b = m_0 \frac{\partial}{\partial y} (\mu_0 J_0 x \hat{\mathbf{y}}) = 0.$$



Problem 6.6

Aluminum, copper, copper chloride, and sodium all have an *odd* number of electrons, so we expect them to be paramagnetic. The rest (having an even number) should be diamagnetic.

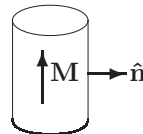
Problem 6.7

$$\mathbf{J}_b = \nabla \times \mathbf{M} = 0; \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = M \hat{\phi}.$$

The field is that of a surface current $\mathbf{K}_b = M \hat{\phi}$,

but that's just a solenoid, so the field

outside is zero, and inside $B = \mu_0 K_b = \mu_0 M$. Moreover, it points upward (in the drawing), so $\boxed{\mathbf{B} = \mu_0 \mathbf{M}}$.



Problem 6.8

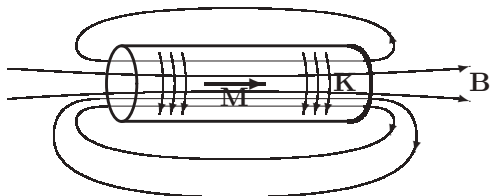
$$\nabla \times \mathbf{M} = \mathbf{J}_b = \frac{1}{s} \frac{\partial}{\partial s} (s k s^2) \hat{\mathbf{z}} = \frac{1}{s} (3k s^2) \hat{\mathbf{z}} = 3k s \hat{\mathbf{z}}, \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = k s^2 (\hat{\phi} \times \hat{\mathbf{s}}) = -k R^2 \hat{\mathbf{z}}.$$

So the bound current flows up the cylinder, and returns down the surface. [Incidentally, the *total* current should be zero... *is* it? Yes, for $\int J_b da = \int_0^R (3ks)(2\pi s ds) = 2\pi k R^3$, while $\int K_b dl = (-kR^2)(2\pi R) = -2\pi k R^3$.] Since these currents have cylindrical symmetry, we can get the field by Ampère's law:

$$B \cdot 2\pi s = \mu_0 I_{\text{enc}} = \mu_0 \int_0^s J_b da = 2\pi k \mu_0 s^3 \Rightarrow \boxed{\mathbf{B} = \mu_0 k s^2 \hat{\phi}} = \mu_0 \mathbf{M}.$$

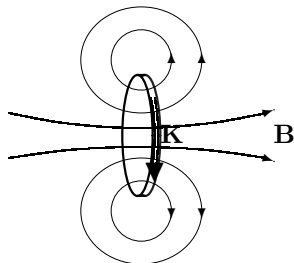
Outside the cylinder $I_{\text{enc}} = 0$, so $\boxed{\mathbf{B} = \mathbf{0}}$.

Problem 6.9

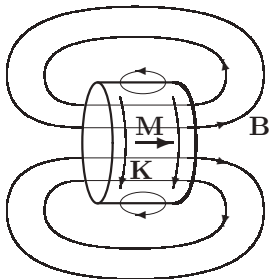


$$\boxed{\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = M \hat{\phi}.$$

(Essentially a long solenoid)



(Essentially a physical dipole)



(Intermediate case)

[The external fields are the same as in the electrical case; the *internal* fields (inside the bar) are completely different—in fact, opposite in direction.]

Problem 6.10

$K_b = M$, so the field inside a *complete* ring would be $\mu_0 M$. The field of a square loop, at the center, is given by Prob. 5.8: $B_{\text{sq}} = \sqrt{2} \mu_0 I / \pi R$. Here $I = Mw$, and $R = a/2$, so

$$B_{\text{sq}} = \frac{\sqrt{2} \mu_0 M w}{\pi(a/2)} = \frac{2\sqrt{2} \mu_0 M w}{\pi a}; \quad \text{net field in gap: } \boxed{\mathbf{B} = \mu_0 \mathbf{M} \left(1 - \frac{2\sqrt{2} w}{\pi a} \right)}.$$

Problem 6.11

As in Sec. 4.2.3, we want the average of $\mathbf{B} = \mathbf{B}_{\text{out}} + \mathbf{B}_{\text{in}}$, where $\mathbf{B}_{\text{out}}$ is due to molecules *outside* a small sphere around point P , and $\mathbf{B}_{\text{in}}$ is due to molecules *inside* the sphere. The average of $\mathbf{B}_{\text{out}}$ is same as field at center (Prob. 5.59b), and for this it is OK to use Eq. 6.10, since the center is “far” from all the molecules in question:

$$\mathbf{A}_{\text{out}} = \frac{\mu_0}{4\pi} \int_{\text{outside}} \frac{\mathbf{M} \times \hat{\mathbf{r}}}{r^2} d\tau$$

The average of $\mathbf{B}_{\text{in}}$ is $\frac{\mu_0}{4\pi} \left(\frac{2\mathbf{m}}{R^3} \right)$ —Eq. 5.93—where $\mathbf{m} = \frac{4}{3}\pi R^3 \mathbf{M}$. Thus the average $\mathbf{B}_{\text{in}}$ is $2\mu_0 \mathbf{M}/3$. But what is *left out* of the integral $\mathbf{A}_{\text{out}}$ is the contribution of a uniformly magnetized sphere, to wit: $2\mu_0 \mathbf{M}/3$ (Eq. 6.16), and this is precisely what $\mathbf{B}_{\text{in}}$ puts back in. So we’ll get the correct macroscopic field using Eq. 6.10. **qed**

Problem 6.12

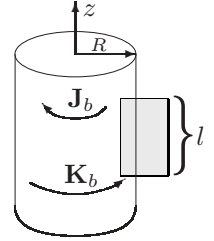
- (a) $\mathbf{M} = ks\hat{\mathbf{z}}$; $\mathbf{J}_b = \nabla \times \mathbf{M} = -k\hat{\phi}$; $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = kR\hat{\phi}$.

$\mathbf{B}$ is in the z direction (this is essentially a superposition of solenoids). So

$\mathbf{B} = 0$ outside. Use the amperian loop shown (shaded)—inner side at radius s :

$$\oint \mathbf{B} \cdot d\mathbf{l} = Bl = \mu_0 I_{\text{enc}} = \mu_0 \left[\int J_b da + K_b l \right] = \mu_0 [-kl(R-s) + kRl] = \mu_0 kls.$$

$$\therefore \mathbf{B} = \mu_0 ks\hat{\mathbf{z}} \text{ inside.}$$



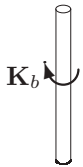
- (b) By symmetry, $\mathbf{H}$ points in the z direction. That same amperian loop gives $\oint \mathbf{H} \cdot d\mathbf{l} = Hl = \mu_0 I_{f\text{enc}} = 0$, since there *is* no free current here. So $\mathbf{H} = 0$, and hence $\mathbf{B} = \mu_0 \mathbf{M}$. Outside $\mathbf{M} = 0$, so $\mathbf{B} = 0$; inside $\mathbf{M} = ks\hat{\mathbf{z}}$, so $\mathbf{B} = \mu_0 ks\hat{\mathbf{z}}$.

Problem 6.13

- (a) The field of a magnetized sphere is $\frac{2}{3}\mu_0 \mathbf{M}$ (Eq. 6.16), so $\mathbf{B} = \mathbf{B}_0 - \frac{2}{3}\mu_0 \mathbf{M}$, with the sphere removed.

$$\text{In the cavity, } \mathbf{H} = \frac{1}{\mu_0} \mathbf{B}, \text{ so } \mathbf{H} = \frac{1}{\mu_0} (\mathbf{B}_0 - \frac{2}{3}\mu_0 \mathbf{M}) = \mathbf{H}_0 + \mathbf{M} - \frac{2}{3}\mathbf{M} \Rightarrow \mathbf{H} = \mathbf{H}_0 + \frac{1}{3}\mathbf{M}.$$

- (b) The field inside a long solenoid is $\mu_0 K$. Here $K = M$, so the field of the bound current on the inside surface of the cavity is $\mu_0 M$, pointing *down*. Therefore



$$\mathbf{B} = \mathbf{B}_0 - \mu_0 \mathbf{M};$$

$$\mathbf{H} = \frac{1}{\mu_0} (\mathbf{B}_0 - \mu_0 \mathbf{M}) = \frac{1}{\mu_0} \mathbf{B}_0 - \mathbf{M} \Rightarrow \mathbf{H} = \mathbf{H}_0.$$

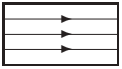
- (c)

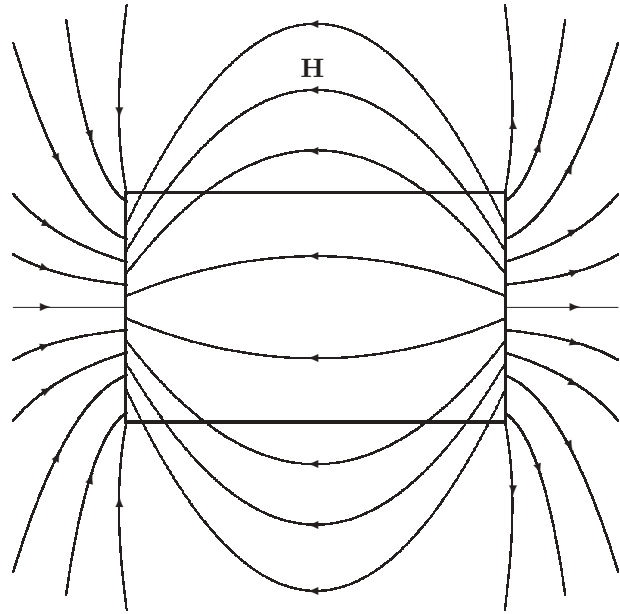
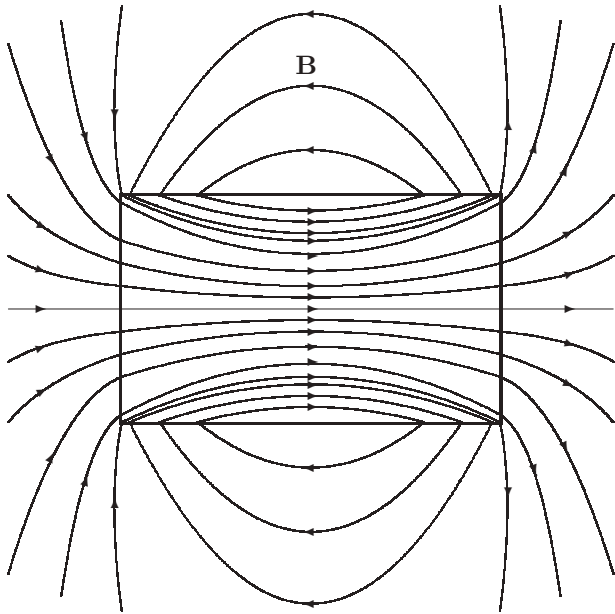


This time the bound currents are small, and far away from the center, so $\mathbf{B} = \mathbf{B}_0$, while $\mathbf{H} = \frac{1}{\mu_0} \mathbf{B}_0 = \mathbf{H}_0 + \mathbf{M} \Rightarrow \mathbf{H} = \mathbf{H}_0 + \mathbf{M}.$

[*Comment:* In the wafer, $\mathbf{B}$ is the field in the medium; in the needle, $\mathbf{H}$ is the $\mathbf{H}$ in the medium; in the sphere (intermediate case) both $\mathbf{B}$ and $\mathbf{H}$ are modified.]

Problem 6.14

$\mathbf{M}$: ; $\mathbf{B}$ is the same as the field of a short solenoid; $\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}$.



Problem 6.15

“Potentials”:

$$\begin{cases} W_{\text{in}}(r, \theta) = \sum A_l r^l P_l(\cos \theta), & (r < R); \\ W_{\text{out}}(r, \theta) = \sum \frac{B_l}{r^{l+1}} P_l(\cos \theta), & (r > R). \end{cases}$$

Boundary Conditions:

$$\begin{cases} \text{(i)} & W_{\text{in}}(R, \theta) = W_{\text{out}}(R, \theta), \\ \text{(ii)} & -\frac{\partial W_{\text{out}}}{\partial r} \Big|_R + \frac{\partial W_{\text{in}}}{\partial r} \Big|_R = M^\perp = M \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = M \cos \theta. \end{cases}$$

(The continuity of W follows from the gradient theorem: $W(\mathbf{b}) - W(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} \nabla W \cdot d\mathbf{l} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{H} \cdot d\mathbf{l}$; if the two points are infinitesimally separated, this last integral $\rightarrow 0$.)

$$\begin{cases} \text{(i)} \Rightarrow A_l R^l = \frac{B_l}{R^{l+1}} \Rightarrow B_l = R^{2l+1} A_l, \\ \text{(ii)} \Rightarrow \sum (l+1) \frac{B_l}{R^{l+2}} P_l(\cos \theta) + \sum l A_l R^{l-2} P_l(\cos \theta) = M \cos \theta. \end{cases}$$

Combining these:

$$\sum (2l+1) R^{l-1} A_l P_l(\cos \theta) = M \cos \theta, \text{ so } A_l = 0 \text{ } (l \neq 1), \text{ and } 3A_1 = M \Rightarrow A_1 = \frac{M}{3}.$$

Thus $W_{\text{in}}(r, \theta) = \frac{M}{3} r \cos \theta = \frac{M}{3} z$, and hence $\mathbf{H}_{\text{in}} = -\nabla W_{\text{in}} = -\frac{M}{3} \hat{\mathbf{z}} = -\frac{1}{3} \mathbf{M}$, so

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M}) = \mu_0 \left(-\frac{1}{3} \mathbf{M} + \mathbf{M} \right) = \boxed{\frac{2}{3} \mu_0 \mathbf{M}}. \quad \checkmark$$

Problem 6.16

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{f_{\text{enc}}} = I, \text{ so } \mathbf{H} = \frac{I}{2\pi s} \hat{\phi}. \quad \mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H} = \boxed{\mu_0(1 + \chi_m) \frac{I}{2\pi s} \hat{\phi}}. \quad \mathbf{M} = \chi_m \mathbf{H} = \boxed{\frac{\chi_m I}{2\pi s} \hat{\phi}}.$$

$$\mathbf{J}_b = \nabla \times \mathbf{M} = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\chi_m I}{2\pi s} \right) \hat{\mathbf{z}} = \boxed{0}. \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = \begin{cases} \frac{\chi_m I}{2\pi a} \hat{\mathbf{z}}, & \text{at } s = a; \\ -\frac{\chi_m I}{2\pi b} \hat{\mathbf{z}}, & \text{at } s = b. \end{cases}$$

Total enclosed current, for an amperian loop between the cylinders:

$$I + \frac{\chi_m I}{2\pi a} 2\pi a = (1 + \chi_m)I, \quad \text{so } \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} = \mu_0(1 + \chi_m)I \Rightarrow \mathbf{B} = \frac{\mu_0(1 + \chi_m)I}{2\pi s} \hat{\phi}. \quad \checkmark$$

Problem 6.17

From Eq. 6.20: $\oint \mathbf{H} \cdot d\mathbf{l} = H(2\pi s) = I_{f_{\text{enc}}} = \begin{cases} I(s^2/a^2), & (s < a); \\ I & (s > a). \end{cases}$

$$H = \left\{ \frac{Is}{2\pi a^2}, \quad (s < a) \right\}, \quad \text{so } B = \mu H = \begin{cases} \frac{\mu_0(1 + \chi_m)Is}{2\pi a^2}, & (s < a); \\ \frac{\mu_0 I}{2\pi s}, & (s > a). \end{cases}$$

$\mathbf{J}_b = \chi_m \mathbf{J}_f$ (Eq. 6.33), and $J_f = \frac{I}{\pi a^2}$, so $\boxed{J_b = \frac{\chi_m I}{\pi a^2}}$ (same direction as I).

$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} = \chi_m \mathbf{H} \times \hat{\mathbf{n}} \Rightarrow \boxed{\mathbf{K}_b = \frac{\chi_m I}{2\pi a} \hat{\mathbf{z}}}$ (opposite direction to I).

$I_b = J_b(\pi a^2) + K_b(2\pi a) = \chi_m I - \chi_m I = \boxed{0}$ (as it *should* be, of course).

Problem 6.18

By the method of Prob. 6.15:

For large r , we want $\mathbf{B}(r, \theta) \rightarrow \mathbf{B}_0 = B_0 \hat{\mathbf{z}}$, so $\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} \rightarrow \frac{1}{\mu_0} B_0 \hat{\mathbf{z}}$, and hence $W \rightarrow -\frac{1}{\mu_0} B_0 z = -\frac{1}{\mu_0} B_0 r \cos \theta$.

“Potentials”:

$$\begin{cases} W_{\text{in}}(r, \theta) = \sum A_l r^l P_l(\cos \theta), & (r < R); \\ W_{\text{out}}(r, \theta) = -\frac{1}{\mu_0} B_0 r \cos \theta + \sum \frac{B_l}{r^{l+1}} P_l(\cos \theta), & (r > R). \end{cases}$$

Boundary Conditions:

$$\begin{cases} \text{(i)} & W_{\text{in}}(R, \theta) = W_{\text{out}}(R, \theta), \\ \text{(ii)} & -\mu_0 \frac{\partial W_{\text{out}}}{\partial r} \Big|_R + \mu \frac{\partial W_{\text{in}}}{\partial r} \Big|_R = 0. \end{cases}$$

(The latter follows from Eq. 6.26.)

$$\text{(ii)} \Rightarrow \mu_0 \left[\frac{1}{\mu_0} B_0 \cos \theta + \sum (l+1) \frac{B_l}{R^{l+2}} P_l(\cos \theta) \right] + \mu \sum l A_l R^{l-1} P_l(\cos \theta) = 0.$$

For $l \neq 1$, (i) $\Rightarrow B_l = R^{2l+1} A_l$, so $[\mu_0(l+1) + \mu l] A_l R^{l-1} = 0$, and hence $A_l = 0$.

For $l = 1$, (i) $\Rightarrow A_1 R = -\frac{1}{\mu_0} B_0 R + B_1/R^2$, and (ii) $\Rightarrow B_0 + 2\mu_0 B_1/R^3 + \mu A_1 = 0$, so $A_1 = -3B_0/(2\mu_0 + \mu)$.

$$W_{\text{in}}(r, \theta) = -\frac{3B_0}{(2\mu_0 + \mu)} r \cos \theta = -\frac{3B_0 z}{(2\mu_0 + \mu)}. \quad \mathbf{H}_{\text{in}} = -\nabla W_{\text{in}} = \frac{3B_0}{(2\mu_0 + \mu)} \hat{\mathbf{z}} = \frac{3\mathbf{B}_0}{(2\mu_0 + \mu)}.$$

$$\mathbf{B} = \mu \mathbf{H} = \frac{3\mu \mathbf{B}_0}{(2\mu_0 + \mu)} = \boxed{\left(\frac{1 + \chi_m}{1 + \chi_m/3} \right) \mathbf{B}_0}.$$

By the method of Prob. 4.23:

Step 1: $\mathbf{B}_0$ magnetizes the sphere: $\mathbf{M}_0 = \chi_m \mathbf{H}_0 = \frac{\chi_m}{\mu_0(1+\chi_m)} \mathbf{B}_0$. This magnetization sets up a field within the sphere given by Eq. 6.16:

$$\mathbf{B}_1 = \frac{2}{3} \mu_0 \mathbf{M}_0 = \frac{2}{3} \frac{\chi_m}{1+\chi_m} \mathbf{B}_0 = \frac{2}{3} \kappa \mathbf{B}_0 \quad (\text{where } \kappa \equiv \frac{\chi_m}{1+\chi_m}).$$

Step 2: $\mathbf{B}_1$ magnetizes the sphere an additional amount $\mathbf{M}_1 = \frac{\kappa}{\mu_0} \mathbf{B}_1$. This sets up an additional field in the sphere:

$$\mathbf{B}_2 = \frac{2}{3} \mu_0 \mathbf{M}_1 = \frac{2}{3} \kappa \mathbf{B}_1 = \left(\frac{2\kappa}{3} \right)^2 \mathbf{B}_0, \quad \text{etc.}$$

The *total* field is:

$$\mathbf{B} = \mathbf{B}_0 + \mathbf{B}_1 + \mathbf{B}_2 + \cdots = \mathbf{B}_0 + (2\kappa/3)\mathbf{B}_0 + (2\kappa/3)^2\mathbf{B}_0 + \cdots = [1 + (2\kappa/3) + (2\kappa/3)^2 + \cdots] \mathbf{B}_0 = \frac{\mathbf{B}_0}{(1 - 2\kappa/3)}.$$

$$\frac{1}{1 - 2\kappa/3} = \frac{3}{3 - 2\chi_m/(1 + \chi_m)} = \frac{3 + 3\chi_m}{3 + 3\chi_m - 2\chi_m} = \frac{3(1 + \chi_m)}{3 + \chi_m}, \quad \text{so} \quad \boxed{\mathbf{B} = \left(\frac{1 + \chi_m}{1 + \chi_m/3} \right) \mathbf{B}_0.}$$

Problem 6.19

$\Delta \mathbf{m} = -\frac{e^2 r^2}{4m_e} \mathbf{B}$; $\mathbf{M} = \frac{\Delta \mathbf{m}}{V} = -\frac{e^2 r^2}{4m_e V} \mathbf{B}$, where V is the volume per electron. $\mathbf{M} = \chi_m \mathbf{H}$ (Eq. 6.29) $= \frac{\chi_m}{\mu_0(1+\chi_m)} \mathbf{B}$ (Eq. 6.30). So $\chi_m = -\frac{e^2 r^2}{4m_e V} \mu_0$. [Note: $\chi_m \ll 1$, so I won't worry about the $(1 + \chi_m)$ term; for the same reason we need not distinguish $\mathbf{B}$ from $\mathbf{B}_{\text{else}}$, as we did in deriving the Clausius-Mossotti equation in Prob. 4.41.] Let's say $V = \frac{4}{3}\pi r^3$. Then $\chi_m = -\frac{\mu_0}{4\pi} \left(\frac{3e^2}{4m_e r} \right)$. I'll use $1 \text{ \AA} = 10^{-10} \text{ m}$ for r . Then $\chi_m = -(10^{-7}) \left(\frac{3(1.6 \times 10^{-19})^2}{4(9.1 \times 10^{-31})(10^{-10})} \right) = \boxed{-2 \times 10^{-5}}$, which is not bad—Table 6.1 says $\chi_m = -1 \times 10^{-5}$. However, I used only *one electron* per atom (copper has 29) and a very crude value for r . Since the orbital radius is smaller for the inner electrons, they count for less ($\Delta m \sim r^2$). I have also neglected competing paramagnetic effects. But never mind... this is in the right ball park.

Problem 6.20

Place the object in a region of zero magnetic field, and heat it above the Curie point—or simply drop it on a hard surface. If it's delicate (a watch, say), place it between the poles of an electromagnet, and magnetize it back and forth many times; each time you reverse the direction, reduce the field slightly.

Problem 6.21

(a) The magnetic force on the dipole is given by Eq. 6.3; to move the dipole in from infinity *we* must exert an opposite force, so the work done is

$$U = - \int_{\infty}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{l} = - \int_{\infty}^{\mathbf{r}} \nabla(\mathbf{m} \cdot \mathbf{B}) \cdot d\mathbf{l} = -\mathbf{m} \cdot \mathbf{B}(\mathbf{r}) + \mathbf{m} \cdot \mathbf{B}(\infty)$$

(I used the gradient theorem, Eq. 1.55). As long as the magnetic field goes to zero at infinity, then, $U = -\mathbf{m} \cdot \mathbf{B}$. If the magnetic field does *not* go to zero at infinity, one must stipulate that the dipole starts out oriented perpendicular to the field.

(b) Identical to Prob. 4.8, but starting with Eq. 5.89 instead of 3.104.

(c) $U = -\frac{\mu_0}{4\pi} \frac{1}{r^3} [3 \cos \theta_1 \cos \theta_2 - \cos(\theta_2 - \theta_1)] m_1 m_2$. Or, using $\cos(\theta_2 - \theta_1) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$,

$$\boxed{U = \frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^3} (\sin \theta_1 \sin \theta_2 - 2 \cos \theta_1 \cos \theta_2).}$$

Stable position occurs at minimum energy: $\frac{\partial U}{\partial \theta_1} = \frac{\partial U}{\partial \theta_2} = 0$

$$\begin{cases} \frac{\partial U}{\partial \theta_1} = \frac{\mu_0 m_1 m_2}{4\pi r^3} (\cos \theta_1 \sin \theta_2 + 2 \sin \theta_1 \cos \theta_2) = 0 \Rightarrow 2 \sin \theta_1 \cos \theta_2 = -\cos \theta_1 \sin \theta_2; \\ \frac{\partial U}{\partial \theta_2} = \frac{\mu_0 m_1 m_2}{4\pi r^3} (\sin \theta_1 \cos \theta_2 + 2 \cos \theta_1 \sin \theta_2) = 0 \Rightarrow 2 \sin \theta_1 \cos \theta_2 = -4 \cos \theta_1 \sin \theta_2. \end{cases}$$

Thus $\sin \theta_1 \cos \theta_2 = \sin \theta_2 \cos \theta_1 = 0$. $\begin{cases} \text{Either } \sin \theta_1 = \sin \theta_2 = 0 : & \xrightarrow{\textcircled{1}} \text{ or } \xrightarrow{\textcircled{2}} \text{ } \\ \text{or } \cos \theta_1 = \cos \theta_2 = 0 : & \begin{matrix} \uparrow \uparrow \\ \textcircled{3} \end{matrix} \text{ or } \begin{matrix} \uparrow \downarrow \\ \textcircled{4} \end{matrix} \end{cases}$

Which of these is the *stable* minimum? Certainly not ② or ③—for these $\mathbf{m}_2$ is not parallel to $\mathbf{B}_1$, whereas we know $\mathbf{m}_2$ will line up along $\mathbf{B}_1$. It remains to compare ① (with $\theta_1 = \theta_2 = 0$) and ④ (with $\theta_1 = \pi/2$, $\theta_2 = -\pi/2$): $U_1 = \frac{\mu_0 m_1 m_2}{4\pi r^3}(-2)$; $U_2 = \frac{\mu_0 m_1 m_2}{4\pi r^3}(-1)$. U_1 is the lower energy, hence the more stable configuration.

Conclusion: They line up parallel, along the line joining them: $\longrightarrow \longrightarrow$

(d) They'd line up the same way: $\longrightarrow \longrightarrow \longrightarrow \longrightarrow \longrightarrow \longrightarrow$

Problem 6.22

$$\mathbf{F} = I \oint d\mathbf{l} \times \mathbf{B} = I \left(\oint d\mathbf{l} \right) \times \mathbf{B}_0 + I \oint d\mathbf{l} \times [(\mathbf{r} \cdot \nabla_0) \mathbf{B}_0] - I \left(\oint d\mathbf{l} \right) \times [(\mathbf{r}_0 \cdot \nabla_0) \mathbf{B}_0] = I \oint d\mathbf{l} \times [(\mathbf{r} \cdot \nabla_0) \mathbf{B}_0]$$

(because $\oint d\mathbf{l} = \mathbf{0}$). Now

$$(d\mathbf{l} \times \mathbf{B}_0)_i = \sum_{j,k} \epsilon_{ijk} dl_j (B_0)_k, \quad \text{and } (\mathbf{r} \cdot \nabla_0) = \sum_l r_l (\nabla_0)_l, \text{ so}$$

$$\begin{aligned} F_i &= I \sum_{j,k,l} \epsilon_{ijk} \left[\oint r_l dl_j \right] [(\nabla_0)_l (B_0)_k] \quad \left\{ \text{Lemma 1 : } \oint r_l dl_j = \sum_m \epsilon_{ljm} a_m \text{ (proof below).} \right\} \\ &= I \sum_{j,k,l,m} \epsilon_{ijk} \epsilon_{ljm} a_m (\nabla_0)_l (B_0)_k \quad \left\{ \text{Lemma 2 : } \sum_j \epsilon_{ijk} \epsilon_{ljm} = \delta_{il} \delta_{km} - \delta_{im} \delta_{kl} \text{ (proof below).} \right\} \\ &= I \sum_{k,l,m} (\delta_{il} \delta_{km} - \delta_{im} \delta_{kl}) a_m (\nabla_0)_l (B_0)_k = I \sum_k [a_k (\nabla_0)_i (B_0)_k - a_i (\nabla_0)_k (B_0)_k] \\ &= I [(\nabla_0)_i (\mathbf{a} \cdot \mathbf{B}_0) - a_i (\nabla_0 \cdot \mathbf{B}_0)]. \end{aligned}$$

But $\nabla_0 \cdot \mathbf{B}_0 = 0$ (Eq. 5.50), and $\mathbf{m} = I\mathbf{a}$ (Eq. 5.86), so $\mathbf{F} = \nabla_0(\mathbf{m} \cdot \mathbf{B}_0)$ (the subscript just reminds us to take the derivatives at the point where $\mathbf{m}$ is located). $\square$

Proof of Lemma 1:

Eq. 1.108 says $\oint (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l} = \mathbf{a} \times \mathbf{c} = -\mathbf{c} \times \mathbf{a}$. The j th component is $\sum_p \oint c_p r_p dl_j = -\sum_{p,m} \epsilon_{jpm} c_p a_m$. Pick $c_p = \delta_{pl}$ (i.e. 1 for the l th component, zero for the others). Then $\oint r_l dl_j = -\sum_m \epsilon_{jlm} a_m = \sum_m \epsilon_{ljm} a_m$. $\square$

Proof of Lemma 2:

$\epsilon_{ijk} \epsilon_{ljm} = 0$ unless ijk and ljm are both permutations of 123. In particular, i must either be l or m , and k must be the other, so

$$\sum_j \epsilon_{ijk} \epsilon_{ljm} = A \delta_{il} \delta_{km} + B \delta_{im} \delta_{kl}.$$

To determine the constant A , pick $i = l = 1$, $k = m = 3$; the only contribution comes from $j = 2$:

$$\epsilon_{123}\epsilon_{123} = 1 = A\delta_{11}\delta_{33} + B\delta_{13}\delta_{31} = A \Rightarrow A = 1.$$

To determine B , pick $i = m = 1$, $k = l = 3$:

$$\epsilon_{123}\epsilon_{321} = -1 = A\delta_{13}\delta_{31} + B\delta_{11}\delta_{33} = B \Rightarrow B = -1.$$

So

$$\sum_j \epsilon_{ijk}\epsilon_{ljm} = \delta_{il}\delta_{km} - \delta_{im}\delta_{kl}. \quad \text{qed}$$

Problem 6.23

(a) $\mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{2m}{z^3} \hat{\mathbf{z}}$ (Eq. 5.88, with $\theta = 0$). So $\mathbf{m}_2 \cdot \mathbf{B}_1 = -\frac{\mu_0}{2\pi} \frac{m^2}{z^3}$. $\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$ (Eq. 6.3) $\Rightarrow \mathbf{F} = \frac{\partial}{\partial z} \left[-\frac{\mu_0}{2\pi} \frac{m^2}{z^3} \right] \hat{\mathbf{z}} = \frac{3\mu_0 m^2}{2\pi z^4} \hat{\mathbf{z}}$. This is the magnetic force *upward* (on the upper magnet); it balances the gravitational force downward ($-m_d g \hat{\mathbf{z}}$):

$$\frac{3\mu_0 m^2}{2\pi z^4} - m_d g = 0 \Rightarrow z = \left[\frac{3\mu_0 m^2}{2\pi m_d g} \right]^{1/4}.$$

(b) The middle magnet is repelled *upward* by lower magnet and *downward* by upper magnet:

$$\frac{3\mu_0 m^2}{2\pi x^4} - \frac{3\mu_0 m^2}{2\pi y^4} - m_d g = 0.$$

The top magnet is repelled *upward* by middle magnet, and attracted *downward* by lower magnet:

$$\frac{3\mu_0 m^2}{2\pi y^4} - \frac{3\mu_0 m^2}{2\pi (x+y)^4} - m_d g = 0.$$

Subtracting: $\frac{3\mu_0 m^2}{2\pi} \left[\frac{1}{x^4} - \frac{1}{y^4} - \frac{1}{y^4} + \frac{1}{(x+y)^4} \right] - m_d g + m_d g = 0$, or $\frac{1}{x^4} - \frac{2}{y^4} + \frac{1}{(x+y)^4} = 0$, so: $2 = \frac{1}{(x/y)^4} + \frac{1}{(x/y+1)^4}$.

Let $\alpha \equiv x/y$; then $2 = \frac{1}{\alpha^4} + \frac{1}{(\alpha+1)^4}$. Mathematica gives the numerical solution $\alpha = \boxed{x/y = 0.850115 \dots}$

Problem 6.24

(a) Forces on the upper charge:

$$\mathbf{F}_q = \frac{1}{4\pi\epsilon_0} \frac{q^2}{z^2} \hat{\mathbf{z}}, \quad \mathbf{F}_m = \nabla(\mathbf{m} \cdot \mathbf{B}) = \nabla \left(m \frac{2\mu_0 m}{4\pi z^3} \right) = \frac{\mu_0 m^2}{2\pi} \left(\frac{-3}{z^4} \right) \hat{\mathbf{z}}.$$

At equilibrium,

$$\frac{1}{4\pi\epsilon_0} \frac{q^2}{z^2} = \frac{3\mu_0 m^2}{2\pi z^4} \Rightarrow z^2 = \frac{6\mu_0 \epsilon_0 m^2}{q^2} \Rightarrow z = \boxed{\sqrt{6} \frac{m}{qc}},$$

where $1/\sqrt{\epsilon_0 \mu_0} = c$, the speed of light.

(b) For electrons, $q = 1.6 \times 10^{-19} \text{ C}$ (actually, it's the *magnitude* of the charge we want in the expression above), and $m = 9.22 \times 10^{-24} \text{ A m}^2$ (the Bohr magneton—see Problem 5.58), so

$$z = \sqrt{6} \frac{9.22 \times 10^{-24}}{(1.6 \times 10^{-19})(3 \times 10^8)} = \boxed{4.72 \times 10^{-13} \text{ m}}.$$

(For comparison, the Bohr radius is 0.5×10^{-10} m, so the equilibrium separation is about 1% of the size of a hydrogen atom.)

(c) Good question! Certainly the answer is no. Presumably this is an unstable equilibrium, so unless you could find a way to maintain the orientation of the dipoles, and keep them on the z axis, the structure would fall apart.

Problem 6.25

(a) The *electric* field inside a uniformly *polarized* sphere, $\mathbf{E} = -\frac{1}{3\epsilon_0}\mathbf{P}$ (Eq. 4.14) translates to $\mathbf{H} = -\frac{1}{3\mu_0}(\mu_0\mathbf{M}) = -\frac{1}{3}\mathbf{M}$. But $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$. So the *magnetic* field inside a uniformly *magnetized* sphere is $\mathbf{B} = \mu_0(-\frac{1}{3}\mathbf{M} + \mathbf{M}) = \frac{2}{3}\mu_0\mathbf{M}$ (same as Eq. 6.16).

(b) The *electric* field inside a sphere of linear *dielectric* in an otherwise uniform *electric* field is $\mathbf{E} = \frac{1}{1+\chi_e/3}\mathbf{E}_0$ (Eq. 4.49). Now χ_e translates to χ_m , for then Eq. 4.30 ($\mathbf{P} = \epsilon_0\chi_e\mathbf{E}$) goes to $\mu_0\mathbf{M} = \mu_0\chi_m\mathbf{H}$, or $\mathbf{M} = \chi_m\mathbf{H}$ (Eq. 6.29). So Eq. 4.49 $\Rightarrow \mathbf{H} = \frac{1}{1+\chi_m/3}\mathbf{H}_0$. But $\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H}$, and $\mathbf{B}_0 = \mu_0\mathbf{H}_0$ (Eqs. 6.31 and 6.32), so the *magnetic* field inside a sphere of linear *magnetic* material in an otherwise uniform *magnetic* field is $\frac{\mathbf{B}}{\mu_0(1 + \chi_m)} = \frac{1}{(1 + \chi_m/3)}\frac{\mathbf{B}_0}{\mu_0}$, or $\mathbf{B} = \left(\frac{1 + \chi_m}{1 + \chi_m/3}\right)\mathbf{B}_0$ (as in Prob. 6.18).

(c) The average *electric* field over a sphere, due to charges within, is $\mathbf{E}_{\text{ave}} = -\frac{1}{4\pi\epsilon_0}\frac{\mathbf{P}}{R^3}$. Let's pretend the charges are all due to the frozen-in polarization of some medium (whatever ρ might be, we can solve $\nabla \cdot \mathbf{P} = -\rho$ to find the appropriate $\mathbf{P}$). In this case there are *no* free charges, and $\mathbf{p} = \int \mathbf{P} d\tau$, so $\mathbf{E}_{\text{ave}} = -\frac{1}{4\pi\epsilon_0}\frac{1}{R^3} \int \mathbf{P} d\tau$, which translates to

$$\mathbf{H}_{\text{ave}} = -\frac{1}{4\pi\mu_0}\frac{1}{R^3} \int \mu_0\mathbf{M} d\tau = -\frac{1}{4\pi R^3}\mathbf{m}.$$

But $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$, so $\mathbf{B}_{\text{ave}} = -\frac{\mu_0}{4\pi}\frac{\mathbf{m}}{R^3} + \mu_0\mathbf{M}_{\text{ave}}$, and $\mathbf{M}_{\text{ave}} = \frac{\mathbf{m}}{\frac{4}{3}\pi R^3}$, so $\mathbf{B}_{\text{ave}} = \frac{\mu_0}{4\pi}\frac{2\mathbf{m}}{R^3}$, in agreement with Eq. 5.93. (We must assume for this argument that all the currents are *bound*, but again it doesn't really matter, since we can model any current configuration by an appropriate frozen-in magnetization. See G. H. Goedecke, *Am. J. Phys.* **66**, 1010 (1998).)

Problem 6.26

$$\text{Eq. 2.15 : } \mathbf{E} = \rho \left\{ \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{\mathbf{r}}}{r^2} d\tau' \right\} \quad (\text{for uniform charge density});$$

$$\text{Eq. 4.9 : } V = \mathbf{P} \cdot \left\{ \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{\mathbf{r}}}{r^2} d\tau' \right\} \quad (\text{for uniform polarization});$$

$$\text{Eq. 6.11 : } \mathbf{A} = \mu_0\epsilon_0\mathbf{M} \times \left\{ \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{\mathbf{r}}}{r^2} d\tau' \right\} \quad (\text{for uniform magnetization}).$$

$$\text{For a uniformly charged sphere (radius } R): \begin{cases} \mathbf{E}_{\text{in}} = \rho \left(\frac{1}{3\epsilon_0} \mathbf{r} \right) & (\text{Prob. 2.12}), \\ \mathbf{E}_{\text{out}} = \rho \left(\frac{1}{3\epsilon_0} \frac{R^3}{r^2} \hat{\mathbf{r}} \right) & (\text{Ex. 2.3}). \end{cases}$$

$$\text{So the scalar potential of a uniformly polarized sphere is: } \begin{cases} V_{\text{in}} = \frac{1}{3\epsilon_0}(\mathbf{P} \cdot \mathbf{r}), \\ V_{\text{out}} = \frac{1}{3\epsilon_0} \frac{R^3}{r^2}(\mathbf{P} \cdot \hat{\mathbf{r}}), \end{cases}$$

$$\text{and the vector potential of a uniformly magnetized sphere is: } \begin{cases} \mathbf{A}_{\text{in}} = \frac{\mu_0}{3}(\mathbf{M} \times \mathbf{r}), \\ \mathbf{A}_{\text{out}} = \frac{\mu_0}{3} \frac{R^3}{r^2}(\mathbf{M} \times \hat{\mathbf{r}}), \end{cases}$$

(confirming the results of Ex. 4.2 and of Exs. 6.1 and 5.11).

Problem 6.27

At the interface, the perpendicular component of $\mathbf{B}$ is continuous (Eq. 6.26), and the parallel component of $\mathbf{H}$ is continuous (Eq. 6.25 with $\mathbf{K}_f = 0$). So $B_1^\perp = B_2^\perp$, $\mathbf{H}_1^\parallel = \mathbf{H}_2^\parallel$. But $\mathbf{B} = \mu\mathbf{H}$ (Eq. 6.31), so $\frac{1}{\mu_1}\mathbf{B}_1^\parallel = \frac{1}{\mu_2}\mathbf{B}_2^\parallel$. Now $\tan\theta_1 = B_1^\parallel/B_1^\perp$, and $\tan\theta_2 = B_2^\parallel/B_2^\perp$, so

$$\frac{\tan\theta_2}{\tan\theta_1} = \frac{B_2^\parallel}{B_2^\perp} \frac{B_1^\perp}{B_1^\parallel} = \frac{B_2^\parallel}{B_1^\parallel} = \frac{\mu_2}{\mu_1}$$

(the same form, though for different reasons, as Eq. 4.68).

Problem 6.28

In view of Eq. 6.33, there is a *bound* dipole at the center: $\mathbf{m}_b = \chi_m \mathbf{m}$. So the *net* dipole moment at the center is $\mathbf{m}_{\text{center}} = \mathbf{m} + \mathbf{m}_b = (1 + \chi_m)\mathbf{m} = \frac{\mu}{\mu_0}\mathbf{m}$. This produces a field given by Eq. 5.89:

$$\mathbf{B}_{\text{center dipole}} = \frac{\mu}{4\pi} \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}].$$

This accounts for the *first* term in the field. The remainder must be due to the bound surface current ($\mathbf{K}_b$) at $r = R$ (since there can be no volume bound current, according to Eq. 6.33). Let us make an educated guess (based either on the answer provided or on the analogous electrical Prob. 4.37) that the field due to the surface bound current is (for interior points) of the form $\mathbf{B}_{\text{surface current}} = A\mathbf{m}$ (i.e. a constant, proportional to $\mathbf{m}$). In that case the magnetization will be:

$$\mathbf{M} = \chi_m \mathbf{H} = \frac{\chi_m}{\mu} \mathbf{B} = \frac{\chi_m}{4\pi} \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}] + \frac{\chi_m}{\mu} A\mathbf{m}.$$

This will produce bound currents $\mathbf{J}_b = \nabla \times \mathbf{M} = \mathbf{0}$, as it should, for $0 < r < R$ (no need to *calculate* this curl—the second term is *constant*, and the first is essentially the field of a dipole, which we know is curl-less, except at $r = 0$), and

$$\mathbf{K}_b = \mathbf{M}(R) \times \hat{\mathbf{r}} = \frac{\chi_m}{4\pi R^3} (-\mathbf{m} \times \hat{\mathbf{r}}) + \frac{\chi_m A}{\mu} (\mathbf{m} \times \hat{\mathbf{r}}) = \chi_m m \left(-\frac{1}{4\pi R^3} + \frac{A}{\mu} \right) \sin\theta \hat{\boldsymbol{\phi}}.$$

But this is exactly the surface current produced by a spinning sphere: $\mathbf{K} = \sigma \mathbf{v} = \sigma \omega R \sin\theta \hat{\boldsymbol{\phi}}$, with $(\sigma \omega R) \leftrightarrow \chi_m m \left(\frac{A}{\mu} - \frac{1}{4\pi R^3} \right)$. So the field it produces (for points inside) is (Eq. 5.70):

$$\mathbf{B}_{\text{surface current}} = \frac{2}{3} \mu_0 (\sigma \omega R) = \frac{2}{3} \mu_0 \chi_m m \left(\frac{A}{\mu} - \frac{1}{4\pi R^3} \right).$$

Everything is consistent, therefore, provided $A = \frac{2}{3} \mu_0 \chi_m \left(\frac{A}{\mu} - \frac{1}{4\pi R^3} \right)$, or $A \left(1 - \frac{2\mu_0}{3\mu} \chi_m \right) = -\frac{2}{3} \frac{\mu_0 \chi_m}{4\pi R^3}$. But $\chi_m = \left(\frac{\mu}{\mu_0} \right) - 1$, so $A \left(1 - \frac{2}{3} + \frac{2}{3} \frac{\mu_0}{\mu} \right) = -\frac{2}{3} \frac{(\mu - \mu_0)}{4\pi R^3}$, or $A \left(1 + \frac{2\mu_0}{\mu} \right) = 2 \frac{(\mu_0 - \mu)}{4\pi R^3}$; $A = \frac{\mu}{4\pi} \frac{2(\mu_0 - \mu)}{R^3(2\mu_0 + \mu)}$, and hence

$$\mathbf{B} = \frac{\mu}{4\pi} \left\{ \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}] + \frac{2(\mu_0 - \mu)\mathbf{m}}{R^3(2\mu_0 + \mu)} \right\}. \quad \text{qed}$$

The *exterior* field is that of the central dipole plus that of the surface current, which, according to Prob. 5.37, is *also* a perfect dipole field, of dipole moment

$$\mathbf{m}_{\text{surface current}} = \frac{4}{3} \pi R^3 (\sigma \omega R) = \frac{4}{3} \pi R^3 \left(\frac{3}{2\mu_0} \mathbf{B}_{\text{surface current}} \right) = \frac{2\pi R^3}{\mu_0} \frac{\mu}{4\pi} \frac{2(\mu_0 - \mu)\mathbf{m}}{R^3(2\mu_0 + \mu)} = \frac{\mu(\mu_0 - \mu)\mathbf{m}}{\mu_0(2\mu_0 + \mu)}.$$

So the *total* dipole moment is:

$$\mathbf{m}_{\text{tot}} = \frac{\mu}{\mu_0} \mathbf{m} + \frac{\mu}{\mu_0} \mathbf{m} \frac{(\mu_0 - \mu)}{(2\mu_0 + \mu)} = \frac{3\mu \mathbf{m}}{(2\mu_0 + \mu)},$$

and hence the field (for $r > R$) is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \left(\frac{3\mu}{2\mu_0 + \mu} \right) \frac{1}{r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{m}].$$

Problem 6.29

The problem is that the field inside a *cavity* is not the same as the field in the material itself.

(a) *Ampère type*. The field deep inside the magnet is that of a long solenoid, $\mathbf{B}_0 \approx \mu_0 \mathbf{M}$. From Prob. 6.13:

$$\begin{cases} \text{Sphere : } \mathbf{B} = \mathbf{B}_0 - \frac{2}{3}\mu_0 \mathbf{M} = \frac{1}{3}\mu_0 \mathbf{M}; \\ \text{Needle : } \mathbf{B} = \mathbf{B}_0 - \mu_0 \mathbf{M} = 0; \\ \text{Wafer : } \mathbf{B} = \mu_0 \mathbf{M}. \end{cases}$$

(b) *Gilbert type*. This is analogous to the *electric* case. The field at the center is approximately that midway between two distant point charges, $\mathbf{B}_0 \approx 0$. From Prob. 4.16 (with $\mathbf{E} \rightarrow \mathbf{B}$, $1/\epsilon_0 \rightarrow \mu_0$, $\mathbf{P} \rightarrow \mathbf{M}$):

$$\begin{cases} \text{Sphere : } \mathbf{B} = \mathbf{B}_0 + \frac{\mu_0}{3} \mathbf{M} = \frac{1}{3}\mu_0 \mathbf{M}; \\ \text{Needle : } \mathbf{B} = \mathbf{B}_0 = 0; \\ \text{Wafer : } \mathbf{B} = \mathbf{B}_0 + \mu_0 \mathbf{M} = \mu_0 \mathbf{M}. \end{cases}$$

In the cavities, then, the fields are the *same* for the two models, and this will be no test at all. Yes. Fund it with \$1 M from the Office of Alternative Medicine.

Chapter 7

Electrodynamics

Problem 7.1

(a) Let Q be the charge on the inner shell. Then $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$ in the space between them, and $(V_a - V_b) = -\int_b^a \mathbf{E} \cdot d\mathbf{r} = -\frac{1}{4\pi\epsilon_0} Q \int_b^a \frac{1}{r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$.

$$I = \int \mathbf{J} \cdot d\mathbf{a} = \sigma \int \mathbf{E} \cdot d\mathbf{a} = \sigma \frac{Q}{\epsilon_0} = \frac{\sigma}{\epsilon_0} \frac{4\pi\epsilon_0(V_a - V_b)}{(1/a - 1/b)} = \boxed{4\pi\sigma \frac{(V_a - V_b)}{(1/a - 1/b)}}.$$

(b) $R = \frac{V_a - V_b}{I} = \boxed{\frac{1}{4\pi\sigma} \left(\frac{1}{a} - \frac{1}{b} \right)}.$

(c) For large b ($b \gg a$), the second term is negligible, and $R = 1/4\pi\sigma a$. Essentially all of the resistance is in the region right around the inner sphere. Successive shells, as you go out, contribute less and less, because the cross-sectional area ($4\pi r^2$) gets larger and larger. For the two submerged spheres, $R = \frac{2}{4\pi\sigma a} = \frac{1}{2\pi\sigma a}$ (one R as the current leaves the first, one R as it converges on the second). Therefore $I = V/R = \boxed{2\pi\sigma a V}.$

Problem 7.2

(a) $V = Q/C = IR$. Because positive I means the charge on the capacitor is *decreasing*, $\frac{dQ}{dt} = -I = -\frac{1}{RC}Q$, so $Q(t) = Q_0 e^{-t/RC}$. But $Q_0 = Q(0) = CV_0$, so $\boxed{Q(t) = CV_0 e^{-t/RC}}.$

Hence $I(t) = -\frac{dQ}{dt} = CV_0 \frac{1}{RC} e^{-t/RC} = \boxed{\frac{V_0}{R} e^{-t/RC}}.$

(b) $W = \boxed{\frac{1}{2} CV_0^2}.$ The energy delivered to the resistor is $\int_0^\infty P dt = \int_0^\infty I^2 R dt = \frac{V_0^2}{R} \int_0^\infty e^{-2t/RC} dt = \frac{V_0^2}{R} \left(-\frac{RC}{2} e^{-2t/RC} \right) \Big|_0^\infty = \frac{1}{2} CV_0^2. \quad \checkmark$

(c) $V_0 = Q/C + IR$. This time positive I means Q is *increasing*: $\frac{dQ}{dt} = I = \frac{1}{RC}(CV_0 - Q) \Rightarrow \frac{dQ}{Q - CV_0} = -\frac{1}{RC} dt \Rightarrow \ln(Q - CV_0) = -\frac{1}{RC}t + \text{constant} \Rightarrow Q(t) = CV_0 + ke^{-t/RC}$. But $Q(0) = 0 \Rightarrow k = -CV_0$, so $\boxed{Q(t) = CV_0 (1 - e^{-t/RC})}.$ $I(t) = \frac{dQ}{dt} = CV_0 \left(\frac{1}{RC} e^{-t/RC} \right) = \boxed{\frac{V_0}{R} e^{-t/RC}}.$

$$(d) \text{ Energy from battery: } \int_0^\infty V_0 I dt = \frac{V_0^2}{R} \int_0^\infty e^{-t/RC} dt = \frac{V_0^2}{R} \left(-RC e^{-t/RC} \right) \Big|_0^\infty = \frac{V_0^2}{R} RC = \boxed{CV_0^2}.$$

Since $I(t)$ is the same as in (a), the energy delivered to the resistor is again $\boxed{\frac{1}{2}CV_0^2}$. The final energy in the capacitor is also $\boxed{\frac{1}{2}CV_0^2}$, so $\boxed{\text{half}}$ the energy from the battery goes to the capacitor, and the other half to the resistor.

Problem 7.3

(a) $I = \oint \mathbf{J} \cdot d\mathbf{a}$, where the integral is taken over a surface enclosing the positively charged conductor. But $\mathbf{J} = \sigma \mathbf{E}$, and Gauss's law says $\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q$, so $I = \sigma \oint \mathbf{E} \cdot d\mathbf{a} = \frac{\sigma}{\epsilon_0} Q$. But $Q = CV$, and $V = IR$, so $I = \frac{\sigma}{\epsilon_0} CIR$, or $\boxed{R = \frac{\epsilon_0}{\sigma C}}$. qed

(b) $Q = CV = CIR \Rightarrow \frac{dQ}{dt} = -I = -\frac{1}{RC} Q \Rightarrow \boxed{Q(t) = Q_0 e^{-t/RC}}$, or, since $V = Q/C$, $V(t) = V_0 e^{-t/RC}$. The time constant is $\tau = RC = \boxed{\epsilon_0/\sigma}$.

Problem 7.4

$I = J(s) 2\pi s L \Rightarrow J(s) = I/2\pi s L$. $E = J/\sigma = I/2\pi s \sigma L = I/2\pi k L$.
 $V = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = -\frac{I}{2\pi k L} (a - b)$. So $\boxed{R = \frac{b - a}{2\pi k L}}$.

Problem 7.5

$$I = \frac{\mathcal{E}}{r + R}; \quad P = I^2 R = \frac{\mathcal{E}^2 R}{(r + R)^2}; \quad \frac{dP}{dR} = \mathcal{E}^2 \left[\frac{1}{(r + R)^2} - \frac{2R}{(r + R)^3} \right] = 0 \Rightarrow r + R = 2R \Rightarrow \boxed{R = r}.$$

Problem 7.6

$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = \boxed{\text{zero}}$ for *all* electrostatic fields. It *looks* as though $\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = (\sigma/\epsilon_0)h$, as would indeed be the case if the field were really just σ/ϵ_0 inside and zero outside. But in fact there is always a “fringing field” at the edges (Fig. 4.31), and this is evidently just right to kill off the contribution from the left end of the loop. The current is $\boxed{\text{zero}}$.

Problem 7.7

(a) $\mathcal{E} = -\frac{d\Phi}{dt} = -Bl \frac{dx}{dt} = -Blv$; $\mathcal{E} = IR \Rightarrow \boxed{I = \frac{Blv}{R}}$. (Never mind the minus sign—it just tells you the *direction* of flow: $(\mathbf{v} \times \mathbf{B})$ is *upward*, in the bar, so *downward* through the resistor.)

(b) $F = IlB = \boxed{\frac{B^2 l^2 v}{R}}$, to the $\boxed{\text{left}}$.

(c) $F = ma = m \frac{dv}{dt} = -\frac{B^2 l^2}{R} v \Rightarrow \frac{dv}{dt} = -\left(\frac{B^2 l^2}{Rm}\right) v \Rightarrow \boxed{v = v_0 e^{-\frac{B^2 l^2}{mR} t}}$.

(d) The energy goes into heat in the resistor. The power delivered to resistor is $I^2 R$, so

$$\frac{dW}{dt} = I^2 R = \frac{B^2 l^2 v^2}{R^2} R = \frac{B^2 l^2}{R} v_0^2 e^{-2\alpha t}, \text{ where } \alpha \equiv \frac{B^2 l^2}{mR}; \quad \frac{dW}{dt} = \alpha m v_0^2 e^{-2\alpha t}.$$

The total energy delivered to the resistor is $W = \alpha m v_0^2 \int_0^\infty e^{-2\alpha t} dt = \alpha m v_0^2 \left. \frac{e^{-2\alpha t}}{-2\alpha} \right|_0^\infty = \alpha m v_0^2 \frac{1}{2\alpha} = \frac{1}{2} m v_0^2$. ✓

Problem 7.8

(a) The field of long wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$, so $\Phi = \int \mathbf{B} \cdot d\mathbf{a} = \frac{\mu_0 I}{2\pi} \int_s^{s+a} \frac{1}{s} (a ds) = \boxed{\frac{\mu_0 I a}{2\pi} \ln \left(\frac{s+a}{s} \right)}$.

(b) $\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{\mu_0 I a}{2\pi} \frac{d}{dt} \ln \left(\frac{s+a}{s} \right)$, and $\frac{ds}{dt} = v$, so $-\frac{\mu_0 I a}{2\pi} \left(\frac{1}{s+a} \frac{ds}{dt} - \frac{1}{s} \frac{ds}{dt} \right) = \boxed{\frac{\mu_0 I a^2 v}{2\pi s(s+a)}}$.

The field points *out* of the page, so the force on a charge in the nearby side of the square is *to the right*. In the far side it's also to the right, but here the field is weaker, so the current flows counterclockwise.

(c) This time the flux is *constant*, so $\mathcal{E} = 0$.

Problem 7.9

Since $\nabla \cdot \mathbf{B} = 0$, Theorem 2(c) (Sect. 1.6.2) guarantees that $\int \mathbf{B} \cdot d\mathbf{a}$ is the same for *all* surfaces with a given boundary line.

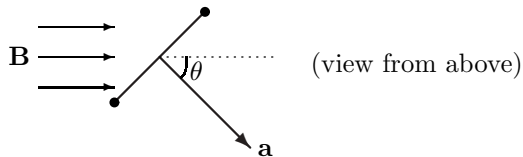
Problem 7.10

$$\Phi = \mathbf{B} \cdot \mathbf{a} = Ba^2 \cos \theta$$

Here $\theta = \omega t$, so

$$\mathcal{E} = -\frac{d\Phi}{dt} = -Ba^2(-\sin \omega t)\omega;$$

$$\boxed{\mathcal{E} = B\omega a^2 \sin \omega t}.$$



Problem 7.11

$\mathcal{E} = Blv = IR \Rightarrow I = \frac{Bl}{R}v \Rightarrow$ upward magnetic force $= IlB = \frac{B^2 l^2}{R}v$. This opposes the gravitational force downward:

$$mg - \frac{B^2 l^2}{R}v = m \frac{dv}{dt}; \quad \frac{dv}{dt} = g - \alpha v, \quad \text{where } \alpha \equiv \frac{B^2 l^2}{mR}. \quad g - \alpha v_t = 0 \Rightarrow v_t = \frac{g}{\alpha} = \boxed{\frac{mgR}{B^2 l^2}}.$$

$$\frac{dv}{g - \alpha v} = dt \Rightarrow -\frac{1}{\alpha} \ln(g - \alpha v) = t + \text{const.} \Rightarrow g - \alpha v = Ae^{-\alpha t}; \quad \text{at } t = 0, v = 0, \text{ so } A = g.$$

$$\alpha v = g(1 - e^{-\alpha t}); \quad v = \frac{g}{\alpha}(1 - e^{-\alpha t}) = \boxed{v_t(1 - e^{-\alpha t})}.$$

At 90% of terminal velocity, $v/v_t = 0.9 = 1 - e^{-\alpha t} \Rightarrow e^{-\alpha t} = 1 - 0.9 = 0.1$; $\ln(0.1) = -\alpha t$; $\ln 10 = \alpha t$;

$$t = \frac{1}{\alpha} \ln 10, \text{ or } \boxed{t_{90\%} = \frac{v_t}{g} \ln 10}.$$

Now the numbers: $m = 4\eta Al$, where η is the mass density of aluminum, A is the cross-sectional area, and l is the length of a side. $R = 4l/A\sigma$, where σ is the conductivity of aluminum. So

$$v_t = \frac{4\eta A l g 4l}{A \sigma B^2 l^2} = \frac{16\eta g}{\sigma B^2} = \frac{16g\eta\rho}{B^2}, \quad \text{and} \quad \left\{ \begin{array}{l} \rho = 2.8 \times 10^{-8} \Omega \text{ m} \\ g = 9.8 \text{ m/s}^2 \\ \eta = 2.7 \times 10^3 \text{ kg/m}^3 \\ B = 1 \text{ T} \end{array} \right\}.$$

$$\text{So } v_t = \frac{(16)(9.8)(2.7 \times 10^3)(2.8 \times 10^{-8})}{1} = \boxed{1.2 \text{ cm/s}}; \quad t_{90\%} = \frac{1.2 \times 10^{-2}}{9.8} \ln(10) = \boxed{2.8 \text{ ms}}.$$

If the loop were cut, it would fall freely, with acceleration g .

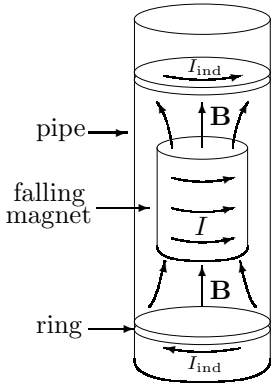
Problem 7.12

$$\Phi = \pi \left(\frac{a}{2} \right)^2 B = \frac{\pi a^2}{4} B_0 \cos(\omega t); \quad \mathcal{E} = -\frac{d\Phi}{dt} = \frac{\pi a^2}{4} B_0 \omega \sin(\omega t). \quad I(t) = \frac{\mathcal{E}}{R} = \boxed{\frac{\pi a^2 \omega}{4R} B_0 \sin(\omega t)}.$$

Problem 7.13

$$\Phi = \int B dx dy = kt^2 \int_0^a dx \int_0^a y^3 dy = \frac{1}{4} kt^2 a^5. \quad \mathcal{E} = -\frac{d\Phi}{dt} = -\boxed{\frac{1}{2} kta^5}.$$

Problem 7.14



Suppose the current (I) in the magnet flows counterclockwise (viewed from above), as shown, so its field, near the ends, points *upward*. A ring of pipe *below* the magnet experiences an increasing upward flux, as the magnet approaches, and hence (by Lenz's law) a current (I_{ind}) will be induced in it such as to produce a *downward* flux. Thus I_{ind} must flow *clockwise*, which is *opposite* to the current in the magnet. Since opposite currents repel, the force on the magnet is *upward*. Meanwhile, a ring *above* the magnet experiences a *decreasing* (upward) flux, so *its* induced current is *parallel* to I , and it *attracts* the magnet upward. And the flux through rings *next* to the magnet is constant, so *no* current is induced in them. *Conclusion:* the delay is due to forces exerted on the magnet by induced eddy currents in the pipe.

Problem 7.15

In the quasistatic approximation, $\mathbf{B} = \begin{cases} \mu_0 n I \hat{\mathbf{z}}, & (s < a); \\ \mathbf{0}, & (s > a). \end{cases}$

Inside: for an “amperian loop” of radius $s < a$,

$$\Phi = B\pi s^2 = \mu_0 n I \pi s^2; \quad \oint \mathbf{E} \cdot d\mathbf{l} = E 2\pi s = -\frac{d\Phi}{dt} = -\mu_0 n \pi s^2 \frac{dI}{dt}; \quad \boxed{\mathbf{E} = -\frac{\mu_0 n s}{2} \frac{dI}{dt} \hat{\phi}}.$$

Outside: for an “amperian loop” of radius $s > a$:

$$\Phi = B\pi a^2 = \mu_0 n I \pi a^2; \quad E 2\pi s = -\mu_0 n \pi a^2 \frac{dI}{dt}; \quad \boxed{\mathbf{E} = -\frac{\mu_0 n a^2}{2s} \frac{dI}{dt} \hat{\phi}}.$$

Problem 7.16

(a) The magnetic field (in the quasistatic approximation) is “circumferential”. This is analogous to the current in a solenoid, and hence the field is longitudinal.

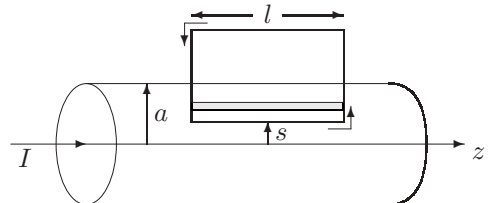
(b) Use the “amperian loop” shown.

Outside, $\mathbf{B} = \mathbf{0}$, so here $\mathbf{E} = \mathbf{0}$ (like $\mathbf{B}$ outside a solenoid).

So $\oint \mathbf{E} \cdot d\mathbf{l} = El = -\frac{d\Phi}{dt} = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a} = -\frac{d}{dt} \int_s^a \frac{\mu_0 I}{2\pi s'} l ds'$

$\therefore E = -\frac{\mu_0}{2\pi} \frac{dI}{dt} \ln\left(\frac{a}{s}\right)$. But $\frac{dI}{dt} = -I_0 \omega \sin \omega t$,

so $\mathbf{E} = \frac{\mu_0 I_0 \omega}{2\pi} \sin(\omega t) \ln\left(\frac{a}{s}\right) \hat{\mathbf{z}}$



Problem 7.17

(a) The field inside the solenoid is $B = \mu_0 n I$. So $\Phi = \pi a^2 \mu_0 n I \Rightarrow \mathcal{E} = -\pi a^2 \mu_0 n (dI/dt)$.

In magnitude, then, $\mathcal{E} = \pi a^2 \mu_0 n k$. Now $\mathcal{E} = I_r R$, so $I_{\text{resistor}} = \frac{\pi a^2 \mu_0 n k}{R}$.

$\mathbf{B}$ is to the right and increasing, so the field of the loop is to the *left*, so the current is counterclockwise, or to the right, through the resistor.

(b) $\Delta\Phi = 2\pi a^2 \mu_0 n I$; $I = \frac{dQ}{dt} = \frac{\mathcal{E}}{R} = -\frac{1}{R} \frac{d\Phi}{dt} \Rightarrow \Delta Q = \frac{1}{R} \Delta\Phi$, in magnitude. So $\Delta Q = \frac{2\pi a^2 \mu_0 n I}{R}$.

Problem 7.18

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a}; \quad \mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}; \quad \Phi = \frac{\mu_0 I a}{2\pi} \int_s^{s+a} \frac{ds'}{s'} = \frac{\mu_0 I a}{2\pi} \ln \frac{s+a}{s};$$

$$\mathcal{E} = I_{\text{loop}} R = \frac{dQ}{dt} R = -\frac{d\Phi}{dt} = -\frac{\mu_0 a}{2\pi} \ln(1+a/s) \frac{dI}{dt}.$$

$$dQ = -\frac{\mu_0 a}{2\pi R} \ln(1+a/s) dI \Rightarrow Q = \frac{I \mu_0 a}{2\pi R} \ln(1+a/s).$$

The field of the wire, at the square loop, is *out of the page*, and *decreasing*, so the field of the induced current must point out of page, within the loop, and hence the induced current flows counterclockwise.

Problem 7.19

In the quasistatic approximation, $\mathbf{B} = \begin{cases} \frac{\mu_0 N I}{2\pi s} \hat{\phi}, & \text{(inside toroid);} \\ 0, & \text{(outside toroid)} \end{cases}$

(Eq. 5.60). The flux around the toroid is therefore

$$\Phi = \frac{\mu_0 N I}{2\pi} \int_a^{a+w} \frac{1}{s} h ds = \frac{\mu_0 N I h}{2\pi} \ln\left(1 + \frac{w}{a}\right) \approx \frac{\mu_0 N h w}{2\pi a} I. \quad \frac{d\Phi}{dt} = \frac{\mu_0 N h w}{2\pi a} \frac{dI}{dt} = \frac{\mu_0 N h w k}{2\pi a}.$$

The electric field is the same as the *magnetic* field of a circular current (Eq. 5.41):

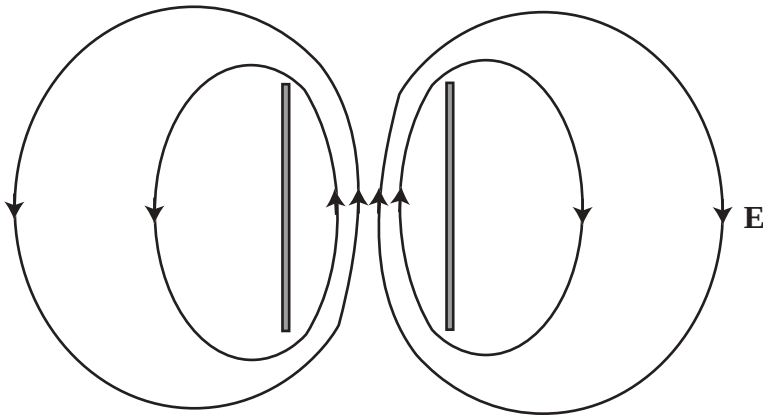
$$\mathbf{B} = \frac{\mu_0 I}{2} \frac{a^2}{(a^2 + z^2)^{3/2}} \hat{\mathbf{z}},$$

with (Eq. 7.19)

$$I \rightarrow -\frac{1}{\mu_0} \frac{d\Phi}{dt} = -\frac{N h w k}{2\pi a}. \quad \text{So } \mathbf{E} = \frac{\mu_0}{2} \left(-\frac{N h w k}{2\pi a} \right) \frac{a^2}{(a^2 + z^2)^{3/2}} \hat{\mathbf{z}} = -\frac{\mu_0}{4\pi} \frac{N h w k a}{(a^2 + z^2)^{3/2}} \hat{\mathbf{z}}.$$

Problem 7.20

$\partial\mathbf{B}/\partial t$ is nonzero along the left and right edges of the shaded rectangle:



The (inward) flux through the strip on the left is *increasing*; the (inward) flux through the strip on the right is *decreasing*. This is analogous to two current sheets under Ampère's law, with $\mathbf{B} \rightarrow \mathbf{E}$ and $\mu_0 I_{\text{enc}} \rightarrow -d\Phi/dt$ (Eq. 7.19). The one on the left is like a current flowing *out* (taking account of the minus sign), so its field is counterclockwise and the one on the right is like a current flowing *in*, so its field is clockwise.

Problem 7.21

The answer is indeterminate, until some boundary conditions are supplied. We know that $\nabla \times \mathbf{E} = -(dB_0/dt)\hat{\mathbf{z}}$ (and $\nabla \cdot \mathbf{E} = 0$), but this is insufficient information to determine $\mathbf{E}$. Ordinarily we would invoke some symmetry of the configuration, or require that the field go to zero at infinity, to resolve the ambiguity, but neither is available in this case. $\mathbf{E} = \frac{1}{2} \frac{dB_0}{dt} (y\hat{\mathbf{x}} - x\hat{\mathbf{y}})$ would do the job, in which case the force would be zero, but we could add any constant vector to this, and make the force anything we like.

Problem 7.22

(a) From Eq. 5.41, the field (on the axis) is $\mathbf{B} = \frac{\mu_0 I}{2} \frac{b^2}{(b^2 + z^2)^{3/2}} \hat{\mathbf{z}}$, so the flux through the little loop (area πa^2)

is $\Phi = \frac{\mu_0 \pi I a^2 b^2}{2(b^2 + z^2)^{3/2}}$.

(b) The field (Eq. 5.88) is $\mathbf{B} = \frac{\mu_0}{4\pi} \frac{m}{r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}})$, where $m = I\pi a^2$. Integrating over the spherical “cap” (bounded by the big loop and centered at the little loop):

$$\Phi = \int \mathbf{B} \cdot d\mathbf{a} = \frac{\mu_0}{4\pi} \frac{I\pi a^2}{r^3} \int (2 \cos \theta) (r^2 \sin \theta d\theta d\phi) = \frac{\mu_0 I a^2}{2r} 2\pi \int_0^{\bar{\theta}} \cos \theta \sin \theta d\theta$$

where $r = \sqrt{b^2 + z^2}$ and $\sin \bar{\theta} = b/r$. Evidently $\Phi = \frac{\mu_0 I \pi a^2}{r} \frac{\sin^2 \theta}{2} \Big|_0^{\bar{\theta}} = \frac{\mu_0 \pi I a^2 b^2}{2(b^2 + z^2)^{3/2}}$, the same as in (a)!!

(c) Dividing off I ($\Phi_1 = M_{12}I_2$, $\Phi_2 = M_{21}I_1$): $M_{12} = M_{21} = \frac{\mu_0 \pi a^2 b^2}{2(b^2 + z^2)^{3/2}}$.

Problem 7.23

$$\mathcal{E} = -\frac{d\Phi}{dt} = -M \frac{dI}{dt} = -Mk.$$